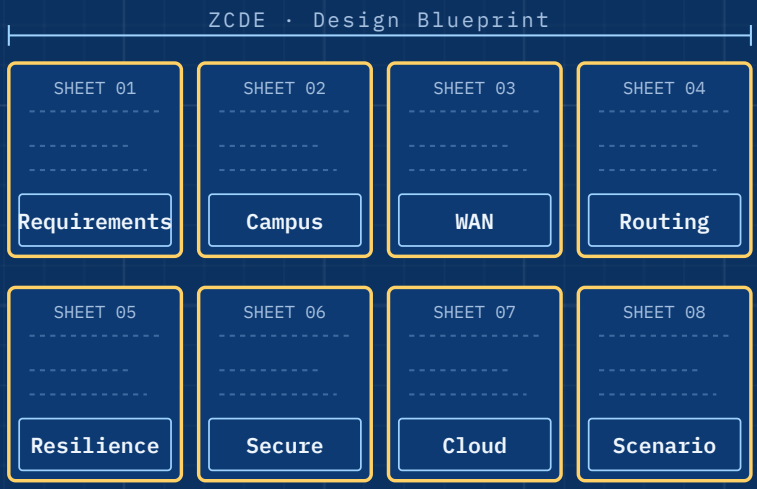
ZCDE

ZyeemZia™ Certified Design Expert

Design for the business, not the box.



COURSE BOOK · WORKBOOK · EXAM PREPARATION

8 MODULES · 34 LESSONS · 60 EXAM QUESTIONS

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BEFORE YOU BEGIN

How to use this book

An expert design certification for architects: turn business requirements into justified network designs across campus, data centre, WAN, routing, resilience, security and hybrid cloud, then defend those decisions in multi-stage scenarios.

01
Learn
Work through 34 lessons. Each ends with a knowledge check.

02
Practise
Fill in the templates, answer the scenarios and case studies, review the glossary.

03
Certify
Take the 45-question official exam at zcde.zyeemzia.com, graded on the ZyeemZia server. Pass mark 85%.

04
Get certified
Download your certificate online. Anyone can verify its credential ID at zcde.zyeemzia.com/#verify. 92%+ and the capstone earn Distinction.

Eight-week study plan

WHEN	MODULE	DO THIS	DONE
Month 1	m1 Requirements	Practise requirement extraction	☐
Month 2	m2 Campus	Campus and DC design studies	☐
Month 3	m3 WAN	WAN and edge design studies	☐
Month 4	m4 Routing	Routing design exercises	☐
Month 5	m5 Resilience	Availability maths and DR	☐
Month 6	m6 Secure	Segmentation design	☐
Month 7	m7 Cloud	Hybrid and automation design	☐
Month 8	m8 Scenario	Multi-stage scenarios, capstone, exam	☐

Learner name .....

Start date ..... Target exam date .....

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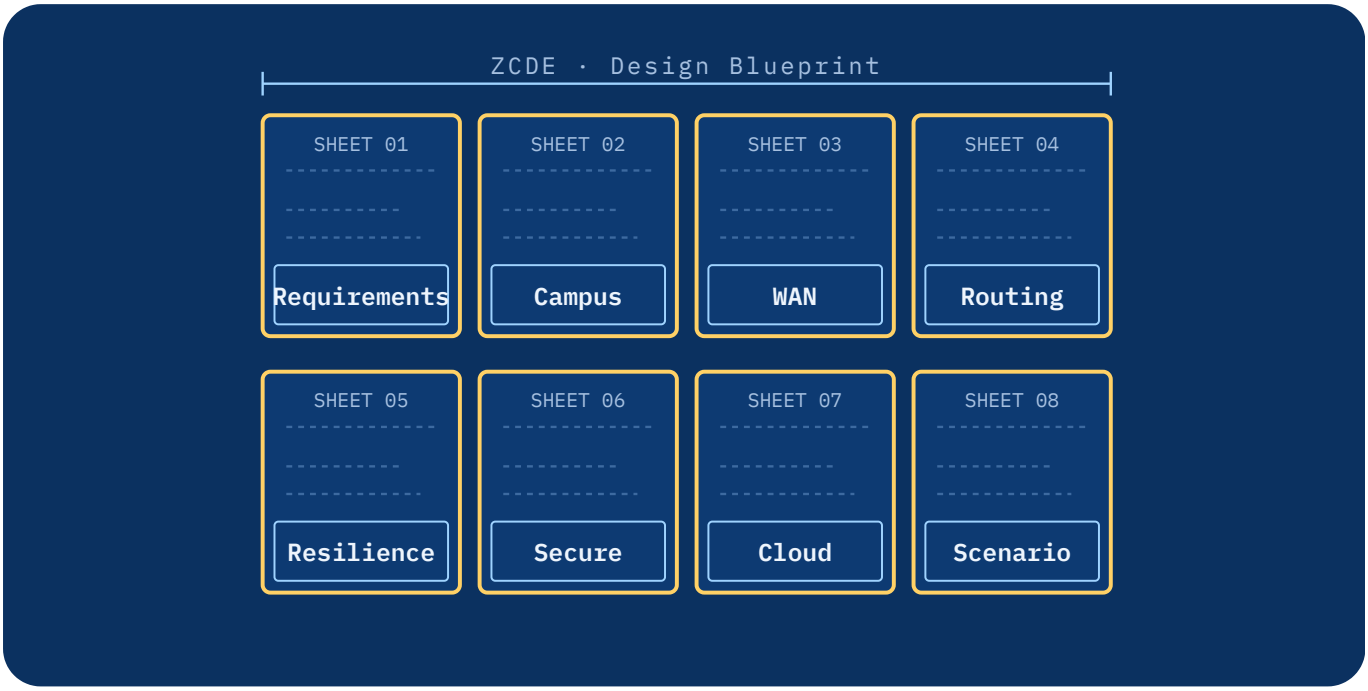
+ Glossary · Capstone guide · Practice exam & answers · Certificate · Notes

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THE FRAMEWORK

The ZCDE Design Blueprint

Eight blueprint sheets every complete network design needs, from requirements to the final scenario. A design is only finished when every sheet is drawn and every decision is justified.



01 Requirements
From business needs to defensible, documented design decisions

02 Campus
Hierarchy, boundaries, fabrics and the arithmetic of scale

03 WAN
Transports, overlays, edges and the routing that ties them to providers

04 Routing
IGP choice, hierarchy, summarisation, redistribution, BGP scale and convergence

05 Resilience
Failure domains, availability maths, convergence, safe change and disaster recovery

06 Secure
Zero trust, segmentation, enforcement placement, identity, encryption, compliance and visibility

07 Cloud
Cloud constructs, hybrid connectivity, addressing and DNS, automation, observability and cost

08 Scenario
Staged scenarios, constraint reading, migration and rollback, executive communication and the capstone

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01

MODULE 1 · REQUIREMENTS

Design Methodology

From business needs to defensible, documented design decisions

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Separate business requirements, technical requirements, constraints and assumptions, and trace each to an owner
- Gather and validate requirements using people, data and documents
- Apply modularity, hierarchy, simplicity and scalability to compare options
- Score options with weighted decision matrices and sensitivity checks
- Write decision records and HLD/LLD documents, and read design scenarios systematically

LESSONS

1. Requirements That Drive Design · 28 min
2. Gathering and Validating Requirements · 28 min
3. Design Principles and Trade-off Analysis · 33 min
4. Design Decisions, Documents and Scenarios · 33 min

MODULE 1 · REQUIREMENTS · LESSON 1 · 28 MIN

Requirements That Drive Design

Separate what the business needs from how the network might deliver it

- Distinguish business requirements, technical requirements, constraints and assumptions
- Map stakeholders to the requirements they own and the risks they carry
- Translate vague business statements into measurable technical requirements
- Record requirements in a register that later decisions can trace back to

Every design decision you will defend in the ZCDE Design Blueprint starts on the **Requirements sheet**. A design is not good or bad in the abstract; it is good or bad *for a given set of requirements and constraints*. Two architects can propose opposite topologies for the same building and both be right, because they were answering different requirements. The expert skill is not knowing the most technology — it is knowing which facts actually decide the design.

The five inputs on the Requirements sheet

BR

Business requirement

What the organisation must achieve, in business language: open 40 branches in 18 months, pass a payment-card audit, cut the cost of connectivity by a fifth. Owned by executives and business units.

TR

Technical requirement

A measurable property the network must have to satisfy a business requirement: 99.95% availability for the trading floor, under 150 ms one-way delay for voice, segmentation between cardholder and corporate systems.

CON

Constraint

A fixed limit you must design within and cannot trade away: budget ceiling, a mandated vendor framework, an existing fibre plant, a regulator's data-residency rule, the power and cooling envelope of a building or data hall, a change freeze, the skills of the operations team.

ASM

Assumption

Something believed true but not yet verified: the carrier can deliver fibre to rural sites, traffic grows 30% a year, the cloud provider supports the required throughput. Every assumption is a risk until validated.

RSK

Risk

An uncertain event that would hurt the outcome if it happened. Risks usually come from unvalidated assumptions, single points of failure, or dependencies on third parties.

Business versus technical requirements

Business requirement

- Stated by sponsors and business owners
- Describes an outcome or a capability
- Often vague: "the network must be reliable"
- Changes when strategy changes
- Judges whether the project succeeded

Technical requirement

- Derived by the architect, agreed with owners
- Describes a measurable property
- Precise: "no single failure isolates more than one floor"
- Changes when the business requirement changes
- Judges whether the design is fit for purpose

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PRO TIP · REQUIREMENT TYPES A 2026 DESIGN MUST NOT MISS

Beyond bandwidth and availability, current briefs routinely hide five decisive inputs: **power and cooling** (a GPU rack or a Wi-Fi 7 floor can exceed what the room or the switch can supply), **sustainability targets** (energy per port, equipment lifetime, reporting duties), **operational-resilience regulation** that demands tested recovery, **IPv6 policy** (some public bodies and providers now expect IPv6-only or IPv6-mostly networks), and **operability** — whether the design can be run from a source of truth with automation and telemetry. Ask about each one explicitly; stakeholders rarely volunteer them.

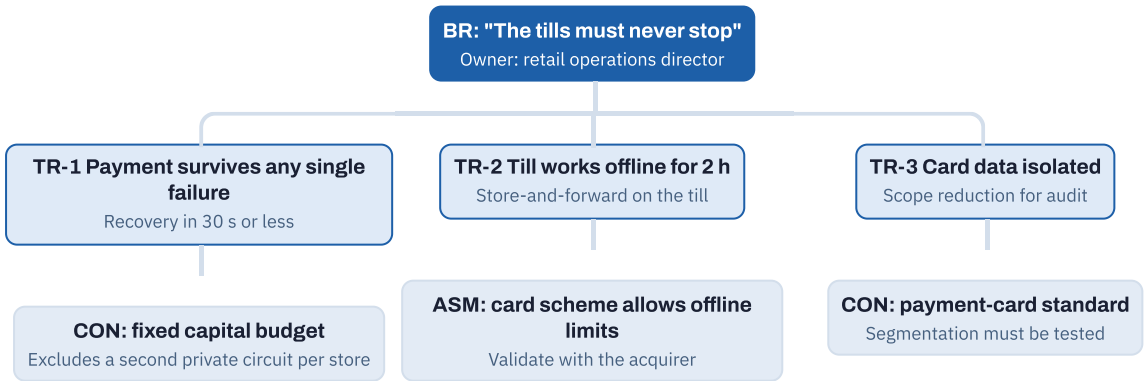
INSIGHT · CONSTRAINTS ARE NOT PREFERENCES

A constraint removes options from the table; a preference only weights them. If the CIO says "we would like to stay with our current switching vendor", that is a preference you score in a decision matrix. If the procurement contract says "all purchases must come from framework X until 2028", that is a constraint and options outside it are not options at all. Misclassifying one as the other is one of the most common causes of a rejected design.

Translating business statements into technical requirements

BUSINESS STATEMENT	CLARIFYING QUESTION	TECHNICAL REQUIREMENT
The tills must never stop	What does a lost minute of trading cost, and how long can a till work offline?	Each store keeps payment service through any single WAN or device failure; recovery within 30 seconds
We are moving to the cloud	Which workloads, which regions, and what share of traffic by when?	Provide private connectivity of 2 x 10 Gbps per region with diverse paths by Q3
Staff should work from anywhere	Which applications, which devices, and which security posture is required?	Identity-based access to internal applications without full-tunnel VPN for SaaS traffic
We must pass the audit	Which standard, which scope, and which controls are network-relevant?	Cardholder systems isolated in their own segment, with logged and filtered access only from named sources

Tracing one business statement to testable requirements



Read the tree top-down: one vague statement becomes several measurable requirements, each tied to the constraint or assumption that shapes it. Every leaf should later trace to a design decision.

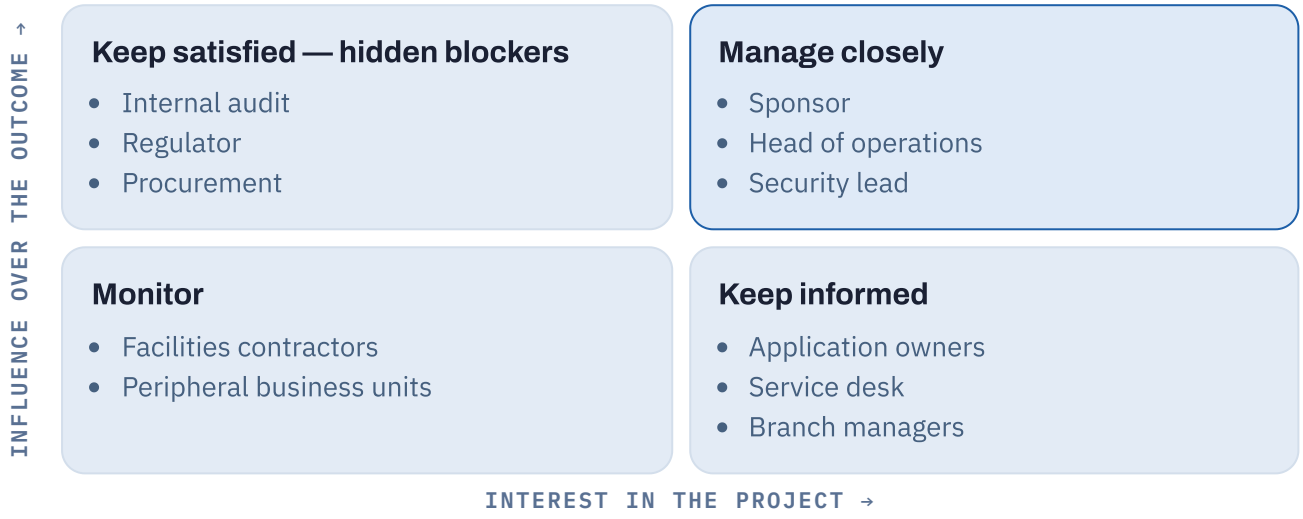
Identifying stakeholders and what they own

1. **List everyone affected** Sponsors, business-unit leads, application owners, security, operations, facilities, finance, procurement, regulators and — easily forgotten — the people who will run the network at 3 a.m.

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2. **Record what each one decides** Who can approve a budget change, who owns the availability target, who can accept a residual risk. A requirement without an owner cannot be negotiated.
3. **Note influence and interest** High-influence stakeholders who are not interested are a hidden risk: they can block late. Engage them early with short, decision-focused briefings.
4. **Capture each stakeholder's success measure** Finance measures cost per site, operations measures mean time to repair, security measures audit findings. Your design must show progress on each measure that matters.

Stakeholder influence and interest



Plot each stakeholder once. The dangerous quadrant is top-left: people who can block the design late but are not paying attention now — brief them early and briefly.

WATCH OUT · REQUIREMENTS DISGUISED AS SOLUTIONS

"We need SD-WAN" or "we need a stretched VLAN between data centres" is a proposed solution, not a requirement. Ask *why* until you reach the underlying need — cheaper branch connectivity, or keeping a virtual machine's address when it moves — because a different solution may meet that need with less risk.

SCENARIO · DECIDE · CLASSIFYING AN INPUT

During a kickoff workshop the head of operations says: "Our team of four has never run BGP in production and we cannot hire before next year." How should the architect record this?

- (A) As a business requirement for staff training
- (B) As a constraint that limits which protocols and designs are realistic
- (C) As an assumption to be validated later
- (D) As a preference to be weighted against cost and performance in the decision matrix

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Requirements register**

Capture every requirement, constraint and assumption with an owner and a way to verify it, so each later design decision can be traced back to the input that justified it.

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ID	TYPE (BR/TR/CON/ASM)	STATEMENT	OWNER	PRIORITY	HOW VERIFIED	TRACES TO DECISION
TR-07	*TR*	*Trading-floor LAN survives any single device or link failure with under 1 s of disruption*	*Head of trading technology*	*Must*	*Failure test in pre-production*	*D-12 dual-homed access, D-15 BFD on uplinks*
ASM-03	*ASM*	*Carrier can deliver diverse fibre into the north site by June*	*Network architect*	*High*	*Carrier survey and duct map*	*D-04 dual WAN at north site*

What separates an expert requirements analysis

- Every technical requirement is measurable and has an owner
- Constraints are listed separately and never weighted like preferences
- Assumptions carry a validation action and a date
- Non-functional needs — operability, manageability, growth — are captured, not only bandwidth and availability
- Conflicts between stakeholders are written down, not resolved silently by the architect

Key terms

Business requirement
An outcome or capability the organisation needs, expressed in business terms and owned by a business stakeholder.

Technical requirement
A measurable property of the network derived from a business requirement, such as an availability target or a delay budget.

Constraint
A fixed limit on the design — budget, policy, regulation, existing assets, skills or time — that removes options rather than weighting them.

Assumption
A statement treated as true for planning but not yet verified; each one is a potential risk until validated.

Non-functional requirement
A requirement about how well the network operates — availability, manageability, security, scalability — rather than what it connects.

Requirements traceability
The ability to link each design decision back to the requirement or constraint that justified it, and forward to how it is verified.

Stakeholder
Any person or group affected by, or able to influence, the design and its outcome.

Success measure
The metric a stakeholder uses to judge the project, such as cost per site, mean time to repair or audit findings.

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REFLECT

Think of the last network change you worked on. Write down one requirement that was really a solution in disguise. What was the underlying need, and would a different design have met it more simply?

KNOWLEDGE CHECK

1. A retail chain's finance director states that the project budget cannot exceed its approved amount under any circumstances. How should this be recorded?

   (A) As a constraint that removes any option costing more than the approved amount

   (B) As a business requirement to reduce the cost of connectivity across all stores

   (C) As a weighted criterion in the decision matrix alongside performance and risk

   (D) As an assumption about funding that must be validated with procurement later

2. Which statement is a well-formed technical requirement?

   (A) The network should always be fast and reliable enough for all of our users and sites

   (B) Voice calls between any two sites must see under 150 ms one-way delay

   (C) We need a modern software-defined WAN to replace the old circuits

   (D) The business must reduce its carbon footprint across every region

3. An architect assumes that a carrier can deliver diverse fibre to a rural site. What is the correct next step?

   (A) Design the site as dual-homed because the carrier has always delivered on time before

   (B) Remove the site from scope until the carrier formally commits to a delivery date

   (C) Log it with an owner and a validation action, and treat it as a risk until confirmed

   (D) Record it as a constraint because the carrier controls whether the fibre exists

4. A stakeholder insists: "We need a stretched VLAN between our two data centres." What should the architect do first?

   (A) Accept it as a requirement because the stakeholder owns the application estate

   (B) Reject it because stretching layer 2 between sites is considered poor practice

   (C) Add it to the decision matrix as one weighted option among several designs

   (D) Ask what outcome the stretched VLAN is meant to achieve

5. Why should the operations team be treated as a key stakeholder in design?

   (A) They usually control the project budget and sign off on all hardware purchases

   (B) They own the network after go-live, so their skills limit which designs succeed

   (C) They are responsible for writing the business case and choosing the vendors

   (D) They decide the availability targets that each business unit is allowed to request

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · REQUIREMENTS · LESSON 2 · 28 MIN

Gathering and Validating Requirements

Turn interviews, data and documents into a trustworthy baseline

- Plan a requirements-gathering approach that combines people, data and documents
- Baseline the current state with measured evidence rather than opinion
- Prioritise requirements and surface conflicts between stakeholders
- Validate requirements before committing to a design

Requirements are gathered, not invented. The most expensive design failures come from requirements that were wrong from the start — a growth forecast nobody checked, a latency-sensitive application nobody mentioned, a regulatory scope that was misunderstood. Gathering is therefore a deliberate process with three sources: **people** (what they need), **data** (what the network actually does), and **documents** (what the organisation has committed to).

Requirements-gathering pipeline

01 Plan — Scope, stakeholders, sources and timebox →

02 Elicit — Interviews, workshops, document review →

03 Baseline — Measure traffic, inventory and incidents →

04 Analyse — Classify, quantify and find conflicts →

05 Prioritise — Agree what is essential versus desirable →

06 Validate — Play back and obtain sign-off

Elicitation techniques and when to use them

TECHNIQUE	BEST FOR	WATCH OUT FOR
One-to-one interview	Senior stakeholders, sensitive topics, hidden constraints	Hearing one view only; confirm with data
Facilitated workshop	Cross-team trade-offs and agreeing priorities	Loudest voice dominates; use structured voting
Document review	Contracts, policies, audit reports, previous designs	Out-of-date documents presented as current
Traffic and flow analysis	Real application mix, peaks, growth trends	Short samples that miss month-end or seasonal peaks
Incident and change history	Recurring failure modes and fragile areas	Poor ticket hygiene hiding the real cause
Site survey	Physical facts: cabling, power, cooling, ducts	Assuming every site matches the standard template

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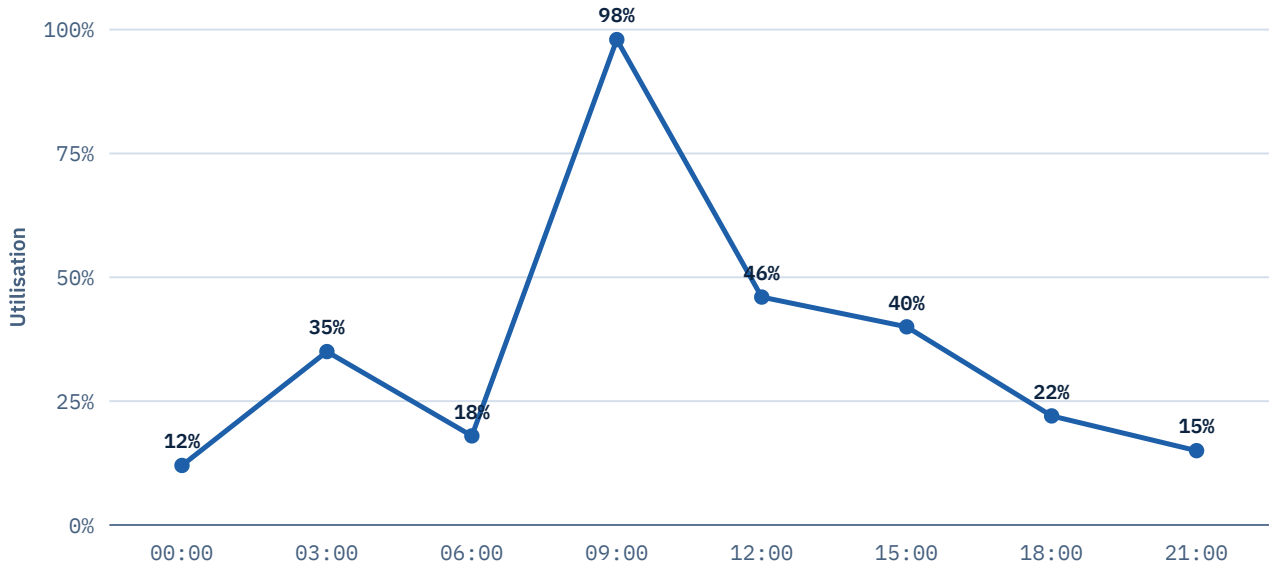
Baselining the current state

1. **Inventory** Devices, software levels, end-of-support dates, circuits and contract terms, plus power draw and the rack, power and cooling headroom at each site. You cannot plan a migration without knowing what exists.
2. **Topology and dependencies** Physical and logical topology, routing domains, shared services (DNS, DHCP, identity, time) and the applications that depend on them.
3. **Traffic** Utilisation at the 95th percentile and the peak, top talkers, application mix, directionality (north-south versus east-west) and growth over at least a full business cycle.
4. **Health and incidents** Error rates, outages and their root causes, change-failure rate. These reveal where the current design hurts the business.
5. **Operations** Tooling, monitoring and telemetry coverage, automation maturity, whether a usable source of truth exists, team size and on-call model. These set realistic limits on new complexity.

PRO TIP · MEASURE PEAKS, NOT AVERAGES

A 30% average utilisation can hide a link that saturates for twenty minutes every morning when backups overlap with log-ins. Size and prioritise from the 95th percentile and the observed peak over a representative period, and ask the business which periods matter most — month-end, exam week, a shipping season.

Why averages mislead: one day on a branch uplink



Illustrative data. The average of these samples is about 36%, yet the link is close to saturation at 09:00, when log-ins overlap with backups. Size from the 95th percentile and the observed peak, not from the average.

MoSCoW prioritisation

M

Must have

Without it the solution fails its purpose or breaks a law or contract. Must-haves are few; if everything is a must, nothing is.

S

Should have

Important and expected, but a workaround exists for a short time.

C

Could have

Desirable if time and budget allow; first to be dropped under pressure.

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W

Won't have (this time)

Explicitly out of scope for this phase — recording it prevents silent scope creep.

Value against effort: a starting point for MoSCoW

BUSINESS VALUE ↑

Quick wins — usually Must or Should

- Guest isolation
- Second transport for tills

Major work — Must only if the driver demands it

- Data-centre interconnect rebuild
- Campus fabric

Could have

- Cosmetic naming clean-up
- Extra dashboards

Won't have this time

- Multi-cloud failover for a minor application

EFFORT AND COST TO DELIVER →

Use the grid in the prioritisation workshop, then let the owners assign MoSCoW labels. Anything legally or contractually required is a Must wherever it lands on the grid.

WATCH OUT · CONFLICTING REQUIREMENTS

Security wants every flow inspected by a central firewall; the trading desk wants the lowest possible latency between two server racks. Both are legitimate. The architect's job is to make the conflict visible, describe the options and their consequences, and let the owners decide — not to resolve it quietly by favouring the requirement the architect personally prefers.

SCENARIO · DECIDE · VALIDATING A GROWTH ASSUMPTION

A logistics company's IT manager tells you WAN traffic "roughly doubles every year". Your 12-month interface data shows 18% annual growth at the 95th percentile, but the company plans to roll out video-based remote inspection next year. What is the best approach?

A. Size for 100% annual growth because the IT manager knows the business best

B. Size from the 18% trend only, because historic data is the most reliable evidence

C. Defer the sizing decision until the video service is live and can be measured

D. Use the measured trend plus a named uplift for video, confirmed with owners

Circle your choice. Answer and feedback: module answer key.

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Requirements validation checklist

☐ Every requirement has an owner who has confirmed it in writing

☐ Technical requirements are measurable and have a stated test

☐ Peaks and growth are based on data over a representative period

☐ Regulatory and contractual obligations have been checked against source documents

☐ Conflicts are recorded with the options and the decision owner

☐ Assumptions have validation actions and target dates

☐ Out-of-scope items are explicitly listed

☐ The requirements were played back to stakeholders and signed off

Key terms

Elicitation
The structured activity of drawing out requirements from stakeholders, documents and data.

Baseline
A measured snapshot of the current network — inventory, traffic, health and operations — used as the reference for design and comparison.

95th percentile
The utilisation value exceeded only 5% of the time in a sample; a common basis for sizing that ignores brief spikes but captures sustained load.

MoSCoW
A prioritisation scheme dividing requirements into Must, Should, Could and Won't-have-this-time.

Playback
Presenting the analysed requirements back to stakeholders to confirm understanding before design begins.

Scope creep
Uncontrolled growth of requirements after the baseline is agreed, usually because exclusions were never written down.

Requirement owner
The stakeholder accountable for a requirement, who can confirm, change or trade it off.

Representative period
A measurement window long enough to include the business cycles, such as month-end or seasonal peaks, that drive demand.

REFLECT

Which data source do you most often skip when gathering requirements — incident history, traffic peaks, contracts, or operations capability? What design mistake could that gap cause in your environment?

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Key takeaways

- Gather from three sources — people, data and documents — and test each against the others
- Baseline with measured evidence over a representative business cycle, including power, telemetry and operations capability
- Size from the 95th percentile and observed peaks, never from averages
- Few Musts, explicit Won't-haves, and conflicts handed to their owners with options
- Play the result back and obtain written sign-off before comparing designs

KNOWLEDGE CHECK

6. Why should WAN sizing use the 95th percentile and observed peaks rather than average utilisation?

   (A) Averages hide sustained busy periods that cause congestion

   (B) Averages always overstate demand and lead to oversized circuits

   (C) Carriers bill only on peak usage, so averages are irrelevant to cost

   (D) The 95th percentile is required by every service level agreement

7. Two stakeholders give conflicting requirements: central inspection of all flows versus minimum latency between two racks. What should the architect do?

   (A) Choose the security requirement because security always outranks performance

   (B) Choose the latency requirement because it is more precisely measurable

   (C) Document the conflict with options and consequences, and ask the owners to decide

   (D) Design for both requirements in full and simply accept the extra cost without any discussion

8. In MoSCoW prioritisation, why is the "Won't have this time" category valuable?

   (A) It lists requirements that will be rejected permanently and never revisited

   (B) It identifies items the architect can deliver later without approval

   (C) It groups low-cost features that can be added if the budget allows

   (D) It records exclusions explicitly, preventing silent scope creep

9. Which baseline activity is MOST likely to reveal realistic limits on how complex the new design can be?

   (A) Assessing the operations team's tooling, skills and on-call model

   (B) Collecting interface utilisation at five-minute intervals across the core

   (C) Listing end-of-support dates for every switch and router in the estate

   (D) Reviewing the physical cabling and duct routes at the largest sites

10. A requirement reads: "The network must support future growth." What is the main problem with it?

    (A) It is a constraint and should not appear in the requirements register

    (B) It describes a solution rather than the business need behind it

    (C) It is a risk and should be recorded in the risk log rather than here

    (D) It cannot be tested, because it has no quantity, horizon or owner

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · REQUIREMENTS · LESSON 3 · 33 MIN

Design Principles and Trade-off Analysis

Use modularity, hierarchy and simplicity — and score options honestly

- Apply the core design principles and explain what each one buys
- Build and use a weighted decision matrix, including sensitivity checks
- Analyse options across cost, complexity and risk rather than features alone

Design principles are not slogans; they are reusable answers to recurring problems. Modularity contains change, hierarchy contains state, simplicity contains operational risk, and planned scalability contains future cost. When a scenario gives you two technically valid options, these principles — weighed against the stated requirements — usually decide which is better.

Core design principles

MOD

Modularity

Build the network from repeatable blocks — an access block, a branch type, a data-centre pod — so that a change or failure in one block is contained and new capacity is added by repeating a known pattern.

HIER

Hierarchy

Give each layer a clear role and aggregate at boundaries. Hierarchy enables summarisation, predictable traffic paths and smaller failure and flooding domains.

SIMP

Simplicity

Prefer the fewest features, protocols and exceptions that meet the requirements. Every feature is a dependency to configure, monitor, upgrade and troubleshoot.

SCAL

Scalability

Know which resource limits growth — ports, routes, adjacencies, power, licences — and design so the next increment is linear, not a redesign.

FD

Failure domains

Bound the impact of any single fault: one failed device, link or misconfiguration should affect as little as possible.

OPS

Operability

Design for the people and tools that run the network: consistent naming, standard templates, telemetry, and a safe way to change things.

SUS

Sustainability

Treat power, cooling and equipment lifetime as design criteria: right-size capacity, prefer efficient optics and platforms, and count energy in the lifetime cost, not as an afterthought.

INSIGHT · COMPLEXITY HAS A COST EVEN WHEN IT WORKS

A design with more moving parts can still meet every functional requirement and pass every test. Its cost appears later: longer troubleshooting, riskier upgrades, harder hiring, and unexpected interactions between features. In trade-off analysis, treat added complexity as a real cost that must be justified by a requirement, not as free capability.

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Trade-off lenses

Cost

- Capital: hardware, licences, build
- Operating: circuits, support, power, staff time
- Energy: power and cooling over the asset life
- Exit: migration and contract penalties
- Opportunity: what else the money could fund

Complexity

- Number of protocols and features
- Exceptions to standard templates
- Skills required to operate
- Interdependencies between components

Risk

- Probability and impact of failure
- Delivery risk: lead times, untested technology
- Vendor and supplier dependency
- Security exposure and audit findings

Requirement fit against complexity

REQUIREMENTS MET →

Simplest sufficient — preferred

- VRF-lite with firewall policy for a small campus

Justified only by a stated requirement

- Overlay fabric for policy at very large scale

Cheap but insufficient — exclude

- Flat network with port ACLs

Avoid — complex and still short

- Three overlays the team cannot run

OPERATIONAL COMPLEXITY →

Options in the top-left meet the requirements with the least machinery; that is where the recommendation usually sits. Top-right options need a requirement that justifies their complexity.

Weighted decision matrix

MEASURE	FORMULA	WHAT IT TELLS YOU
Weighted score	$S(option) = \Sigma\ (w_i \times s_i)$	Multiply each criterion's weight by the option's score on that criterion, then add them up. Weights should sum to 100% (or 1.0).
Example	$0.4 \times 4 + 0.3 \times 3 + 0.2 \times 5 + 0.1 \times 2 = 3.7$	Criteria: meets availability (40%), cost (30%), operability (20%), time to deliver (10%), each scored 1–5.
Sensitivity check	Change each weight by ±10 points and re-rank	If a small, plausible change in weighting flips the winner, the decision is fragile — present both options and the deciding factor to the owners.

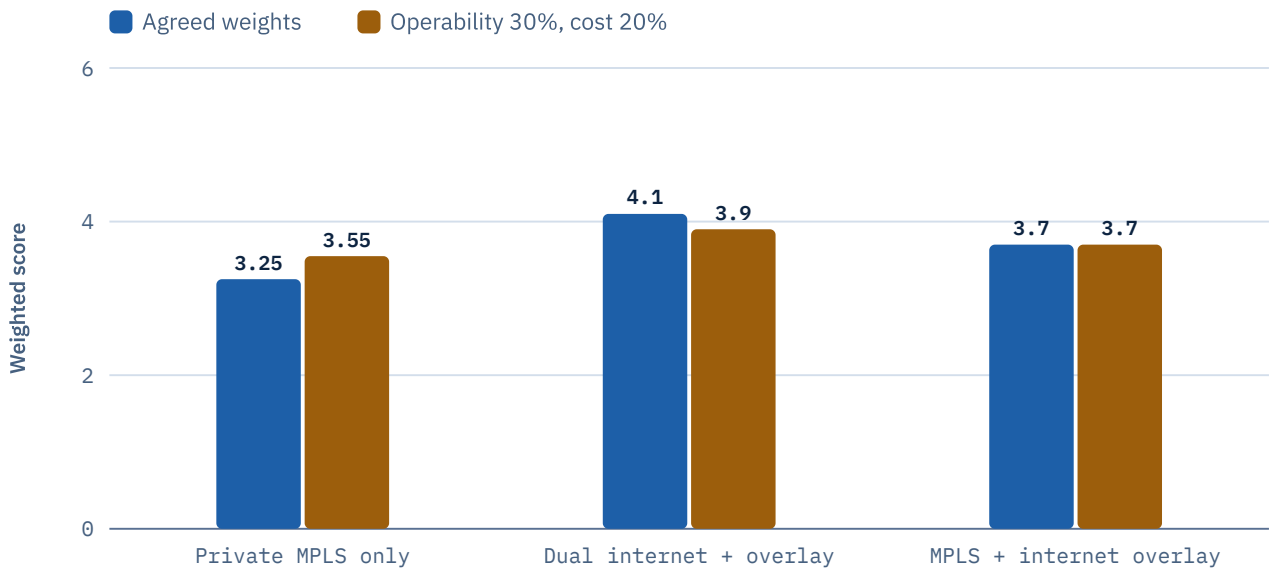
Worked decision matrix: branch WAN options for 80 sites

CRITERION (WEIGHT)	PRIVATE MPLS ONLY	DUAL INTERNET WITH OVERLAY	MPLS PLUS INTERNET OVERLAY
Meets availability target (35%)	3	4	5
Total 5-year cost (30%)	2	5	3
Operational fit for a team of six (20%)	5	3	3

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CRITERION (WEIGHT)	PRIVATE MPLS ONLY	DUAL INTERNET WITH OVERLAY	MPLS PLUS INTERNET OVERLAY
Time to deliver within 9 months (15%)	4	4	3
Weighted total	3.25	4.10	3.70

Sensitivity check on the branch WAN matrix



Computed from the worked matrix above. Shifting ten points of weight from cost to operability narrows the gap, but dual internet with an overlay still wins, so the recommendation is reasonably robust.

EXAMPLE · READING THE MATRIX

Dual internet with an overlay wins at 4.10, but its operability score is the weakest. Before recommending it, the architect should check sensitivity: if operability were weighted 30% and cost 20%, the totals become 3.55, 3.90 and 3.70 — the same winner, so the recommendation is reasonably robust. The matrix also shows *where* to mitigate: training and a managed overlay service would raise the weakest score.

WATCH OUT · MATRICES CAN LAUNDER BIAS

Scores and weights look objective but are judgements. Agree the criteria and weights with stakeholders *before* scoring options, keep any hard constraint out of the matrix (a failing option is excluded, not scored low), and record the evidence behind each score.

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SCENARIO · DECIDE · SIMPLICITY VERSUS CAPABILITY

A mid-sized manufacturer needs guest isolation and separation of production machinery from office users across a single-building campus with six access switches. One proposal uses a campus overlay fabric with a dedicated control plane; another uses VRF-lite on a collapsed core with firewall-controlled inter-VRF routing. The team has no fabric experience. Which recommendation best fits?

- (A) VRF-lite on the collapsed core, with firewall policy between segments
- (B) The overlay fabric, because it is the more modern and future-proof architecture
- (C) Separate physical switches for production, office and guest networks
- (D) A single flat VLAN with access-control lists on every access port

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Weighted decision matrix**

Compare design options against agreed, weighted criteria. Exclude options that break a constraint before scoring.

CRITERION	WEIGHT (%)	OPTION A SCORE (1-5)	OPTION B SCORE (1-5)	OPTION C SCORE (1-5)	EVIDENCE / RATIONALE
Meets 99.95% availability	*35*	*3*	*4*	*5*	*Failure analysis; dual-path design for B and C*
5-year total cost	*30*	*2*	*5*	*3*	*Carrier quotes Q1; support uplift included*

Key terms

Modularity
Building the network from repeatable, self-contained blocks so change and failure are contained and growth is predictable.

Failure domain
The part of the network affected when a single component fails or is misconfigured.

Sensitivity analysis
Varying weights or scores to see whether the preferred option changes; a fragile result needs explicit discussion.

Total cost of ownership
All costs over the life of a solution — capital, operating, support, staff and exit — not just the purchase price.

Hierarchy
Assigning clear roles to layers and aggregating at their boundaries to control state, traffic paths and failure domains.

Weighted decision matrix
A table that scores options against criteria multiplied by agreed weights to support a transparent choice.

Operability
How easily and safely a network can be monitored, changed and repaired by the team that runs it.

Trade-off analysis
Comparing options by what each gains and gives up across cost, complexity and risk, rather than by features alone.

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REFLECT

Pick a design you inherited that you consider too complex. Which principle was traded away, and what requirement — if any — justified that trade at the time?

Key takeaways

- Principles are reusable answers: modularity contains change, hierarchy contains state, simplicity contains risk
- Count complexity, energy and lifetime operations as real costs alongside capital
- Agree criteria and weights before scoring, and exclude constraint-breaking options rather than scoring them low
- Run a sensitivity check; if a plausible re-weighting flips the winner, give the choice to the owners
- Prefer the simplest option that meets every requirement

KNOWLEDGE CHECK

11. In a weighted decision matrix, how should an option that breaks a hard constraint be handled?

(A) Score it 1 on the cost criterion and let the totals decide

(B) Increase the weight of the constraint so the option loses

(C) Exclude it before scoring, since it is not a viable option

(D) Keep it in the matrix and add a footnote describing the breach

12. Criteria weights are 50% availability, 30% cost and 20% operability. Option X scores 4, 3 and 2. What is its weighted total?

(A) 3.0

(B) 3.3

(C) 3.6

(D) 2.9

13. A sensitivity check shows that moving 10 points of weight from cost to operability changes the winning option. What should the architect conclude?

(A) The original weighting was wrong and should be replaced by the new one

(B) The matrix is invalid and the decision should be made on experience alone

(C) The decision is fragile; present both options and the deciding factor

(D) The option with the higher cost score should be chosen to protect budget

14. Which principle MOST directly enables route summarisation and smaller flooding domains?

(A) Simplicity

(B) Operability

(C) Modularity

(D) Hierarchy

15. Why should added complexity be counted as a cost in trade-off analysis even when the design meets every functional requirement?

(A) It raises the effort and risk of operating, upgrading and troubleshooting

(B) It always increases hardware cost because more devices are required

(C) It makes the design harder to document in the high-level design

(D) It causes the network to fail acceptance tests more often than simpler designs

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MODULE 1 · REQUIREMENTS · LESSON 4 · 33 MIN

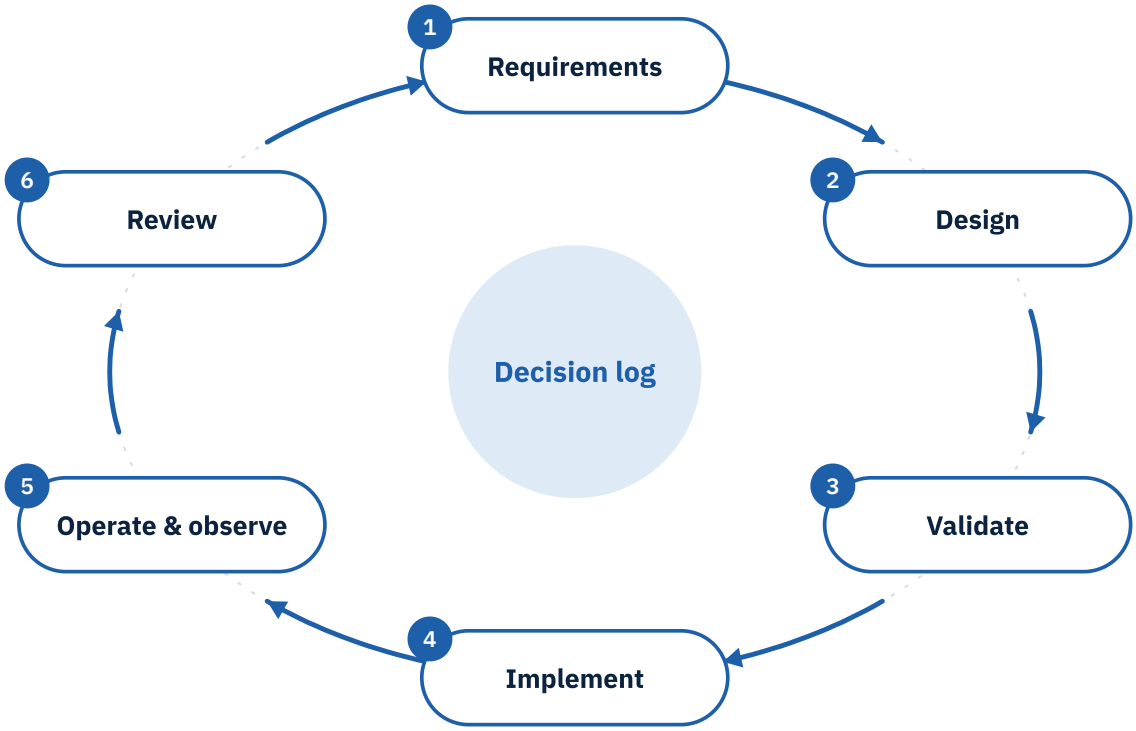
Design Decisions, Documents and Scenarios

Write decisions others can audit and read scenarios like an examiner

- Write a design decision record that captures context, options and consequences
- Distinguish the content and audience of high-level and low-level design documents
- Read a design scenario systematically to find the facts that decide the answer

A design that cannot be explained cannot be defended, reviewed or safely changed. Experts therefore record **decisions**, not just diagrams. Each decision states the context, the options considered, the choice and its consequences — so that a reviewer, an auditor or the next architect understands not only *what* was built but *why*, and knows what would have to change for the decision to be revisited.

The design lifecycle



1. **Requirements** — Register, owners, priorities
2. **Design** — Options, decisions, HLD and LLD
3. **Validate** — Review, model or digital twin, pilot
4. **Implement** — Pipeline-driven build in waves
5. **Operate & observe** — Telemetry against objectives
6. **Review** — Triggers met? Reopen decisions

Decisions are made in the design stage but checked at every turn of the loop. Observed behaviour in operation feeds the next round of requirements, which is why each decision record carries a review trigger.

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Anatomy of a design decision record

ID

Identifier and title

A short, unique reference such as D-12 "Routed access for office floors" so documents and changes can cite it.

CTX

Context and drivers

The requirements, constraints and assumptions that make this decision necessary, cited by register ID.

OPT

Options considered

Two to four realistic options, each with a one-line summary — including "do nothing" where it is genuinely viable.

DEC

Decision and rationale

The chosen option and the reasons it best satisfies the drivers, referring to the decision matrix where one was used.

CON

Consequences

What becomes easier, what becomes harder, new risks, and follow-on decisions this one triggers.

REV

Status and review trigger

Proposed, accepted or superseded, plus the condition that would reopen it — for example, if branch count exceeds 300. Keeping records in version control next to the source of truth lets reviewers see how each decision evolved.

High-level versus low-level design

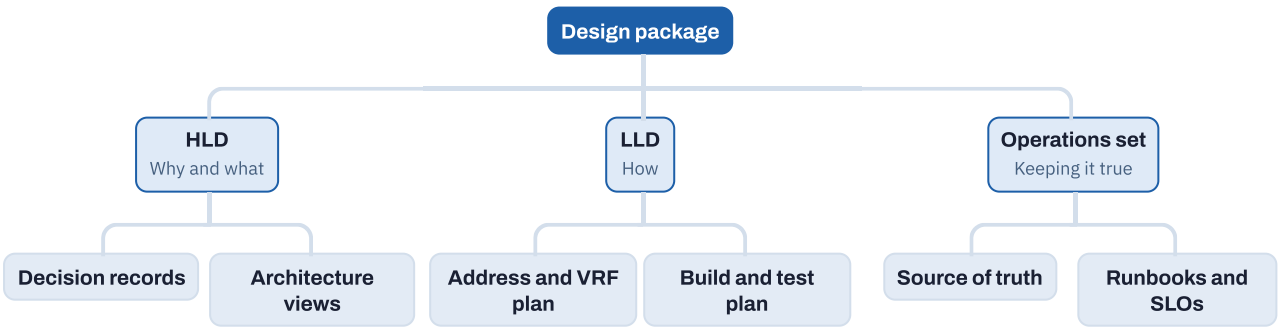
High-level design (HLD)

- Audience: sponsors, architects, security, operations leads
- Answers why and what: architecture, principles, major decisions
- Topology at block level, traffic flows, resilience model
- Technology choices with rationale and trade-offs
- Stable: changes only when requirements change

Low-level design (LLD)

- Audience: implementers and operations engineers
- Answers how: exact configuration standards and values
- Addressing tables, VLAN/VRF IDs, protocol timers, port maps
- Build, test and migration procedures
- Detailed: updated as the build progresses

A complete design document set



The HLD explains why, the LLD explains how, and the operations set keeps the design true after go-live. The source of truth is the machine-readable form of the LLD.

TEMPLATE ## High-level design outline

A reusable skeleton for an HLD so that every design covers the same ground and reviewers know where to look.

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SECTION	CONTENT	KEY QUESTIONS ANSWERED	OWNER / REVIEWER
1. Context and scope	*Business drivers, in/out of scope, phases*	*Why is this design needed and what does it cover?*	*Lead architect / sponsor*
4. Design decisions	*Decision records D-01 to D-n with matrices*	*What did we choose, what did we reject and why?*	*Lead architect / design authority*

Typical HLD sections

- Context, scope and business drivers
- Requirements, constraints and assumptions (by reference to the register)
- Design principles applied
- Logical and physical architecture overview
- Key design decisions with options and rationale
- Resilience, security, management and operations model, including source of truth, automation and telemetry
- Power, cooling and sustainability considerations where they constrain the design
- Migration approach, risks and open issues

PRO TIP · WRITE THE CONSEQUENCES HONESTLY
Reviewers trust a decision record that admits its downsides. "Routed access means VLANs cannot span floors; wireless roaming will rely on the controller tunnelling client traffic" tells the next architect exactly what to watch. A record that lists only benefits invites the question of what was hidden.

Reading a design scenario like an examiner

1. **Read the question first** Know whether you are asked for the BEST design, the FIRST action, the MOST important requirement, or the BIGGEST risk. The same facts support different answers to different questions.
2. **Mark the requirements and constraints** Highlight availability targets, budgets, timelines, skills, regulations and existing assets. Phrases such as "must", "cannot", "limited staff" and "within six months" are deciding facts.
3. **Spot the hidden priority** Scenarios often contain one requirement that outranks the others — regulatory isolation, a hard deadline, a tiny team. Identify it before comparing options.
4. **Eliminate options that break a constraint** An option that is technically excellent but exceeds the budget, needs skills the team lacks, or misses the deadline is wrong, however attractive.

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5. **Choose the simplest option that meets everything** Between remaining options, prefer the one that meets all stated requirements with the least added complexity and risk — not the one with the most features.
6. **Check against the exact question** Re-read the question stem. If it asked what to do FIRST, a perfect final design may still be the wrong answer.

SCENARIO · DECIDE · WHAT TO DO FIRST

A hospital group asks you to redesign its WAN. In the first meeting the CIO says the new design must be live before the current carrier contract ends in five months, and also mentions that a new clinical imaging system will "probably" be deployed across all sites next year. What should you do FIRST?

- (A) Start comparing carrier quotes immediately to protect the five-month deadline
- (B) Design for today's traffic and treat the imaging system as a future phase
- (C) Confirm the imaging system's scope, timing and traffic profile
- (D) Recommend extending the existing carrier contract to remove the time pressure

Circle your choice. Answer and feedback: module answer key.

CASE STUDY · NATIONAL UNIVERSITY NETWORK REFRESH

A national university has 42,000 students across a main campus, three regional campuses 80–300 km away, and a teaching hospital it shares with the health ministry. The campus network is 11 years old and flat at layer 2 in many buildings. Research groups transfer multi-terabyte datasets to international partners and complain that the central firewall throttles them. The security office failed an audit finding that student-residence traffic shares segments with administrative systems. The board has approved a fixed three-year capital budget, procurement must use a public tender, and the network team has nine engineers with strong routing skills but little automation experience. The research council expects a 10x increase in data transfers over five years.

1. Classify the key inputs into business requirements, technical requirements, constraints and assumptions.

2. Which requirement is the hidden priority, and why?

3. What design decisions would you expect to see recorded in the HLD, and which ones conflict?

4. What should be validated before options are compared?

Model answer: module answer key.

Key terms

Design decision record
A short, structured document capturing one design decision: context, options, choice, rationale, consequences and review trigger.

High-level design (HLD)
A document describing the architecture, principles, major decisions and their rationale for sponsors, architects and operations leads.

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Low-level design (LLD)
A detailed document specifying configuration standards, values and procedures for those who build and operate the network.

Review trigger
A condition recorded with a decision that, if met, requires the decision to be revisited.

Design exception
A documented, approved deviation from a standard design, recorded with its owner and justification.

Design authority
The person or board that reviews and approves design decisions and their exceptions.

Hidden priority
The requirement in a scenario that outranks the others — often regulatory, deadline-driven or skill-limited — and decides between otherwise valid options.

Superseded decision
A decision record replaced by a newer one; kept for history and linked to the decision that replaced it.

REFLECT

Write a one-paragraph decision record for a choice you made recently. Did you list the options you rejected and the consequences you accepted? What review trigger would you add now?

Key takeaways

- Record decisions, not just diagrams: context, options, choice, consequences and a review trigger
- The HLD answers why and what for decision-makers; the LLD answers how for implementers
- Admit negative consequences — it builds trust and tells operations what to watch
- Read scenarios question-first, mark the binding constraint, eliminate violators, then choose the simplest sufficient option
- A FIRST question is about sequence: validate a decisive unknown before designing

KNOWLEDGE CHECK

16. Which section of a design decision record tells a future architect when the decision should be reopened?

(A) Options considered
(B) Status and review trigger
(C) Context and drivers
(D) Decision, rationale and evidence

17. Which content belongs in a low-level design rather than a high-level design?

(A) The rationale for choosing routed access over layer 2 access in each campus block
(B) Interface addressing tables, VRF identifiers and protocol timer values
(C) The resilience model and how failure domains are bounded
(D) The business drivers and the scope of each project phase

18. A scenario asks what the architect should do FIRST. One option is a complete, excellent target design. Why might it still be wrong?

(A) Complete designs are never accepted as answers in scenario-based assessments
(B) An excellent design always exceeds the budget given in a scenario
(C) The target design should be written by the operations team, not the architect
(D) The question asks for the first step, which is often to validate a key fact

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19. Why should a decision record list its negative consequences?

(A) It builds trust and tells others which risks and limits to manage

(B) Auditors reject any design document that omits a risk section entirely

(C) It allows the architect to avoid responsibility if problems occur

(D) Negative consequences increase the weighted score of the chosen option

20. When two remaining options both meet every stated requirement and constraint, which should usually be preferred?

(A) The one with the richest feature set for possible future needs

(B) The one using the most recent technology on the market

(C) The one that meets them with the least added complexity and risk

(D) The one that the largest number of stakeholders already know about

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MODULE 1 · REQUIREMENTS

Answer key

Knowledge checks

1. A A non-negotiable budget ceiling is a constraint: options above it are not viable at all. Weighting it would allow an expensive option to win on other criteria, and it is a known fact rather than an assumption.

2. B Only the delay target is measurable and testable. The others are a vague quality, a solution in disguise, and a business goal with no network property attached.

3. C Assumptions are acceptable for planning only if they are visible and validated. Designing as if it were fact hides the risk; removing the site or calling it a constraint misrepresents what is known.

4. D The request is a solution. Discovering the underlying need — for example address mobility for workload migration — lets the architect consider options that meet it with less shared failure risk, or confirm the stretch is truly needed.

5. B A design the operations team cannot run safely will fail in production regardless of its elegance. Budget, business cases and availability targets usually sit with other stakeholders.

6. A Averages smooth out the busy periods that users actually feel. The 95th percentile captures sustained load while ignoring short spikes; it is a sizing convention, not a universal contract term.

7. C Resolving conflicts belongs to the requirement owners, informed by the architect's analysis. Arbitrary precedence rules and silent cost increases both remove the business from a business decision.

8. D Writing down what is out of scope for this phase sets expectations and protects the baseline. The items may return in later phases, and "could have" is the category for optional extras.

9. A Complexity is limited by the people and tools that must run the network. Traffic, lifecycle and physical data are essential but address capacity, timing and physical feasibility instead.

10. D Growth is a legitimate requirement, but it must be quantified — for example, double the access ports in five years without forklift replacement — to be designed and verified.

11. C A constraint removes options rather than weighting them. Scoring a non-viable option risks it winning on other criteria and confuses the decision record.

12. B $0.5 \times 4 + 0.3 \times 3 + 0.2 \times 2 = 2.0 + 0.9 + 0.4 = 3.3$.

13. C A fragile result means the choice depends on how the business values cost versus operability. That is a stakeholder decision, so the architect makes it visible rather than picking a weighting.

14. D Hierarchy creates aggregation boundaries where addressing and routing state can be summarised. Modularity helps by making blocks repeatable, but the aggregation point comes from the hierarchical role.

15. A Complexity is mainly an operational and risk cost that appears after go-live. It does not necessarily add hardware or fail tests, which is why it is easy to overlook.

16. B The review trigger states the condition — growth, a contract change, a new regulation — that would make the decision worth revisiting.

17. B Exact values and build details are implementation information for the LLD. Rationale, resilience models and drivers are architectural and belong in the HLD.

18. D Scenario questions test sequencing as well as knowledge. If a decisive requirement is uncertain, validating it comes before any design.

19. A Honest consequences make the trade-off explicit and guide operations and future designers. The aim is transparency and safe change, not blame avoidance.

20. C Unstated future needs and novelty are not requirements. Simplicity reduces operational risk, which is a real cost the business carries.

Scenarios & case studies

> **Scenario · Classifying an input**
> **B** is best. Correct. Team size and skills cannot be traded away in the short term, so they limit which designs are realistic. The design can still include BGP, but only with a plan to manage that gap, such as a managed service or a phased introduction with training.

> **Scenario · Validating a growth assumption**
> **D** is best. Correct. Measured data anchors the baseline, while the known new service is modelled separately as a named driver. Playing the result back makes the assumption visible and agreed.

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Scenario · Simplicity versus capability

A is best. Correct. It meets both segmentation requirements with well-understood technology, fits the size of the campus, and matches the team's skills. A fabric becomes attractive when scale, mobility or policy automation needs justify its complexity.

Scenario · What to do first

C is best. Correct. A large, uncertain demand driver discovered at the start must be pinned down before options are compared; designing without it risks a WAN that is obsolete within a year. The deadline makes early validation more urgent, not less.

Case study · National university network refresh

Classification. Business requirements: support research collaboration, pass the security audit, refresh ageing infrastructure, serve the regional campuses and hospital. Technical requirements (to be quantified): segment residences from administrative systems; high-throughput research transfers that do not traverse a stateful bottleneck; availability targets per user group. Constraints: fixed three-year capital budget, public tender procurement and its timelines, a team strong in routing but weak in automation, shared governance of the hospital. Assumptions: 10x research growth, regional campus circuit availability, hospital ministry cooperation.

Hidden priority. The audit finding is effectively a must-have with an external deadline; it drives segmentation design early and may need an interim fix before the full refresh.

Expected decisions and conflicts. Campus model per site (collapsed core at regional campuses, full hierarchy at the main campus); L2/L3 boundary; segmentation method (VRF-based macro-segmentation is a natural fit for the team's routing skills); a research data path that bypasses the general-purpose firewall and is protected with filtering and monitoring instead; WAN design for regional sites. The research data path conflicts with the security office's preference for central inspection — record it, quantify the risk, and let the owners decide.

Validate first. The research growth figure (data, not anecdote), the audit scope and deadline, the tender timeline against the budget years, the hospital's responsibilities and network boundary, and fibre or carrier options to regional campuses.

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02

MODULE 2 · CAMPUS

Campus & Data Centre Design

Hierarchy, boundaries, fabrics and the arithmetic of scale

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Choose between three-tier, collapsed-core, routed-access and fabric campus designs from requirements
- Place the L2/L3 boundary to balance mobility against failure-domain size
- Size spine-leaf fabrics using port counts and oversubscription ratios
- Design data centre interconnects that keep sites as separate failure domains

LESSONS

1. Campus Hierarchy Models · 28 min
2. The L2/L3 Boundary and Campus Fabrics · 33 min
3. Data Centre Spine-Leaf Design · 38 min
4. Data Centre Interconnect and Choosing a Design · 33 min

MODULE 2 · CAMPUS · LESSON 1 · 28 MIN

Campus Hierarchy Models

Three-tier, collapsed core and when each one earns its place

- Explain the role of the access, distribution and core layers
- Choose between a three-tier and a collapsed-core campus from requirements
- Calculate and judge access and distribution oversubscription

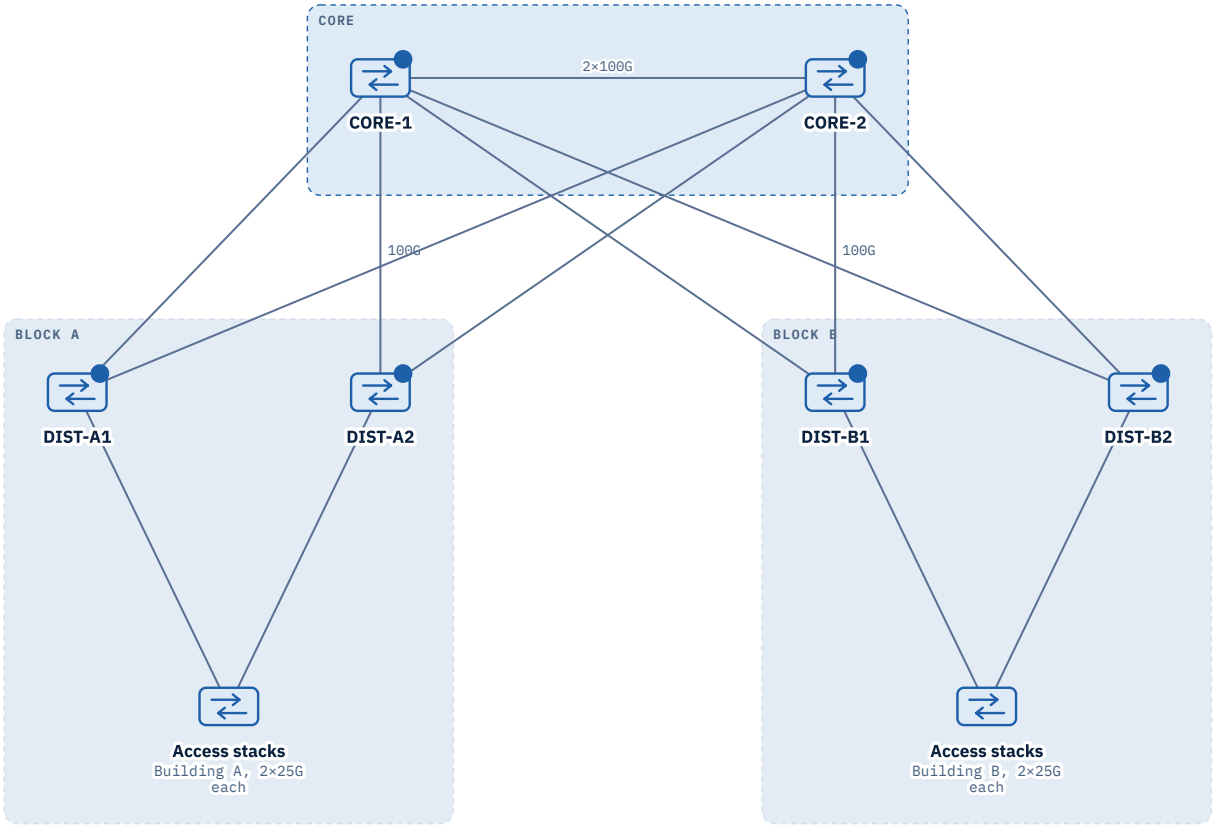
The **Campus sheet** of the ZCDE Design Blueprint begins with the shape of the network. Hierarchy is not tradition for its own sake: each layer exists to solve a specific problem. Access connects endpoints and enforces edge policy; distribution aggregates access blocks, hosts the layer 2/layer 3 boundary in many designs and summarises routes; the core moves traffic between distribution blocks quickly and does as little else as possible. The design question is how many of these layers a particular campus actually needs.

Three-tier campus roles

Core	`High-speed transport between blocks` `Minimal policy, minimal state` `Fast, predictable convergence` `No direct endpoint connections`
Distribution	`Aggregates access switches into a block` `Layer 2/3 boundary (in L2-access designs)` `Route summarisation and policy` `First-hop redundancy for endpoint subnets`
Access	`Endpoint and wireless access-point ports` `Power over Ethernet (up to about 90 W per port), port security, 802.1X` `QoS marking and trust boundary` `VLAN or segment assignment`

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A three-tier campus with two distribution blocks



Each building is a distribution block with its own pair; the core only joins blocks. Adding a third building means two new uplink pairs to the core, not links to every existing block.

Three-tier versus collapsed core

Three-tier

- Separate core joins many distribution blocks
- Scales to large, multi-building campuses
- Failure or change in one block stays local
- Core links grow linearly as blocks are added
- More devices, power, cost and management

Collapsed core (two-tier)

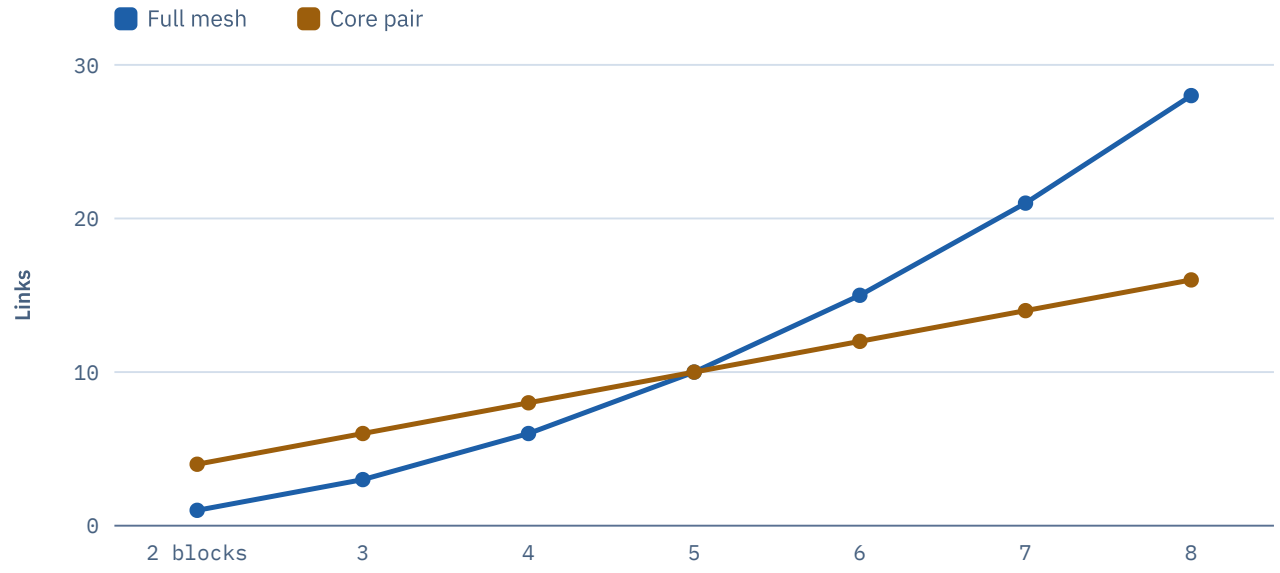
- Core and distribution combined in one pair
- Fits a single building or small campus
- Fewer devices and lower cost
- Simple to operate for small teams
- Becomes a scaling and change bottleneck as blocks grow

INSIGHT · WHY A SEPARATE CORE APPEARS

Without a core, every distribution pair must connect to every other pair to avoid hair-pinning traffic, and the number of inter-block links grows roughly with the square of the number of blocks. A dedicated core turns that mesh into a star: each new block needs only its uplinks to the core. When a campus reaches several distribution blocks, or buildings with separate fibre plants, the core usually pays for itself in simplicity.

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Inter-block links: full mesh versus a core pair



Computed with one link per block pair in the mesh and one uplink from each block to each of two core switches. The mesh grows with the square of the block count; the core grows linearly.

Choosing a campus model

SITUATION	LIKELY FIT	DECIDING FACTORS
Single building, under ~10 access stacks	Collapsed core	Cost, small team, little east-west growth
Multiple buildings, 3+ distribution blocks	Three-tier	Fibre layout, change isolation, link count
Small branch office	Single switch stack or access-only to a router	Cost and remote manageability
Campus with plans to double in 3 years	Collapsed core designed to add a core later	Reserve core ports, address plan by block
Large campus with heavy segmentation needs	Three-tier or fabric	Policy scale and mobility (see next lesson)

Campus oversubscription

MEASURE	FORMULA	WHAT IT TELLS YOU
Oversubscription ratio	`OSR = total downstream capacity ÷ total upstream capacity`	How much more traffic the edge could offer than its uplinks can carry if every port transmitted at full rate at the same time.
Access example	`48 × 2.5 Gbps = 120 Gbps; 2 × 25 Gbps = 50 Gbps; OSR = 2.4:1`	A multigigabit access switch serving Wi-Fi access points with two 25 Gbps uplinks.

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MEASURE	FORMULA	WHAT IT TELLS YOU
Distribution example	`8 access switches × 50 Gbps = 400 Gbps; 2 × 100 Gbps = 200 Gbps; OSR = 2:1`	Uplink capacity from a distribution pair to the core, counting both uplinks as active (routed ECMP).
Usable capacity after failure	`Capacity(n−1) = (n − 1) ÷ n × total uplink capacity`	With two uplinks, a single failure halves upstream capacity — check that the post-failure ratio still meets the requirement.

PRO TIP · OVERSUBSCRIPTION IS A DECISION, NOT A FIXED RATIO

Older guidelines quoted ratios such as 20:1 at access and 4:1 at distribution, from an era of light, bursty office traffic. Wi-Fi 6/7 access points, video and large file sync change the picture. Size from measured peaks and the applications in each block, check the ratio *after a single uplink failure*, and pick uplink speeds that keep the post-failure ratio acceptable.

INSIGHT · POWER IS A CAMPUS DESIGN CONSTRAINT

Wi-Fi 7 access points, cameras and building systems draw more Power over Ethernet than older designs assumed. Check the PoE budget of each access switch with every port at its planned class, the power and cooling of each wiring closet, and whether the switch keeps powering devices during a supervisor restart. Energy-efficient Ethernet and powering down unused ports reduce running cost without touching resilience.

SCENARIO · DECIDE · A GROWING HEADQUARTERS

A company's headquarters is one 8-floor building with 14 access stacks and a collapsed core. The board has approved a second building on the same site within two years, and a third is likely later. The current core pair has no spare 100 Gbps ports. What is the best design direction for the refresh now under way?

- (A) Add a core pair now and treat each building as a distribution block
- (B) Keep the collapsed core and connect the new building's access stacks directly to it
- (C) Build a full mesh of links between the two buildings' access stacks
- (D) Defer any change until the second building is complete and then reassess

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Campus block design sheet**

Record the sizing and resilience decisions for each campus block so blocks stay consistent and capacity is planned.

BLOCK	ACCESS STACKS / 2.5G PORTS	UPLINKS PER STACK	ACCESS OSR (NORMAL / N-1)	DISTRIBUTION UPLINKS TO CORE	L2/L3 BOUNDARY	NOTES
Building A floors 1-4	*8 stacks / 384 ports*	*2 × 25G*	*2.4:1 / 4.8:1*	*2 × 100G per distribution switch*	*Distribution*	*Wi-Fi 7 APs on floors 2-3*
Building B (new)	*6 stacks / 288 ports*	*2 × 25G*	*2.4:1 / 4.8:1*	*2 × 100G per distribution switch*	*Access (routed)*	*Reserve 2 stack positions*

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BLOCK	ACCESS STACKS / 2.5G PORTS	UPLINKS PER STACK	ACCESS OSR (NORMAL / N-1)	DISTRIBUTION UPLINKS TO CORE	L2/L3 BOUNDARY	NOTES

Key terms

Access layer
The campus layer that connects endpoints and wireless access points and applies edge policy such as authentication and QoS marking.

Core layer
The high-speed transport layer that interconnects distribution blocks with minimal policy and state.

Campus block
A repeatable unit of access and distribution switches serving a building or zone.

Trust boundary
The point at which the network starts trusting QoS markings, usually the access port or the endpoint.

Distribution layer
The layer that aggregates access switches into a block, often hosts the L2/L3 boundary and summarises routing information.

Collapsed core
A two-tier design where core and distribution functions run on the same pair of switches.

Oversubscription ratio
The ratio of total downstream port capacity to total upstream capacity at an aggregation point.

Uplink
A link from a lower layer to a higher one, such as access to distribution, whose capacity sets the oversubscription ratio.

REFLECT

Count the distribution blocks in a campus you know. Would a collapsed core still serve it if the estate doubled? Where would the first bottleneck appear — ports, links, or change risk?

Key takeaways

- Each layer solves a problem: access attaches and enforces, distribution aggregates and summarises, the core joins blocks
- A collapsed core suits one building and a small team; a separate core appears when blocks multiply or buildings are added
- State oversubscription for normal and single-uplink-failure conditions
- PoE budgets and closet power and cooling are part of access-layer sizing

KNOWLEDGE CHECK

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1. What is the PRIMARY reason to add a dedicated core layer to a campus?

   (A) To provide Power over Ethernet to high-density wireless areas

   (B) To host the first-hop gateway for every endpoint subnet

   (C) To apply detailed security policy to traffic between floors

   (D) To avoid a growing mesh of links between distribution blocks

2. An access switch has 48 × 1 Gbps ports and 2 × 10 Gbps uplinks. What is its oversubscription ratio, and what does it become after one uplink fails?

   (A) 4.8:1 normally and 2.4:1 after the failure

   (B) 2.4:1 normally and 4.8:1 after the failure

   (C) 2.4:1 normally and 2.4:1 after the failure

   (D) 1.2:1 normally and 2.4:1 after the failure

3. For which campus is a collapsed core MOST appropriate?

   (A) A university with twelve buildings on separate fibre runs

   (B) A hospital campus adding four new clinical wings, each with its own riser, over three years

   (C) A single building with eight access stacks and a small operations team

   (D) A campus with many distribution blocks and heavy east-west traffic

4. Why should campus oversubscription be checked after a single uplink failure?

   (A) Because users experience the reduced capacity during the failure

   (B) Because oversubscription is only defined when an uplink is down

   (C) Because failed uplinks increase the downstream port capacity

   (D) Because most standards require a 1:1 ratio during failures

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MODULE 2 · CAMPUS · LESSON 2 · 33 MIN

The L2/L3 Boundary and Campus Fabrics

Where routing starts decides your failure domains, mobility and segmentation

- Compare layer 2 access, routed access and overlay fabric designs
- Explain the failure-domain and mobility consequences of each boundary choice
- Decide when a campus fabric's complexity is justified

Where layer 2 ends and layer 3 begins is one of the most consequential campus decisions. It determines how far broadcasts and spanning-tree events travel, whether a VLAN can exist on more than one access switch, how fast the network recovers from a failure, and where segmentation policy can be enforced. There are three broad answers: route at distribution, route at access, or build an overlay fabric that routes everywhere but still offers layer 2 semantics where needed.

Three boundary choices

Layer 2 access, L3 at distribution

- VLANs can span access switches in a block
- Needs spanning tree and first-hop redundancy
- Simple access switches
- Larger failure domain; loops are possible
- Familiar to most teams

Routed access

- L3 point-to-point uplinks from every access switch
- No spanning tree on uplinks; ECMP to both distribution switches
- Fast, deterministic routed convergence
- Each VLAN confined to one access switch or stack
- More subnets and routing adjacencies to manage

Overlay fabric

- Routed underlay with an encapsulating overlay
- Anycast gateway on every edge switch
- Subnets can appear anywhere without spanning tree
- Built-in macro and micro segmentation
- Highest complexity; needs tooling and skills

WATCH OUT · THE COST OF SPANNING VLANS

A VLAN that spans many switches is a single broadcast and failure domain. A loop, a misbehaving endpoint or a spanning-tree recalculation anywhere in that domain affects everything in it. If the requirement behind a spanning VLAN is wireless roaming or a legacy application, look for a narrower answer — controller-based roaming, or a dedicated small segment — before stretching layer 2 across a block.

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Routed access uplinks and a local gateway

ACC-3F · IOS-style CLI

```
! Access switch: routed uplinks, local gateway for the floor subnet
interface Ethernet1/49
 description Uplink to DIST-A
 no switchport
 ip address 10.10.0.1 255.255.255.254
 ip ospf network point-to-point
 ip ospf 1 area 10
interface Ethernet1/50
 description Uplink to DIST-B
 no switchport
 ip address 10.10.0.3 255.255.255.254
 ip ospf network point-to-point
 ip ospf 1 area 10
interface Vlan110
 description Floor 3 staff
 ip address 10.110.3.1 255.255.255.0
 ip ospf 1 area 10
router ospf 1
 passive-interface default
 no passive-interface Ethernet1/49
 no passive-interface Ethernet1/50
```

Each uplink is a /31 point-to-point link, so there is no spanning tree and no first-hop redundancy protocol on the uplinks; both are used through equal-cost paths. The trade-off: VLAN 110 lives only on this switch.

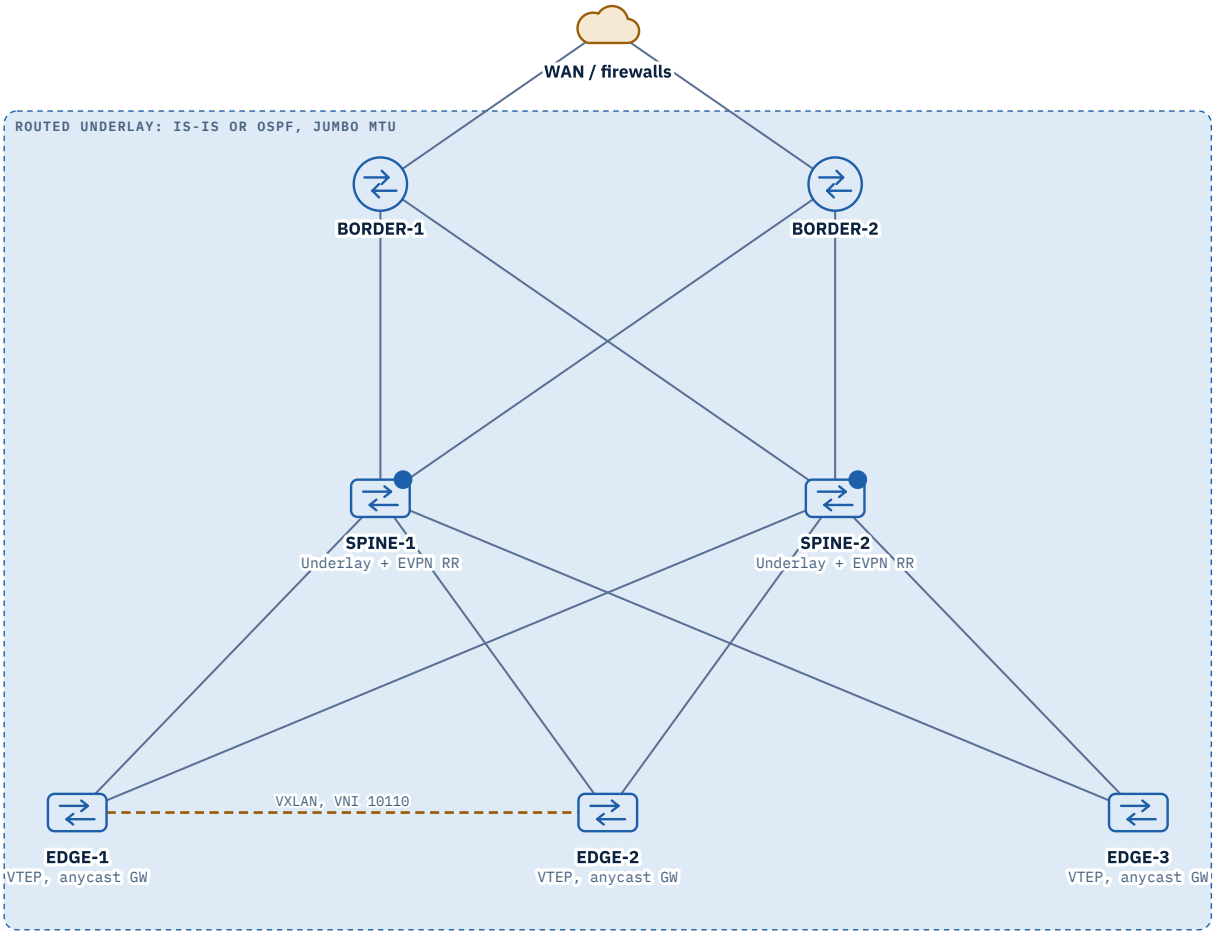
Campus overlay fabric building blocks

Policy	`Identity-based group assignment` `Group-to-group access rules` `Central policy definition and automation`
Overlay control plane	`Endpoint location database or BGP EVPN` `Anycast gateway on every edge` `Virtual networks (VRFs) for macro-segmentation`
Overlay data plane	`VXLAN or similar encapsulation` `Group tag carried in the header` `Border nodes to the rest of the network`
Underlay	`Routed point-to-point links` `Link-state IGP for loopback reachability` `ECMP, BFD, jumbo MTU for encapsulation overhead`

Most new campus fabrics in 2026 use a standards-based **BGP EVPN** control plane with **VXLAN** encapsulation, the same family of protocols used in data centres; some use a location-mapping control plane (such as LISP) instead. Either way the shape is the same: a routed underlay, edge switches acting as tunnel endpoints with an anycast gateway, and border nodes that hand traffic to the rest of the network.

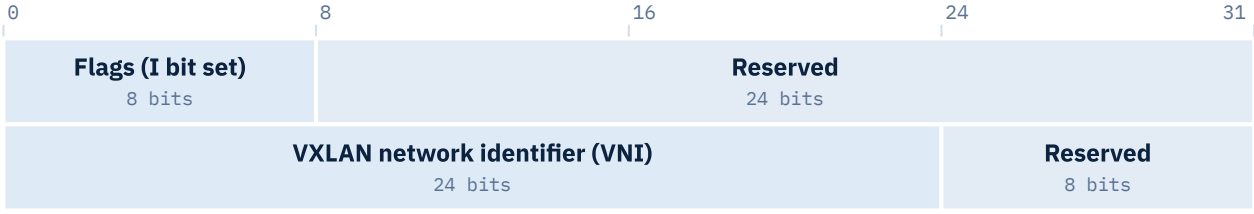
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An EVPN-VXLAN campus fabric



Edge switches are VXLAN tunnel endpoints with the same anycast gateway, so a subnet can appear on any floor without spanning tree. The spines carry the underlay and act as route reflectors for the EVPN overlay; borders join the fabric to the WAN and firewalls.

The VXLAN header



VXLAN adds 8 bytes of its own; with the outer Ethernet (14), IPv4 (20) and UDP (8) headers the total overhead is 50 bytes, or 70 with an IPv6 underlay. The 24-bit VNI allows about 16 million segments, against 4,094 usable VLAN IDs.

When is a campus fabric justified?

REQUIREMENT	WITHOUT A FABRIC	WITH A FABRIC
Same subnet available anywhere on a large campus	Large spanned VLANs or many subnets with re-addressing	Native: anycast gateway and overlay

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REQUIREMENT	WITHOUT A FABRIC	WITH A FABRIC
Consistent group-based policy at scale	ACLs and firewall rules per subnet, hard to keep in sync	Group tags with central policy
Many isolated tenants or departments	VRF-lite configured hop by hop	Virtual networks provisioned centrally
Small site, few segments, small team	Simple routed or L2 access is enough	Usually more complexity than value
Strong automation and telemetry practice	Manual operation remains viable	Fabric tooling can be fully exploited

INSIGHT · FABRICS MOVE COMPLEXITY, THEY DO NOT REMOVE IT

A fabric hides per-switch configuration behind a controller or automation, but adds an underlay, an overlay, a control plane, encapsulation MTU, border handoffs and a dependency on the management system. The design is sound when the requirements — mobility, policy scale, tenant count — are large enough that this concentrated complexity is cheaper than the scattered complexity it replaces.

SCENARIO · DECIDE · CHOOSING THE BOUNDARY

A hospital is refreshing a 6-floor clinical building. Requirements: sub-second recovery from any uplink failure, no spanning-tree loops in clinical areas, and wireless medical devices that must roam across all floors without changing IP address. Wireless is controller-based, with client traffic tunnelled to the controller. The team is experienced with OSPF but has no fabric experience. Which design best fits?

- (A) Layer 2 access with VLANs spanning all six floors, protected by rapid spanning tree
- (B) A campus overlay fabric so wireless subnets can exist on every floor
- (C) Routed access with a separate wired VLAN per floor for the medical devices
- (D) Routed access with OSPF and BFD; roaming handled by the controller tunnels

Circle your choice. Answer and feedback: module answer key.

Boundary design rules of thumb

- Route as close to the edge as the mobility requirement allows
- If a VLAN must span switches, keep that domain small and loop-protected
- Use routed links between layers wherever possible to gain ECMP and deterministic convergence
- Plan underlay MTU for any encapsulation before choosing a fabric
- Choose a fabric for policy scale and mobility at scale, not for novelty

Key terms

L2/L3 boundary
The point where switched (layer 2) forwarding ends and routing begins, usually where the endpoint's default gateway lives.

Routed access
A campus design in which access switches route, using layer 3 point-to-point uplinks instead of trunked VLANs.

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First-hop redundancy protocol
A protocol, such as VRRP, that lets two or more routers share a virtual gateway address for endpoints.

Anycast gateway
The same gateway IP and MAC address configured on every edge switch so endpoints find their gateway locally wherever they connect.

Underlay
The physical routed network that provides reachability between fabric nodes.

Overlay
A logical network built on top of the underlay using encapsulation, such as VXLAN.

Macro-segmentation
Separation of traffic into isolated routing domains, such as VRFs or virtual networks.

Micro-segmentation
Policy between groups of endpoints within the same routing domain, often based on group tags or identity.

VXLAN
A MAC-in-UDP encapsulation that carries layer 2 frames across a routed network, identifying segments with a 24-bit network identifier.

REFLECT

In your current campus, where is the L2/L3 boundary and why was it put there? Which requirement would have to change for you to move it?

KNOWLEDGE CHECK

5. What is the main trade-off of routed access compared with layer 2 access?

   (A) Uplink failures converge more slowly than with spanning tree
   (B) A VLAN cannot span multiple access switches
   (C) Only one uplink can forward traffic at a time
   (D) Endpoint subnets must be routed by the core layer

6. Why does a VXLAN-based fabric require attention to underlay MTU?

   (A) Encapsulation adds header bytes, so full-size frames need a larger underlay MTU
   (B) VXLAN only operates when every link uses an MTU of exactly 1500 bytes
   (C) The overlay control plane negotiates MTU and fails on mismatches
   (D) Anycast gateways must fragment every packet that exceeds the endpoint's configured MTU

7. Which requirement MOST strongly justifies a campus overlay fabric?

   (A) Connecting a single floor of 200 users with a small operations team
   (B) Providing a default gateway for users that survives the failure of one distribution switch
   (C) Consistent group-based policy and subnet mobility across a very large campus
   (D) Reducing the purchase price of access switches in a small office

8. Wireless clients are tunnelled to a central controller. How does this affect the L2/L3 boundary decision for wired access?

   (A) Wired access must use layer 2 so the controller tunnels can terminate
   (B) A campus overlay fabric becomes mandatory so that the controller's tunnels can be carried
   (C) Every access switch must host the wireless client gateway locally
   (D) Roaming no longer requires spanning campus VLANs, so routed access becomes viable

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9. What is the best description of a campus fabric's effect on complexity?

(A) It removes complexity because switches no longer need configuration

(B) It concentrates complexity into the control plane and tooling

(C) It adds complexity only during the initial deployment phase

(D) It eliminates the need for a routed underlay in the campus

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MODULE 2 · CAMPUS · LESSON 3 · 38 MIN

Data Centre Spine-Leaf Design

Clos fabrics, east-west traffic and the arithmetic of scale

- Explain why spine-leaf replaced tree-based data centre designs
- Size a two-tier fabric: leaves, spines, ports and oversubscription
- Identify the scaling limit of a fabric and when a third stage is needed
- Recognise the distinct requirements of back-end fabrics for GPU and AI clusters

Modern applications talk to each other far more than to users: a single web request can trigger dozens of calls between services, databases and storage. This **east-west** traffic overwhelmed tree-shaped data centre designs built for **north-south** client-server flows, where traffic between racks climbed to an aggregation layer and was limited by spanning tree to a single active path. The spine-leaf (folded Clos) topology answers this with a simple rule: every leaf connects to every spine, and every leaf-to-leaf path is exactly two hops through any spine.

Traditional three-tier DC versus spine-leaf

Three-tier tree

- Access, aggregation and core layers
- Built for north-south client-server traffic
- Spanning tree blocks redundant paths
- Inconsistent latency between racks
- Scale-up: bigger aggregation boxes

Spine-leaf (Clos)

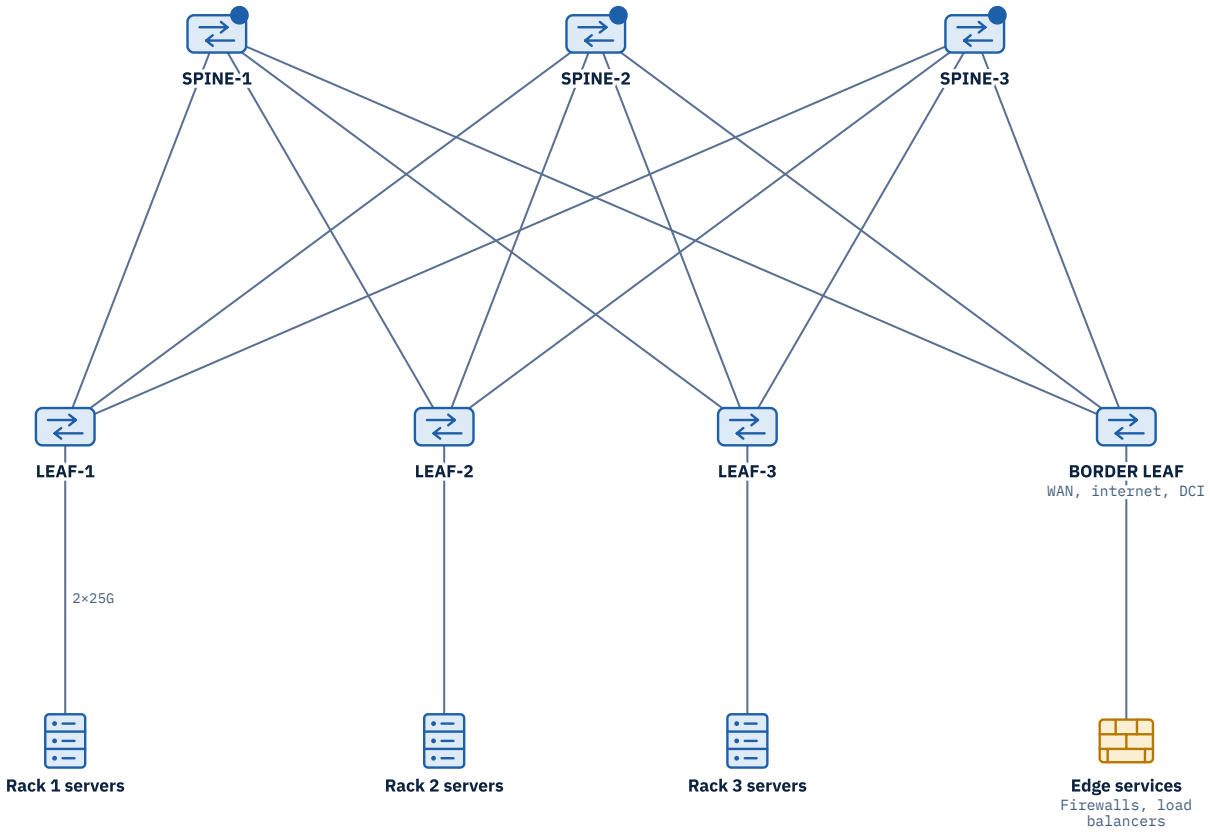
- Two tiers: leaves and spines
- Built for east-west service-to-service traffic
- All links active using routed ECMP
- Predictable, equal hop count between any two leaves
- Scale-out: add spines for bandwidth, leaves for ports

Spine-leaf fabric roles

Border / edge	`Border leaves to WAN, internet and DCI` `Firewalls and load balancers attached` `Route exchange with the rest of the enterprise`
Spine	`Interconnects all leaves` `Pure transport: no endpoints attached` `Often route reflector or route server for the overlay`
Leaf	`Server and storage connections` `Gateway for workloads (anycast)` `Overlay tunnel endpoints`
Underlay protocols	`eBGP or link-state IGP between leaf and spine` `ECMP across all spines` `BFD for fast failure detection`

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A two-tier spine-leaf fabric



Every leaf connects to every spine and leaves never connect to each other, so any two servers on different leaves are exactly two hops apart. Losing one of three spines removes a third of fabric bandwidth.

Fabric sizing arithmetic (two-tier, one link per leaf-spine pair)

MEASURE	FORMULA	WHAT IT TELLS YOU
Maximum leaves	`Leaves(max) = ports per spine`	Each leaf consumes one port on every spine, so the spine's port count caps the number of leaves.
Maximum spines	`Spines(max) = uplink ports per leaf`	Each spine consumes one uplink on every leaf.
Leaf oversubscription	`OSR = (server ports × server speed) ÷ (uplinks × uplink speed)`	Example: 48 × 25 Gbps = 1,200 Gbps down; 8 × 100 Gbps = 800 Gbps up; OSR = 1.5:1.
Fabric server ports	`Ports = leaves × server ports per leaf`	Example: 64-port spines and 48-port leaves give up to 64 × 48 = 3,072 server ports.
Capacity lost per spine failure	`Loss = 1 ÷ number of spines`	With 4 spines a spine failure removes 25% of fabric bandwidth; with 8 spines, 12.5%.

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EXAMPLE · SIZING A FABRIC

Requirement: 2,000 dual-attached servers at 25 Gbps, at most 3:1 oversubscription, tolerate one spine failure at no worse than 4:1. Each server uses one port on each of two leaves, so 4,000 server ports are needed. With 48-port server leaves that is 84 leaves (42 pairs). Each leaf with 4 × 100 Gbps uplinks gives 1,200 ÷ 400 = 3:1, and 4:1 after losing a spine — just inside the limit. With 6 uplinks the ratios become 2:1 and 2.4:1, buying headroom. Spines need at least 84 ports plus border leaves, so 128-port spines (or a third stage) are required.

East-west versus north-south: design consequences

ASPECT	EAST-WEST DOMINATED	NORTH-SOUTH DOMINATED
Traffic path	Leaf → spine → leaf inside the fabric	Leaf → spine → border leaf → WAN or internet
Bottleneck to watch	Leaf uplink oversubscription	Border leaf and edge capacity
Security placement	Distributed or micro-segmentation near workloads	Perimeter firewalls at the border
Growth lever	More spines (bandwidth) and leaves (ports)	Border capacity and WAN/internet circuits

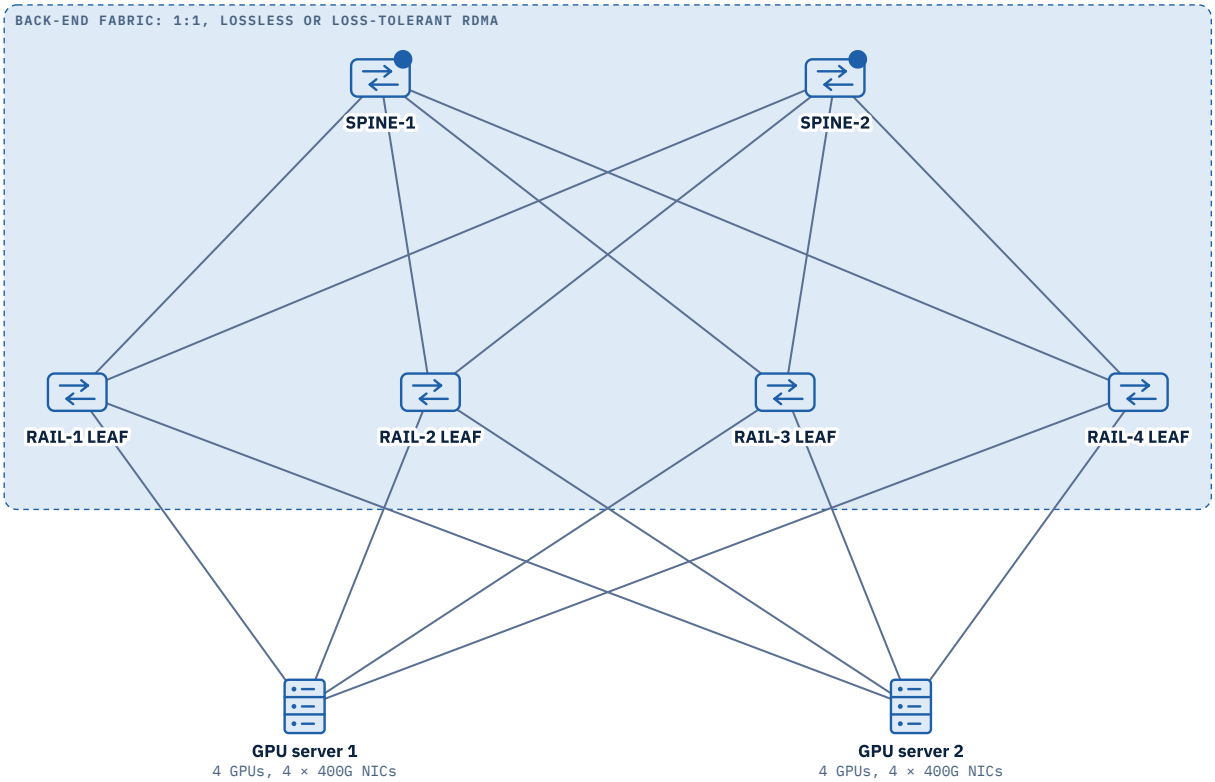
INSIGHT · WHEN TWO TIERS ARE NOT ENOUGH

A two-tier fabric stops growing when the spine runs out of ports. The answer is a three-stage (five-hop) design: multiple spine-leaf pods connected by a super-spine layer. It adds a hop and another oversubscription point between pods, so place workloads that talk heavily to each other in the same pod.

Fabrics for AI and machine-learning clusters are now a distinct design problem. Training jobs synchronise GPUs with huge, bursty, all-to-all transfers, and the slowest flow sets the pace of the whole job, so the **back-end GPU fabric** is usually built non-blocking (1:1), at 400 or 800 Gbps per port, and kept separate from the front-end, storage and management networks. Many designs are **rail-optimised**: GPU number n in every server connects to leaf (rail) n, so most traffic between servers stays one hop away. Ethernet back-ends carry RDMA (RoCEv2), which needs either a lossless configuration (priority flow control with ECN-based congestion control) or newer transports that tolerate loss; a few large, low-entropy flows also defeat simple hash-based ECMP, which is why adaptive routing and packet spraying appear in these fabrics. Industry work such as the Ultra Ethernet specification (version 1.0 published in 2025) targets exactly these needs — check product maturity before relying on it.

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Rail-optimised back-end fabric for a GPU cluster



Simplified to four rails and two servers. Each server's GPU NICs land on different rail leaves, so a collective operation between the same-numbered GPUs crosses only one leaf; traffic between rails climbs to the spines.

WATCH OUT · POWER AND COOLING DECIDE AI FABRIC SCALE

A rack of current GPU servers can draw tens of kilowatts, often needing liquid cooling, and high-speed optics add measurable power per port. In practice the hall's power and cooling, not switch radix, often caps cluster size — capture it as a constraint on the Requirements sheet before sizing the fabric.

CASE STUDY · STOCK EXCHANGE TRADING DATA CENTRE

A regional stock exchange is rebuilding its primary trading data centre. The matching engines must process orders with consistent latency measured in microseconds. Member firms colocate their servers in the same hall and require demonstrably equal access to the matching engines. Market data is distributed by multicast and must survive the loss of any single network device without gaps. Back-office, surveillance, settlement and corporate systems also run in the building and generate heavy east-west traffic. Regulators require time-stamping of every order with traceable precision and full separation of trading and corporate networks. The architecture team proposes a single large EVPN-VXLAN fabric for everything.

1. Does a single overlay fabric meet the trading requirements? Why or why not?

2. How should market-data resilience be designed?

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3. How would you meet the fairness requirement for colocated members?

4. Where does a spine-leaf fabric belong in this design?

Model answer: module answer key.

SCENARIO · DECIDE · FABRIC OVERSUBSCRIPTION FOR AN ANALYTICS CLUSTER

A bank is adding a 40-rack analytics cluster that shuffles large datasets between servers continuously. Each rack has 32 servers at 100 Gbps. Leaves have 32 × 100 Gbps server ports and up to 8 × 400 Gbps uplinks. Budget allows either 4 or 8 spines. Which option best fits the workload?

- (A) 4 spines with 4 × 400 Gbps uplinks per leaf, giving 2:1 oversubscription to save cost
- (B) 8 spines, 8 × 400 Gbps uplinks per leaf: 1:1, non-blocking
- (C) 4 spines with 8 × 400 Gbps uplinks, two links from each leaf to each spine
- (D) Connect the cluster through the existing general-purpose fabric's border leaves

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Data centre fabric sizing sheet**

Size a spine-leaf fabric and record the assumptions so growth and failure behaviour are explicit.

POD / TIER	LEAVES (SERVER / BORDER)	SERVER PORTS × SPEED	UPLINKS × SPEED PER LEAF	OSR NORMAL / ONE SPINE DOWN	SPINES × PORTS	GROWTH HEADROOM
Pod 1 general compute	*40 / 2*	*48 × 25G*	*6 × 100G*	*2:1 / 2.4:1*	*6 × 64*	*22 spine ports free*
Pod 2 analytics	*40 / 0*	*32 × 100G*	*8 × 400G*	*1:1 / 1.14:1*	*8 × 64*	*24 spine ports free*

Key terms

Spine-leaf
A two-tier folded Clos topology in which every leaf connects to every spine, giving equal-cost paths between any two leaves.

Clos network
A multistage switching topology that builds large, non-blocking capacity from many smaller switches.

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East-west traffic
Traffic between servers, services or storage inside the data centre.

ECMP
Equal-cost multipath: forwarding flows across several equal-cost routes at once, typically by hashing packet headers.

Super-spine
An additional tier that interconnects multiple spine-leaf pods to scale beyond one spine's port count.

North-south traffic
Traffic entering or leaving the data centre, to users, the WAN or the internet.

Border leaf
A leaf that connects the fabric to external networks and services rather than to servers.

Non-blocking
A design whose uplink capacity equals or exceeds downlink capacity (1:1), so any traffic pattern can be carried at full rate in theory.

REFLECT

For a data centre you know, estimate the leaf oversubscription and the bandwidth lost when one spine fails. Is that loss acceptable for its most demanding application?

Key takeaways

- Spine-leaf gives equal two-hop paths and scales out: spines add bandwidth, leaves add ports
- Spine port count caps leaves; leaf uplinks cap spines; a super-spine joins pods when ports run out
- State oversubscription normally and after a spine failure, and match it to the workload
- GPU back-end fabrics are separate, non-blocking and often rail-optimised, with power and cooling as the real limit

KNOWLEDGE CHECK

10. In a two-tier spine-leaf fabric with one link between each leaf and each spine, what limits the maximum number of leaves?

    (A) The number of uplink ports on each leaf
    (B) The number of server ports on each leaf
    (C) The number of routing adjacencies per spine
    (D) The number of ports on each spine

11. A leaf has 48 × 25 Gbps server ports and 6 × 100 Gbps uplinks. What is its oversubscription ratio?

    (A) 1.5:1
    (B) 3:1
    (C) 2:1
    (D) 4:1

12. Why did spine-leaf replace tree-based designs in most data centres?

    (A) East-west traffic needed many active, equal-length paths
    (B) North-south traffic to users grew faster than server traffic
    (C) Spanning tree could not run on high-speed data centre links
    (D) Tree designs could not connect firewalls and load balancers

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13. A fabric has 4 spines. Moving to 8 spines of the same speed, with leaf uplinks doubled to match, mainly improves what?

    (A) The number of server-facing ports available on each individual leaf switch

    (B) Fabric bandwidth and the share of capacity lost per spine failure

    (C) The maximum number of leaves the fabric can connect

    (D) The latency between two servers on the same leaf

14. When does a fabric need a super-spine tier?

    (A) When the fabric needs more than two equal-cost paths

    (B) When servers require 100 Gbps instead of 25 Gbps

    (C) When the required leaves exceed the spine's port count

    (D) When north-south traffic exceeds east-west traffic

15. A team is designing the back-end network for a GPU training cluster. Which design choice BEST fits the workload?

    (A) Share the general-purpose fabric at 3:1 oversubscription to reuse existing spines

    (B) Route GPU traffic through the border leaves so firewalls can inspect every flow

    (C) A separate non-blocking fabric, often rail-optimised, tuned for RDMA traffic

    (D) Stretch one VLAN across all GPU racks so the servers share a broadcast domain

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MODULE 2 · CAMPUS · LESSON 4 · 33 MIN

Data Centre Interconnect and Choosing a Design

Connect sites without merging their failure domains

- Compare data centre interconnect options and their failure-domain impact
- Calculate the latency cost of distance and judge synchronous replication limits
- Select a campus or data centre design from requirements using a structured comparison

Organisations run more than one data centre for resilience, capacity or proximity. The interconnect between them (DCI) is where many designs quietly undermine their own goals: a second site built for resilience is linked to the first in a way that lets one failure — a broadcast storm, a control-plane bug, a split brain — take down both. The design principle is simple to state and hard to keep: **connect the sites, but keep them separate failure domains**.

Data centre interconnect options

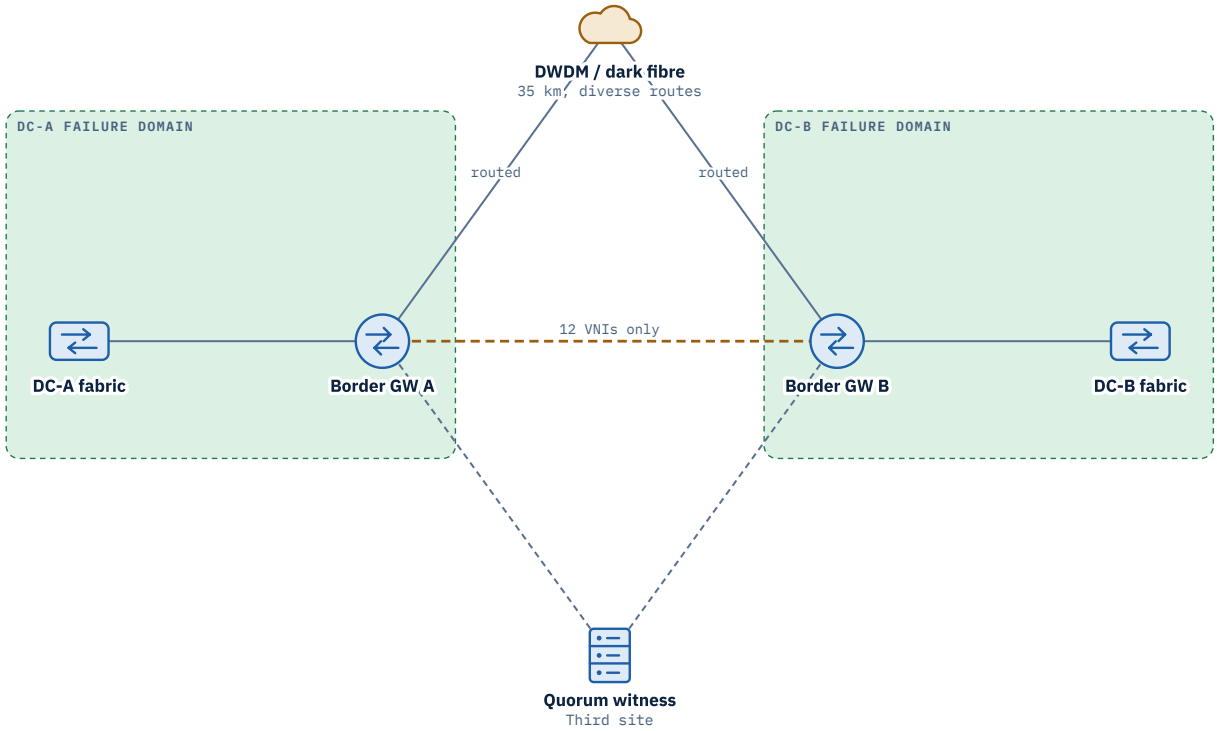
OPTION	WHAT IT PROVIDES	STRENGTHS	WATCH OUT FOR
Layer 3 routed DCI	IP routing between sites over WAN, MPLS or fibre	Independent failure domains; simple; scales	Applications must tolerate different subnets per site
Overlay multi-site (e.g. EVPN-VXLAN with border gateways)	Selective layer 2 extension over a routed core	Stretches only chosen segments; BUM control; local gateways	Control-plane and operational complexity
Native layer 2 stretch (VLAN trunk, pseudowire)	Same VLANs at both sites	Simple to understand; supports any L2 application	Shared broadcast and failure domain; loop and split-brain risk
Dark fibre / DWDM	Physical capacity carrying any of the above	Very high bandwidth; low latency; protocol-agnostic	Cost, distance limits, need for diverse routes

WATCH OUT · WHY STRETCHED LAYER 2 IS RISKY

Extending a VLAN across sites joins their broadcast domains. A loop or storm at one site reaches the other; a failure of the inter-site link can leave both sites believing they own the same gateway or cluster resource (split brain); and traffic can trombone across the DCI when a workload moves but its gateway or storage does not. Stretch only what a validated requirement demands, and use an overlay that confines flooding and localises gateways.

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Routed DCI with selective layer 2 extension



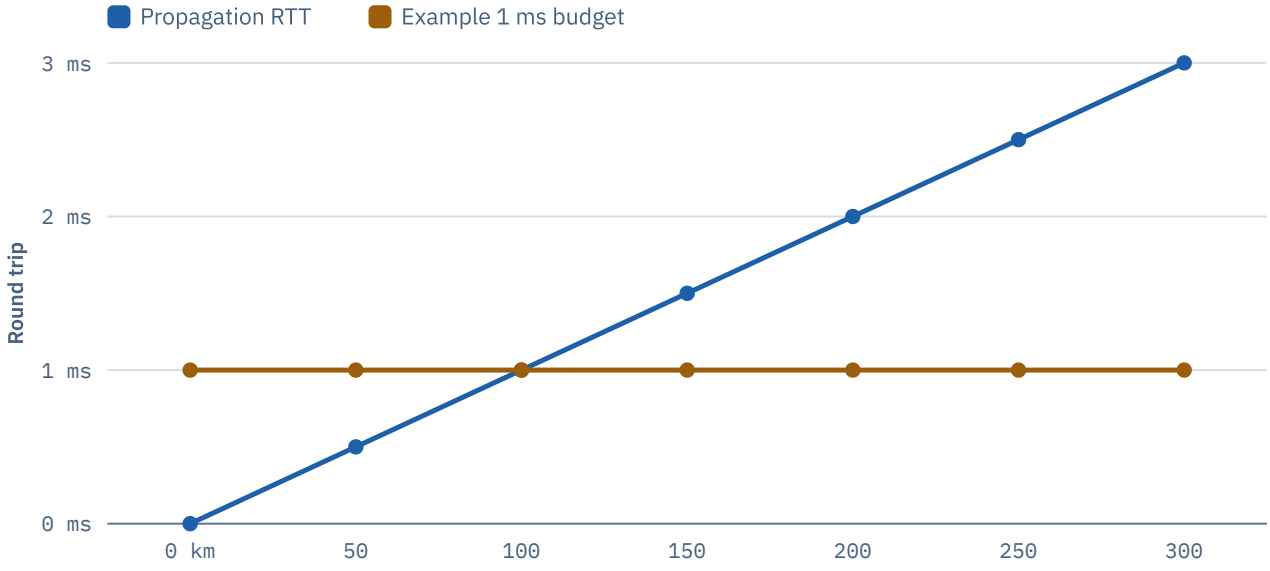
The sites stay separate failure domains joined by routing; only the validated segments ride an EVPN multi-site tunnel between border gateways, each site keeps its own anycast gateway, and a witness at a third location breaks ties for clusters.

Distance and latency

MEASURE	FORMULA	WHAT IT TELLS YOU
Propagation delay in fibre	`≈ 5 µs per km, one way`	Light travels at roughly two-thirds of its vacuum speed in glass, about 200,000 km per second.
Round-trip time from distance	`RTT ≈ 2 × route-km × 5 µs + equipment delay`	A 100 km fibre route gives about 1 ms RTT from propagation alone; the real fibre route is usually longer than the straight line.
Synchronous write penalty	`Write latency ≥ local write + RTT to the remote copy`	Every synchronous write waits for the remote acknowledgement, so the application's write-latency tolerance sets the maximum distance.

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Round-trip propagation delay against fibre route length



Computed at about 5 µs per km one way, before any equipment delay. With an illustrative 1 ms synchronous-write budget, propagation alone uses the whole budget at 100 km of route, so the practical limit is lower once equipment delay is added.

PRO TIP · LET THE APPLICATION SET THE DISTANCE

Synchronous replication and active-active clusters are limited by the latency the application tolerates, not by what the network can carry. Ask the storage and application owners for their write-latency budget, convert it to route kilometres, and compare that with the fibre route — not the map distance. If the second site is too far, the design is asynchronous replication with a stated recovery point objective.

Active-active versus active-standby data centres

Active-active

- Both sites serve production simultaneously
- Capacity in use, not idle
- Needs global load balancing or anycast for ingress
- Risk of tromboning and state-synchronisation issues
- Demanding on DCI latency and application design

Active-standby

- One site serves; the other is ready to take over
- Simpler routing: one preferred entry point
- Spare capacity sits idle until failover
- Failover must be tested regularly to be trusted
- Tolerates longer distances with asynchronous replication

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SCENARIO · DECIDE · A REQUEST TO STRETCH

An insurer has two data centres 35 km apart on diverse fibre. The virtualisation team wants all 400 VLANs stretched so that any VM can move between sites. Investigation shows only 12 VLANs host clustered applications that genuinely need the same subnet at both sites; the rest are ordinary workloads restored from backup in a disaster. Which DCI design best fits?

(A) Stretch all 400 VLANs over the fibre so the virtualisation team has full flexibility

(B) Pure routed DCI and re-engineer the 12 clustered applications before migration

(C) Routed DCI, plus an overlay stretching only the 12 segments with local gateways

(D) A layer 2 pseudowire per VLAN for the 12 clustered segments only

Circle your choice. Answer and feedback: module answer key.

Choosing between designs: a structured comparison

QUESTION	POINTS TOWARD SIMPLER DESIGN	POINTS TOWARD MORE ADVANCED DESIGN
How many blocks, pods or sites?	One or two, little growth	Many, or rapid repeatable growth
Does policy need to follow users or workloads?	Policy by subnet is enough	Group or identity policy at scale
Do applications need the same subnet in many places?	No, or only a few	Widespread, validated mobility needs
How mature are operations and automation?	Manual, small team	Automated pipelines and telemetry in place
What is the cost of a shared failure?	Low; outages tolerable	High; must keep domains separate

Campus and data centre design review checklist

☐ Campus model justified by block count, buildings and growth plan

☐ L2/L3 boundary chosen from mobility and failure-domain requirements

☐ Oversubscription stated for normal and single-failure conditions

☐ Fabric spine and leaf port counts allow the planned growth

☐ Underlay MTU accounts for any overlay encapsulation

☐ DCI keeps each site a separate failure domain unless a requirement says otherwise

☐ Any stretched segment is listed with its owner and justification

☐ Synchronous replication distance checked against the application's latency budget

☐ Border and north-south capacity sized separately from east-west

Key terms

Data centre interconnect (DCI)
The network connecting two or more data centres for replication, workload mobility and resilience.

Split brain
A failure in which two sites or cluster members each believe they own the same resource after losing contact with each other.

Traffic tromboning
Inefficient traffic that crosses the DCI and back because a workload and its gateway, service or storage are at different sites.

BUM traffic
Broadcast, unknown-unicast and multicast traffic that must be flooded in a layer 2 segment.

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Synchronous replication
Replication in which a write completes only after the remote copy acknowledges it, so distance adds directly to write latency.

Recovery point objective (RPO)
The maximum acceptable amount of data loss, measured in time, after a failure.

Active-active
A design in which multiple sites serve production traffic simultaneously.

Active-standby
A design in which one site serves production while another is held ready to take over after failover.

REFLECT

If your organisation's two data centres lost their interconnect right now, what would each site believe about the other? Is there any resource both would try to own?

Key takeaways

- Connect data centres but keep them separate failure domains
- Stretch layer 2 only for validated needs, over an overlay with flooding control and local gateways
- Fibre adds about 1 ms of round trip per 100 km of route; the application's write budget sets the distance for synchronous replication
- Active-active needs ingress steering and symmetric state handling; active-standby needs regular failover tests

KNOWLEDGE CHECK

16. What is the main design risk of stretching many VLANs between two data centres?

(A) Routing protocols cannot run over the interconnect once VLANs are stretched
(B) Stretched VLANs prevent the use of any dark fibre between sites
(C) Storage replication must switch from synchronous to asynchronous
(D) The sites become one failure domain for loops, storms and split brain

17. Two data centres are connected by a 120 km fibre route. Approximately what round-trip time does propagation alone add?

(A) About 1.2 ms
(B) About 0.12 ms
(C) About 0.6 ms
(D) About 12 ms

18. Which factor should set the maximum distance for synchronous replication between sites?

(A) The maximum distance supported by the routing protocol
(B) The number of VLANs that need to be stretched
(C) The oversubscription ratio of the leaf uplinks
(D) The application's tolerance for added write latency

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19. In an active-active data centre design, what problem do localised (anycast) gateways at each site help to prevent?

(A) Traffic crossing the interconnect to reach a gateway at the other site

(B) Duplicate IP addresses appearing between workloads hosted at the two different sites

(C) Loss of synchronous replication during a DCI failure

(D) The need for any ingress load balancing between sites

☐ Lesson complete · Date _____ · Knowledge check score ____ / 4

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MODULE 2 · CAMPUS

Answer key

Knowledge checks

1. D The core turns an n-squared mesh of distribution interconnections into a star and isolates blocks. It should carry minimal policy; gateways and PoE belong at lower layers.

2. B 48 Gbps ÷ 20 Gbps = 2.4:1. With one uplink left, 48 ÷ 10 = 4.8:1. Designs should check the post-failure ratio, not just the normal one.

3. C Collapsed core suits small, single-building sites where the cost and complexity of a separate core are not justified. The other cases have many blocks or planned growth that favour three tiers.

4. A Resilience is not only about connectivity; the surviving capacity must still meet the requirement. With two uplinks, a failure halves upstream capacity and doubles the ratio.

5. B Routed access confines each subnet to one access switch or stack. In return it gains ECMP across both uplinks and fast routed convergence without spanning tree.

6. A VXLAN adds roughly 50 bytes of headers. Without a larger underlay MTU, full-size endpoint frames would be fragmented or dropped.

7. C Fabrics pay off where policy scale and mobility at scale are required. Gateway redundancy and small sites are solved more simply without one.

8. D Tunnelling anchors client addresses at the controller, decoupling wireless mobility from the wired topology. The wired access layer can then be routed.

9. B The fabric automates and hides per-switch configuration but adds an underlay, overlay, control plane and management dependency. That trade only pays at sufficient scale.

10. D Each leaf uses one port on every spine, so spine port count caps leaf count. Leaf uplink count caps the number of spines.

11. C Downstream is 48 × 25 = 1,200 Gbps; upstream is 6 × 100 = 600 Gbps; 1,200 ÷ 600 = 2:1.

12. A Service-to-service traffic needs high bisection bandwidth and predictable latency. Spine-leaf provides many active ECMP paths with an equal hop count.

13. B More spines add bandwidth and reduce each spine's share of capacity from 25% to 12.5%. Leaf count is limited by spine ports, and same-leaf traffic never crosses a spine.

14. C A super-spine connects multiple pods once a single spine layer runs out of ports. It adds a hop and an oversubscription point between pods.

15. C Training traffic is bursty, all-to-all and paced by the slowest flow, so a dedicated non-blocking back-end fabric suits it. Oversubscription, firewall hairpins and large layer 2 domains all slow or endanger the job.

16. D Layer 2 extension joins broadcast and failure domains, so a fault at one site can reach the other. Routing and replication can still work over a stretched design.

17. A At roughly 5 µs per km one way, 120 km is 0.6 ms one way and about 1.2 ms round trip, before equipment delay.

18. D Each synchronous write waits at least one round trip, so the application's write-latency budget determines how far apart the copies can be.

19. A With a local gateway at each site, a moved workload routes out locally instead of tromboning across the DCI to a remote gateway.

Scenarios & case studies

> **Scenario · A growing headquarters**
> **A** is best. Correct. Growth to multiple buildings is a stated plan, not speculation. A core turns each new building into a repeatable block, isolates change, and avoids a later disruptive re-architecture.

> **Scenario · Choosing the boundary**
> **D** is best. Correct. Routed access gives fast, loop-free recovery. Because wireless clients are tunnelled to the controller, their addresses are anchored there and roaming does not depend on campus VLANs spanning floors.

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Case study · Stock exchange trading data centre

Single fabric. It would serve back-office and corporate workloads well, but the trading path needs the fewest possible hops, no encapsulation or control-plane dependency in the order path, and strict separation. A dedicated, minimal trading network — few hops, low-latency switching, simple routing — is a better fit, physically separate from the corporate fabric.

Market-data resilience. Use two independent distribution networks (an A feed and a B feed) carrying the same multicast data over separate devices and paths, so receivers can arbitrate between them and a single failure causes no gap. Fast failover inside one network cannot guarantee zero loss; duplication across independent paths can.

Fairness. Give every member the same connection type, the same number of hops and equal cable lengths to the matching-engine network, and publish the method so it can be audited. Precise time synchronisation (for example PTP, IEEE 1588) supports traceable time-stamping.

Spine-leaf placement. A spine-leaf (possibly EVPN-VXLAN) fabric is the right choice for back-office, surveillance analytics and corporate workloads where east-west scale and flexibility matter, connected to the trading environment only through controlled, inspected interfaces.

Scenario · Fabric oversubscription for an analytics cluster

B is best. Correct. A continuous all-to-all shuffle is the case where non-blocking design is justified. Eight spines also limit the bandwidth lost on a spine failure to 12.5%.

Scenario · A request to stretch

C is best. Correct. It meets the validated need while keeping most of the estate in separate failure domains. Localised gateways limit tromboning, and flooding controls stop one site's storms reaching the other.

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03

MODULE 3 · WAN

WAN, Internet Edge & Provider Design

Transports, overlays, edges and the routing that ties them to providers

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Match WAN transports to site requirements using delay, bandwidth and diversity budgets
- Make SD-WAN topology, tiering, breakout and control-plane resilience decisions
- Design a layered internet edge, DMZ and upstream DDoS strategy
- Engineer BGP multihoming and provider connectivity without loops or accidental transit

LESSONS

1. WAN Transport Options and Budgets · 33 min
2. SD-WAN and WAN Topology Design · 38 min
3. Internet Edge and DMZ Design · 33 min
4. BGP Multihoming and Provider Connectivity · 38 min

MODULE 3 · WAN · LESSON 1 · 33 MIN

WAN Transport Options and Budgets

Match each site to transports by SLA, latency, cost and diversity

- Compare private, public and wireless WAN transports on the attributes that matter for design
- Build delay and bandwidth budgets for latency-sensitive applications
- Explain why transport diversity must be physical, not just contractual

The **WAN sheet** of the ZCDE Design Blueprint starts with transport. Every transport is a bundle of trade-offs: guaranteed performance versus cost, speed of delivery versus control, reach versus latency. The expert does not pick one transport for the whole estate; they match each site class to a combination of transports whose weaknesses do not overlap, and then prove with a budget that the critical applications will work over the result.

WAN transport characteristics

TRANSPORT	PERFORMANCE ASSURANCE	TYPICAL LATENCY PROFILE	COST AND DELIVERY	BEST USE
Private MPLS L3VPN	End-to-end SLA, QoS classes honoured	Low and stable	Higher cost; weeks to months to deliver	Sites with strict real-time or regulatory needs
Carrier Ethernet / L2 service	SLA; customer controls routing	Low and stable in metro areas	Moderate to high; limited reach	Metro DC and campus interconnect
Dedicated internet access	SLA to provider edge only	Good but internet path varies	Moderate; symmetric bandwidth	Hubs, data centres, larger branches
Business broadband	Best effort	Variable; contention at peak	Low cost; fast to deliver; often asymmetric	Branch primary or secondary under an overlay
4G / 5G cellular	Best effort (some slicing offers emerging)	Variable; depends on signal and load	Fast to deploy; data plans and coverage vary	Backup, temporary and mobile sites
5G fixed wireless access	Best effort; some operators offer business SLAs	Variable; shared cell capacity at peak	Fast to deploy; no trenching	Branch primary where fixed lines are poor, or a diverse secondary
LEO satellite	Best effort	Typically tens of milliseconds	Fast to deploy; needs clear sky view	Remote sites, diverse backup
GEO satellite	Varies by service	Roughly 500–700 ms round trip	Wide reach; limited bandwidth	Very remote sites, last-resort backup

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Private versus public transport

Private (MPLS, carrier Ethernet)

- Performance and QoS guaranteed by contract
- Traffic does not cross the public internet
- Provider involved in routing (L3VPN)
- Long lead times and contract lock-in
- Higher cost per megabit

Public (internet, cellular, satellite)

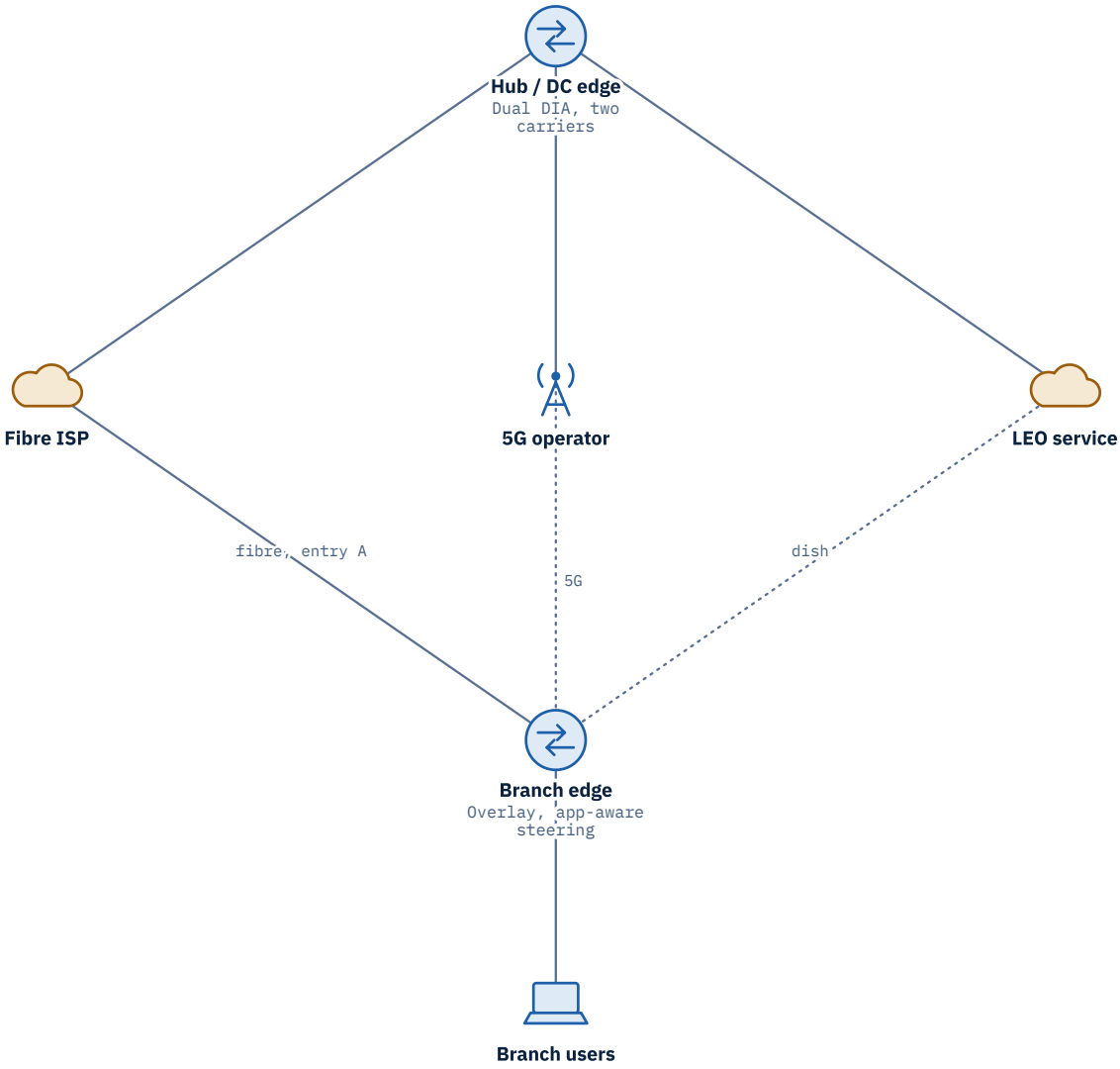
- Best effort beyond the provider edge
- Requires encryption and an overlay for privacy and steering
- Customer controls the overlay end to end
- Quick to order and change
- Much lower cost per megabit

WATCH OUT · DIVERSITY YOU CANNOT SEE

Two circuits from two carriers can still share a duct under the same street, the same building entry, or the same upstream exchange. Contractual diversity is not physical diversity. Ask for route maps, require separate building entries where the requirement justifies it, and combine wired with wireless transports to make a shared physical failure much less likely.

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A branch with three physically independent transports



Each path leaves the building by a different medium and reaches the hub through a different provider network, so no single duct, exchange or operator failure isolates the site. The overlay treats all three as interchangeable transport.

WAN budget arithmetic

MEASURE	FORMULA	WHAT IT TELLS YOU
One-way delay budget	`D = propagation + serialisation + queuing + processing (+ jitter buffer)`	Add every component on the path and compare with the application's budget; a widely used planning target for interactive voice is no more than 150 ms one way.
Serialisation delay	`t = frame size in bits ÷ link rate`	A 1,500-byte frame takes 1.2 ms to clock onto a 10 Mbps link but 12 µs on 1 Gbps — significant only on slow links.
Bandwidth-delay product	`BDP = bandwidth × RTT`	The data in flight needed to fill a path: 1 Gbps × 80 ms = 80 Mbit (10 MB). A sender's window smaller than this cannot use the full link.

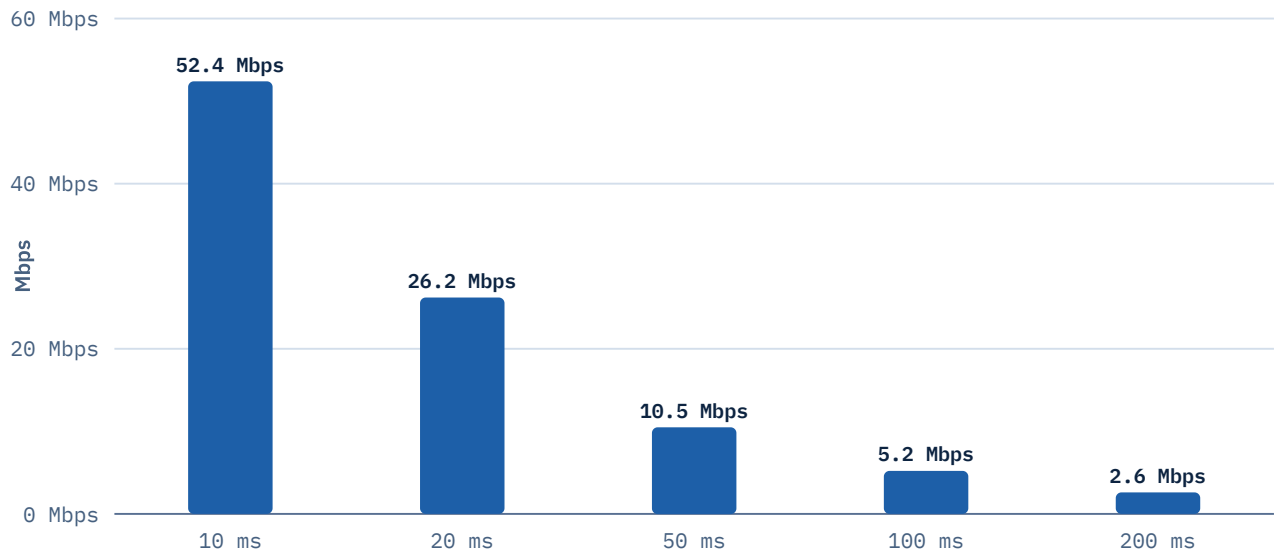
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MEASURE	FORMULA	WHAT IT TELLS YOU
Single-flow TCP ceiling	`Throughput ≤ window size ÷ RTT`	A 64 KB window over 100 ms RTT limits one flow to about 5.2 Mbps regardless of circuit size.
Voice call bandwidth	`(payload + 40 B IP/UDP/RTP) × 8 × packets per second`	A 64 kbps codec with 20 ms packets: (160 + 40) × 8 × 50 = 80 kbps per direction at layer 3, before tunnel overhead.

EXAMPLE · WHY THE LINK UPGRADE DID NOT HELP

A research group complains that copying data to a partner 9,000 km away crawls at under 10 Mbps on a new 1 Gbps circuit. The route RTT is about 110 ms. With a 64 KB window, one TCP flow is capped near 4.8 Mbps. The fix is not more bandwidth but window scaling, parallel streams or a transfer tool built for high-latency paths — a reminder that latency, not just capacity, limits throughput.

Single-flow TCP ceiling with a 64 KB window



Computed from throughput ≤ window ÷ RTT with a 64 KB (524,288-bit) window. Above about 50 ms of round trip, one flow cannot exceed about 10 Mbps whatever the circuit size — window scaling or parallel streams are the fix.

SCENARIO · DECIDE · CHOOSING TRANSPORTS FOR REMOTE SITES

A mining company has 25 remote sites with no fixed-line service, each running safety telemetry (small, latency-tolerant), voice calls and video safety briefings. Requirement: no single failure isolates a site; voice quality must be good. Which transport combination best fits?

- (A) GEO satellite as primary because it has the widest coverage, with no secondary path
- (B) LEO primary, 4G/5G where available, GEO as last resort for telemetry
- (C) Two LEO terminals from the same provider for diversity
- (D) Private MPLS circuits to every site for guaranteed quality

Circle your choice. Answer and feedback: module answer key.

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Key terms

MPLS L3VPN
A provider service that carries customer routes in a private routed network, keeping each customer's traffic separate and supporting QoS end to end.

Dedicated internet access (DIA)
An internet circuit with guaranteed bandwidth and an SLA up to the provider edge.

LEO satellite
Low-Earth-orbit satellite service with much lower latency than geostationary systems, typically tens of milliseconds.

GEO satellite
Geostationary satellite service with wide coverage but long round-trip delay because of orbital distance.

Serialisation delay
The time needed to place a frame's bits onto a link; it depends on frame size and link rate.

Bandwidth-delay product
The amount of data in flight required to keep a path full: bandwidth multiplied by round-trip time.

Physical diversity
Separation of circuits along their entire physical route — ducts, building entries and exchanges — not just different providers.

Jitter
Variation in packet delay; real-time applications absorb it with a jitter buffer that adds delay.

REFLECT

For one of your critical applications, write its delay budget component by component. Which component dominates, and could a transport choice change it?

Key takeaways

- Match each site class to transports whose weaknesses do not overlap
- Prove critical applications with a delay and bandwidth budget, component by component
- Latency limits single-flow throughput: check the bandwidth-delay product before buying capacity
- Physical diversity must be verified on route maps; mixing fibre, 5G and LEO makes shared failures rare

KNOWLEDGE CHECK

1. A file transfer over a 500 Mbps link with 120 ms RTT achieves only about 4 Mbps per flow. What is the MOST likely cause?

   (A) The link is oversubscribed by other traffic during busy hours

   (B) Serialisation delay on the 500 Mbps link is excessive

   (C) The TCP window is too small for the bandwidth-delay product

   (D) The provider is shaping traffic below the contracted rate

2. Why are two circuits from two different carriers not automatically diverse?

   (A) Carriers are required by regulation to route all circuits through one exchange

   (B) They may share ducts, building entry points or upstream facilities

   (C) Two carriers cannot both terminate at the same site router

   (D) Diversity only exists when both circuits use the same technology

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3. What is the approximate serialisation delay of a 1,500-byte frame on a 2 Mbps link?

(A) 6 ms
(B) 0.6 ms
(C) 60 ms
(D) 1.2 ms

4. Which transport is LEAST suitable as the primary path for interactive voice?

(A) LEO satellite
(B) Private MPLS L3VPN
(C) GEO satellite
(D) Dedicated internet access

5. What is the main reason to combine a wired transport with a cellular one at a branch?

(A) Cellular service always offers lower latency than wired broadband
(B) Their failure causes are largely independent, so one event rarely breaks both
(C) Cellular circuits include an end-to-end SLA that wired services lack
(D) Combining them removes the need for an overlay or encryption

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 3 · WAN · LESSON 2 · 38 MIN

SD-WAN and WAN Topology Design

Overlay decisions: topology, site tiers, steering and failure behaviour

- Choose a WAN topology — hub-and-spoke, regional hubs, partial or full mesh — from traffic patterns
- Define site tiers that standardise transports and resilience
- Make SD-WAN design decisions on steering, internet breakout, segmentation and control-plane resilience

A software-defined WAN builds an encrypted overlay across any mix of transports and steers each application over the best path according to policy and measured performance. The technology is mature; the design decisions are where projects succeed or fail. Which sites talk to which? Where does internet traffic leave? What happens when the controller is unreachable? How many site types will you support? The answers come from requirements, not from the product's feature list.

SD-WAN architecture planes

Management / orchestration	`Templates and policy definition` `Zero-touch provisioning` `Monitoring and analytics`
Control plane	`Controllers distribute routes, keys and policy` `Topology and tunnel decisions` `Usually redundant and geographically spread`
Data plane	`Edge devices at sites, hubs and clouds` `Encrypted tunnels over each transport` `Per-application path selection and probing`
Transport	`MPLS, broadband, DIA, cellular, satellite` `Any IP connectivity` `Diversity chosen per site tier`

WAN topology options

TOPOLOGY	TRAFFIC PATTERN IT SUITS	STRENGTHS	WEAKNESSES
Hub-and-spoke	Branch to data centre	Simple policy; central inspection	Branch-to-branch hairpins through the hub
Regional hubs	Large geographic spread	Shorter paths; regional failure isolation	More hubs to build and operate
Partial mesh / dynamic tunnels	Some branch-to-branch (voice, video)	Direct paths created on demand	More state at edges; policy complexity
Full mesh	Heavy any-to-any traffic, few sites	Optimal paths everywhere	Tunnel count grows with the square of sites

Overlay scaling

MEASURE	FORMULA	WHAT IT TELLS YOU
Full-mesh tunnel count	`Tunnels = n × (n − 1) ÷ 2 per transport pairing`	200 sites in one transport give 19,900 tunnels; most of them would carry almost no traffic.

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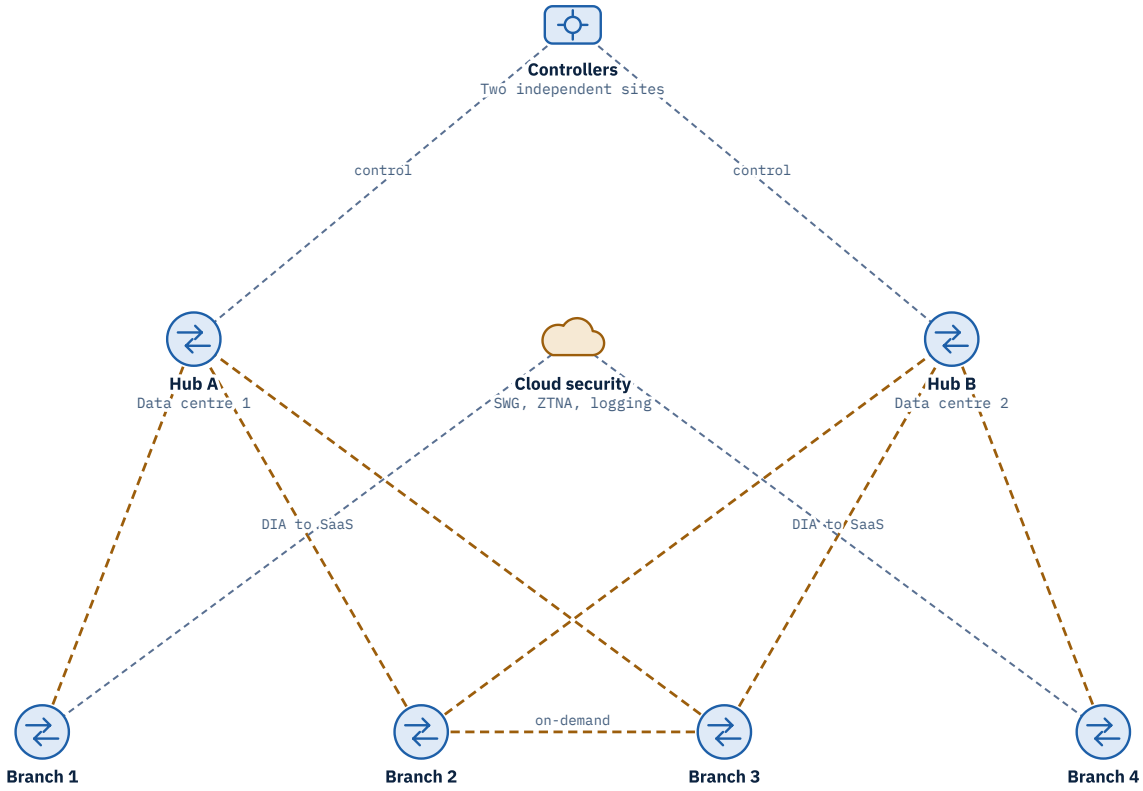
MEASURE	FORMULA	WHAT IT TELLS YOU
Tunnels per edge	`Up to (n − 1) × t² with t transports and all cross-transport pairings`	With 200 sites and 2 transports, each edge could hold 796 tunnels — check the platform's limit and restrict pairings.
Hub tunnel load	`Tunnels at hub ≈ spokes × transports per spoke × hub transports`	Hub sizing is driven by tunnel count and encrypted throughput, not just bandwidth.

Example site tiers

TIER	SITES	EDGE DEVICES	TRANSPORTS	TARGET
Tier 1 — hub / data centre	4	2, separate power and rooms	2 × DIA from different carriers + MPLS	No single failure affects service
Tier 2 — large branch	30	2	MPLS or DIA + broadband, diverse entry	No single failure isolates the site
Tier 3 — standard branch	120	1	Broadband + 4G/5G	Survive any single transport failure
Tier 4 — micro site / kiosk	60	1 (cellular-integrated)	4G/5G (+ broadband where available)	Best effort; next-day hardware swap

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SD-WAN with two hubs, on-demand tunnels and cloud security



Every branch builds encrypted tunnels to both hubs; store-to-store media gets an on-demand tunnel only while a call is active. SaaS and internet traffic breaks out locally to a cloud security service, while controllers sit in two independent locations.

PRO TIP · STANDARDISE TIERS, THEN DEFEND EVERY EXCEPTION

Four or five site tiers with fixed transport, device and policy templates make the WAN operable at scale. When a site does not fit, record a design exception with its owner and reason. A WAN with dozens of one-off sites loses most of the operational benefit of the overlay.

Internet traffic from branches

Backhaul to central hub

- Uses existing central inspection stack
- Consistent policy and logging
- Adds latency to SaaS and cloud
- Consumes hub and WAN bandwidth

Local breakout (direct internet access)

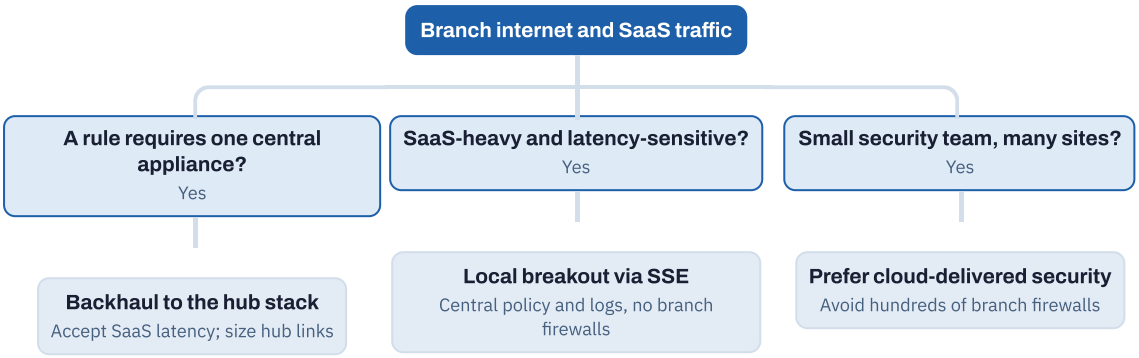
- Lowest latency to SaaS and cloud
- Offloads hubs and private WAN
- Needs security at the branch or in a cloud security service
- More enforcement points to govern

DEFINITION · SASE AND SSE

Secure access service edge (SASE) combines SD-WAN with cloud-delivered security. The security half, often called **security service edge (SSE)**, typically bundles a secure web gateway, cloud access security, zero-trust network access and firewall as a service, delivered from the provider's points of presence. The design questions are where those points of presence are, what happens to branches when the service is unreachable, how logs reach your own tools, and whether one supplier for both halves is worth the reduced choice.

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Where should branch internet traffic leave?



Work down the tree with the branch tier's requirements. Most estates end up with a mix: local breakout through a cloud security service for SaaS, and backhaul for the few flows a rule or a central system really needs.

INSIGHT · CONTROL-PLANE FAILURE IS A DESIGN INPUT

Most overlays keep forwarding on established tunnels for a period when controllers become unreachable, but cannot build new tunnels, re-key indefinitely or accept policy changes. Ask the vendor for the exact behaviour and timers, place controllers in at least two independent locations, and make sure a regional outage cannot isolate every site from every controller at once.

CASE STUDY · REGIONAL BANK WAN MODERNISATION

A regional bank has 150 branches, two data centres 60 km apart, and a call centre. All branches use a single MPLS L3VPN whose contract ends in 14 months. Teller and core-banking applications run in the data centres; email and collaboration have moved to SaaS, and branch staff watch weekly video training. The regulator requires encryption of customer data over any network the bank does not control, documented resilience testing, and logging of all internet access. Thirty rural branches can only obtain 30 Mbps broadband, and some have weak cellular coverage. ATMs at each branch must keep working through any single failure. The network team has six engineers and limited weekend change windows.

1. Which topology and site tiers would you propose?

2. Would you keep MPLS? Justify the decision for different branch classes.

3. How will internet traffic leave branches while meeting the logging requirement?

4. What migration risks must the design address?

Model answer: module answer key.

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SCENARIO · DECIDE · BRANCH-TO-BRANCH VOICE

A retailer's 400 stores connect to two hubs in a hub-and-spoke overlay. The business is introducing store-to-store video calls for remote expert support, peaking at 30 concurrent calls estate-wide. Which design change best fits?

- (A) Convert the overlay to a permanent full mesh between all 400 stores
- (B) Keep hub-and-spoke and upgrade hub bandwidth to carry the extra calls
- (C) Route store-to-store calls over cellular so they bypass the hubs
- (D) On-demand dynamic tunnels between stores for real-time traffic

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **WAN design sheet**

Standardise each site tier and record its transports, resilience, breakout and exceptions.

SITE TIER	NUMBER OF SITES	EDGES	TRANSPORTS (PRIMARY / SECONDARY)	INTERNET BREAKOUT	RESILIENCE TARGET	EXCEPTIONS
Tier 3 standard branch	*120*	*1*	*Broadband 300M / 5G*	*Local via cloud security service*	*Survive any one transport failure*	*8 rural sites use LEO instead of 5G*

Key terms

SD-WAN
A software-defined overlay that builds encrypted tunnels across multiple transports and steers applications by policy and measured path quality.

Application-aware routing
Selecting a path per application based on measured loss, latency and jitter against policy thresholds.

Dynamic tunnel
An overlay tunnel created on demand between two sites when traffic needs a direct path, and removed after it is idle.

Zero-touch provisioning
Automatic onboarding in which a new edge device obtains its configuration from the orchestrator when first connected.

Overlay
A logical network of tunnels built on top of one or more underlying transports.

Local internet breakout
Sending internet-bound traffic directly from a branch rather than backhauling it to a central site.

Site tier
A standard site design defining edge count, transports, resilience target and policy for a class of locations.

Hub site
A data centre or regional site where many spoke tunnels terminate and shared services or inspection are provided.

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REFLECT

How many distinct site designs does your WAN really have today? Which of them could be merged into a tier, and which deserve to stay as documented exceptions?

Key takeaways

- Choose topology from traffic patterns: hub-and-spoke for branch-to-DC, on-demand tunnels for occasional branch-to-branch
- A small set of site tiers with templates makes a large WAN operable; exceptions are documented
- Full meshes grow with the square of the sites — check tunnel limits per edge
- Local breakout through a cloud security service shortens SaaS paths while keeping central policy and logs
- Spread controllers so one regional outage cannot strand every site, and know the headless behaviour

KNOWLEDGE CHECK

6. A WAN of 300 sites runs a single transport in a full mesh. How many tunnels does the overlay contain?
   - (A) 90,000
   - (B) 299
   - (C) 89,700
   - (D) 44,850

7. What usually happens to an established SD-WAN data plane when all controllers become unreachable?
   - (A) All tunnels drop immediately and sites fall back to the underlay only
   - (B) Edges elect a new controller among themselves and then continue operating normally
   - (C) Traffic is forwarded unencrypted until a controller returns
   - (D) Existing tunnels keep forwarding for a period but new changes cannot be made

8. Which requirement MOST favours local internet breakout over backhaul?
   - (A) Low latency to SaaS applications from widely spread branches
   - (B) A rule that all internet access must pass through one appliance
   - (C) A shortage of security skills to manage many enforcement points
   - (D) Branches whose only transport is a private MPLS service

9. What is the main operational benefit of defining a small number of site tiers?
   - (A) Each site can be optimised individually for its exact traffic
   - (B) Tiers remove the need for any per-site documentation
   - (C) Standard templates make many sites predictable to deploy and support
   - (D) Tiers guarantee the lowest possible cost for every location

10. A hub-and-spoke overlay is adding store-to-store video. What is the main drawback of keeping all traffic through the hubs?
    - (A) The hubs cannot encrypt traffic between two stores
    - (B) Each call hairpins through a hub, adding delay
    - (C) Stores lose their path to the data centres during calls
    - (D) Video traffic cannot be classified in a hub-and-spoke design

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MODULE 3 · WAN · LESSON 3 · 33 MIN

Internet Edge and DMZ Design

Resilient, secure entry points for users, services and partners

- Design a layered internet edge with defined roles for routing, DDoS protection and inspection
- Choose a DMZ pattern that fits the services and risk profile
- Plan DDoS mitigation that works when the access link itself is saturated

The internet edge carries three different jobs: letting users out, letting customers and partners in to published services, and protecting both from attack. Mixing those jobs on one undifferentiated stack makes it hard to scale, hard to secure and hard to change. A clear edge design separates them into layers, each with a defined responsibility and failure behaviour.

Layered internet edge

Upstream protection	`Provider or cloud DDoS scrubbing`
	`Remotely triggered blackholing via provider communities`
	`Diverse ISPs and points of presence`

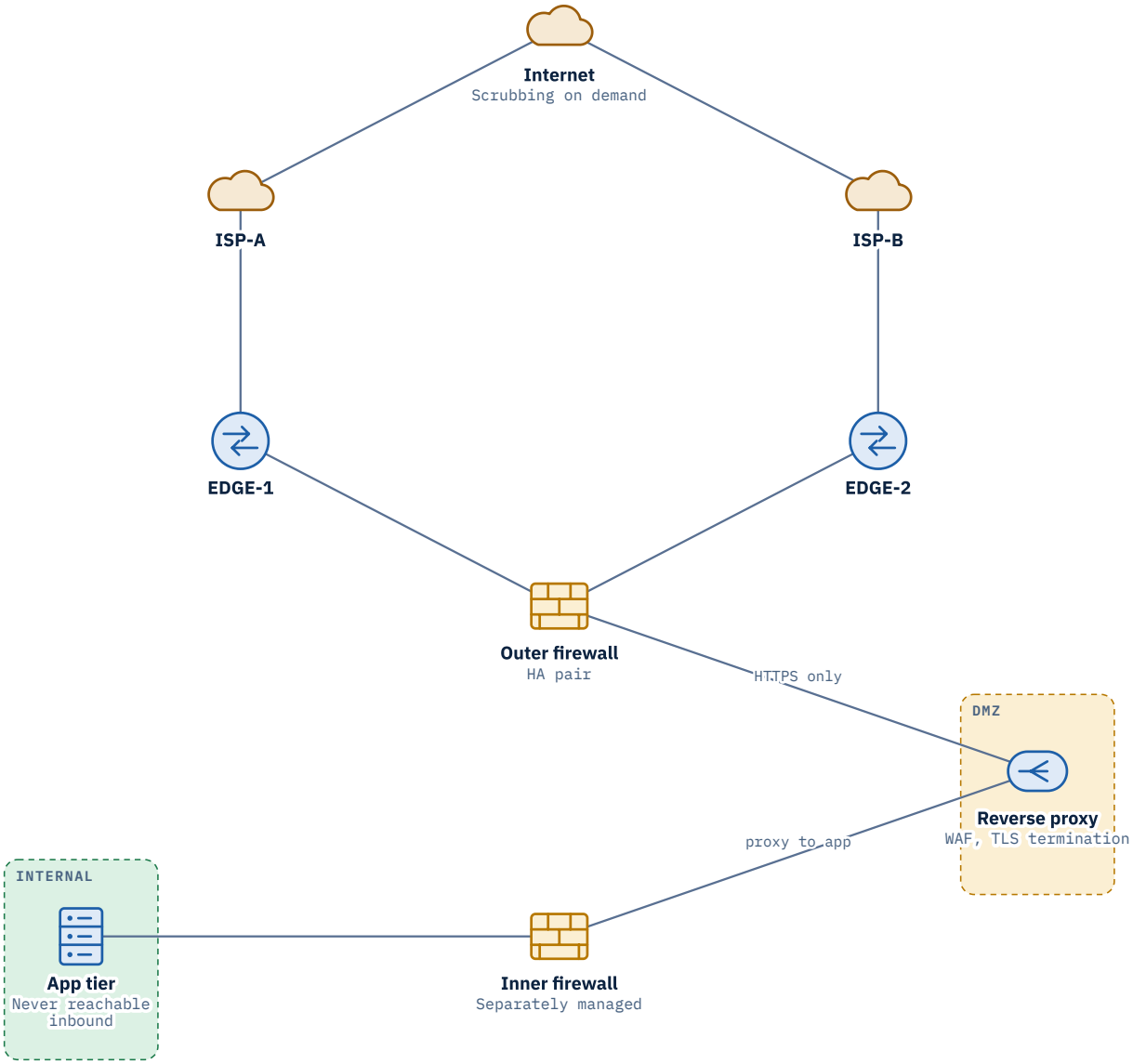
Edge routing	`BGP with ISPs` `Inbound and outbound route filtering with RPKI origin validation` `Anti-spoofing filters on ingress`
Perimeter firewall	`Stateful filtering between zones` `NAT where required` `High-availability pair or cluster`

DMZ services	`Reverse proxies and web application firewalls`
	`Public DNS, mail relays, partner gateways`
	`No direct path from internet to internal hosts`

Internal and outbound control	`Secure web gateway or cloud security service` `Internal firewalls and segmentation` `Logging and monitoring`

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A layered internet edge with a dual-tier DMZ



Traffic from the internet meets upstream protection, diverse ISPs and edge routers before the outer firewall. Published services live in the DMZ between the outer and inner firewalls, and only the reverse proxy may open sessions to the internal application tier.

DMZ patterns

Three-legged firewall

- One firewall pair with outside, DMZ and inside zones
- Lower cost and simpler to operate
- A firewall compromise or misconfiguration affects all zones
- Fits small and mid-sized estates

Back-to-back (dual-tier) firewalls

- DMZ between an outer and an inner firewall pair
- Defence in depth; inner pair protects internal networks
- Optionally different technologies at each tier
- More cost, rules and operational effort

Cloud-hosted front door

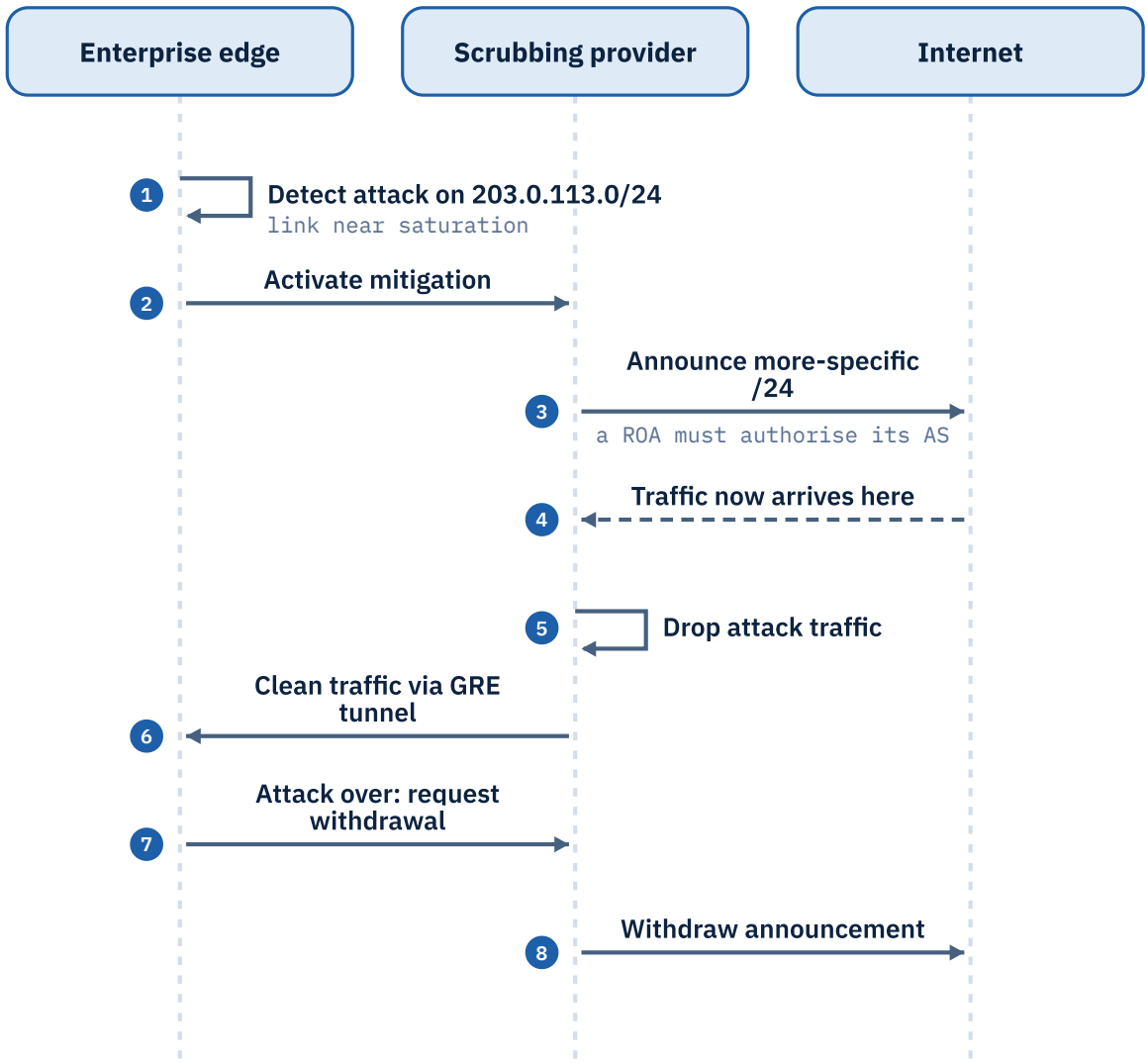
- Public services fronted by a cloud proxy, WAF and DDoS service
- Absorbs attacks far from your links
- Origin reachable only from the provider
- Dependence on the provider; data-handling review needed

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WATCH OUT · ON-PREMISES APPLIANCES CANNOT STOP A FULL PIPE

A volumetric attack that fills your 10 Gbps access link has succeeded before traffic reaches any on-site device. Mitigation must happen upstream — in the ISP's network or a cloud scrubbing service — with traffic diverted by BGP or DNS. On-site devices still matter for application-layer attacks and for attacks small enough to fit through the link.

On-demand cloud scrubbing by BGP diversion



Diversion works only if the more-specific announcement, the return tunnel and the runbook were staged and tested beforehand; the gap between steps 1 and 4 is the diversion time the business must accept.

DDoS mitigation options

OPTION	HOW IT WORKS	STRENGTHS	LIMITS
Remotely triggered blackhole (RTBH)	Signal a prefix to the ISP to drop all traffic to it	Fast; protects the rest of the link	Completes the attack for the targeted address
ISP scrubbing service	ISP filters attack traffic in its network	Large capacity close to the source	Per-ISP; multi-ISP edges need coverage on all

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OPTION	HOW IT WORKS	STRENGTHS	LIMITS
Cloud scrubbing, on demand (BGP diversion)	Advertise a more-specific prefix via the scrubbing provider, receive clean traffic over a tunnel	Very large capacity; ISP-independent	Diversion time; needs pre-staged routing and tunnels
Always-on cloud proxy	Public services always reached through a cloud front door	No diversion delay; also gives WAF and caching	Only for proxied services (mainly web)

Designing a resilient internet edge

1. **Classify flows** Separate user outbound, published inbound services, partner and B2B connections, and remote access. Each has different capacity and security needs.
2. **Choose provider diversity** Two ISPs with physically diverse paths, ideally into two edge sites, with provider-independent or portable addressing if inbound services must survive a provider failure.
3. **Pick the DMZ pattern** Match the pattern to the risk of the published services and the regulatory requirement for defence in depth.
4. **Plan DDoS response before you need it** Pre-stage scrubbing, blackhole communities and runbooks; test diversion at least annually.
5. **Size for the worst case** Size firewalls and links for single-failure conditions and peak inspection load, including TLS decryption cost if used.

SCENARIO · DECIDE · EDGE DESIGN FOR A PORT AUTHORITY

A port authority publishes a vessel-booking portal used by shipping agents worldwide and runs operational systems that must never be reachable from the internet. It has one data centre, two ISPs, a small security team, and a regulator that requires defence in depth for critical infrastructure. Last year a DDoS attack saturated its 5 Gbps link for six hours. Which edge design best fits?

- (A) Always-on cloud front door, dual-tier DMZ for the origin, no inbound path to operations
- (B) Add a larger on-premises DDoS appliance and upgrade both links to 10 Gbps
- (C) A single three-legged firewall with the portal and operational systems in separate DMZ legs
- (D) Remotely triggered blackholing of the portal address whenever an attack starts

Circle your choice. Answer and feedback: module answer key.

Internet edge design checklist

☐ Flows classified: outbound, inbound services, partners, remote access
☐ Two ISPs with verified physical diversity
☐ Addressing strategy supports inbound failover between providers
☐ Inbound and outbound BGP filtering defined
☐ Anti-spoofing filters on all external interfaces
☐ DMZ pattern chosen and justified against the risk profile
☐ No direct internet path to internal or operational systems
☐ DDoS mitigation pre-staged upstream and tested
☐ Firewall and link capacity sized for single-failure and peak inspection load
☐ Outbound access logged to meet regulatory needs

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Key terms

DMZ
A network zone for services reachable from untrusted networks, separated by filtering from both the internet and internal networks.

Back-to-back firewalls
A DMZ placed between an outer and an inner firewall tier for defence in depth.

Scrubbing
Filtering attack traffic out of a traffic stream in a provider or cloud network and forwarding the clean traffic.

Anti-spoofing filter
A filter that drops packets whose source addresses could not legitimately arrive on that interface.

Three-legged firewall
A single firewall (or pair) with separate interfaces for outside, DMZ and inside zones.

Volumetric DDoS
An attack that aims to exhaust link or network capacity with sheer traffic volume.

RTBH
Remotely triggered blackholing: signalling a provider, typically with a BGP community, to discard all traffic to a specific prefix.

Web application firewall
A filter that inspects traffic to published web applications for attacks such as injection and automated abuse.

REFLECT

If your main internet link were saturated by an attack right now, what is the first action, who takes it, and how long would it take? Is that written down anywhere?

Key takeaways

- Separate the edge jobs: users out, services in, and protection for both
- Volumetric DDoS must be absorbed upstream; pre-stage scrubbing, routing and runbooks and test diversion
- Choose the DMZ pattern by risk: dual-tier or brokered where defence in depth is required
- Provider-independent space advertised to two ISPs keeps inbound services reachable through either

KNOWLEDGE CHECK

11. Why can an on-premises DDoS appliance NOT stop a volumetric attack that saturates the access link?

(A) The appliance cannot inspect encrypted traffic at line rate

(B) Volumetric attacks always use spoofed IPv6 source addresses

(C) Appliances only work when traffic is first sent to a cloud proxy

(D) The congestion occurs upstream of the appliance, before traffic reaches it

12. What is the main drawback of remotely triggered blackholing during an attack?

(A) All traffic to the targeted address is dropped, including legitimate users

(B) It requires replacing the existing edge routers with dedicated scrubbing hardware first

(C) It only works when the attacker uses a single source address

(D) It forces all outbound traffic onto the secondary ISP

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13. When is a back-to-back (dual-tier) firewall DMZ MOST justified over a three-legged design?

(A) When the organisation has a very small security team to manage rules

(B) When regulation or risk demands defence in depth for internal networks

(C) When the only published service is a static marketing website

(D) When the cost of the edge must be kept as low as possible

14. Which addressing approach best supports inbound services surviving the loss of one ISP?

(A) Separate provider-assigned blocks with no DNS changes

(B) Private addressing translated only at the primary ISP

(C) Provider-independent space advertised to both ISPs

(D) A single provider-assigned block advertised to that ISP only

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MODULE 3 · WAN · LESSON 4 · 38 MIN

BGP Multihoming and Provider Connectivity

Control inbound and outbound paths, and design around provider routing models

- Select a multihoming model from availability and traffic requirements
- Choose the right inbound and outbound traffic-engineering tools and predict their effect
- Design PE-CE routing and dual-provider WANs without loops or accidental transit

Multihoming is about two different directions. **Outbound** traffic engineering is easy because you control it: your routers choose the exit. **Inbound** traffic engineering is hard because other networks choose how to reach you; you can only influence their choice with the attributes and prefixes you advertise. Good designs set realistic expectations for inbound control and use the strongest tool that is acceptable to the internet community.

Multihoming models

MODEL	DESCRIPTION	PROTECTS AGAINST	DOES NOT PROTECT AGAINST
Single-homed	One link, one ISP	Nothing beyond the ISP's own resilience	Link, router or ISP failure
Dual-linked, one ISP	Two links, one router or two, same provider	Link and router failure	Provider-wide outage or routing incident
Dual-homed, one ISP, two PoPs	Two links to different provider points of presence	Local and PoP failure	Provider-wide outage
Multihomed, two ISPs	Links to two independent providers, own AS and portable addressing	Provider failure, most single failures	Shared physical paths if diversity is not verified

Inbound traffic-engineering tools

MSP

More-specific prefixes

Advertise longer prefixes through the preferred path; longest match wins everywhere. Most effective, but adds routes to the global table and is limited by maximum accepted prefix length (commonly /24 for IPv4, /48 for IPv6).

COM

Provider communities

Many providers let customers set communities that change local preference or limit propagation inside their network — precise control where offered.

PRP

AS-path prepending

Lengthen the AS path on the less-preferred link. Coarse and only a tie-breaker: networks that set local preference ignore it.

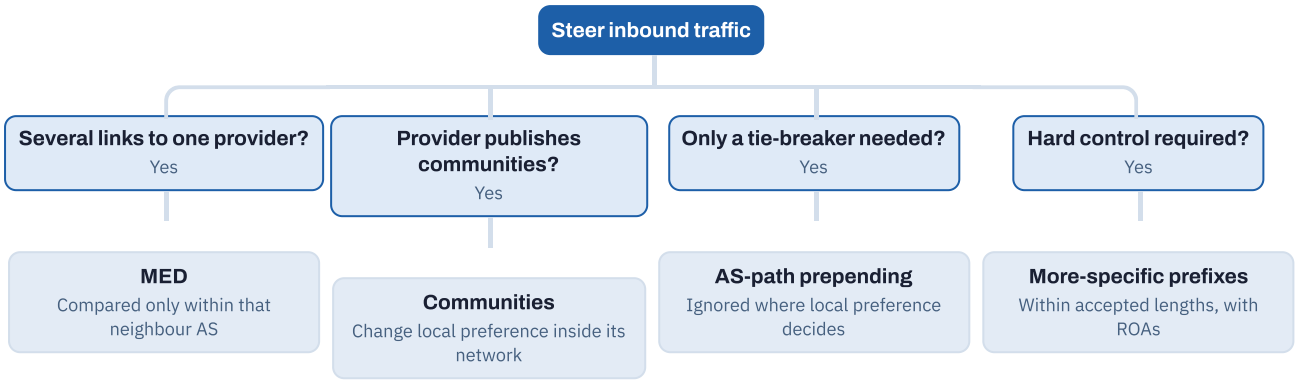
MED

Multi-exit discriminator

Suggests an entry point to a single neighbouring AS with multiple links. Not compared across different providers by default.

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Choosing an inbound traffic-engineering tool



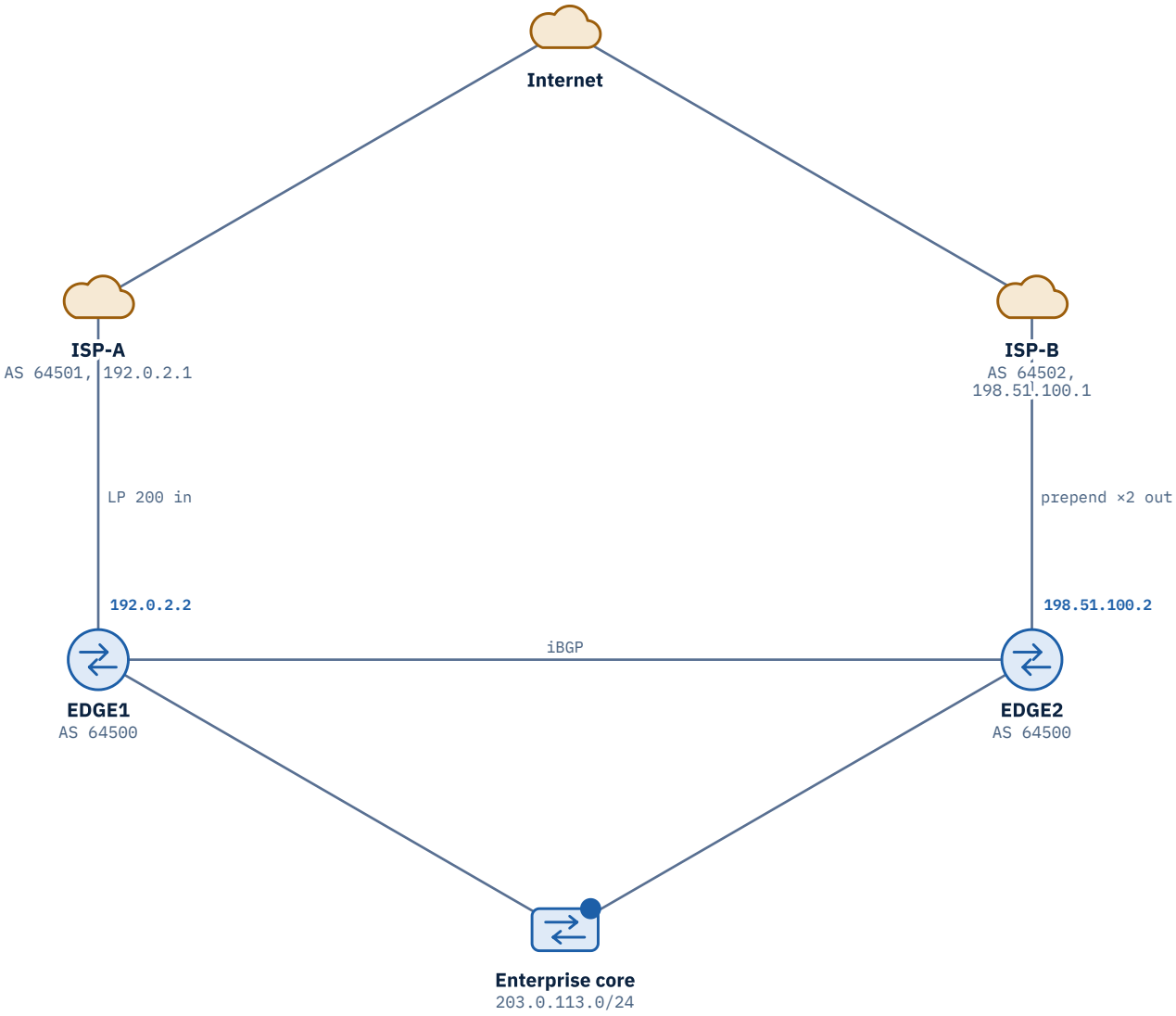
Start with the gentlest tool that can work. More-specific prefixes always win by longest match, but they add global routes and are capped at /24 for IPv4 and /48 for IPv6.

INSIGHT · OUTBOUND IS LOCAL PREFERENCE

To choose your exit, set local preference on routes received from each provider. Taking a default route from both providers plus partial routes (the providers' own customers) is often enough: it gives sensible exits for well-connected destinations without the memory and churn of full internet tables on every edge router.

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Dual-ISP multihoming for 203.0.113.0/24



Outbound, local preference 200 on ISP-A routes chooses the exit. Inbound, prepending towards ISP-B only nudges remote networks; ISP-B's customer communities would be stronger. Both sessions advertise only 203.0.113.0/24 so the enterprise never becomes transit.

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Dual-ISP edge: prefer ISP-A outbound, de-prefer ISP-B inbound

EDGE1 · IOS-style CLI

```
! Single router shown for clarity; production designs often use two
router bgp 64500
 neighbor 192.0.2.1 remote-as 64501
 neighbor 192.0.2.1 description ISP-A primary
 neighbor 192.0.2.1 prefix-list OUR-PREFIXES out
 neighbor 192.0.2.1 route-map FROM-ISP-A in
 neighbor 198.51.100.1 remote-as 64502
 neighbor 198.51.100.1 description ISP-B secondary
 neighbor 198.51.100.1 prefix-list OUR-PREFIXES out
 neighbor 198.51.100.1 route-map TO-ISP-B out
 network 203.0.113.0 mask 255.255.255.0
ip route 203.0.113.0 255.255.255.0 Null0
ip prefix-list OUR-PREFIXES seq 10 permit 203.0.113.0/24
route-map FROM-ISP-A permit 10
 set local-preference 200
route-map TO-ISP-B permit 10
 set as-path prepend 64500 64500
```

The outbound prefix list is the most important line: advertising only your own prefixes is what prevents your network becoming transit between the two ISPs. Production inbound policies would also filter bogons and your own prefixes.

WATCH OUT · ACCIDENTAL TRANSIT

If an edge router re-advertises routes learned from ISP-A to ISP-B, your network can start carrying traffic between two providers — saturating your links and potentially disrupting parts of the internet. Always apply an explicit outbound filter that permits only your own prefixes, and use implementations that refuse to advertise external routes without a policy.

Provider WAN routing models

MPLS L3VPN (provider routes)

- CE peers with the provider PE (static, BGP or an IGP)
- Provider carries your routes between sites
- Simple for the customer; QoS end to end
- If every site reuses one private AS, the provider must override the AS path or sites must accept their own AS
- Path control limited to what the provider supports

Layer 2 service (customer routes)

- Provider delivers Ethernet; you run routing end to end
- Full control of IGP or BGP design
- Many sites on one multipoint segment form many adjacencies
- Provider is not involved in convergence
- You own all routing complexity

SCENARIO · DECIDE · TWO MPLS PROVIDERS

A logistics company uses two MPLS L3VPN providers for resilience. Each of its 60 sites connects to both. After a provider maintenance event, traffic between two depots took a path in through provider A, across a third depot's LAN, and out through provider B, congesting that depot. What design change best addresses the root cause?

(A) Remove one provider and accept the reduced resilience to simplify routing

(B) Increase the bandwidth at the third depot so that it can carry the transit traffic comfortably

(C) Set provider preference, tag routes by provider and block re-advertisement between them

(D) Ask both providers to shorten their maintenance windows

Circle your choice. Answer and feedback: module answer key.

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Designing a BGP multihomed edge

1. **Confirm the requirement** Is the goal resilience, load sharing, or control of specific flows? Each leads to different tools and expectations.
2. **Secure addressing and AS number** Provider-independent space and a public AS for true multi-provider designs; register route-origin authorisations so others can validate your announcements.
3. **Define outbound policy** Local preference per provider; default plus partial routes or full tables based on router capacity and need.
4. **Define inbound policy** Start with communities and prepending; use more-specifics only when necessary and within accepted prefix lengths.
5. **Filter both directions** Outbound: only your prefixes. Inbound: drop bogons, your own space, overly specific routes and RPKI-invalid origins; set maximum-prefix limits. Where peers support it, configure BGP roles so route leaks are detected automatically; newer path-validation work (such as AS-path authorisation) is still being standardised and deployed — check its status before relying on it.
6. **Test failure and recovery** Verify convergence time, path symmetry for stateful devices, and behaviour when a provider withdraws routes.

Key terms

Multihoming
Connecting a network to more than one provider or provider location for resilience or traffic control.

Local preference
A BGP attribute, local to an AS, that sets the preferred exit; higher is preferred.

MED
Multi-exit discriminator: a BGP attribute suggesting the preferred entry among several links to the same neighbouring AS.

Transit
Carrying traffic between two other networks; an enterprise should not provide it between its providers.

PE-CE routing
The routing protocol or static routing between a customer edge router and the provider edge in a provider VPN.

Provider-independent addressing
Address space assigned directly to an organisation so it can be announced through any provider.

AS-path prepending
Adding extra copies of your AS number to an advertisement to make a path look longer and less attractive.

BGP community
A tag attached to routes that providers and peers use to apply agreed policy, such as changing preference or blackholing.

Route origin authorisation (ROA)
A signed record stating which AS may originate a prefix, used by others to validate route origins.

REFLECT

Which direction of traffic does your organisation actually need to control at the internet edge — inbound, outbound or both? Which tool are you relying on, and how much control does it really give?

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Key takeaways

- Outbound is yours to choose with local preference; inbound can only be influenced
- Prefer communities and prepending; use more-specifics only when necessary and within accepted lengths
- An explicit outbound filter permitting only your prefixes prevents accidental transit
- Publish ROAs and validate origins; drop bogons, your own space and invalid routes inbound
- With two MPLS providers, tag and filter so sites never become transit between the VPNs

KNOWLEDGE CHECK

15. Which tool gives an enterprise direct control over which ISP it uses for outbound traffic?

(A) Prepending its AS number on advertisements to the backup ISP
(B) Sending a lower MED to the preferred ISP on every link
(C) Advertising more-specific prefixes through the preferred ISP
(D) Setting local preference on routes received from each ISP

16. Why is AS-path prepending considered a weak inbound traffic-engineering tool?

(A) Prepending is filtered by most providers and never reaches the internet
(B) Prepending only works when both links connect to the same provider
(C) Remote networks often set local preference, which is compared before path length
(D) Prepending changes outbound traffic rather than inbound traffic

17. What is the MOST important control to prevent an enterprise from becoming transit between two ISPs?

(A) A maximum-prefix limit on the sessions with both providers
(B) An outbound filter that advertises only the enterprise's own prefixes
(C) Taking only a default route from each of the providers
(D) Setting a higher local preference on the primary provider

18. All sites of an MPLS L3VPN customer use the same private AS. Routes from other sites are being rejected. Why?

(A) BGP loop prevention drops routes containing the site's own AS number
(B) Private AS numbers cannot be used on PE-CE sessions at all
(C) The provider is summarising routes before sending them to CE routers
(D) The CE routers require full internet tables to accept VPN routes

19. Which inbound tool is generally the MOST effective but adds routes to the global routing table?

(A) AS-path prepending
(B) Advertising more-specific prefixes
(C) Multi-exit discriminator
(D) Provider local-preference communities

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 3 · WAN

Answer key

Knowledge checks

1. C A 64 KB window over 120 ms caps a flow near 4.4 Mbps. The BDP is 7.5 MB, so the window, not the circuit, limits throughput.

2. B Physical diversity depends on the actual route. Different carriers often lease the same infrastructure for the last mile.

3. A 1,500 × 8 = 12,000 bits; 12,000 ÷ 2,000,000 = 0.006 s = 6 ms.

4. C GEO round-trip delay of roughly half a second or more makes conversation difficult. The others normally keep voice within its delay budget.

5. B Mixing physical media avoids shared failures such as a cut duct. Cellular is usually best effort and still needs encryption.

6. D n(n − 1) ÷ 2 = 300 × 299 ÷ 2 = 44,850 tunnels. Each edge maintains 299 of them.

7. D Typical overlays keep forwarding on current state for a limited time. Exact behaviour and timers vary, so they must be confirmed and designed for.

8. A Breakout shortens the path to SaaS. The other conditions favour backhaul or make breakout impossible.

9. C Standardisation is what makes a large WAN operable. Exceptions are allowed but should be documented and justified.

10. B Hairpinning lengthens the path and adds latency for real-time media. Dynamic tunnels give a direct path when needed.

11. D Once the link is full, legitimate traffic is already being dropped upstream. Mitigation must happen in the provider or a scrubbing network.

12. A RTBH protects the rest of the network by sacrificing the target. It is useful but makes the targeted service unavailable.

13. B Dual tiers add protection at the cost of more rules and devices. Small teams and low-risk services usually favour the simpler pattern.

14. C Provider-independent space can be announced to both providers, so the same addresses stay reachable through either.

15. D Outbound path choice is made by your own routers, and local preference is the standard, AS-wide attribute for choosing an exit. The other tools influence inbound traffic.

16. C AS-path length is only considered after local preference and other attributes. Providers that prefer customer routes will ignore a longer path.

17. B Transit happens when routes from one provider are advertised to another. An explicit outbound prefix filter stops that regardless of what is received.

18. A Seeing its own AS in the path, each site treats the route as a loop. Providers solve this with AS override, or customers allow their own AS in received paths.

19. B Longest-prefix match beats any BGP attribute, so more-specifics always win — at the cost of extra routes, and only up to commonly accepted prefix lengths.

Scenarios & case studies

Scenario · Choosing transports for remote sites

B is best. Correct. LEO gives latency suitable for voice and video, cellular adds a physically different path where available, and GEO keeps telemetry flowing if both fail — its high delay is acceptable for that traffic but not for voice.

Case study · Regional bank WAN modernisation

Topology and tiers. An SD-WAN overlay with both data centres as hubs (hub-and-spoke for core banking, with dynamic branch-to-branch tunnels only if voice requires it). Tier 1: data centres with dual DIA from different carriers plus any retained private circuits. Tier 2: large branches with two edges and two diverse transports. Tier 3: standard branches with one edge, broadband plus cellular. Rural exceptions: broadband plus LEO satellite where cellular is weak.

MPLS. Keeping a reduced private service at the data centres and the largest branches can ease migration and offer predictable QoS, but the ATM and teller requirements can be met by two independent transports under an encrypted overlay. Present both options in a decision matrix weighted on cost, resilience and the team's capacity; do not keep MPLS by default.

Internet traffic. Local breakout for approved SaaS through a cloud security service that logs all access centrally, with other internet traffic backhauled to the data-centre stack. Both paths must produce logs that meet the regulator's requirement.

Migration risks. Encryption must be on before any public transport carries customer data; ATM paths must be tested for failover per branch; the 14-month contract end leaves little slack for 150 sites with limited windows — plan waves by tier, run MPLS and the overlay in parallel during migration, and negotiate a short MPLS extension as a contingency.

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Scenario · Branch-to-branch voice

D is best. Correct. Calls get a direct path with lower delay, while tunnel state stays small because only a handful of store pairs are active at any time.

Scenario · Edge design for a port authority

A is best. Correct. The cloud front door absorbs volumetric attacks before they reach the links, the dual-tier DMZ provides the required defence in depth, and operational systems stay unreachable from outside.

Scenario · Two MPLS providers

C is best. Correct. The problem is uncontrolled mutual redistribution between the two VPNs through customer sites. Preference plus tagging and filtering keeps transit between providers from happening through branch LANs.

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04

MODULE 4 · ROUTING

Routing Protocol Design

IGP choice, hierarchy, summarisation, redistribution, BGP scale and convergence

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Select an interior routing protocol from requirements and justify keeping or changing it
- Design OSPF areas and IS-IS levels that contain churn and support summarisation
- Design summarisation and redistribution that avoid black holes and loops
- Scale BGP with redundant route reflection and meet application convergence budgets

LESSONS

1. Choosing an Interior Routing Protocol · 28 min
2. OSPF Area and IS-IS Level Design · 38 min
3. Summarisation and Redistribution Design · 38 min
4. BGP in the Enterprise, Route Reflection and Convergence · 43 min

MODULE 4 · ROUTING · LESSON 1 · 28 MIN

Choosing an Interior Routing Protocol

Select the IGP from requirements, not habit

- Compare OSPF, IS-IS, BGP-only and static designs against design criteria
- Identify the requirements that genuinely decide between link-state protocols
- Recognise when changing an existing IGP is — and is not — justified

The **Routing sheet** of the ZCDE Design Blueprint begins with the interior gateway protocol (IGP). For most enterprises the realistic choice is between two link-state protocols — OSPF and IS-IS — with BGP increasingly used as the only routing protocol in large data-centre fabrics, and static routing still right for small stub sites. Both link-state protocols are mature, fast and scalable; the decision is rarely about raw capability and usually about topology, address families, extensibility, operations and existing skills.

IGP decision criteria

CRITERION	OSPF	IS-IS	BGP-ONLY FABRIC	STATIC
Hierarchy model	Areas around a mandatory backbone area 0	Two levels; L2 backbone, flexible boundaries on links	None in the IGP sense; policy per session	None
IPv6	Separate OSPFv3 instance or address families	Same instance carries IPv4 and IPv6	Multiprotocol by design	Manual per prefix
Extensibility	LSA types, opaque LSAs	TLV-based; easily extended	Attributes and address families	Not applicable
Transport	Runs over IP	Runs directly over layer 2, not IP	Runs over TCP	Not applicable
Typical home	Enterprise campus, WAN and branches	Service provider and large backbone networks	Large data-centre Clos fabrics	Small stub sites, edge defaults
Team familiarity (typical enterprise)	High	Lower	Medium	High

What actually decides the IGP

TOPO

Topology and hierarchy

Hub-and-spoke WANs and campus blocks map naturally to OSPF areas; large flat backbones often suit single-level IS-IS.

AF

Address families

Running IPv4 and IPv6 in one IS-IS instance avoids operating two OSPF processes or depending on OSPFv3 address-family support.

EXT

Extensions and services

Segment routing, traffic engineering and multi-topology are supported by both link-state protocols; check exact platform support across every vendor in the estate.

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OPS

Operations and skills

The protocol your team can troubleshoot at 3 a.m. is a requirement, not a preference — unless training and time are funded.

INT

Interoperability

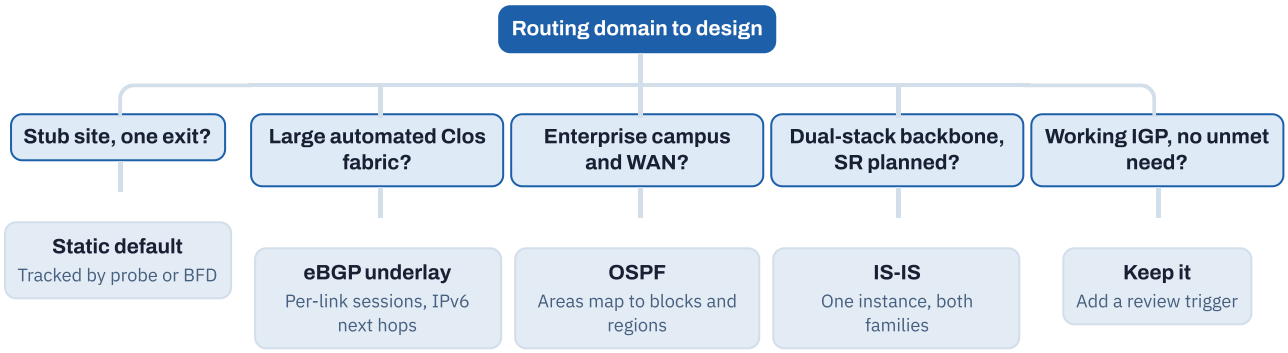
Standards-based protocols across all vendors in the estate; avoid single-vendor protocols where multi-vendor is required or likely.

MIG

Migration cost

Replacing a working IGP touches every router; the benefit must outweigh the risk and effort.

An IGP decision tree



A starting point, not a rule: each branch still has to be checked against skills, vendor support and migration cost. Note that keeping a working protocol is a legitimate outcome of the tree.

INSIGHT · BGP AS THE ONLY PROTOCOL IN A DATA-CENTRE FABRIC

In large Clos fabrics, eBGP on every leaf-spine link — with spines sharing one AS and each leaf or leaf pair its own — gives simple, policy-rich routing with failure impact limited to the affected links. It suits fabrics with thousands of links and automated configuration. For a small fabric, a link-state IGP underlay with BGP only in the overlay is equally valid; choose by scale and automation maturity. Many such fabrics also run BGP over IPv6 link-local addresses and carry IPv4 routes with IPv6 next hops (RFC 8950), so individual fabric links need no IPv4 addressing at all — a small example of the IPv6-first underlays now common in new builds.

Link-state versus static at the edge

Dynamic IGP to small sites

- Automatic failover across multiple links
- No manual updates as prefixes change
- Adds adjacencies and flooding to the domain
- Needs hierarchy to stop branch churn spreading

Static routing at stub sites

- No routing protocol state to manage
- Predictable and secure
- Failover needs tracking (e.g. probes or BFD)
- Error-prone at scale without automation

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SCENARIO · DECIDE · TO MIGRATE OR NOT

A manufacturing enterprise runs OSPF across 380 routers in a well-designed multi-area hierarchy. It is IPv4-only with no IPv6 plans in the next three years, uses two router vendors, and has a five-person team with deep OSPF experience. A consultant recommends migrating to IS-IS "because large networks use it". What should the architect recommend?

A. Migrate to IS-IS to align with large service-provider networks
B. Migrate to BGP everywhere because it scales furthest
C. Run IS-IS alongside OSPF permanently so the team can learn both protocols safely
D. Stay on OSPF and record a review trigger for IPv6 or segment routing

Circle your choice. Answer and feedback: module answer key.

IGP selection principles

- Choose from requirements: topology, address families, extensions, skills and interoperability
- Prefer standards-based protocols in multi-vendor estates
- Do not change a working IGP without a requirement it cannot meet
- Keep one IGP per routing domain; avoid running two for convenience
- Use static routing where it is simpler and failover can be tracked

Key terms

IGP
Interior gateway protocol: a routing protocol used within one administrative domain, such as OSPF or IS-IS.

OSPF
Open Shortest Path First: an IP-based link-state IGP using areas around a backbone area 0.

TLV
Type-length-value encoding that lets a protocol carry new kinds of information without changing its structure.

Stub site
A site with a single path or exit to the rest of the network, where a default or static route is sufficient.

Link-state protocol
A protocol in which routers flood descriptions of their links so every router builds the same topology map and computes shortest paths.

IS-IS
Intermediate System to Intermediate System: a link-state IGP that runs directly over layer 2 and uses two levels of hierarchy.

BGP-only fabric
A data-centre design using eBGP on every fabric link instead of an IGP.

Routing domain
A set of routers running one routing protocol instance under a single administrative policy.

REFLECT

If you had to justify your network's current IGP to a design review tomorrow, which requirement would you cite? Is it a real requirement or a historical habit?

KNOWLEDGE CHECK

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1. Which requirement MOST favours IS-IS over OSPF in a new backbone design?

   (A) Carrying IPv4 and IPv6 in one instance with simple extensibility

   (B) A hub-and-spoke WAN with many small stub branch areas attached to regional hubs

   (C) A team that has deep experience only with OSPF

   (D) A need for an IP-based protocol that every vendor supports

2. A working OSPF network has no unmet requirements. When is migrating to another IGP justified?

   (A) When larger organisations in the same industry use a different IGP

   (B) When the routers are refreshed to a newer software release

   (C) When a new requirement appears that the current IGP cannot meet well

   (D) When the team wants to gain experience with another protocol

3. Why is eBGP often chosen as the only routing protocol in very large data-centre fabrics?

   (A) It always converges faster than any link-state IGP, in every topology and under every failure

   (B) It removes the need for any equal-cost multipath forwarding

   (C) It is the only protocol able to carry IPv6 in a Clos fabric

   (D) It scales to many links with simple per-session policy and contained failure impact

4. What is the main design drawback of static routing to many stub sites?

   (A) It floods every topology change across the whole routing domain

   (B) It cannot forward traffic for more than one prefix per site

   (C) It requires a separate routing protocol for each address family

   (D) It needs extra mechanisms for failover and becomes error-prone at scale

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MODULE 4 · ROUTING · LESSON 2 · 38 MIN

OSPF Area and IS-IS Level Design

Use hierarchy to contain flooding, state and failure

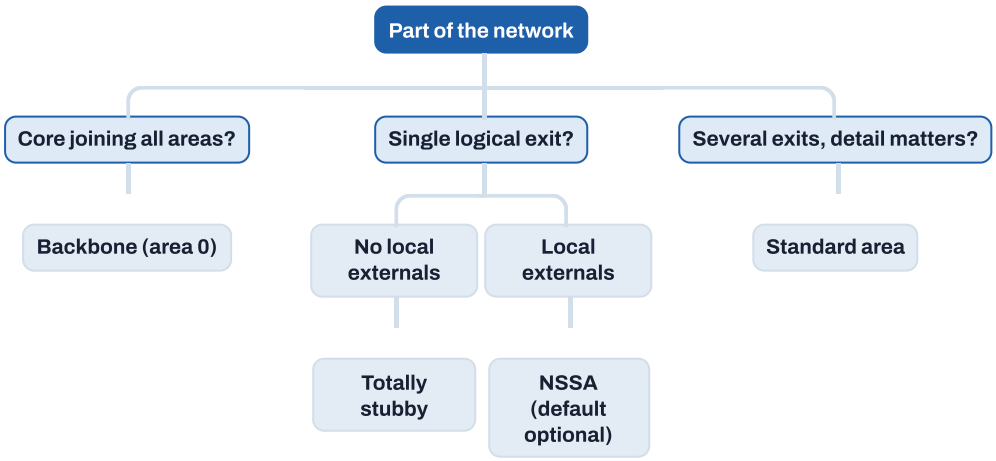
- Design OSPF areas and choose area types for each part of the network
- Design IS-IS level boundaries and route leaking for optimal exits
- Place area or level boundaries where addressing and topology aggregate

Link-state protocols scale by dividing the network into flooding domains. Inside a domain every router knows the full topology; between domains only summarised reachability passes. The design questions are where to draw the boundaries, what to let through them, and how to avoid the classic traps: a partitioned backbone, a boundary that cannot summarise, or an exit that becomes suboptimal because detail was hidden.

OSPF area types

AREA TYPE	RECEIVES	ALLOWS LOCAL EXTERNAL ROUTES?	TYPICAL USE
Backbone (area 0)	Everything	Yes	Core interconnecting all other areas
Standard area	Intra-area, inter-area and external routes	Yes	Large blocks needing full detail
Stub area	Intra-area, inter-area and a default; no external routes	No	Blocks with one logical exit
Totally stubby area	Intra-area routes and a default only	No	Branch or access areas with a single exit
Not-so-stubby area (NSSA)	Intra-area and inter-area routes plus its own externals; default optional	Yes, converted at the ABR	A stub-like area that must inject local external routes

Choosing an area type



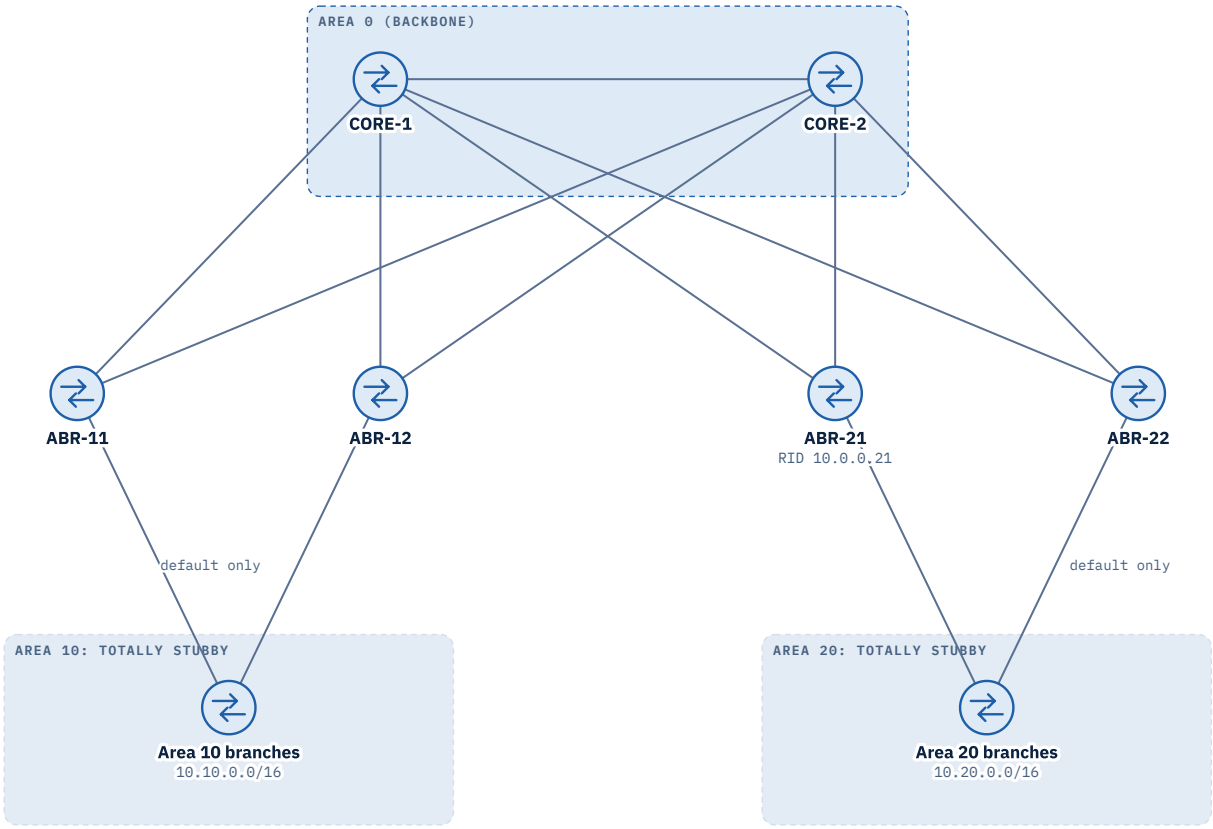
Ask the questions in order. A single logical exit makes a default enough; local external routes are what push a stub-like area to NSSA.

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Designing OSPF areas

1. **Start with the backbone** Area 0 should be contiguous, well connected and contain the core. Every other area attaches to it through area border routers (ABRs).
2. **Align areas with address blocks** Assign each area a contiguous address block so its ABRs can summarise it with one or a few ranges.
3. **Choose ABR placement** Place ABRs at natural aggregation points — distribution switches or regional WAN hubs — with at least two per area for resilience.
4. **Minimise what crosses the boundary** Use stub or totally stubby areas wherever an area has a single logical exit, NSSA where a local external source exists.
5. **Bound churn** Put unstable parts of the network, such as WAN branches, in their own areas so their flaps do not trigger full SPF runs elsewhere.

A multi-area OSPF design with stubby branch areas



Each area has two ABRs at an aggregation point and one contiguous address block. The backbone sees two stable summaries; branches in the totally stubby areas see only their own area plus a default.

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ABR for a totally stubby branch area with summarisation

ABR-21 · IOS-style CLI

```
! ABR between the backbone and branch area 20
router ospf 1
 router-id 10.0.0.21
 area 20 stub no-summary
 area 20 range 10.20.0.0 255.255.0.0
 passive-interface default
 no passive-interface Ethernet1/1
 no passive-interface Ethernet1/2
interface Ethernet1/1
 description To core (area 0)
 ip ospf 1 area 0
interface Ethernet1/2
 description To branch aggregation (area 20)
 ip ospf 1 area 20
```

Branch routers in area 20 see only their own area plus a default; the backbone sees one /16 for the entire area. Every router in area 20 must also be configured as stub.

IS-IS levels

Level 1 (L1)

- Routing within an area
- L1 routers know only their area's topology
- Reach other areas via the nearest L1/L2 router (default from the attached bit)
- Like a totally stubby OSPF area by default

Level 2 (L2)

- The backbone between areas
- Must be contiguous across L2-capable routers
- Carries inter-area reachability
- Many networks run a single L2-only domain

INSIGHT · HIDDEN DETAIL CAN COST YOU THE BEST EXIT

An L1 area with two L1/L2 exits sends traffic to the nearest exit by default — not necessarily the best one for each destination. If the business needs optimal paths to specific destinations, leak those prefixes from L2 into L1. The same trade-off applies to OSPF totally stubby areas: less state, but a default route cannot express which exit is closer to a given destination.

PRO TIP · FLAT IS FINE UNTIL IT IS NOT

Modern routers handle large single-area or single-level domains, and flat designs avoid summarisation side effects. Introduce hierarchy when you need to contain churn from unstable regions, limit failure propagation, reduce state on small devices, or meet a policy boundary — not because a textbook area-size figure was exceeded.

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SCENARIO · DECIDE · A WAN THAT KEEPS FLAPPING

A national retailer runs a single OSPF area containing its core, two data centres and 600 branches connected by broadband overlays. Branch links flap several times a day, causing frequent SPF runs and CPU spikes on core routers. Each branch has two tunnels to two regional hubs. Which design best fixes this?

- (A) Regional totally stubby areas with hub ABRs summarising to the backbone
- (B) Increase SPF throttle timers on the core so it recalculates less often
- (C) Move each branch into its own area to isolate it completely
- (D) Replace OSPF on the branches with static default routes towards a single hub and no failover

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **IGP design sheet**

Record the protocol, hierarchy and boundary decisions for each routing domain so they stay consistent as the network grows.

DOMAIN / AREA OR LEVEL	ADDRESS BLOCK(S)	AREA TYPE / LEVEL	BORDER ROUTERS	SUMMARIES ADVERTISED	EXTERNALS ALLOWED	GROWTH LIMIT / REVIEW TRIGGER
Area 20 — north branches	*10.20.0.0/16*	*Totally stubby*	*HUB-N1, HUB-N2*	*10.20.0.0/16 to area 0*	*No*	*Review at 250 branches or if local internet breakout needs routes*
IS-IS L2 backbone	*10.0.0.0/20 (loopbacks, links)*	*Level 2 only*	*All core routers*	*None (loopbacks kept specific)*	*Default only*	*Review if L1 areas are needed for regional churn*

Key terms

Area border router (ABR)
An OSPF router connecting the backbone to one or more other areas, where inter-area summarisation is configured.

Backbone area
OSPF area 0, to which all other areas must connect and through which inter-area traffic passes.

Stub area
An OSPF area that receives no external routes, using a default route to reach them.

Totally stubby area
An OSPF area that receives only intra-area routes and a default route.

NSSA
Not-so-stubby area: a stub-like OSPF area that may inject its own external routes, which the ABR translates for the rest of the domain.

Attached bit
An IS-IS flag set by L1/L2 routers that tells L1 routers to use them as the default exit from the area.

Route leaking
Advertising selected prefixes from the IS-IS level 2 backbone into a level 1 area so L1 routers choose better exits.

Flooding domain
The set of routers that receive each other's link-state advertisements and share one topology database.

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REFLECT

Where are the flooding-domain boundaries in your network today? Do they coincide with the address blocks, or does summarisation fight the topology?

Key takeaways

- Hierarchy contains flooding, state and failure: draw boundaries where topology and addresses aggregate
- Give each area or level a contiguous block and at least two border routers
- Use totally stubby areas for single-exit branches and NSSA where local externals must enter
- Hidden detail can cost the best exit; leak specific prefixes where optimal paths matter
- Stay flat until churn, failure containment or device limits justify hierarchy

KNOWLEDGE CHECK

5. A branch area has a single logical exit through two ABRs, and no local external routes. Which OSPF area type MOST reduces its routing state?

   (A) Totally stubby area
   (B) Not-so-stubby area
   (C) Standard area
   (D) Stub area

6. When should an OSPF area be configured as an NSSA rather than a stub area?

   (A) When the area has more than one ABR connecting it to the backbone
   (B) When the area must inject external routes from a local source
   (C) When the area needs to receive all external routes from the backbone
   (D) When the area contains the data-centre core and the internet edge

7. IS-IS L1 routers in an area with two L1/L2 exits send traffic for some destinations through the farther exit. What fixes this while keeping the hierarchy?

   (A) Clear the attached bit on both L1/L2 routers in the area
   (B) Configure every router in the area as level 2 only
   (C) Leak the relevant prefixes from level 2 into level 1
   (D) Summarise the area's prefixes more aggressively at level 2

8. Why should OSPF areas be aligned with contiguous address blocks?

   (A) So that every router in the area can share one router ID
   (B) So ABRs can summarise each area into a few ranges
   (C) So that inter-area traffic avoids the backbone entirely
   (D) So that the area can be configured as an NSSA if needed

9. What is the main reason to place unstable WAN branches in their own OSPF areas?

   (A) To allow the branches to run a different routing protocol from the one used in the core
   (B) To give each branch a path to the internet without the core
   (C) To remove the need for ABRs at regional hub sites
   (D) To keep their flaps from triggering SPF runs across the rest of the network

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MODULE 4 · ROUTING · LESSON 3 · 38 MIN

Summarisation and Redistribution Design

Hide detail deliberately and move routes between domains without loops

- Decide where and what to summarise, and what must never be summarised
- Explain the black-hole and suboptimal-routing side effects of summarisation
- Design redistribution with tagging and filtering that prevents loops

Summarisation and redistribution are where routing designs most often go wrong — not because the mechanics are difficult, but because their side effects appear only under failure. A summary hides a failed subnet from the rest of the network, which is precisely its benefit and its risk. Redistribution between two protocols at two points creates a loop opportunity that may remain invisible for years until a link fails in the wrong place.

Summarisation trade-offs

Benefits

- Smaller routing tables and databases
- Topology changes hidden behind the summary
- Fewer SPF runs and updates elsewhere
- Clearer, more stable routing domains

Costs and risks

- Traffic can follow a suboptimal path
- Failed components may attract traffic into a black hole
- Requires a hierarchical address plan
- Can break services that need specific routes

Summary arithmetic

MEASURE	FORMULA	WHAT IT TELLS YOU
Block size	`Addresses in a /p = 2^(32 − p)`	A /16 summary covers 65,536 addresses; a /20 covers 4,096.
Components covered	`Number of /q prefixes in a /p = 2^(q − p)`	A /20 summary covers 16 contiguous /24s — only if they start on a /20 boundary.
Summary boundary check	`Network address of block mod block size = 0`	10.20.32.0/20 is valid (32 is a multiple of 16 in the third octet); 10.20.40.0/20 is not.

WATCH OUT · NEVER SUMMARISE WHAT SERVICES NEED INDIVIDUALLY

Loopbacks used as BGP next hops, tunnel endpoints or label-switched-path endpoints often need to be known as specific /32 (or /128) routes. Summarising them can hide a failed router from protocols that depend on next-hop tracking, or break label-switched paths that require an exact match. Keep these addresses in a dedicated block and leave them unsummarised.

INSIGHT · THE DISCARD ROUTE IS A FEATURE

When a router advertises a summary it normally installs a matching route to a null interface. Packets for unused or failed parts of the summary are dropped locally instead of following a less-specific route — often a default — back towards the network and looping. The trade-off: during a partial failure, traffic for the failed component is discarded at the summarising router, so ensure redundant summarising routers each have full component reachability.

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Safe redistribution design

01 **Avoid it if possible**
Use a default, a summary or a single protocol instead

→

02 **Minimise points**
One point, or two with explicit control

→

03 **Filter**
Redistribute only the prefixes needed

→

04 **Tag**
Mark every redistributed route with its source domain

→

05 **Block re-entry**
Deny tagged routes from returning to their source

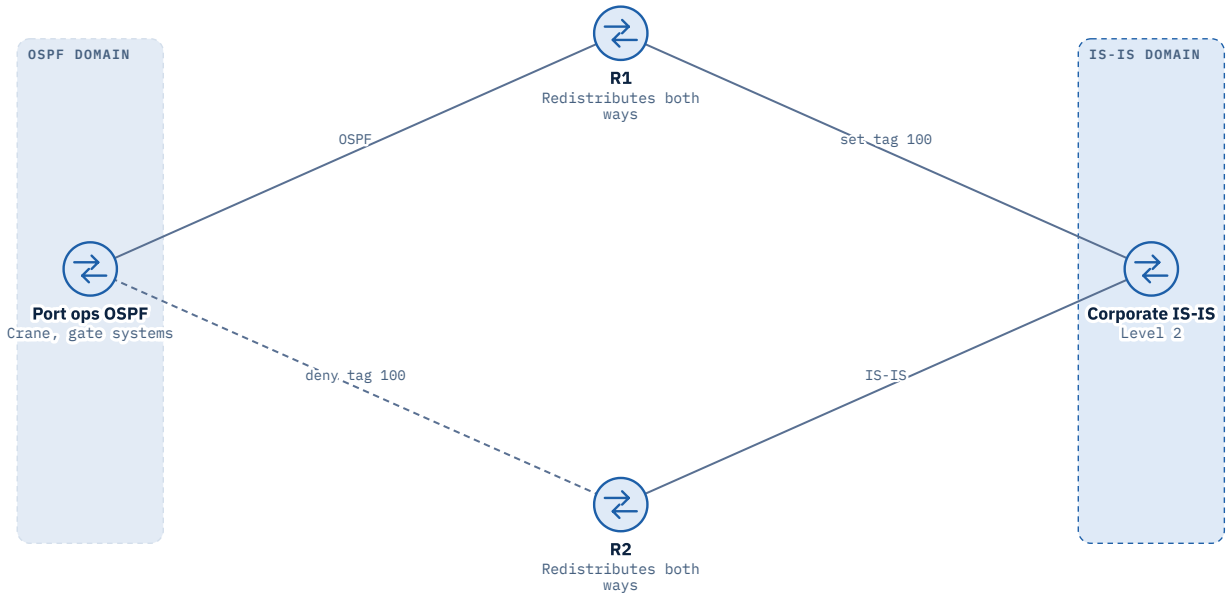
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06 **Set metrics and preference**
Make the intended path win deterministically

EXAMPLE · HOW A TWO-POINT REDISTRIBUTION LOOP FORMS

Domain A and domain B are joined by routers R1 and R2, both redistributing in both directions. A route from A enters B at R1; R2 learns it from B and, because the protocol in B has a better administrative preference on R2 than the original A route, R2 prefers the B-learned copy and redistributes it back into A. Traffic can now circle between the domains, or the route can persist after the original disappears. Tagging at R1 and denying that tag at R2 breaks the cycle.

Two-point mutual redistribution and the tag that stops a loop



R1 tags port-operations routes 100 as they enter IS-IS; R2 refuses to send tag-100 routes back into OSPF, and the reverse applies to corporate routes tagged 200. Without the tags, a route can leave through R1 and re-enter through R2.

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Tag on the way in, deny on the way back

R2 · IOS-style CLI

```
! Routes from OSPF into IS-IS are tagged 100; routes from IS-IS into OSPF are tagged 200
route-map OSPF-TO-ISIS deny 10
 match tag 200
route-map OSPF-TO-ISIS permit 20
 match ip address prefix-list PORT-OPS
 set tag 100
route-map ISIS-TO-OSPF deny 10
 match tag 100
route-map ISIS-TO-OSPF permit 20
 match ip address prefix-list CORPORATE
 set tag 200
! Prefix lists PORT-OPS and CORPORATE are defined separately
router isis CORE
 metric-style wide
 redistribute ospf 1 route-map OSPF-TO-ISIS
router ospf 1
 redistribute isis CORE level-2 subnets route-map ISIS-TO-OSPF
```

The same policy is applied on both redistribution routers. A route that has already crossed in one direction carries a tag that stops it crossing back. IS-IS needs wide metrics to carry route tags.

CASE STUDY · PORT AUTHORITY NETWORK MERGER

A port authority is merging its corporate network with the network of the terminal operations company it has acquired. Corporate runs IS-IS level 2 across 60 routers; terminal operations runs OSPF across 90 routers, including crane control and gate automation systems that must remain available during any single failure. The two networks will be joined at two sites — the main office and the container terminal — for resilience. Address space overlaps in two /24 ranges. The operations team insists that corporate routing problems must never affect crane control. The board wants a single network within 18 months.

1. Should the networks be merged into one IGP immediately? What are the risks?

2. How would you design the interim interconnection?

3. How do the overlapping ranges affect the design?

4. What must be true before a full merger is attempted?

Model answer: module answer key.

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SCENARIO · DECIDE · TWO-POINT REDISTRIBUTION

A company redistributes between OSPF and IS-IS at two routers for resilience. After a link failure, some prefixes oscillate and traceroutes show traffic looping between the two redistribution routers. What is the best design correction?

- (A) Remove redistribution from one of the two routers so that only a single point remains
- (B) Tag routes at redistribution and deny the other domain's tag on the way back
- (C) Lower the administrative preference of OSPF on both routers
- (D) Increase the metric of all redistributed routes to the maximum value

Circle your choice. Answer and feedback: module answer key.

Key terms

Route summarisation
Advertising one aggregate prefix in place of many more specific prefixes it contains.

Black hole
A condition in which traffic is attracted to a router that cannot deliver it, usually because a summary hides a failed component.

Route tag
A value attached to a route, often at redistribution, used by policy to recognise its origin.

Mutual redistribution
Redistributing in both directions between two routing domains.

Discard route
A route to a null interface installed for a summary, so packets for unused parts of it are dropped rather than looping.

Redistribution
Injecting routes learned from one routing protocol or source into another.

Administrative preference
A router-local value (often called administrative distance) that ranks routes to the same prefix from different sources.

Address plan
A hierarchical allocation of address blocks to regions, sites and functions that makes summarisation possible.

REFLECT

Identify every redistribution point in a network you know. For each, could you prove that a route cannot re-enter the domain it came from? What would you add to make that provable?

Key takeaways

- Summaries hide detail on purpose — and can attract traffic for failed components
- Keep next-hop and tunnel-endpoint loopbacks in a dedicated, unsummarised block
- Each summarising router must keep full reachability to the components it advertises
- Avoid redistribution where a default or summary will do; otherwise filter, tag and block re-entry at every point

KNOWLEDGE CHECK

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10. Which addresses should normally be kept out of summaries?

(A) Loopbacks used as BGP next hops or tunnel endpoints

(B) User subnets on access switches in a campus block

(C) Point-to-point links between distribution and core

(D) Branch LAN subnets inside a totally stubby area

11. What is the purpose of the discard (null) route installed with a summary?

(A) To advertise the summary to neighbours in other routing domains

(B) To send traffic for failed components to a backup router

(C) To drop traffic for unused parts of the summary instead of looping it

(D) To prevent the summary from being installed in the routing table

12. Is 10.20.40.0/21 a valid summary boundary?

(A) No, because 40 is not a multiple of 16

(B) No, because a /21 must start at a multiple of 32

(C) Yes, because 40 is a multiple of 8

(D) Yes, because any /21 can start at an even value

13. What is the PRIMARY loop-prevention technique for mutual redistribution at two points?

(A) Running both routing protocols with identical hello and dead timers on both redistribution routers

(B) Redistributing only the default route in both directions at each point

(C) Configuring a different router ID on each of the redistribution routers

(D) Tagging routes on redistribution and denying the other domain's tag on the way back

14. Why can a summary create a black hole during a partial failure?

(A) The summary is withdrawn whenever any component route fails

(B) Traffic for a failed component is still attracted by the summary

(C) Summaries are always advertised with an infinite metric

(D) The discard route forwards packets to the backbone area

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MODULE 4 · ROUTING · LESSON 4 · 43 MIN

BGP in the Enterprise, Route Reflection and Convergence

Use BGP where it earns its place, scale it, and design for fast recovery

- Decide when an enterprise needs BGP internally and where it belongs
- Design redundant route reflection and avoid hidden-path problems
- Build a convergence budget and choose techniques to meet it
- Plan growth by identifying the resource that will run out first

BGP is the protocol of policy and scale. Inside an enterprise it earns its place where the IGP should not carry certain routes (internet tables, large external sets), where independent domains must exchange routes with policy, or where an overlay — an MPLS VPN, an EVPN fabric, an SD-WAN — needs a scalable control plane. Using BGP where an IGP would do adds a second protocol, a second set of failure modes and a policy language the team must master.

When BGP belongs inside the enterprise

SITUATION	WHY BGP	ALTERNATIVE IF NOT JUSTIFIED
Internet edge with multiple providers	Policy control and full or partial tables	Default routes with tracking
Large route sets the IGP should not carry	Keeps the IGP small and fast	Summarise or filter at the edge
Separate administrative domains (subsidiaries, partners)	Policy and route control between AS boundaries	Static routes or filtered redistribution
Overlay control planes (EVPN, L3VPN, some SD-WAN)	Carries VPN, MAC and tenant routes at scale	Not usually replaceable
Large data-centre fabric underlay	Scale, simplicity, failure containment	Link-state IGP for smaller fabrics

iBGP scaling

MEASURE	FORMULA	WHAT IT TELLS YOU
Full-mesh sessions	`Sessions = n × (n − 1) ÷ 2`	40 iBGP routers need 780 sessions, 39 per router; every new router adds a session on every existing one.
With route reflectors	`Sessions ≈ clients × RRs per client + RR mesh`	40 clients with 2 RRs each need 80 client sessions plus one RR-to-RR session.
BFD detection time	`Detection = transmit interval × multiplier`	50 ms × 3 = 150 ms to declare a neighbour down, versus tens of seconds with default protocol hold timers.

Route reflection rules

ROUTE LEARNED FROM	REFLECTED TO CLIENTS?	REFLECTED TO NON-CLIENTS?	NOTES
An eBGP peer	Yes	Yes	Normal BGP behaviour

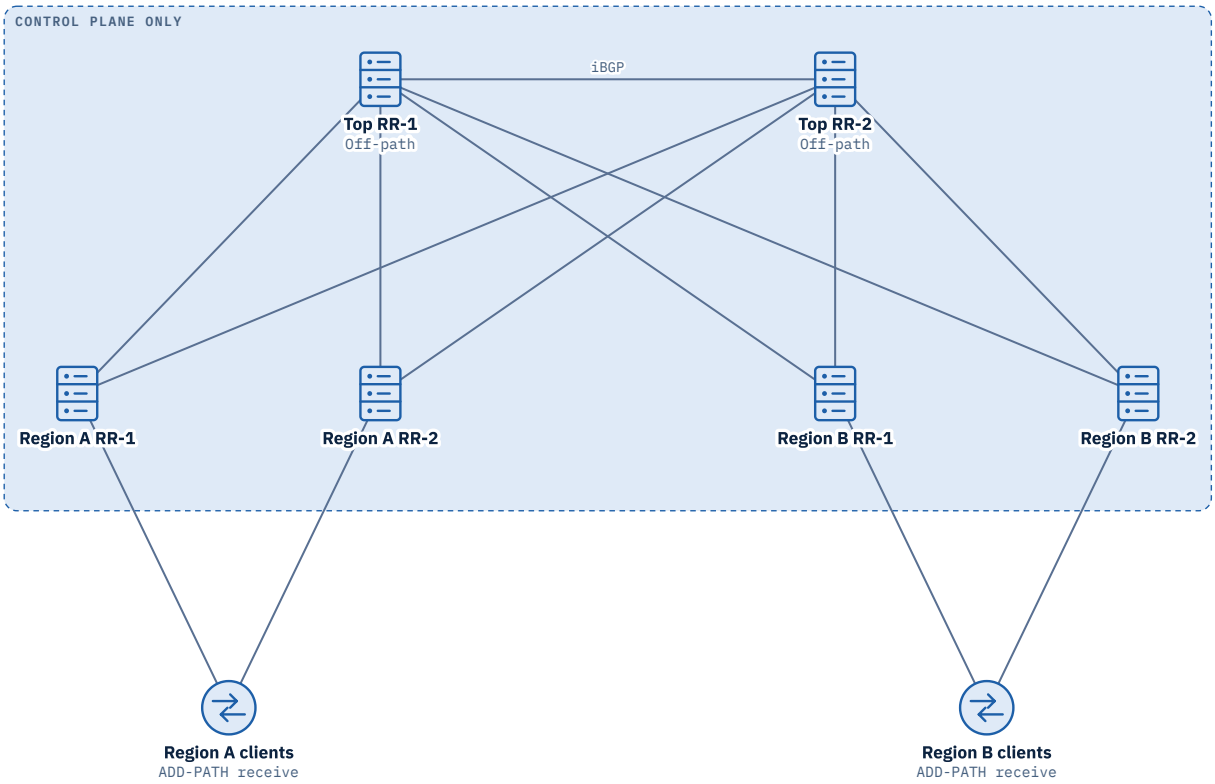
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ROUTE LEARNED FROM	REFLECTED TO CLIENTS?	REFLECTED TO NON-CLIENTS?	NOTES
A client	Yes (all except the sender)	Yes	This is what removes the full mesh
A non-client iBGP peer	Yes	No	Non-clients still need a full mesh among themselves

Designing route reflection

1. **Make it redundant** Every client peers with at least two reflectors in different failure domains (power, room, site).
2. **Decide on placement** Reflectors do not change the next hop by default, so they need not be in the forwarding path. Off-path or virtual reflectors keep control-plane load away from busy forwarding routers — but they compute best paths from their own position.
3. **Choose cluster IDs deliberately** Shared cluster IDs on a reflector pair save memory; distinct IDs let each reflector accept the other's reflected routes, adding resilience at the cost of more paths stored.
4. **Preserve path diversity** A reflector advertises only its best path, hiding alternatives. Where fast failover or multipath matters, advertise additional paths (ADD-PATH) or use diverse next-hop designs.
5. **Consider hierarchy** Very large networks use regional reflectors that are clients of a top-level pair, keeping session counts and update fan-out bounded.

Hierarchical route reflection with off-path reflectors



Regional reflector pairs serve their clients and are themselves clients of a top-level pair. Every client peers with two reflectors in different failure domains, and ADD-PATH gives clients a backup path for each internet exit.

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WATCH OUT · THE HIDDEN-PATH PROBLEM

If two exits exist for a prefix but the reflector sends clients only its single best path, clients have no backup: when that exit fails they must wait for the reflector to compute and send a new path. They also cannot load-share. Advertising more than one path, or ensuring each exit uses a distinguishable route, restores diversity — at the cost of more memory and updates.

Anatomy of convergence

01 Detect — Loss of signal, BFD, protocol hellos →

02 Propagate — Flood LSAs or send BGP updates →

03 Compute — SPF or best-path recalculation →

04 Install — Update RIB and program the FIB →

05 Protect — Pre-installed backup paths switch locally

What happens after a protected link fails

0 MS — Link fails
Far end stops answering; the local port may stay up

≈150 MS — BFD declares the neighbour down
Three missed packets at 50 ms

≈160 MS — Local switch to the backup next hop
Pre-installed alternate; traffic flows again

≈200–500 MS — Flooding and SPF elsewhere
Other routers learn and recompute

< 1 S — FIB updated across the domain
Post-convergence paths installed

Illustrative timings for a link protected by BFD (50 ms × 3) and a pre-installed loop-free alternate. Traffic loss ends at the local switchover; the network-wide steps that follow only tidy up the forwarding paths.

Convergence budget

MEASURE	FORMULA	WHAT IT TELLS YOU
Total convergence	`T ≈ T(detect) + T(propagate) + T(compute) + T(install)`	Estimate each component; the largest one is where design effort pays off.
Loop-free alternate condition	`D(N,D) < D(N,S) + D(S,D)`	Neighbour N is a loop-free backup for source S toward destination D if N's own shortest path to D does not come back through S. Backups that meet this can be pre-installed for local switchover in tens of milliseconds.
FIB update for many prefixes	`T(install) grows with prefixes unless next hops are shared`	Hierarchical FIB designs that point many prefixes at one shared next-hop object update in constant time when that next hop changes.

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PRO TIP · SET CONVERGENCE TARGETS BY APPLICATION

"Fast convergence" is not a requirement. "Voice calls survive any single link failure without dropping" is. Translate application tolerance into a budget — for example under 1 s for voice, under 50 ms for some industrial or trading flows — and spend design effort only where the budget is tight: BFD for detection, fast reroute for protection, tuned throttles for computation. Where the backbone uses segment routing — over MPLS labels (SR-MPLS) or IPv6 (SRv6), increasingly in place of older label-distribution and tunnel-signalling protocols — **TI-LFA** can pre-compute a loop-free backup for almost any topology, closing the coverage gaps of plain LFA; adopt it where a requirement justifies the change and every platform supports it.

SCENARIO · DECIDE · REFLECTOR PLACEMENT

A company runs iBGP across 120 routers in three regions to carry internet partial routes and VPN routes. Its two core routers act as route reflectors for all clients and are also the busiest forwarding devices; during a recent software upgrade, reflector CPU spikes delayed convergence. Several clients need fast failover between two internet exits. Which design best fits?

- (A) Return to a full iBGP mesh among all 120 routers to remove reflectors
- (B) Keep the two core reflectors but increase their BGP timers to reduce the CPU load during upgrades
- (C) Dedicated off-path regional reflector pairs under a top-level pair, with ADD-PATH
- (D) Add a single dedicated reflector to replace both core reflectors

Circle your choice. Answer and feedback: module answer key.

Growth planning: what runs out first?

RESOURCE	SIGNAL TO WATCH	DESIGN RESPONSE
Routing table / FIB capacity	Prefix count trend versus hardware limit	Summarise, filter, default plus partial routes
Link-state database and SPF load	SPF runs per hour, LSDB size, CPU	Add areas or levels, contain churn
Adjacencies and sessions	Neighbours per device, session count	Route reflection, hierarchy, fewer meshed links
Control-plane CPU and memory	Peaks during failures and upgrades	Dedicated reflectors, staged upgrades
Operational capacity	Change failure rate, time to troubleshoot	Simplify, standardise, automate

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Routing design review checklist

☐ IGP choice justified by requirements, with a review trigger
☐ Flooding-domain boundaries align with address blocks
☐ Summaries defined, with next-hop and endpoint loopbacks excluded
☐ Every redistribution point filtered and tagged against re-entry
☐ BGP used only where policy, scale or overlays require it
☐ Route reflectors redundant and placed in separate failure domains
☐ Path diversity preserved where fast failover or multipath is required
☐ Convergence budget stated per critical application and tested
☐ Growth thresholds defined for tables, sessions and control-plane load

Key terms

iBGP full mesh
The requirement that every iBGP speaker peer with every other, because routes learned via iBGP are not re-advertised to iBGP peers by default.

Route reflector
An iBGP speaker allowed to re-advertise iBGP routes to its clients, removing the need for a full mesh.

Cluster ID
An identifier used by route reflectors to detect and discard reflected routes that loop back to their cluster.

ADD-PATH
A BGP capability allowing more than one path for the same prefix to be advertised to a peer.

BFD
Bidirectional Forwarding Detection: a lightweight protocol that detects forwarding failures between neighbours in milliseconds.

Fast reroute
Pre-computed backup paths installed in advance so a router can switch locally as soon as a failure is detected.

Loop-free alternate (LFA)
A neighbour that can be used as a backup next hop because its shortest path to the destination does not loop back through the failing router.

Convergence
The process by which all routers reach a consistent, correct forwarding state after a change.

REFLECT

Pick one critical application and write its convergence budget. Which component — detection, propagation, computation or installation — is your biggest unknown, and how would you measure it?

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Key takeaways

- Use BGP inside the enterprise where policy, scale or an overlay control plane needs it — not by default
- Route reflectors remove the full mesh; deploy them in pairs across failure domains, often off-path
- Reflectors hide alternative paths: advertise additional paths where fast failover or multipath matters
- Build convergence budgets per application and spend effort on the largest term: BFD, fast reroute or prefix-independent convergence
- Plan growth by finding the resource that runs out first

KNOWLEDGE CHECK

15. How many iBGP sessions does a full mesh of 25 routers require?

(A) 300 (B) 600
(C) 625 (D) 250

16. A route reflector receives a route from a non-client iBGP peer. To whom does it reflect it?

(A) To its clients and its other non-client peers (B) To its non-client iBGP peers only
(C) To nobody, as iBGP routes are never reflected (D) To its route-reflector clients only

17. Why might route reflectors be deployed out of the forwarding path?

(A) They do not change next hops by default, so they need not forward traffic (B) Reflectors in the forwarding path cannot run any IGP at all
(C) Off-path reflectors always choose the optimal exit for every one of their clients automatically (D) Placing reflectors in the forwarding path disables ADD-PATH support

18. BFD is configured with a 100 ms interval and a multiplier of 3. What is the detection time?

(A) 100 ms (B) 300 ms
(C) 33 ms (D) 3 s

19. Which technique MOST directly reduces convergence time after failure detection for a protected link?

(A) Increasing SPF throttle timers to batch more topology changes (B) Advertising every prefix as a separate summary to all neighbours
(C) Pre-installing loop-free backup next hops for local switchover (D) Moving route reflectors into the forwarding path of the core

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 4 · ROUTING

Answer key

Knowledge checks

1. A A single IS-IS instance carries both families and is easily extended with TLVs. The other factors point toward OSPF or are equal.

2. C Migration touches every router and carries risk. It should be driven by an unmet requirement, which is what a review trigger in the decision record captures.

3. D BGP's per-session model, policy control and limited failure propagation suit automated fabrics with thousands of links. It is not inherently faster, and ECMP is still used.

4. D Static routes add no protocol state, but failover must be tracked and manual changes multiply with site count.

5. A Totally stubby areas receive only intra-area routes and a default, which is all a single-exit branch area needs.

6. B NSSA keeps stub behaviour for incoming externals while allowing local external routes, which the ABR translates for the rest of the domain.

7. C Route leaking gives L1 routers specific reachability so they can pick the better exit for those destinations, without flattening the design.

8. B Summarisation depends on addressing. If an area's prefixes are scattered, the ABR must advertise many specific routes.

9. D Area boundaries contain topology changes. With summarisation, the backbone sees stable summaries while the area absorbs local churn.

10. A Services that track next hops or build paths to specific endpoints need the exact routes; summarising them can hide failures or break label-switched paths.

11. C Without it, packets for non-existent components could follow a less specific route such as a default and loop.

12. C A /21 spans 8 values of the third octet (2,048 addresses), so it must start at a multiple of 8. 40 ÷ 8 = 5, so the boundary is valid.

13. D Tags record where a route came from so policy can stop it returning. Defaults alone may not meet reachability needs and timers do not affect loops.

14. B The summary stays advertised while any component exists, so traffic for the failed part still arrives and is dropped by the discard route.

15. A $n(n-1) \div 2 = 25 \times 24 \div 2 = 300$ sessions.

16. D Routes from non-clients are reflected to clients only; non-clients must still be fully meshed with each other.

17. A Because next hops are preserved, reflectors can be dedicated control-plane devices. Their best-path choice reflects their own position, which is a trade-off to manage.

18. B Detection time is interval × multiplier: 100 ms × 3 = 300 ms.

19. C Fast reroute switches to a pre-computed backup without waiting for propagation or computation. Longer throttles slow convergence.

Scenarios & case studies

> **Scenario · To migrate or not**
> **D** is best. Correct. No current requirement is unmet by OSPF, and migration would touch every router and the team's skills for no measurable gain. A review trigger keeps the decision honest if requirements change.

> **Scenario · A WAN that keeps flapping**
> **A** is best. Correct. Branch flaps stay inside regional areas, the core sees only stable summaries, and branches need only a default because every exit is via their hubs.

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Case study · Port authority network merger

Immediate merger. Unifying the IGPs at once would put crane control in the same flooding domain as corporate routing, conflicting with the operations requirement, and would require resolving overlaps first. A staged approach is lower risk.

Interim interconnection. Mutual redistribution at both sites, filtered to the specific prefixes each side needs, with route tags blocking re-entry and metrics making one site preferred for each direction. Consider exchanging only summaries or a default toward operations, and placing the operational networks in their own VRF so corporate changes cannot reach them. A firewall at each interconnection enforces policy between the organisations.

Overlaps. Overlapping ranges cannot be routed between the domains. Re-address the smaller side (preferably corporate, to avoid touching crane systems), or use NAT temporarily at the interconnect with a planned removal date.

Before full merger. Overlaps removed; a target addressing plan with summarisable blocks; agreement on the final IGP (likely keeping operations as a separate area, level or VRF); tested failure scenarios at both interconnect sites; operations' sign-off that its availability requirement is protected.

Scenario · Two-point redistribution

B is best. Correct. Tagging prevents routes that originated in one domain from re-entering it via the other redistribution point — the root cause of the loop — while keeping both points for resilience.

Scenario · Reflector placement

C is best. Correct. Dedicated reflectors separate control-plane load from forwarding, regional hierarchy bounds fan-out, and advertising additional paths gives clients pre-known backups for the internet exits.

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05

MODULE 5 · RESILIENCE

High Availability & Resilience Design

Failure domains, availability maths, convergence, safe change and disaster recovery

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Identify failure domains and shared fate, and shrink blast radius to targets the business accepts
- Calculate availability for serial and parallel designs, including common-cause effects and MTTR
- Engineer convergence budgets and choose between fast reroute, graceful restart and non-stop routing
- Design maintenance and change processes that never reduce protection below target
- Translate RPO and RTO into replication, addressing and failover designs

LESSONS

1. Failure Domains & Blast Radius · 27 min
2. Redundancy Models & Availability Mathematics · 33 min
3. Convergence Budgets & Fast Recovery · 31 min
4. Maintenance & Change Design · 27 min
5. Disaster Recovery Design · 31 min

MODULE 5 · RESILIENCE · LESSON 1 · 27 MIN

Failure Domains & Blast Radius

Decide what is allowed to break together, and make everything else break apart

- Define failure domains and blast radius and locate them in an existing design
- Identify hidden shared fate across physical, logical, software and human dependencies
- Select containment techniques that limit blast radius at a cost the business accepts

Every resilience conversation begins with one question: **what breaks together?** Redundancy only helps when the redundant parts fail independently. An expert designer therefore spends less time adding boxes and more time drawing boundaries — deciding which failures are allowed to spread, how far, and who notices when they do.

DEFINITION · FAILURE DOMAIN AND BLAST RADIUS

A **failure domain** is the set of components and services that are impaired by a single fault. The **blast radius** is the business-visible consequence of that fault: how many users, transactions, sites or safety functions are affected, and for how long. Good design keeps failure domains small and aligned with boundaries the business can tolerate.

The failure-domain hierarchy

Region / metro	`Natural disaster, regional power grid` `Shared carrier backbone` `Regional regulatory event`
Site / data centre	`Utility feed and generators` `Building entry points for fibre` `Site-wide control or management cluster`
Hall / pod / power zone	`UPS string or PDU` `Fabric pod and its spines` `Cooling zone`
Device	`Chassis, backplane, route processor` `Operating system image and its bugs`
Component	`Optic, line card, power supply` `A single process such as a routing daemon`

Six sources of shared fate

SF1

Physical

Two 'diverse' circuits in the same duct, two power feeds from the same substation, two racks under the same sprinkler zone.

SF2

Logical

One Layer 2 broadcast domain, one spanning-tree instance, one routing flooding domain or one overlay control plane spanning everything.

SF3

Control and management

A single controller cluster, orchestrator, identity store, DNS or NTP service that every site depends on to function or to recover.

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SF4

Software

Both members of a pair running the same image: one bug, one malformed update or one time-handling defect hits both at once.

SF5

Operational

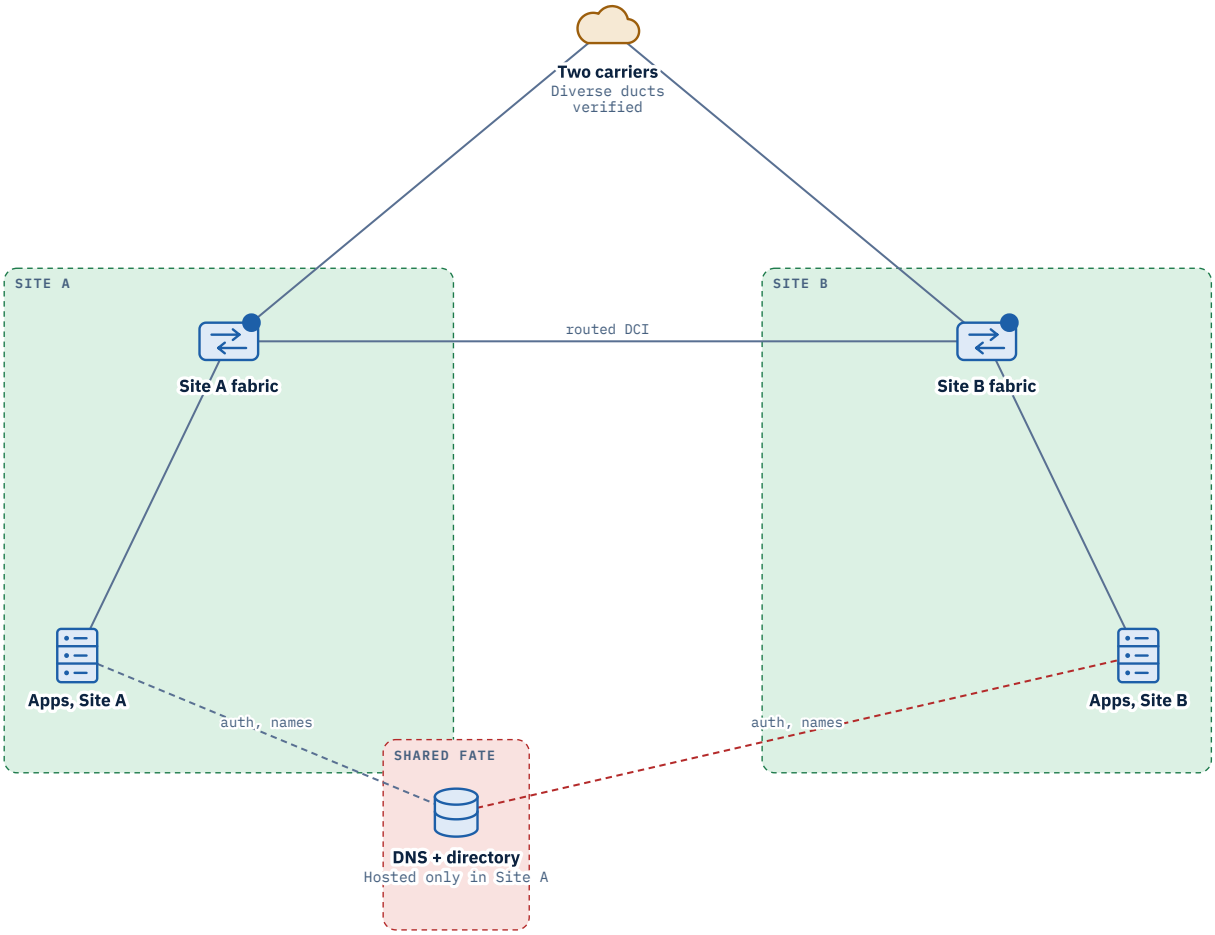
The same change pushed to every device by the same pipeline in the same minute, or the same engineer working on both halves of a pair.

SF6

Provider

Two carriers that share a last-mile segment, an exchange building or an upstream wholesale provider.

Two independent sites with one hidden shared dependency



Every network element is paired across two sites, yet both sites resolve names and authenticate against services that live only in Site A. The red zone is a single point of failure for the whole service, however many boxes are duplicated.

Layer 2 is the classic oversized failure domain. A loop, a broadcast storm or a flapping MAC address can consume every link and every control plane in the VLAN, wherever it is stretched. Routed boundaries stop that propagation, and summarisation at area or level borders stops routing instability from flooding beyond its origin. In the control plane, separate route-reflector clusters or regional instances keep a software fault in one region from reaching the next.

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Containment techniques — decision matrix

TECHNIQUE	WHAT IT CONTAINS	TRADE-OFF	USE WHEN
Routed access / Layer 3 boundary	Loops, storms, MAC instability	Less VLAN mobility; needs L3-aware applications	Campus and DC edges where stretched VLANs are not essential
Summarisation at area/level borders	Routing churn and SPF load	Possible sub-optimal paths; black-hole risk without careful design	Large link-state domains, hub sites
Cell or pod architecture	Any fault inside one cell	Duplicated infrastructure; cross-cell traffic needs explicit design	Large platforms, multi-tenant services, retail estates
Separate control-plane instances	Control-plane bugs, bad policy pushes	More systems to operate and keep consistent	Multi-region networks, SD-WAN and fabric controllers
Staggered software versions	Common-mode software defects	Wider test matrix; feature parity must be managed	Critical pairs where a shared bug is the dominant risk
Protective limits (max-prefix, storm control, control-plane policing)	Overload from a neighbour or a flood	Limits must be tuned; wrong values cause self-inflicted outages	Every external peering and every access edge

WATCH OUT · REDUNDANCY THAT SHARES FATE IS DECORATION

A pair of firewalls in a cluster synchronises configuration and state, so a bad rule or a state-table bug reaches both members within seconds. A cluster protects against **hardware** failure, not against **software or human** failure. For the latter you need a second, independently managed failure domain.

Mapping failure domains in a design review

1. **List dependencies** For each critical service, list every dependency: power, fibre paths, carriers, controllers, DNS, NTP, identity, certificates, licensing servers.
2. **Group shared fate** Mark components that would fail together under one physical, logical, software or human fault.
3. **Find single shared dependencies** Any dependency that appears under every redundant path is a single point of failure, even if every box is paired.
4. **Size the blast radius** Express each fault in business terms: users, revenue per hour, safety functions, regulatory exposure.
5. **Decide** For each fault choose: contain (shrink the domain), duplicate (independent second domain) or accept (documented, with an owner).
6. **Record** Write the decision and its rationale into the design document and risk register so it survives staff turnover.

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Contain, duplicate or accept

BLAST RADIUS (BUSINESS IMPACT) →

Duplicate: build an independent second domain

- Site loss
- Controller cluster failure

Design priority: contain and duplicate

- Bad change pushed everywhere
- Shared-duct cut

Accept with a named owner

- Single kiosk switch failure

Contain: shrink the domain, automate recovery

- Access-port loops
- Branch link flaps

LIKELIHOOD OF THE FAULT →

Place each mapped fault by how likely it is and how much of the business it stops. The quadrant suggests the default response; the decision and its owner still go into the risk register.

EXAMPLE · CELLS IN A RETAIL ESTATE

A retailer with 900 stores splits its estate into nine cells of about 100 stores, each with its own controller pair, policy instance and change wave. A bad policy push now affects at most one cell — roughly 11% of stores — and the rollout process promotes changes cell by cell. The cost is nine sets of management infrastructure; the benefit is that no single action can close every store at once.

SCENARIO · DECIDE · STRETCHED VLAN, SHARED DISASTER

A data centre stretches a server VLAN across two halls so that virtual machines can move freely. A cabling error in Hall A creates a loop; the resulting storm saturates the inter-hall links and both halls lose service for 40 minutes. The CIO asks for a design change that prevents a repeat without stopping VM mobility for the few applications that genuinely need it.

(A) Tune spanning-tree timers and add BPDU guard everywhere, keeping the stretched VLAN as it is.

(B) Make each hall its own routed failure domain; carry Layer 2 only where proven necessary over an overlay with storm control and ARP suppression, and move other workloads to Layer 3 mobility.

(C) Double the inter-hall link capacity so that a storm cannot saturate it.

(D) Consolidate into a single hall to remove the inter-hall links entirely.

Circle your choice. Answer and feedback: module answer key.

INSIGHT · BLAST RADIUS IS A BUSINESS METRIC

Executives rarely care that a pod has four spines. They care that a single fault can stop at most 10% of branches, or at most one of three payment channels. State blast-radius targets in those terms in the requirements, and test designs against them.

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Failure-domain review checklist

☐ Every critical service has a dependency map that includes DNS, NTP, identity and certificates

☐ Diverse circuits have been verified for physical path diversity, not just different carriers

☐ No Layer 2 domain spans more than one site or power zone without a documented exception

☐ Controller and management clusters are scoped so one failure cannot affect the whole estate

☐ Paired devices have a strategy against common-mode software defects

☐ Changes are rolled out in waves aligned with failure domains

☐ Blast-radius targets are stated in business terms and traced to design decisions

REFLECT

Pick one service you support today. Which single fault — physical, logical, software or human — would hurt it most, and is that dependency visible in your current design documents?

Key terms

Failure domain
The set of components and services impaired by a single fault.

Blast radius
The business-visible scope of a failure: users, sites, transactions or functions affected and for how long.

Shared fate
A condition in which supposedly independent components fail together because of a common dependency.

Shared risk link group (SRLG)
A set of links that share a physical resource, such as a duct or fibre, and therefore fail together.

Cell architecture
A design that partitions a service into independent, self-contained units so a fault affects only one unit.

Common-mode failure
A single cause, such as a software defect or a bad change, that defeats all redundant members at once.

Control-plane policing
Rate-limiting traffic destined to a device's control plane so floods cannot exhaust its CPU.

Containment
Design choices that stop a fault from propagating beyond its origin, such as routed boundaries or summarisation.

Key takeaways

- Redundancy only helps when the redundant parts fail independently — ask what breaks together
- Look for shared fate in six places: physical, logical, control and management, software, operational and provider
- Any dependency under every redundant path is a single point of failure
- Cells, routed boundaries and separate control-plane instances cap blast radius
- Express blast-radius targets in business terms and test designs against them

KNOWLEDGE CHECK

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1. Two WAN circuits from different carriers both fail when road works cut a single duct outside the building. What design concept was missed?

   (A) Insufficient bandwidth on the secondary circuit for the full failover load

   (B) Routing protocol timers too slow to detect the loss of either of the links

   (C) An active/standby model that should have been an active/active one instead

   (D) Shared fate: both circuits sat in one shared risk link group

2. Why is a firewall cluster that synchronises configuration not sufficient protection against a bad rule change?

   (A) A bad change replicates to every member, so the cluster shares one failure domain

   (B) Clusters always fail over too slowly for any stateful sessions to survive the event

   (C) The standby member does not process traffic and so cannot detect the bad rule

   (D) Configuration synchronisation is disabled by default during every maintenance window

3. A designer proposes nine regional cells, each with its own controller pair and change wave, for a 900-store estate. What is the main benefit?

   (A) Overall capital cost falls because each cell uses smaller and cheaper controllers

   (B) No single fault or change can affect more than about one ninth of stores

   (C) Troubleshooting gets simpler because every cell shares one consolidated policy

   (D) Bandwidth to the data centre drops because each cell caches its own applications

4. Which technique contains routing instability within one part of a large link-state network?

   (A) Raising hello and dead timers on all core links

   (B) Extending the backbone area across all remote sites

   (C) Summarisation at area or level borders

   (D) Mutual redistribution between BGP and the IGP

5. How should blast-radius targets be expressed in design requirements?

   (A) In device terms, such as the number of redundant chassis installed per site

   (B) In business terms, such as the maximum share of branches or transactions affected

   (C) In protocol terms, such as the number of routing areas used in the design

   (D) In supplier terms, such as the support contract level and response time for each platform

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 5 · RESILIENCE · LESSON 2 · 33 MIN

Redundancy Models & Availability Mathematics

Turn 'it must never go down' into numbers you can design, cost and defend

- Compare N+1, 2N and active/active versus active/standby redundancy models
- Calculate serial, parallel and MTBF/MTTR-based availability and convert it to downtime
- Explain why common-cause failures and failover coverage cap real-world availability

Availability requirements arrive as adjectives — 'always on', 'mission critical' — and an expert's first job is to turn them into numbers. Numbers let you compare designs, price them, and show a sponsor exactly what an extra nine costs. They also expose where effort is wasted: doubling a component that is already reliable while a single unprotected dependency dominates the result.

Redundancy models

N+1

- N units carry peak load; one spare absorbs a single failure
- Cost-efficient for pools of identical units (power supplies, cluster nodes, links in a bundle)
- During maintenance of one unit you are at N — a further failure becomes an outage

2N

- Two complete, independent systems, each able to carry the full load
- Highest isolation: separate power, paths and often management
- Roughly double the cost; each half runs at 50% or less in normal operation

N+2 / 2N+1

- Tolerates a failure while another unit is in maintenance
- Used for services where maintenance must never reduce protection
- Cost justified only where the price of downtime is very high

Active/active versus active/standby

Active/active

- Both members carry traffic; capacity is used every day
- Failover is partial — only the failed share moves
- Each member must stay under about 50% load, or loss of one overloads the other
- State sharing and asymmetric paths need careful design

Active/standby

- One member carries all traffic; the other waits
- Simple, predictable state handling and symmetric paths
- The standby can hide latent faults until the day it is needed
- Failover moves 100% of traffic at once

WATCH OUT · THE 50% RULE NOBODY MONITORS

Active/active pairs are often sized correctly on day one and silently grow past 50% each. The first failure then becomes an overload outage. Track **pair utilisation against the survivable ceiling**, not individual link utilisation.

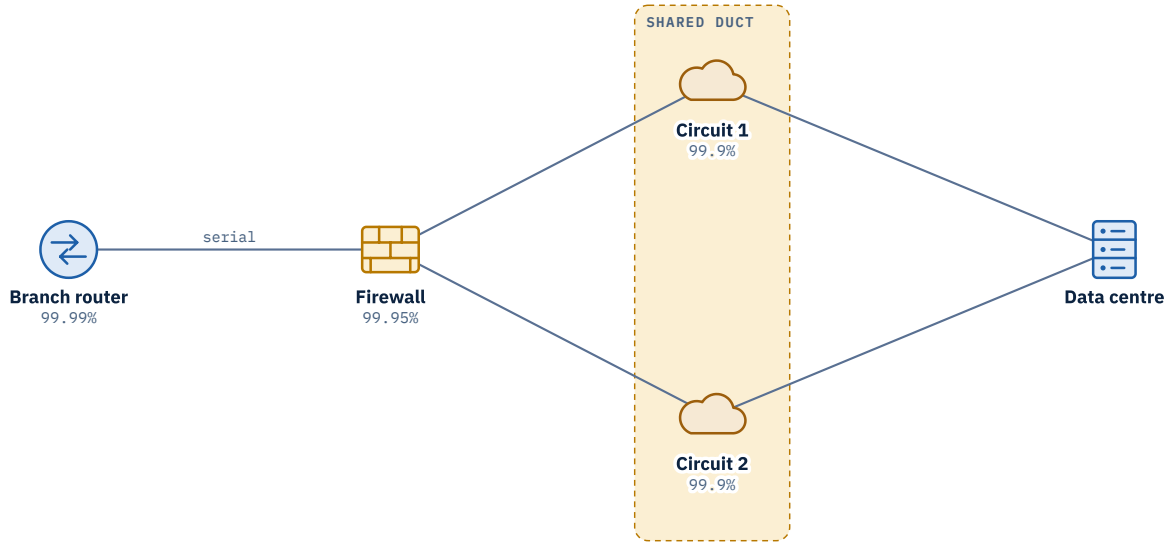
Availability formulas

MEASURE	FORMULA	WHAT IT TELLS YOU
Steady-state availability	`A = MTBF ÷ (MTBF + MTTR)`	The fraction of time a component is working. MTBF is mean time between failures, MTTR is mean time to restore.
Serial chain	`A_total = A1 × A2 × … × An`	Every component must work. The chain is always less available than its weakest link.

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MEASURE	FORMULA	WHAT IT TELLS YOU
Parallel (independent)	`A_total = 1 − (1 − A1) × (1 − A2) × …`	At least one path must work. Assumes independent failures and perfect, instant failover.
Common-cause adjustment (simplified beta-factor model)	`U_pair ≈ β × U + ((1 − β) × U)²`	U = 1 − A for one member; β is the fraction of failures that hit both members together. The first term usually dominates.
Annual downtime	`Downtime = (1 − A) × 525,600 minutes`	Converts availability into minutes per year that a sponsor can understand.
Survivable utilisation	`Max load per unit = (n − f) ÷ n`	With n units sharing load and f tolerated failures, each unit must stay below this share of its capacity.

A branch-to-data-centre path: serial and parallel elements



The router and firewall are serial — each must work. The two circuits are parallel, but only as independent as their routes: the shaded zone marks the shared duct whose common-cause failures (β) cap what the pair can deliver.

The nines in minutes

AVAILABILITY	DOWNTIME PER YEAR	DOWNTIME PER 30-DAY MONTH	TYPICAL MEANING
99.9%	≈ 8.8 hours	≈ 43 minutes	Good single-site enterprise service
99.95%	≈ 4.4 hours	≈ 22 minutes	Redundant single site, disciplined operations
99.99%	≈ 53 minutes	≈ 4.3 minutes	Redundant paths and sites, automated failover
99.999%	≈ 5.3 minutes	≈ 26 seconds	Carrier-grade; needs independent domains and very fast recovery

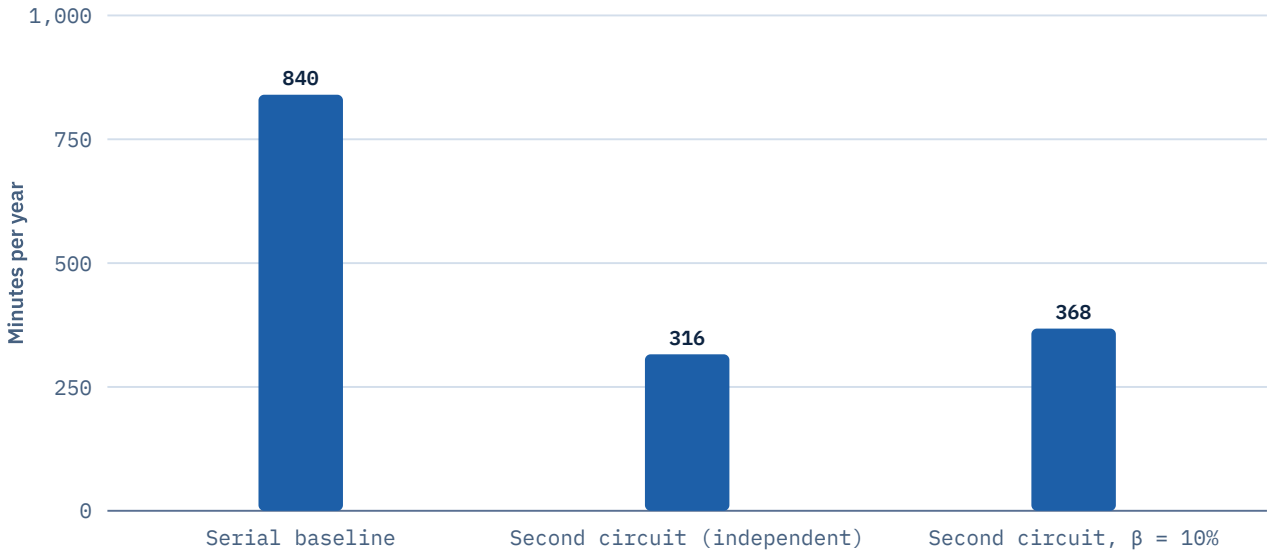
Worked example — a branch-to-data-centre path

1. **Serial baseline** Router 99.99%, firewall 99.95%, single WAN circuit 99.9%. A = 0.9999 × 0.9995 × 0.999 ≈ 0.9984, about **99.84%**, or roughly 14 hours of downtime a year. The WAN circuit dominates.

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2. **Add a second circuit (naïve)** Two circuits in parallel: 1 − (0.001 × 0.001) = 0.999999. The chain becomes 0.9999 × 0.9995 × 0.999999 ≈ **99.94%**. The bottleneck has moved to the firewall.
3. **Apply common cause** If 10% of circuit failures are shared (same duct, same exchange), U_pair ≈ 0.1 × 0.001 + (0.9 × 0.001)² ≈ 0.0001, so the pair is about **99.99%**, not 99.9999%. Shared fate erased two nines.
4. **Fix the real bottleneck** Pair the firewalls and routers as well, and verify circuit diversity to push β down. The result is now limited by common-cause factors and failover behaviour, which is exactly where design attention should go.

Annual downtime at each step of the worked example



Computed from the worked example. A naive second circuit appears to cut downtime by more than half, but with 10% common-cause failures part of that gain disappears — and the serial firewall then dominates what remains.

EXAMPLE · MTTR IS A DESIGN VARIABLE

A switch with an MTBF of 200,000 hours and a four-hour on-site spare MTTR gives A = 200,000 ÷ 200,004 ≈ **99.998%**. The same switch with a 48-hour courier replacement gives 200,000 ÷ 200,048 ≈ **99.976%** — about twelve times the downtime. Spares strategy, remote hands and automation are availability design, not just operations.

PRO TIP · FAILOVER COVERAGE

The parallel formula assumes every failure is detected and every failover succeeds. In practice a share of failures are 'grey' — packet loss, a hung process, a link that stays up but drops traffic. Quote a **coverage factor** (the proportion of failures that are detected and recovered automatically) and invest in detection, because an undetected failure behaves like a serial component.

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SCENARIO · DECIDE · WHERE SHOULD THE NEXT DOLLAR GO?

A trading firm's order path runs through dual internet-edge routers (each 99.99%), a single firewall cluster (99.95%, both members in one rack on one PDU pair) and dual diverse carriers (each 99.9%, diversity verified). The target is 99.99%. The budget allows one improvement this year.

- (A) Split the firewall members across independent racks and power, and test failover under load.
- (B) Add a third carrier to the edge.
- (C) Replace the edge routers with higher-MTBF models.
- (D) Negotiate a stricter service credit with the firewall support provider.

Circle your choice. Answer and feedback: module answer key.

TEMPLATE Availability calculator sheet

Model each element of a service path, its arrangement and its common-cause exposure, then convert the result into annual downtime for the sponsor.

ELEMENT	AVAILABILITY (A)	ARRANGEMENT	COMMON-CAUSE B	EFFECTIVE A	DOWNTIME/YR	NOTES / EVIDENCE
WAN circuits (2 carriers)	*99.9% each*	*Parallel*	*0.10*	*≈ 99.99%*	*≈ 53 min*	*Diversity verified to building entry; shared exchange remains*
Firewall cluster	*99.95%*	*Serial in chain*	*n/a (single domain)*	*99.95%*	*≈ 4.4 h*	*Members share rack and PDU — action item*

REFLECT

Which service you support has an availability target that has never been converted into minutes per year? What would the number tell the sponsor that the adjective does not?

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Key terms

MTBF
Mean time between failures: the average operating time between failures of a repairable component.

MTTR
Mean time to restore: the average time from failure to restored service, including detection, travel, repair and verification.

N+1
A redundancy model with one spare unit beyond what peak load requires.

2N
A redundancy model with two complete, independent systems, each able to carry the full load.

Active/active
A redundancy model in which all members carry traffic in normal operation.

Active/standby
A redundancy model in which one member carries traffic and another waits to take over.

Beta factor (β)
The fraction of a component's failures that are common-cause and affect redundant members simultaneously.

Coverage factor
The proportion of failures that are detected and recovered automatically by the redundancy mechanism.

Survivable ceiling
The maximum utilisation per member that still lets the survivors carry the load after the tolerated failures.

Key takeaways

- Turn adjectives into numbers: availability, minutes of downtime per year and the cost of each extra nine
- Serial chains multiply; parallel paths help only as far as failures are independent and detected
- Common-cause failures and failover coverage usually dominate real availability
- MTTR is a design lever: spares, remote hands and automation can beat duplication
- Keep load-sharing members under their survivable ceiling, $(n - f) \div n$

KNOWLEDGE CHECK

6. Three components with availabilities of 99.99%, 99.95% and 99.9% are in series. What is the approximate end-to-end availability?

   (A) About 99.84%
   (B) About 99.90%
   (C) About 99.95%
   (D) About 99.99%

7. A component has an MTBF of 50,000 hours. Reducing its MTTR from 24 hours to 4 hours mainly changes what?

   (A) The failure rate falls, so outages happen far less often
   (B) Nothing, because availability depends only on the MTBF
   (C) Annual downtime falls by roughly a factor of six
   (D) The component no longer needs any redundant partner

8. Two load-sharing firewalls each run at 65% of capacity. What is the design problem?

   (A) Active/active pairs must always become active/standby above 50% load
   (B) Utilisation above 60% stops stateful synchronisation from working
   (C) Load sharing makes a pair less available than a single firewall
   (D) Neither can absorb the other's load, so one failure causes overload

9. Why does a parallel pair rarely reach the availability the independent formula predicts?

   (A) The formula ignores bandwidth differences between the two members
   (B) Parallel members run at half capacity and so fail much more often
   (C) Common-cause failures and imperfect failure detection dominate
   (D) The members share one MTTR, which cancels the benefit of the pair

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10. A service is at 99.99%. How much downtime per year does that allow?

(A) Roughly 5 minutes
(B) Roughly 53 minutes
(C) Roughly 8.8 hours
(D) Roughly 4.4 hours

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MODULE 5 · RESILIENCE · LESSON 3 · 31 MIN

Convergence Budgets & Fast Recovery

Engineer recovery time component by component instead of hoping the protocol is fast enough

- Decompose convergence into detection, propagation, computation and FIB update
- Choose between fast detection, precomputed backups and prefix-independent convergence
- Weigh graceful restart and non-stop routing against fast reroute in a given topology

Applications define recovery requirements in their own language: a voice call tolerates a few hundred milliseconds of loss, a storage replication session may time out after a few seconds, a payment authorisation has an end-to-end timeout. The designer's job is to translate that into a **convergence budget** and allocate it across the stages of recovery, exactly as a project manager allocates money.

The stages of convergence

01 Detect — Notice the failure: loss of light, protocol timers, BFD

→

02 Propagate — Tell the rest of the domain: LSA/LSP flooding, BGP updates

→

03 Compute — Run SPF or best-path selection, including throttling delays

→

04 Update FIB — Program new forwarding entries into hardware

→

05 Recover — Traffic flows on the new path; transport and applications resume

Convergence budget

MEASURE	FORMULA	WHAT IT TELLS YOU
Total convergence time	`T = T_detect + T_propagate + T_compute + T_FIB`	Each term can be engineered separately; the largest term is usually where to invest.
BFD detection time	`T_detect ≈ negotiated interval × multiplier`	For 50 ms × 3, detection takes about 150 ms; for 300 ms × 3, about 900 ms.
FIB update (prefix-dependent)	`T_FIB ≈ affected prefixes × per-prefix programming time`	Grows with table size. Prefix-independent designs update a shared next-hop object instead, making this term roughly constant.

Detection options

MECHANISM	TYPICAL SPEED	WHEN IT WORKS	WATCH OUT FOR
Loss of signal on a direct fibre	Milliseconds	Point-to-point links with no intermediate device	Does not work through carrier transponders or media converters that keep the link up
Routing protocol hellos	Seconds to tens of seconds	Everywhere, as a backstop	Too slow on its own for most real-time traffic
BFD	Tens to hundreds of milliseconds	Through Layer 2 services, carrier Ethernet, multi-hop paths	CPU load at scale; aggressive timers on a busy or virtual platform cause false positives
Ethernet OAM / link fault signalling	Sub-second	Carrier Ethernet services that support it	Requires provider support and correct configuration on both ends

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Fast detection and SPF throttling

R1 · IOS-style CLI

```
interface TenGigabitEthernet0/0/1
 bfd interval 50 min_rx 50 multiplier 3
 ip ospf bfd
!
router ospf 10
 timers throttle spf 50 200 5000
 timers throttle lsa 50 200 5000
 timers lsa arrival 50
```

BFD detects a failure in about 150 ms. The throttle timers react quickly to the first event (50 ms) but back off towards 5 s if the network keeps changing, which protects the control plane during instability.

Precomputed protection versus reactive convergence

Reactive convergence

- Recompute after the failure is detected
- Recovery time is the full budget: detect + propagate + compute + FIB
- Simple, nothing extra to configure or verify
- Typically hundreds of milliseconds to seconds

Fast reroute (LFA, remote LFA, TI-LFA)

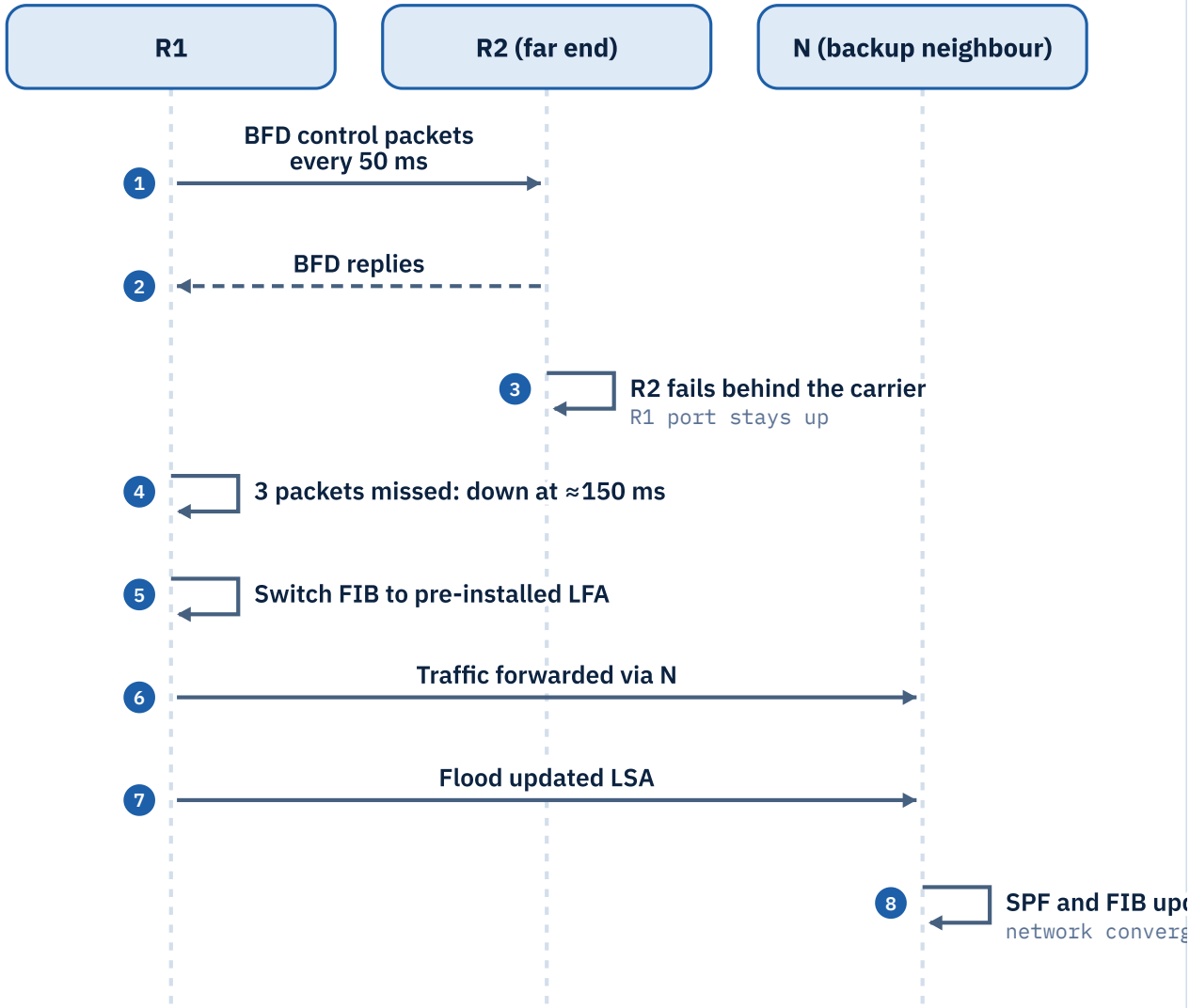
- A backup next hop is computed and installed in advance
- On local detection, traffic switches immediately; the network converges behind it
- TI-LFA can provide full coverage in most topologies using segment routing
- Typically tens of milliseconds after detection

Prefix-independent convergence

- Many prefixes share one next-hop object in the FIB
- One pointer change moves all affected prefixes at once
- Essential for edge routers carrying large BGP tables
- Requires a precomputed backup path (for example a second BGP path)

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BFD detection and a pre-installed backup across a carrier service



The carrier keeps R1's port up, so only BFD notices the far-end failure. Traffic moves to the precomputed alternate at about 160 ms; flooding and SPF happen afterwards without adding to the loss.

INSIGHT · MICRO-LOOPS

While routers converge at slightly different moments, packets can loop briefly between a router that has updated and one that has not. On busy links this causes measurable loss even after the protected failure itself is handled. Ordered FIB updates and segment-routing-based micro-loop avoidance are design options where the loss budget is tight.

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Graceful restart versus non-stop routing

Graceful restart (GR) with non-stop forwarding

- Forwarding continues on the old FIB while the control plane restarts
- Neighbours act as helpers and keep routes during the restart
- Risk: if the topology changes during the restart, traffic follows stale state and may be black-holed
- Interacts with fast detection: BFD must be able to survive a control-plane restart, or it tears the session down

Non-stop routing (NSR)

- A standby route processor keeps protocol sessions synchronised
- Neighbours do not notice the switchover; no helper needed
- Costs memory and CPU, and the synchronisation itself adds complexity
- Suits single-homed critical nodes such as provider edges

PRO TIP · THE DESIGN RULE OF THUMB

Where the topology offers **redundant paths**, it is usually better to detect the failure fast and reroute around the failed node than to keep forwarding through it on stale state. Where a node is a **single point of attachment** — a single-homed customer edge, a lone aggregation router — preserving forwarding with NSR or GR is often the better trade-off. Decide per role, not per network.

Preserve forwarding or reroute around the node?

EXPOSURE TO CONTROL-PLANE RESTARTS →

NSR first; GR as a fallback
- Single-homed provider edge
- Lone aggregation router

Fast detection + FRR; drain before planned work
- Core routers
- Spines

GR helper is usually enough
- Small stub router

BFD + FRR; GR optional
- Redundant leaf pairs

ALTERNATIVE PATHS AROUND THE NODE →

Decide per role. Where alternative paths exist, fast detection and reroute is usually safer than forwarding on stale state; where a node is the only attachment, preserving forwarding through a restart is the priority.

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SCENARIO · DECIDE · A VOICE SERVICE WITH A 300 MS BUDGET

A regional network carries real-time voice between two contact centres. The application tolerates up to 300 ms of loss before calls drop. Links between core sites are carrier Ethernet services that stay up even when the far end fails. Current convergence relies on routing hellos and takes about 4 seconds.

(A) Enable graceful restart on all core routers so forwarding continues during failures.

(B) Tune routing hellos to one second and dead intervals to three seconds.

(C) Deploy BFD at about 50 ms × 3 on the carrier links, with precomputed loop-free alternates and modest SPF throttling.

(D) Replace the carrier Ethernet services with dark fibre to get loss-of-signal detection.

Circle your choice. Answer and feedback: module answer key.

Convergence design checklist

☐ Recovery-time requirements are captured per application class, in milliseconds

☐ Each link type has a detection mechanism appropriate to whether loss of signal is visible

☐ BFD timers have been validated against platform scale and virtual-machine scheduling jitter

☐ Precomputed backup coverage has been measured, not assumed, for every protected link

☐ Edge routers with large tables use prefix-independent convergence

☐ Graceful restart and non-stop routing choices are made per node role

☐ Convergence has been tested under load, including micro-loop behaviour

REFLECT

For your most time-sensitive application, write down the convergence budget and estimate each of the four terms in your current network. Which term dominates?

Key terms

Convergence
The process by which a network restores consistent forwarding after a topology change.

BFD
Bidirectional Forwarding Detection: a lightweight hello protocol that detects path failures in milliseconds, independent of the routing protocol.

SPF throttling
Timers that delay and back off shortest-path computations to balance fast reaction against stability.

LFA
Loop-free alternate: a precomputed backup next hop that will not loop traffic back through the failed element.

TI-LFA
Topology-independent loop-free alternate: a segment-routing technique that provides precomputed backup paths along the post-convergence path.

Prefix-independent convergence
A FIB structure in which many prefixes share a next-hop object, so one update reroutes them all.

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Graceful restart
A mechanism in which neighbours keep forwarding through a restarting router while its control plane recovers.

Micro-loop
A transient forwarding loop caused by routers updating their FIBs at slightly different times.

Non-stop routing
A mechanism in which a standby route processor keeps routing sessions alive so neighbours never see a restart.

Key takeaways

- Translate application tolerance into a convergence budget, then allocate it to detect, propagate, compute and FIB update
- BFD detects failures that carrier services hide from the port; validate timers against the platform
- Precomputed backups (LFA, TI-LFA) and prefix-independent convergence remove most of the budget from the loss window
- Choose NSR or graceful restart for single points of attachment, fast reroute where paths are redundant

KNOWLEDGE CHECK

11. Core links run over a carrier Layer 2 service that keeps the local port up when the far end fails. Which detection approach fits best?

(A) BFD on the routed adjacency across the service
(B) Loss-of-signal detection on the local router port
(C) Longer routing dead intervals for added stability
(D) Graceful restart on the routers at both of the ends

12. An internet edge router carries a full routing table. Failover to a second upstream takes many seconds because every prefix must be reprogrammed. What addresses this?

(A) More aggressive BFD timers on both of the upstream sessions
(B) SPF throttle timers configured with a shorter initial wait
(C) Graceful restart helper mode enabled on both upstream peers
(D) Prefix-independent convergence with a precomputed backup path

13. When is graceful restart most likely to be the better choice than fast detection and reroute?

(A) On a core router with three diverse paths to every destination
(B) On any link that already has a loop-free alternate installed
(C) On nodes whose topology is likely to change during a restart
(D) On a single-homed node where no alternative path exists

14. BFD is configured at 300 ms with a multiplier of 3. What is the approximate detection time?

(A) About 100 ms
(B) About 900 ms
(C) About 300 ms
(D) About 3 seconds

15. What causes micro-loops during convergence?

(A) BFD sessions flapping because their timers are too aggressive
(B) Summarisation hiding the failed link from all remote areas
(C) Routers updating forwarding tables at slightly different times
(D) Graceful restart helpers holding stale routes for too long

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MODULE 5 · RESILIENCE · LESSON 4 · 27 MIN

Maintenance & Change Design

Most outages are self-inflicted — design the network so that change is routine and safe

- Design for maintenance so that planned work never reduces protection below target
- Use drain, cost-out and graceful shutdown techniques to remove traffic before work
- Structure change waves, canaries and verification to limit the blast radius of change

Post-incident reviews across the industry keep finding the same pattern: a large share of major outages start with a planned change. Hardware is now reliable; people, pipelines and software updates are less so. A resilient design therefore treats **change as a failure mode** and gives operators safe, boring ways to take things out of service and put them back.

WATCH OUT · N+1 BECOMES N DURING MAINTENANCE

With N+1, taking one unit out for maintenance leaves zero spare. A failure elsewhere in that window is an outage. Either keep maintenance windows short and low-risk, design critical services for N+2, or schedule work so that the paired domain is never touched at the same time.

Design-for-maintenance principles

DM1

Drain before touch

Every device can be gracefully removed from the traffic path before any work starts, and the drain is verified by counters, not assumed.

DM2

Independent halves

Paired systems can be upgraded one half at a time without depending on each other's software version.

DM3

Small, ordered waves

Changes move through failure domains in a defined order: lab, canary, one domain, then the rest.

DM4

Automated verification

Pre-checks and post-checks compare routing, forwarding and application health before and after each step.

DM5

Fast, rehearsed rollback

Every change has a tested way back, and a clear trigger for using it.

DM6

Maintenance is visible

Monitoring knows when a node is drained, so alerts reflect real risk instead of planned work.

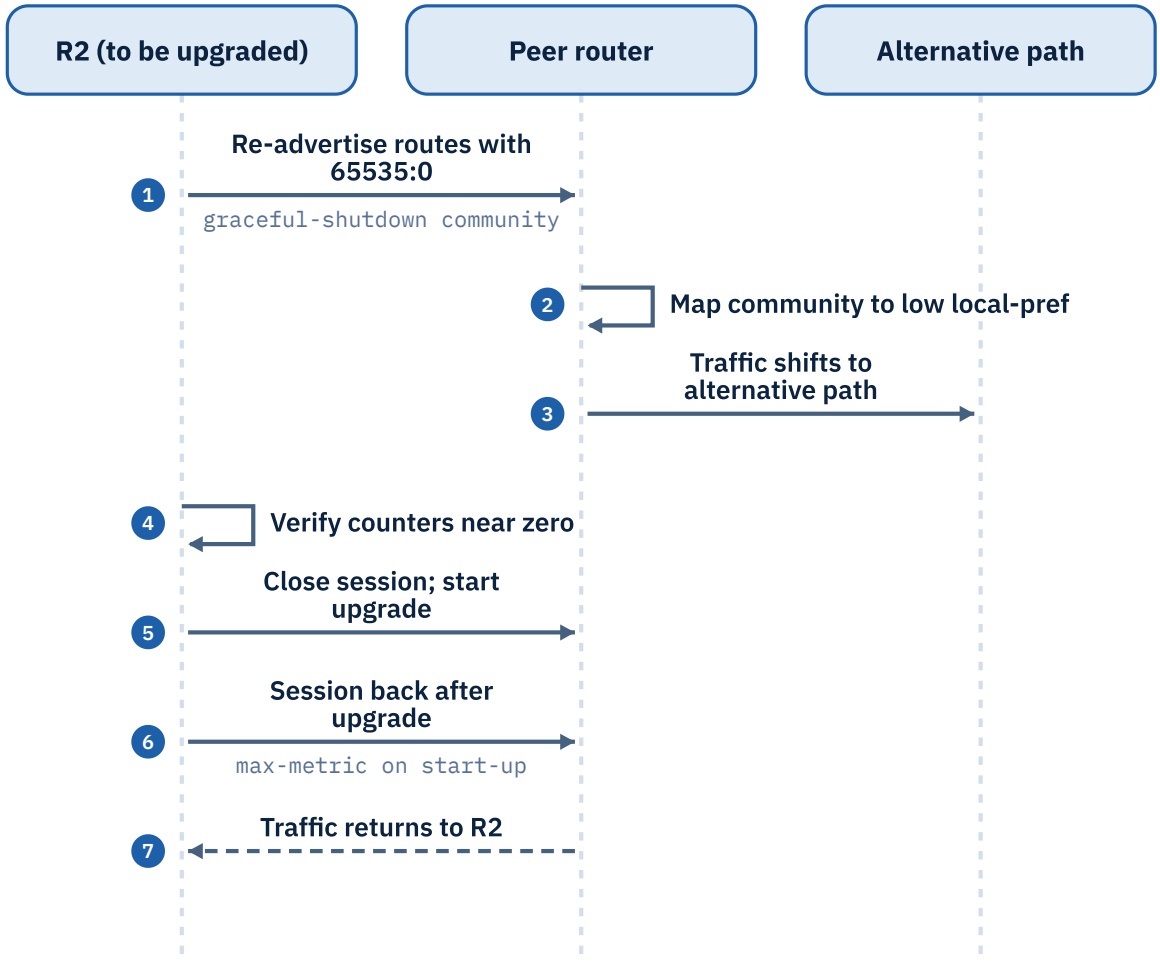
Drain techniques by protocol

PROTOCOL / LAYER	DRAIN TECHNIQUE	EFFECT	DESIGN NOTE
OSPF	Advertise maximum metric for transit links	Router stays reachable but is used only as a last resort	Also useful on start-up so traffic arrives only after the FIB is ready
IS-IS	Set the overload bit	Router is not used for transit by other routers	A similar on-start-up option exists
BGP	Graceful shutdown community, lower local preference or prepend	Peers prefer alternative paths before the session is closed	Needs receivers to honour the community — agree it in peering policy

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PROTOCOL / LAYER	DRAIN TECHNIQUE	EFFECT	DESIGN NOTE
First-hop redundancy	Lower priority so the peer becomes active	Hosts move to the other gateway without changing configuration	Check preemption behaviour before and after
Link aggregation / fabric	Shut member links or put a leaf into maintenance mode	Traffic shifts to remaining members or leaves	Confirm the survivors stay under the survivable ceiling

Draining a BGP edge router with graceful shutdown



The drain is complete only when counters prove it, not when the command is entered. On return, max-metric on start-up keeps transit traffic away until the forwarding table is ready.

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Draining a node before maintenance

R2 · IOS-style CLI

```
! Link-state: stop being a transit path
router ospf 10
 max-metric router-lsa on-startup 300
 max-metric router-lsa
!
! BGP: signal graceful shutdown to peers (well-known community 65535:0)
ip bgp-community new-format
route-map DRAIN-OUT permit 10
 set community 65535:0 additive
router bgp 65001
 neighbor 192.0.2.1 route-map DRAIN-OUT out
```

The on-startup option keeps the router out of the transit path for 300 seconds after a reload, so traffic is not attracted before the forwarding table is complete. Peers must map the graceful-shutdown community to a low local preference for the BGP drain to work.

In-service software upgrade versus rolling upgrade

In-service upgrade on a redundant chassis

- Upgrades one route processor at a time while forwarding continues
- Attractive where a chassis is a single point of attachment
- Complex process with its own failure modes; supported upgrade paths are narrow
- Still one device and one software failure domain

Rolling upgrade across independent devices

- Drain one device, upgrade, verify, return to service, repeat
- Simple, testable, works with any platform
- Needs genuine path redundancy and the capacity to run on N
- Mixed versions run side by side for a while — check interoperability

A change-wave design

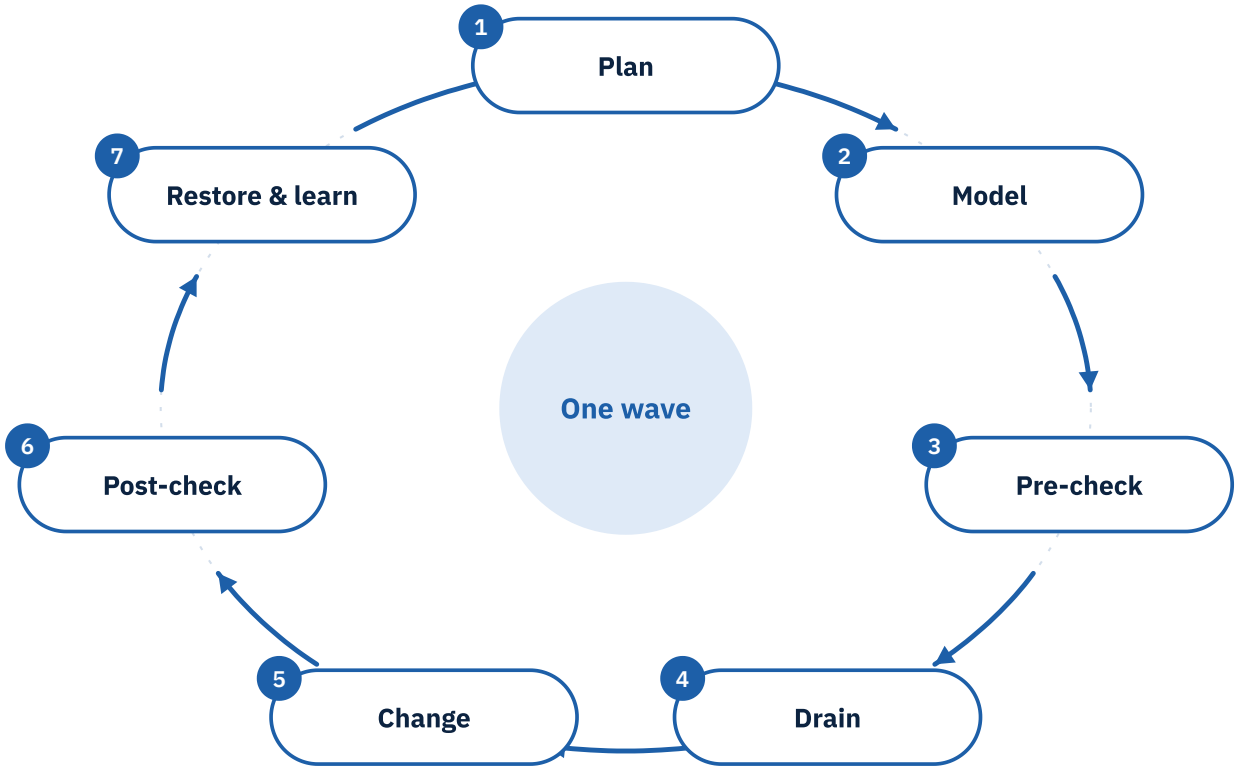
1. **Lab and digital twin** Validate the change and the rollback against a model of the production topology.
2. **Canary** Apply to one low-impact device or site with full monitoring; hold for a defined soak time.
3. **First failure domain** Apply to one cell, region or pod; compare health indicators against untouched domains.
4. **Remaining domains in order** Roll forward domain by domain, never touching both halves of a redundant pair in the same wave.
5. **Close** Confirm the estate is consistent, update the source of truth and record lessons.

PRO TIP · WHAT A NETWORK DIGITAL TWIN CAN AND CANNOT PROVE

A digital twin — a model built from the real configurations and topology, sometimes with emulated devices — can check reachability, policy intent, failure scenarios and the exact effect of a change before it touches production. It cannot prove hardware behaviour, software bugs under real load or the timing of convergence. Use it to catch logic errors early, and keep canaries and post-checks to catch what the model cannot see.

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A safe change loop



1. **Plan** — Scope, wave order, rollback
2. **Model** — Digital twin or simulation
3. **Pre-check** — Baseline routing and health
4. **Drain** — Move traffic, verify counters
5. **Change** — Automated, rate-limited
6. **Post-check** — Compare with baseline
7. **Restore & learn** — Return traffic, update records

Each pass through the loop is one wave. The model check and pre-checks come before any traffic moves, and every wave ends by updating the source of truth so the next change starts from reality.

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SCENARIO · DECIDE · THE UPGRADE WEEKEND

A university core has two routers in an active/active pair, each running at 40% of capacity. A security advisory requires a software upgrade on both. The operations lead proposes upgrading both routers in the same two-hour window 'to get it over with'.

(A) Approve; the pair is redundant, so one router will always be up.

(B) Postpone the upgrade until a spare chassis can be bought and pre-staged.

(C) Upgrade both at once but with the supplier's support engineer on the call.

(D) Drain and upgrade one router, soak it under production traffic for an agreed period, then drain and upgrade the other in a later window.

Circle your choice. Answer and feedback: module answer key.

INSIGHT · MAINTENANCE WINDOWS ARE A DESIGN INPUT

Some services — hospitals, payment switches, emergency networks — have no acceptable maintenance window. That requirement pushes the design towards independent halves, rolling procedures and capacity that can run on N for long periods. Capture it in requirements, because it changes topology and budget.

Change-safety design checklist

☐ Every device role has a documented, tested drain procedure

☐ Survivors can carry full load while any one device is drained

☐ Paired devices can run different software versions temporarily

☐ Change waves are aligned with failure domains, starting with a canary

☐ Pre-checks and post-checks are automated and compare against a baseline

☐ Rollback is rehearsed and has a measurable trigger

☐ Monitoring suppresses noise for drained nodes but still alerts on new risk

REFLECT

Think of the last change that caused an incident in your environment. Which of the six design-for-maintenance principles would have prevented or contained it?

Key terms

Drain
Gracefully moving traffic away from a device or link before work so that removing it causes no loss.

Max-metric
An OSPF technique that advertises the highest metric on transit links so other routers avoid the sender.

Overload bit
An IS-IS flag that tells other routers not to use the sender for transit traffic.

Graceful shutdown community
A well-known BGP community (65535:0) that asks receivers to lower the preference of routes before a session closes.

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Canary
The first small target of a change, monitored closely before the change spreads.

Rolling upgrade
Upgrading members of a redundant system one at a time while the others carry traffic.

Change wave
An ordered group of devices or domains that receives a change together.

Soak time
A deliberate waiting period after a change, under production load, before proceeding.

Key takeaways

- Treat change as a failure mode: most major outages start with planned work
- Drain before touch, verify by counters, and keep survivors under their survivable ceiling
- Move changes through failure domains: model, canary, one domain, then the rest — never both halves together
- Digital twins catch logic errors before production; canaries and post-checks catch the rest
- Services with no maintenance window need N+2 or independent halves

KNOWLEDGE CHECK

16. Why is OSPF max-metric configured with an on-startup option?

(A) So a reloaded router attracts transit traffic only once its FIB is ready
(B) So the router never carries any transit traffic again after the reload completes
(C) So BFD sessions are negotiated before any OSPF adjacency is formed
(D) So the router's loopback stays unreachable for the whole upgrade

17. A service uses N+1 redundancy and has no acceptable maintenance window. What design change follows?

(A) Remove redundancy and rely on fast replacement of failed spares instead
(B) Switch from active/active to active/standby so maintenance becomes simpler to plan
(C) Move to N+2 or independent halves so maintenance never removes the last spare
(D) Increase BFD timers so that maintenance work never triggers failovers

18. What makes a BGP graceful-shutdown drain effective?

(A) Peers honour the community by lowering local preference before the session closes
(B) The local router withdraws every prefix at once, just before closing the session itself
(C) The graceful-restart capability is negotiated on every BGP session first
(D) BFD is disabled on the session so that the shutdown is never detected

19. Which change sequence best limits the blast radius of a fleet-wide software upgrade?

(A) All core devices first, because they are the most redundant part of the network
(B) All devices in one window, so the network spends as little time as possible on mixed versions
(C) Alphabetical device order, so the sequence is easy for operators to audit later
(D) Lab, canary, one failure domain, then the rest — never both halves of a pair together

20. An active/active pair runs at 40% each. What does that mean for a rolling upgrade?

(A) Neither member can be drained because the pair is above its survivable ceiling
(B) One member can be drained safely because the survivor stays within capacity
(C) The pair must be converted to active/standby before any upgrade can begin
(D) Both members must be upgraded together to avoid running mixed versions

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MODULE 5 · RESILIENCE · LESSON 5 · 31 MIN

Disaster Recovery Design

Design for the day a whole site is gone: RPO, RTO and the network that makes them real

- Translate RPO and RTO into network requirements for replication and failover
- Compare DR patterns and the network mechanisms that move users to a surviving site
- Avoid split-brain and stretched-Layer-2 traps in multi-site designs

High availability handles component failures inside a site. **Disaster recovery** assumes the site itself is gone — fire, flood, power, regional outage, ransomware — and asks how quickly service returns elsewhere and how much data is lost. The network is often the hardest part of DR: replication needs bandwidth and latency guarantees, and users must find the surviving site without manual reconfiguration.

Cyber recovery is a different problem from site recovery. Ransomware and destructive attacks leave the building standing but corrupt data and the systems that manage it, and synchronous replication copies the damage in milliseconds. Design immutable or logically isolated backups, a clean recovery environment that can be rebuilt without trusting production identity and management systems, and an out-of-band management network. Operational-resilience regulation — for example the EU's DORA, applicable since January 2025 — increasingly expects such scenarios to be tested, not just documented.

> DEFINITION · RPO AND RTO
>
> **Recovery point objective (RPO)** is the maximum acceptable data loss, measured in time: an RPO of 5 minutes means losing up to the last 5 minutes of transactions is tolerable. **Recovery time objective (RTO)** is the maximum acceptable time to restore service after a declared disaster. RPO drives replication design; RTO drives failover design.

DR patterns

PATTERN	TYPICAL RPO	TYPICAL RTO	NETWORK IMPLICATIONS
Backup and restore	Hours to a day	Hours to days	Bulk transfer bandwidth; DR network can be built on demand
Pilot light	Minutes	Hours	Core data replicated continuously; network at DR site pre-built but scaled down
Warm standby	Seconds to minutes	Minutes to an hour	Full network ready, asynchronous replication, tested failover of addressing and DNS
Active/active (multi-site)	Near zero	Near zero to minutes	Both sites serve users; global load balancing, state replication and careful latency budgets

Replication and distance

MEASURE	FORMULA	WHAT IT TELLS YOU
Propagation delay in fibre	`One-way delay ≈ 5 µs per km of fibre path`	100 km of fibre route adds about 1 ms round trip — before any equipment or queuing delay. Fibre routes are usually longer than the straight-line distance.

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MEASURE	FORMULA	WHAT IT TELLS YOU
Synchronous write penalty	`Write latency ≈ local write + round-trip time to the remote site`	Every committed write waits for the remote acknowledgement, so distance directly slows the application.
Replication bandwidth	`Required rate ≥ peak change rate × (1 + protocol overhead)`	Size for the peak change rate, not the average, or the replication lag — and the effective RPO — grows during busy periods.

Synchronous versus asynchronous replication

Synchronous

- RPO near zero: writes commit at both sites
- Practical only over short distances, often metro scale
- Both sites typically share a regional risk
- Link latency and jitter directly affect application performance

Asynchronous

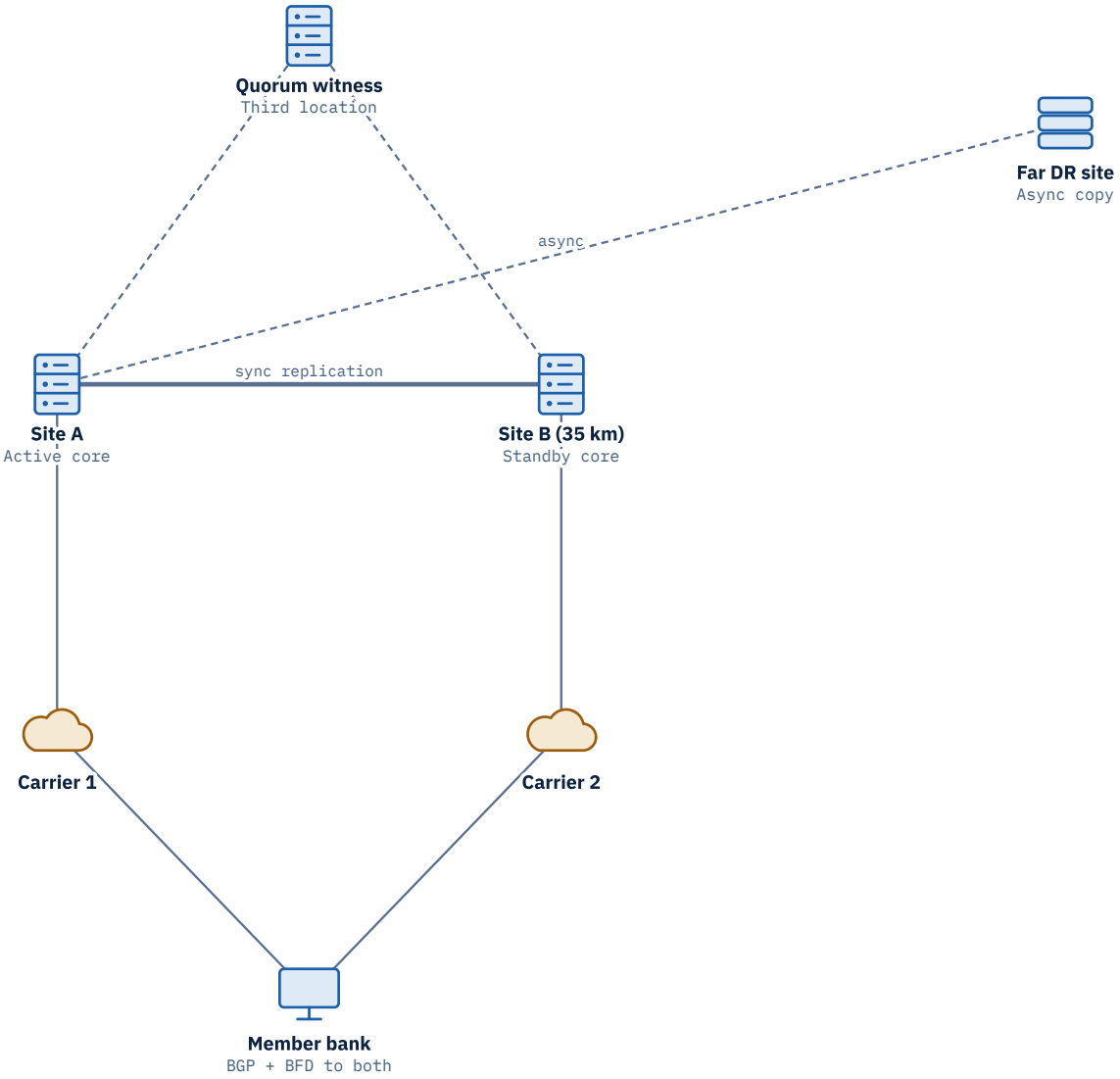
- RPO equals replication lag: seconds to minutes
- Works across long distances and different regions
- Gives true geographic separation
- Lag must be monitored; a burst of changes can widen it

Moving users to the surviving site

MECHANISM	HOW IT WORKS	STRENGTHS	WEAKNESSES
DNS / global server load balancing	Health-checked answers point clients to the live site	Works across any distance; application-aware	Caching and client behaviour delay the switch; TTLs must be designed
Route advertisement (anycast or prefix move)	The surviving site advertises the service prefix	Fast, transparent to clients, no DNS dependency	Needs portable addressing and careful health-to-route coupling
Stretched Layer 2 with the same subnet	Workloads keep IP addresses at the other site	No re-addressing	Joins both sites into one failure domain; split-brain and traffic tromboning risks
Re-addressing with automation	Workloads come up with DR-site addresses; DNS updated	Clean separation of sites	Requires disciplined automation and applications without hard-coded addresses

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Two-site payment switch with a far DR copy



The metro pair replicates synchronously for an RPO of zero; the far site receives an asynchronous copy for regional disasters. The witness sits at a third location, and each bank reaches each site through a different carrier.

WATCH OUT · SPLIT-BRAIN

When two sites lose contact with each other but both remain up, each may believe the other has failed and become active. With stateful systems that means two primaries accepting conflicting writes. Use a **witness or quorum at a third location** with independent connectivity, and decide in advance which site wins when the inter-site link fails.

INSIGHT · STRETCHED LAYER 2 IS NOT DR

Extending a VLAN between two data centres makes mobility easy but merges both sites into one Layer 2 failure domain. A loop, a storm or a control-plane bug can take down both — the exact event DR exists to survive. If adjacency is unavoidable, contain it with an overlay, strict flooding controls and per-site gateways, and keep it to the applications that truly need it.

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From disaster to service restored (illustrative 15-minute RTO)

T+0
Site A lost
Power, fire or regional event

T+1 MIN
Monitoring and witness confirm
Site B wins quorum

T+3 MIN
Disaster declared
Pre-agreed criteria, named authority

T+5 MIN
Site B core promoted
Committed data already replicated

T+7 MIN
Service prefixes advertised from B
Banks follow routing, no DNS wait

T+12 MIN
Transaction tests pass
Service declared restored inside RTO

Illustrative timings. Most of the RTO is spent detecting and deciding, not switching; automating the declaration criteria and the route move is what makes a 15-minute target realistic.

CASE STUDY · NATIONAL PAYMENT SWITCH HA REDESIGN

A national payment switch connects 40 banks and processes card and instant-payment transactions. The regulator requires **99.99% monthly availability**, an **RPO of zero for committed transactions** and an **RTO of 15 minutes** for a full site loss. Today everything runs in one data centre: a single firewall cluster in one rack, dual core switches, and bank connections arriving over one carrier's MPLS service with two circuits per bank. A secondary site 35 km away holds nightly backups. Last year a power-distribution fault in the main hall caused a 70-minute outage, and a carrier exchange fire cut 18 banks off for six hours.

1. Identify the main failure domains and shared-fate risks in the current design.

2. Which DR pattern and replication mode satisfy the RPO and RTO, and is the 35 km site suitable?

3. How should bank connectivity be redesigned?

4. How will banks reach the surviving site after a disaster?

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Model answer: module answer key.

SCENARIO · DECIDE · RPO ZERO ACROSS 600 KM?

A bank wants an RPO of zero and a DR site 600 km away to survive regional disasters. The application team asks for synchronous replication between the sites.

(A) Approve synchronous replication; modern links make 600 km fast enough.

(B) Use asynchronous replication to the far site and record the RPO as zero.

(C) Stretch a VLAN to the far site so the database cluster can fail over transparently.

(D) Use a near site for synchronous replication and a far site for asynchronous replication, and agree a non-zero RPO for the regional-disaster case.

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **DR network requirements sheet**

Capture, per service, the recovery targets and the network mechanisms that must deliver them, so that replication, addressing and failover are designed together.

SERVICE	RPO / RTO	REPLICATION MODE & BANDWIDTH	INTER-SITE LATENCY LIMIT	USER REDIRECTION METHOD	SPLIT-BRAIN CONTROL	LAST DR TEST
Card authorisation	*0 / 15 min*	*Synchronous, 2 × 10G diverse*	*< 2 ms RTT*	*Prefix advertised from active site*	*Witness at third site*	*March — passed in 11 min*

DR design checklist

☐ RPO and RTO are agreed per service and signed off by the business

☐ Replication mode is consistent with distance and latency limits

☐ Replication bandwidth is sized for peak change rate, with lag monitored

☐ A third-location witness or quorum prevents split-brain

☐ User and partner redirection does not depend on manual changes

☐ DNS TTLs and client behaviour are tested for the failover path

☐ DR is rehearsed under realistic load at least annually

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REFLECT

If your primary site disappeared at 10:00 tomorrow, what would be the first thing users noticed, and how long would it take them to reach the surviving site?

Key terms

RPO
Recovery point objective: the maximum tolerable data loss, expressed as a period of time.

RTO
Recovery time objective: the maximum tolerable time to restore service after a disaster.

Synchronous replication
Replication in which a write is acknowledged only after it is stored at both sites.

Asynchronous replication
Replication in which writes are acknowledged locally and copied to the remote site afterwards.

Split-brain
A condition in which isolated sites both act as primary, producing conflicting state.

Witness / quorum
A third-party vote that decides which site remains active when the sites lose contact.

Global server load balancing
Health-aware DNS that directs clients to the best available site.

Warm standby
A DR pattern with a fully built but scaled-down secondary environment ready to take load.

Key takeaways

- RPO drives replication design; RTO drives failover design
- Synchronous replication gives zero data loss only over short, metro-scale distances; use a far asynchronous copy for regional events
- A witness at a third location with independent connectivity prevents split-brain
- Move users by routing or health-checked DNS, not by stretching layer 2
- Plan cyber recovery separately: isolated backups and a clean environment, tested regularly

KNOWLEDGE CHECK

21. A service needs an RPO of zero. What does that require of replication?

(A) Asynchronous replication with lag alerting thresholds
(B) Nightly backups copied to a site with integrity checks
(C) A stretched VLAN so the database cluster can fail over
(D) Synchronous replication, which constrains site distance

22. Approximately how much round-trip delay does 100 km of fibre path add?

(A) About 0.1 ms
(B) About 10 ms
(C) About 1 ms
(D) About 5 ms

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23. What is the primary purpose of a witness at a third site?

(A) To decide which site stays active when the two sites lose contact

(B) To hold a third copy of all replicated data for use in regional disasters

(C) To balance user load evenly between both of the active sites

(D) To add replication bandwidth during peak business periods

24. Why is a stretched Layer 2 domain a weak foundation for disaster recovery?

(A) It prevents asynchronous replication from working

(B) It merges both sites into a single failure domain

(C) It forces every application to change IP addresses

(D) It limits sites to under 10 km apart in practice

25. Partners must reach the DR site without depending on DNS caches. Which mechanism fits?

(A) Lowering DNS record TTLs to one minute for every service

(B) Advertising the service prefix from the surviving site

(C) Emailing a new DR-site address to every partner in turn

(D) A stretched VLAN so that clients keep the same gateways

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 5 · RESILIENCE

Answer key

Knowledge checks

1. D Different carriers do not guarantee physical diversity. Both circuits shared one duct, so they were one failure domain. Bandwidth, timers and redundancy mode do not help when both paths are cut together.

2. A Synchronisation is the point of a cluster, which is why a bad rule reaches every member. Hardware redundancy does not protect against common-mode configuration or software faults.

3. B Cells cap the blast radius. They usually cost more, not less, and they deliberately avoid a single shared policy instance.

4. C Summarisation hides topology changes inside an area from the rest of the domain. Slower timers delay detection, a larger backbone enlarges the flooding domain, and mutual redistribution adds risk.

5. B Business stakeholders accept or reject risk in business terms. Device counts and protocol structure are means, not targets.

6. A Serial availability multiplies: 0.9999 × 0.9995 × 0.999 ≈ 0.9984. A serial chain is always below its weakest member, never at it or above.

7. C A = MTBF ÷ (MTBF + MTTR). Failures are just as frequent, but each one is about six times shorter, so annual downtime drops roughly sixfold.

8. D With two members and one tolerated failure, the survivable ceiling is (2 − 1) ÷ 2 = 50%. At 65% each, the survivor would need 130% capacity.

9. C The independent formula assumes no shared fate and perfect coverage. A small β or a few undetected grey failures quickly become the dominant term.

10. B (1 − 0.9999) × 525,600 ≈ 52.6 minutes. Five minutes corresponds to 99.999%, 4.4 hours to 99.95% and 8.8 hours to 99.9%.

11. A The port stays up, so loss of signal never triggers. BFD tests the forwarding path end to end and detects the failure in milliseconds.

12. D The delay is in the FIB update term. Sharing a next-hop object lets one change move every affected prefix; faster detection or SPF tuning does not shrink FIB programming time.

13. D With no alternative path, preserving forwarding is the only way to avoid an outage during a control-plane restart. With redundant paths, rerouting around the node is usually safer.

14. B Detection time is the interval multiplied by the multiplier: 300 ms × 3 = 900 ms.

15. C Micro-loops are transient and arise from unsynchronised FIB updates between neighbours. Ordered updates or segment-routing avoidance techniques mitigate them.

16. A The on-startup option advertises maximum metric for a period after boot, preventing black-holing while BGP and the FIB are still converging. The router still becomes a normal transit path afterwards.

17. C During maintenance N+1 becomes N, so a further failure is an outage. A service that can never be at risk needs additional redundancy or fully independent halves.

18. A The community only signals intent. Peers must act on it, typically by setting a very low local preference, so they move traffic to alternatives while the path still exists.

19. D Ordering by failure domain with an early canary exposes defects while impact is small and always leaves a known-good path.

20. B At 40% each, the survivor carries 80% when its partner is drained, which is within capacity. Above 50% each, a drain would overload the survivor.

21. D Only synchronous replication guarantees that committed data exists at both sites. The latency penalty limits practical distance.

22. C Light in fibre travels at roughly 5 µs per km one way, so 100 km is about 0.5 ms one way and 1 ms round trip.

23. A A witness breaks ties so that only one site becomes primary, preventing split-brain.

24. B DR depends on the sites failing independently. A shared broadcast domain lets a single loop or bug affect both.

25. B Moving the route moves traffic regardless of client DNS caching. Low TTLs help but many clients ignore them, and the other options need manual action or create shared fate.

Scenarios & case studies

Scenario · Stretched VLAN, shared disaster

B is best. Best. It shrinks the default failure domain to one hall, keeps a controlled exception for applications that need adjacency, and gives the business a path to retire the exception.

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Scenario · Where should the next dollar go?
A is best. Best. The firewall cluster is the serial bottleneck and its members share a rack and power — a common-cause failure domain. Separating them raises the chain closest to the target.

Scenario · A voice service with a 300 ms budget
C is best. Best. BFD detects failures that the carrier hides in about 150 ms, and a precomputed backup switches traffic immediately, keeping total loss inside the 300 ms budget.

Scenario · The upgrade weekend
D is best. Best. Each step leaves an untouched, known-good path, the survivor stays within capacity, and the soak period exposes defects before the second half is touched.

Case study · National payment switch HA redesign
Failure domains: the whole service is one site (power, hall, firewall rack), one carrier (both circuits per bank shared an exchange), and one firewall cluster (software and human common mode). A single 70-minute outage already exceeds the roughly 4.3 minutes a month that 99.99% allows. **DR pattern:** an RPO of zero needs synchronous replication; at 35 km (perhaps 50 km of fibre route) the round trip is about half a millisecond, so synchronous replication is feasible. Move to **two live sites with synchronous data replication** — active/standby for the transaction core, active/active for stateless front ends — with a quorum witness at a third location to prevent split-brain. A 35 km pair shares regional risks, so keep an **asynchronous copy at a distant third site** for regional disasters, with a documented, longer RTO and non-zero RPO agreed with the regulator. **Bank connectivity:** each bank gets two circuits from **two carriers**, one terminating at each site, with physical diversity verified to the building; BGP with BFD towards each bank, and graceful-shutdown policy agreed for maintenance. **Firewalls:** independent clusters per site, changed through a staged pipeline. **Finding the service:** the switch's service prefixes are advertised from the active site and withdrawn automatically by health checks, so banks follow routing without a DNS dependency; failover is rehearsed at least twice a year under load.

Scenario · RPO zero across 600 km?
D is best. Best. It gives zero data loss for the common site-level event and geographic protection for the rare regional event, with an honest, documented trade-off.

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06

MODULE 6 · SECURE

Security & Segmentation Design

Zero trust, segmentation, enforcement placement, identity, encryption, compliance and visibility

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Apply zero-trust principles and combine macro- and micro-segmentation into a coherent plan
- Place enforcement points and design secure edges and DMZs that fail safely
- Design identity-driven access and choose between MACsec, IPsec and TLS
- Use segmentation to reduce regulatory scope and prove isolation
- Design a visibility architecture that security operations can actually use

LESSONS

1. Zero-Trust Principles & Segmentation Strategy · 31 min
2. Enforcement Placement, Secure Edge & DMZ Patterns · 31 min
3. Identity-Driven Access & Encryption Choices · 31 min
4. Compliance Zones & Security Visibility Design · 33 min

MODULE 6 · SECURE · LESSON 1 · 31 MIN

Zero-Trust Principles & Segmentation Strategy

Stop trusting location — design access around identity, context and containment

- Apply zero-trust design principles and the logical architecture of NIST SP 800-207
- Choose between macro- and micro-segmentation and combine them coherently
- Plan a segmentation programme that moves from observation to enforcement safely

The traditional perimeter model trusted anything inside the firewall. Once an attacker landed on one laptop or one unpatched server, the flat internal network let them move sideways to the crown jewels. Ransomware has turned that design flaw into a board-level risk. **Zero trust** is the design response: no implicit trust based on network location, every access decision made per request from identity, device state and context, and every resource protected as if the network around it were hostile.

DEFINITION · ZERO TRUST IN DESIGN TERMS

NIST SP 800-207 describes zero trust as a set of principles in which access to each resource is granted **per session**, based on **dynamic policy** that considers the identity of the subject, the state of the device and other context, with **no implicit trust** granted because of network location. For a network designer, it means the network still matters — it carries policy, enforces boundaries and provides telemetry — but it is no longer the sole source of trust.

ZCDE zero-trust design principles

ZT1

Verify explicitly

Every access decision uses identity, device posture and context — not just a source subnet.

ZT2

Least privilege per session

Grant the narrowest access that lets the task succeed, for as long as it is needed.

ZT3

Assume breach

Design so that a compromised host can reach as little as possible; containment is a primary requirement.

ZT4

Protect resources, not segments

Policy follows the workload or data, wherever it runs — campus, data centre or cloud.

ZT5

Evaluate continuously

Re-assess trust when posture, behaviour or risk signals change, and be able to revoke access mid-session.

ZT6

Telemetry feeds policy

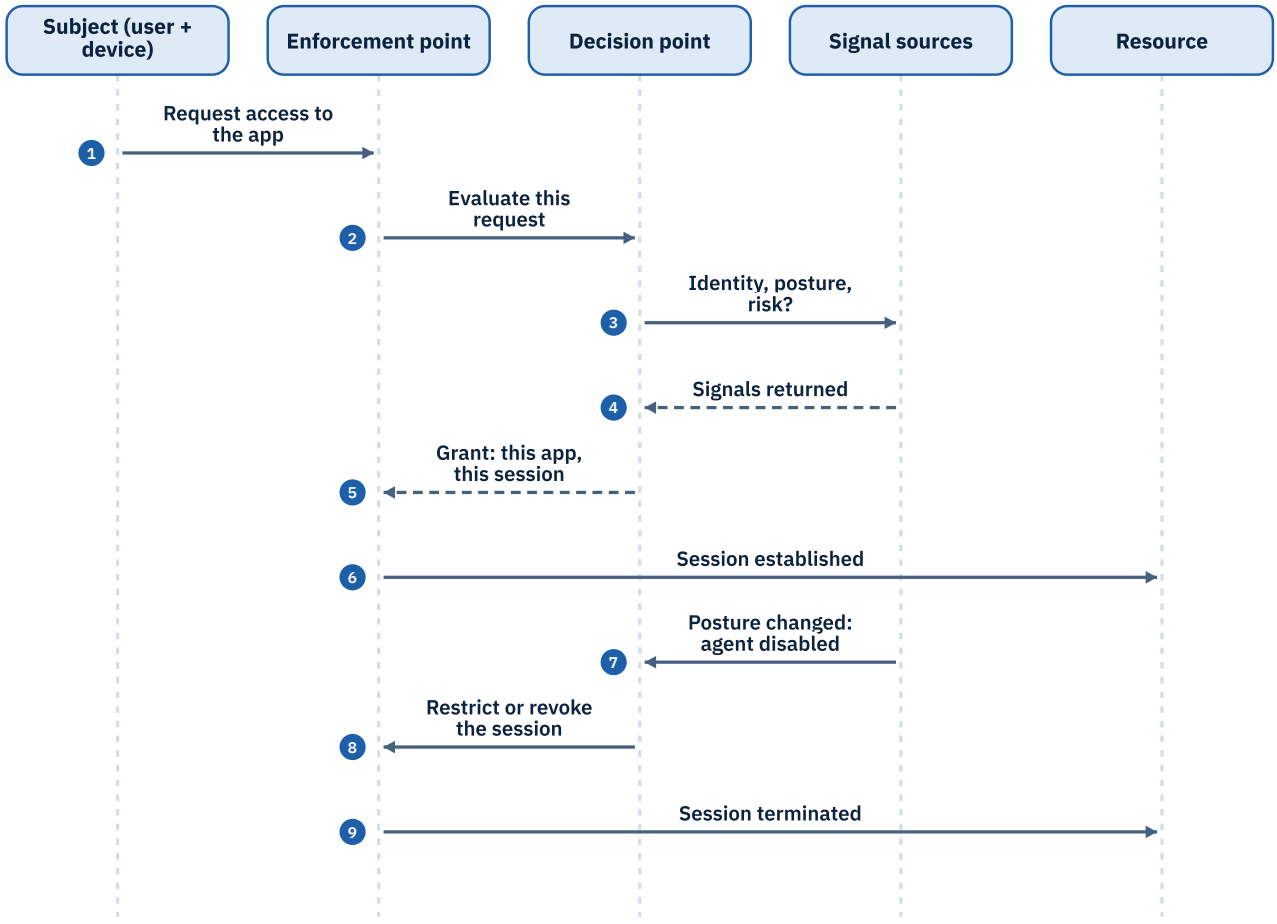
Logs, flows and identity events are designed in from day one so policy can be built, tuned and audited.

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Zero-trust logical architecture (after NIST SP 800-207)

Policy decision point	`Policy engine: decides grant, deny or revoke` `Policy administrator: establishes or tears down the session path`
Policy enforcement points	`Switch or wireless edge, firewall, gateway, proxy` `Host agent, service-mesh sidecar, cloud security group`
Subjects and resources	`Users, devices, workloads and services requesting access` `Applications, data and infrastructure being protected`
Signal sources	`Identity provider and directory` `Device inventory and posture` `Threat intelligence, activity logs, PKI, compliance data`

A zero-trust access decision, then a revocation



The enforcement point never decides on its own: it asks the decision point, which weighs identity, device state and risk signals. Because evaluation continues, a posture change later in the session can restrict or end access without waiting for the user to reconnect.

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Macro- versus micro-segmentation

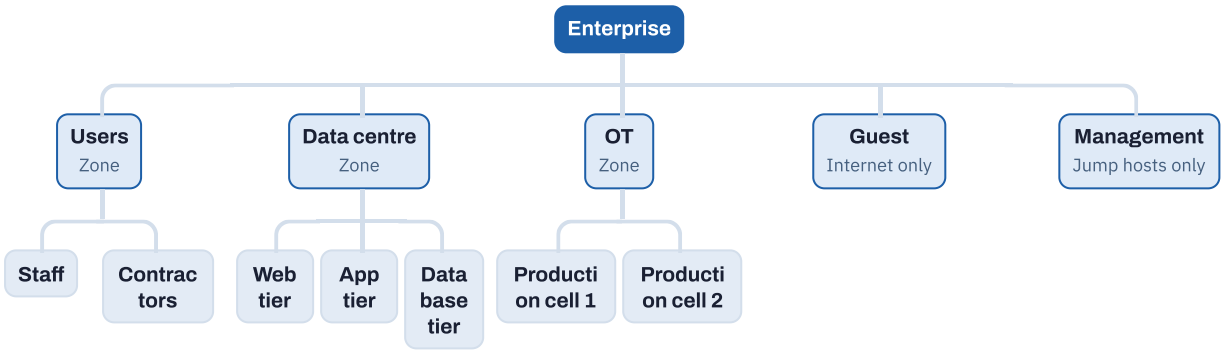
Macro-segmentation

- Separates large zones: users, servers, OT, guests, partners, management
- Usually virtual routing instances or virtual networks with firewalls between them
- Coarse, stable, easy to explain to auditors
- Does not stop lateral movement inside a zone

Micro-segmentation

- Separates groups or individual workloads inside a zone
- Enforced by group tags, distributed firewalls, host agents or cloud security groups
- Directly limits lateral movement and ransomware spread
- Needs accurate flow knowledge and strong operational tooling

Macro zones first, micro groups inside them



Macro zones are separated by stateful boundaries and match trust levels and compliance scope; micro groups inside each zone limit lateral movement. An example estate, not a template.

Segmentation approaches — decision matrix

APPROACH	GRANULARITY	ENFORCEMENT POINT	OPERATIONAL LOAD	BEST FIT
Virtual routing instance / virtual network per zone	Zone	Inter-zone firewall	Low to medium	Macro boundaries, compliance zones, multi-tenant separation
VLAN plus access lists	Subnet	Gateway or switch	High at scale; IP-based rules drift	Small, stable environments
Identity group tags	Group of users or devices	Access switch, wireless, fabric edge, firewall	Medium; needs identity integration	Campus and branch user and device segmentation
Distributed or host-based firewall	Workload	Hypervisor or host agent	Medium to high; agent lifecycle	Data-centre east-west, application tiers
Cloud security groups	Workload or interface	Cloud fabric	Low per rule, high at sprawl	Cloud-native workloads
Service mesh with mutual TLS	Service identity	Sidecar or node proxy	High; platform skills	Container platforms with many microservices

A segmentation programme that does not break the business

1. **Classify** Inventory assets and data, and classify them by sensitivity, criticality and regulatory scope.

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2. **Observe** Collect flow and identity telemetry for long enough to see monthly and quarterly cycles.
3. **Define macro zones** Draw zones by trust level and compliance scope; put firewalls or policy boundaries between them.
4. **Define groups** Within each zone, group workloads and users by role or application tier.
5. **Write the policy matrix** Start from default deny between groups and add only observed, justified flows with an owner.
6. **Monitor, then enforce** Run policy in log-only mode, fix gaps, then enforce group by group with rollback ready.
7. **Review** Re-certify rules on a schedule; remove flows that no longer have an owner.

WATCH OUT · THE BIG-BANG DENY

Switching on default-deny micro-segmentation without flow discovery is the fastest way to discredit a zero-trust programme. Month-end batch jobs, backup agents and licence servers will break first. Discover, model, monitor, then enforce — and start with the groups whose flows you understand best.

SCENARIO · DECIDE · A FLAT MANUFACTURING PLANT

A manufacturer runs office PCs, engineering workstations and production controllers on one flat network. A ransomware outbreak at a peer company has alarmed the board. Many controllers run old operating systems that cannot take agents and must not be patched without the machine builder's approval. You have six months for a first phase.

(A) Deploy host-based micro-segmentation agents on every device, including controllers, in one programme.

(B) Separate OT from IT with a firewalled industrial DMZ, allow only brokered flows between them, then segment OT internally by production cell using network enforcement.

(C) Buy a next-generation antivirus product for all office PCs and leave the network unchanged.

(D) Put every controller on its own VLAN with a full IP access list on the core switch.

Circle your choice. Answer and feedback: module answer key.

INSIGHT · ZERO TRUST IS A DESIGN PROPERTY, NOT A PRODUCT

No single purchase delivers zero trust. It emerges from identity, device health, enforcement points, telemetry and process working together. A useful design test: if an attacker owns one host in any zone tomorrow, what can that host reach, and how quickly would you know?

PRO TIP · REFERENCE MODELS WORTH KNOWING

NIST SP 800-207 remains the common vocabulary for zero-trust architecture, and NIST has since published practice guides with worked example builds. Maturity models such as the US CISA Zero Trust Maturity Model (version 2.0) help stage a programme across identity, devices, networks, applications and data. Use them to structure the roadmap, not as a checklist of products to buy.

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Zero-trust segmentation checklist

☐ Assets and data are classified by sensitivity and regulatory scope
☐ Macro zones are defined and enforced at stateful boundaries
☐ Micro-segmentation targets the highest-risk lateral paths first
☐ Policy is written in terms of identities and groups, not only IP addresses
☐ Flows were observed before enforcement, across business cycles
☐ Every allowed flow has an owner and a review date
☐ Telemetry from enforcement points reaches security operations

REFLECT

If one user laptop in your network were compromised right now, list the systems it could reach. Which one of those paths would you close first, and with what enforcement point?

Key terms

Zero trust
A security model that grants access per session based on identity, device state and context, with no implicit trust from network location.

Policy decision point
The logical component that evaluates policy and decides whether access is granted, denied or revoked.

Policy enforcement point
The component in the traffic path that enables, monitors and terminates connections as instructed by the decision point.

Macro-segmentation
Separation of large trust zones, typically with virtual routing instances and inter-zone firewalls.

Micro-segmentation
Fine-grained separation of groups or individual workloads inside a zone to limit lateral movement.

Lateral movement
An attacker's progression from an initially compromised host to other systems inside the network.

Group tag
A label carried with traffic or bound to an endpoint that lets policy be written by group rather than by IP address.

Default deny
A policy stance in which anything not explicitly permitted is blocked.

Key takeaways

- Zero trust removes implicit trust from network location; every access is decided per session from identity, device and context
- The network still carries and enforces policy and supplies telemetry
- Combine macro zones with micro-segmentation where lateral movement matters most
- Discover flows, run policy in monitor mode, then enforce group by group
- Test the design by asking what one compromised host can reach and how fast you would know

KNOWLEDGE CHECK

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1. Which statement best captures the network designer's role in a zero-trust architecture?

   (A) The network is no longer relevant because all trust decisions move into the applications

   (B) The network grants full trust to any host that sits inside a properly firewalled internal network zone

   (C) The network must encrypt every single packet, which makes segmentation unnecessary

   (D) The network carries and enforces policy and supplies telemetry, but is not the source of trust

2. What is the main limitation of macro-segmentation on its own?

   (A) It does not stop lateral movement inside a zone

   (B) It cannot be enforced with stateful firewalls

   (C) It requires agents on every single workload

   (D) It is impossible to explain to auditors simply

3. A team plans to enable default-deny between all application groups next weekend. Flow data has not been collected. What should the designer recommend?

   (A) Proceed as planned, because default deny is the core principle that defines a zero-trust design

   (B) Enable default deny only on weekdays so that weekend batch jobs are not affected

   (C) Collect flows across business cycles, run policy in log-only mode, then enforce in stages

   (D) Replace the plan with a single perimeter firewall rule set that is easier to audit

4. In the NIST SP 800-207 model, which component actually allows or terminates the connection in the data path?

   (A) The policy engine in the decision point

   (B) The identity provider

   (C) The policy enforcement point

   (D) The threat-intel feed

5. Which segmentation approach fits production controllers that cannot run software agents?

   (A) Network-based enforcement such as zone firewalls or group policy at the switch

   (B) Host-based firewall agents installed on every controller in the plant

   (C) Service mesh sidecars with mutual TLS between all of the controllers

   (D) Antivirus software paired with daily vulnerability scanning of every plant controller

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 6 · SECURE · LESSON 2 · 31 MIN

Enforcement Placement, Secure Edge & DMZ Patterns

Put policy where it protects most and costs least, and build edges that fail safely

- Decide where to place enforcement for north-south and east-west traffic
- Compare DMZ patterns and secure internet-edge layering
- Avoid stateful-inspection pitfalls such as asymmetric paths and hairpinning

Every security control sits somewhere in the traffic path, and its location decides what it can see, what it costs and what breaks when it fails. Centralised enforcement is simple to audit but can become a bottleneck and a single failure domain. Distributed enforcement scales and stays close to the resource, but is harder to operate. The expert answer is almost always a deliberate mix, chosen per traffic pattern.

Centralised, distributed and cloud-delivered enforcement

Centralised firewalls

- All inter-zone traffic hairpins through a firewall cluster
- One place to audit and log
- Throughput, latency and failure-domain concerns at scale
- Natural fit for macro boundaries and compliance zones

Distributed enforcement

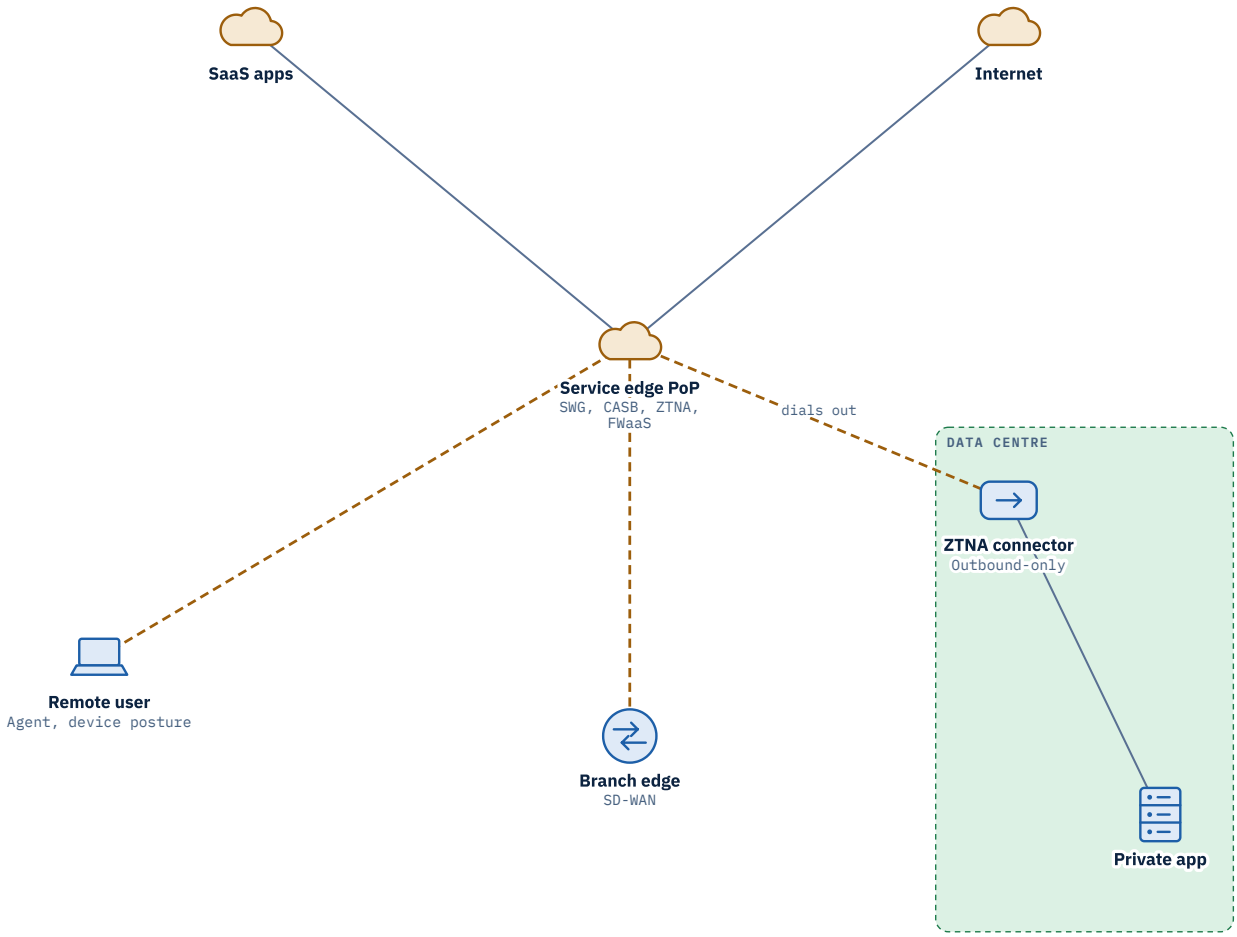
- Policy applied at the access edge, fabric leaf, hypervisor or host
- Scales with the network; no hairpin
- Many enforcement points to keep consistent
- Natural fit for east-west micro-segmentation

Cloud-delivered security edge

- Users connect to nearby service points that apply web, data and access policy
- Consistent policy for remote and branch users without backhaul
- Depends on the provider's availability and points of presence
- Natural fit for internet access and private-application access for mobile users

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Cloud-delivered security edge with application-level access



Remote users and branches connect to a nearby service edge that applies web, data and access policy. Private applications are reached through a connector that dials out from the data centre, so nothing internal listens on the internet.

Enforcement placement — decision matrix

PLACEMENT	SEES	STRENGTHS	WEAKNESSES	USE WHEN
Internet perimeter	North-south traffic	Clear choke point, mature tooling	Blind to internal movement	Always, as one layer among several
Data-centre zone firewall	Traffic between zones	Stateful inspection, compliance evidence	Hairpin cost and capacity	Between macro zones and around regulated data
Access or fabric edge (group policy)	Traffic as it enters the network	Stops threats before they travel	Needs identity integration	Campus, branch and fabric east-west
Host or hypervisor	Every flow to and from the workload	Follows the workload anywhere	Agent lifecycle; compromised host may disable it	Application tiers, cloud and hybrid workloads
Application or API gateway	Requests, users and tokens	Understands the application protocol	Only protects what is published through it	Exposed web services and partner APIs

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Secure internet-edge layering (outside first)

Upstream protection	`Provider-based volumetric DDoS scrubbing` `Remotely triggered black-hole as a last resort`
Edge routing	`Diverse upstreams with BGP` `Anti-spoofing ingress filters` `Prefix limits and route filtering`
Edge firewall	`Stateful policy for inbound and outbound` `Separate management plane`
DMZ services	`Reverse proxy and web application firewall` `Mail relay, external DNS, remote-access brokers`
Internal boundary	`Second firewall layer or zone policy` `Only brokered flows to internal zones`

DMZ patterns

Three-legged single firewall

- Inside, outside and DMZ on one firewall pair
- Cheapest and simplest
- One policy error or one software flaw exposes both boundaries

Dual-firewall (back-to-back)

- Outer and inner firewalls, ideally managed as separate domains
- Defence in depth: an attacker must defeat two independent layers
- More cost and operational effort; rules must be coordinated

Brokered access

- No direct inbound connection reaches internal zones
- Reverse proxies and brokers in the DMZ initiate or relay every session
- Strong containment; requires applications that work through brokers

Anti-spoofing at the internet edge

EDGE1 · IOS-style CLI

```
ip access-list extended EDGE-IN
 remark Drop packets claiming our own or private source addresses
 deny ip 203.0.113.0 0.0.0.255 any
 deny ip 10.0.0.0 0.255.255.255 any
 deny ip 172.16.0.0 0.15.255.255 any
 deny ip 192.168.0.0 0.0.255.255 any
 permit ip any any
!
interface GigabitEthernet0/0/0
 description Upstream provider A
 ip access-group EDGE-IN in
```

203.0.113.0/24 stands for the organisation's own public block. The design point is that edge filtering removes obviously forged traffic before it reaches stateful devices, saving their capacity for real sessions.

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Inbound request through a brokered DMZ

01 **Client**
Resolves the public name

→

02 **Scrubbing**
Volumetric attacks absorbed upstream

→

03 **Edge firewall**
Allows only HTTPS to the proxy

→

04 **Reverse proxy / WAF**
Terminates TLS, inspects, authenticates

→

05 **Inner firewall**
Allows proxy-to-application flows only

→

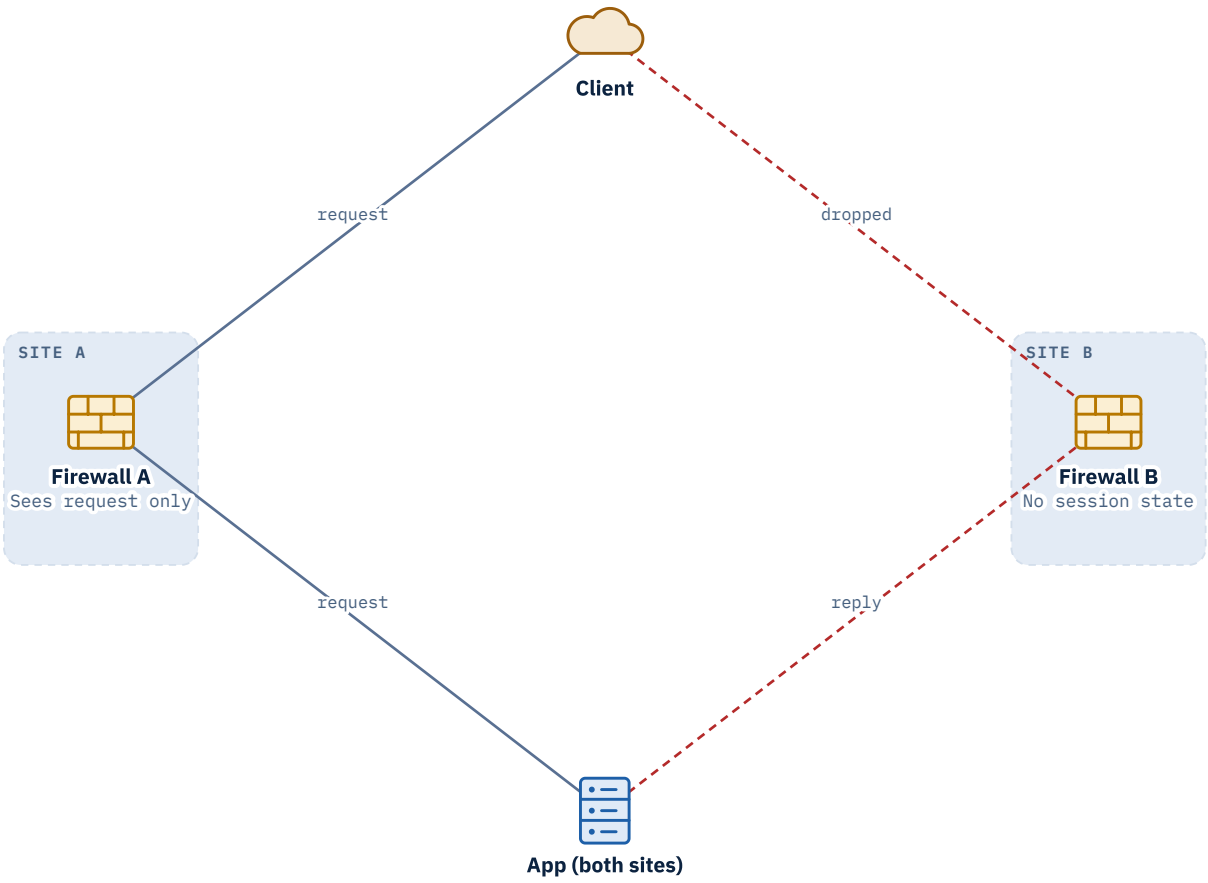
06 **Application tier**
Never directly reachable from outside

WATCH OUT · ASYMMETRIC PATHS THROUGH STATEFUL DEVICES

Stateful firewalls must see both directions of a session. In active/active multi-site designs, routing can send the request through Site A's firewall and the reply through Site B's, and the session is dropped. Design symmetric paths (per-site egress, consistent routing preference, source NAT) or use firewall clustering that shares state across sites — knowing that the latter enlarges the failure domain.

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Why asymmetric routing breaks stateful inspection



The request is inspected by Site A's firewall, but the reply leaves through Site B's, which has no session state and drops it. Per-site egress or source NAT at each site forces both directions through the same firewall.

PRO TIP · REPLACING NETWORK VPNS WITH APPLICATION ACCESS

Traditional remote-access VPNs put a user's device on the internal network. Brokered, application-level access (often called zero-trust network access) connects a verified user and device to a named application only, hiding everything else. For third parties and unmanaged devices it shrinks the blast radius dramatically.

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SCENARIO · DECIDE · THIRD-PARTY ACCESS TO PLANT SYSTEMS

Twelve machine builders need occasional remote access to specific production controllers for diagnostics. Today they use a shared VPN account that lands them on the plant network with broad reach. The security team wants accountability and minimum exposure; the plant manager wants support to stay fast.

(A) Issue each machine builder its own VPN account on the existing concentrator with the same network access.

(B) Block all remote access and require machine builders to visit the site for every fault.

(C) Provide brokered, application-level access through a jump service in the industrial DMZ, with per-vendor policy to named assets, approval workflow and session recording.

(D) Open inbound firewall rules from each vendor's office IP address directly to its controllers.

Circle your choice. Answer and feedback: module answer key.

Enforcement and edge design checklist

☐ Each traffic pattern (north-south, east-west, user-to-app) has a named enforcement point

☐ Stateful devices see both directions of every session they inspect

☐ Hairpin traffic through central firewalls is sized and has a failure-domain plan

☐ The internet edge has upstream DDoS protection and anti-spoofing filters

☐ DMZ services broker all inbound sessions; nothing internal is directly reachable

☐ Third-party and remote access is per-application, attributed and recorded

☐ Management interfaces of security devices sit in a separate, protected zone

REFLECT

Trace one inbound user request to your most important application. How many enforcement points does it cross, and which one would you remove if you had to — and why?

Key terms

North-south traffic
Traffic entering or leaving a site or data centre, such as between users or the internet and servers.

East-west traffic
Traffic between workloads inside a site, data centre or cloud.

DMZ
A semi-trusted zone between the internet and internal networks that hosts externally reachable services.

Reverse proxy
A server that accepts client connections on behalf of back-end applications and relays requests to them.

Web application firewall
A control that inspects HTTP traffic for application-layer attacks.

Zero-trust network access
Brokered access that connects verified users and devices to specific applications rather than to a network.

Hairpinning
Sending traffic to a central device and back, often across the same links, to apply a control.

Asymmetric routing
A condition in which the two directions of a flow take different paths.

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Key takeaways

- Place enforcement per traffic pattern: perimeter for north-south, zone firewalls for macro boundaries, distributed points for east-west
- Cloud-delivered security edges suit remote users and SaaS-heavy branches; check points of presence and failure behaviour
- Brokered DMZs and application-level access keep internal systems unreachable from outside
- Stateful devices must see both directions; design symmetric paths per site

KNOWLEDGE CHECK

6. An active/active two-site design drops sessions intermittently. Requests leave through Site A's firewall and replies return through Site B's. What is the best design fix?

   (A) Add more bandwidth between the two sites so replies arrive much faster

   (B) Make paths symmetric per site, for example with per-site egress and source NAT

   (C) Disable stateful inspection on both firewalls for all of the affected applications

   (D) Shorten the firewall session timeouts so stale state is cleaned up sooner

7. Which DMZ pattern provides the strongest protection against a single policy error exposing internal systems?

   (A) A three-legged firewall with careful rule review each quarter

   (B) A single firewall pair using the supplier's recommended default rule set

   (C) A flat DMZ with host firewalls installed on each public server

   (D) Dual firewalls managed as separate domains, with brokered access

8. Where is micro-segmentation for east-west data-centre traffic usually most scalable?

   (A) At the internet perimeter firewall cluster

   (B) Distributed at the hypervisor, host or fabric edge

   (C) At a central firewall cluster that hairpins all east-west flows

   (D) At the provider's upstream scrubbing centre

9. Why is a remote-access VPN that places users on the internal network a weak fit for third parties?

   (A) It cannot encrypt traffic strongly enough to meet modern security standards

   (B) It requires each third party to buy hardware tokens for every single user

   (C) It grants network-level reach far beyond the applications each party needs

   (D) It always routes traffic through the internet perimeter twice, adding delay

10. What is the main design purpose of anti-spoofing filters at the internet edge?

    (A) Drop forged traffic early so stateful devices spend capacity on real sessions

    (B) Replace the edge firewall so that the DMZ can be simplified and removed

    (C) Encrypt inbound traffic before it reaches any of the DMZ service tiers

    (D) Load-balance inbound sessions evenly across the diverse upstream service providers

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 6 · SECURE · LESSON 3 · 31 MIN

Identity-Driven Access & Encryption Choices

Let identity decide who connects, and choose the right layer to encrypt

- Design identity-driven network access for users, devices and headless endpoints
- Plan a safe rollout of port-based authentication
- Choose between MACsec, IPsec and TLS based on requirements and trade-offs
- Plan crypto-agility for the move to post-quantum key exchange

In a zero-trust design the first question at the network edge is not 'which port?' but 'who or what is this, and is it healthy?'. Identity-driven access answers that question at connection time and hands the result to enforcement points. Encryption then protects traffic wherever the path cannot be trusted. Both are design choices with real operational cost, and both fail badly when introduced in a hurry.

Identity-driven access sequence

01 Connect
Device attaches to a wired port, wireless network or VPN

→

02 Authenticate
802.1X with certificates or credentials; MAC-based fallback for headless devices

→

03 Profile and posture
Device type, ownership and health are assessed

→

04 Decide
The policy server returns an authorisation result

→

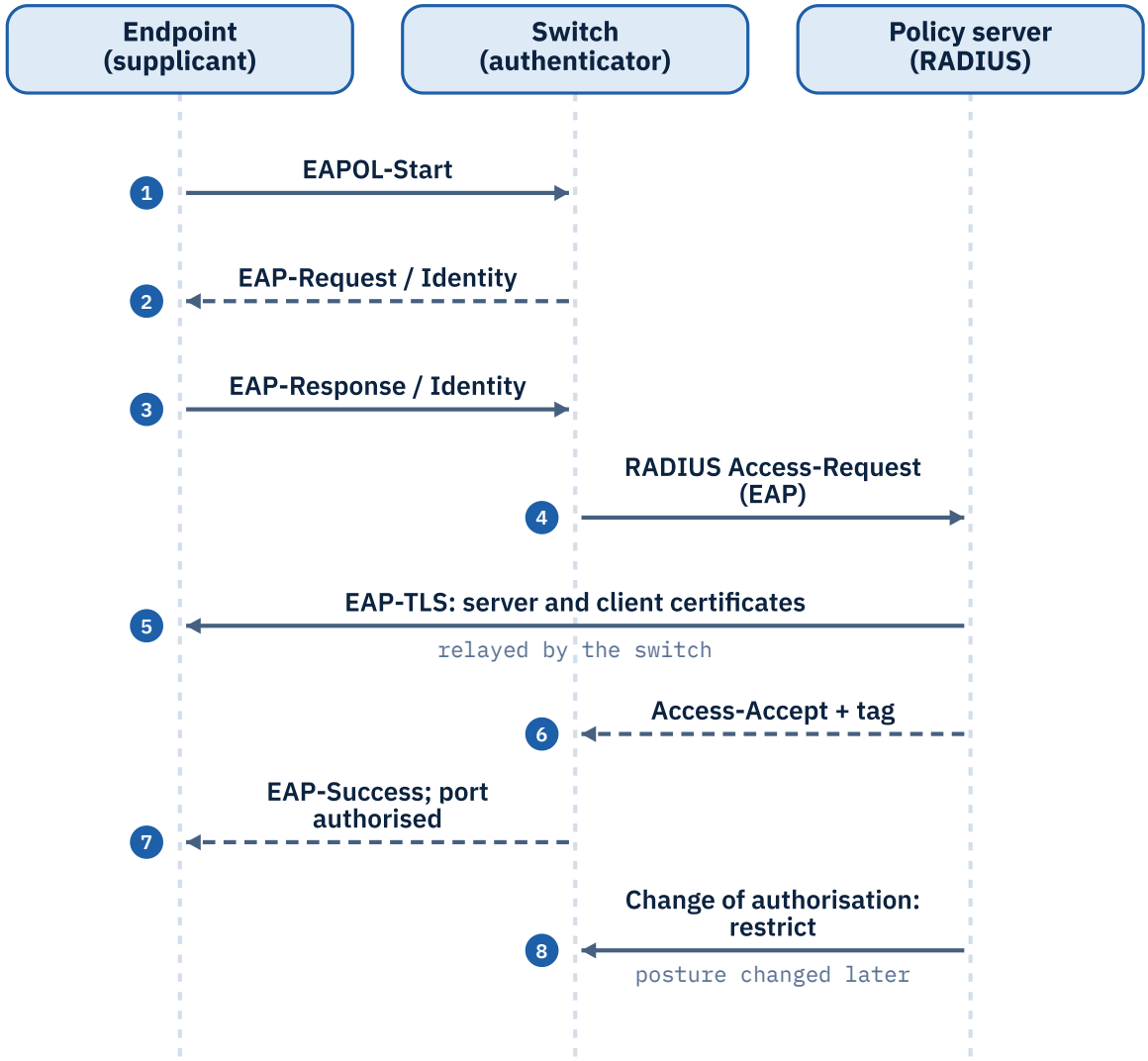
05 Enforce
Group tag, downloadable access list or VLAN applied at the edge

→

06 Reassess
Change of authorisation if posture or risk changes

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802.1X with certificate-based authentication



The switch relays EAP between the endpoint and the policy server and opens the port only after Access-Accept, applying the returned group tag or access list. Change of authorisation lets the server tighten access mid-session.

Authorisation results — trade-offs

RESULT	HOW IT WORKS	STRENGTHS	WEAKNESSES
Dynamic VLAN	Endpoint is placed in a VLAN per role	Simple, widely supported	Ties policy to IP subnets; VLAN sprawl; re-addressing on change
Downloadable access list	Per-session list pushed to the port	Granular, no re-addressing	Rule scale on access switches; IP-based rules
Group tag	Endpoint is labelled; policy written group to group	Policy independent of IP address; scales across sites	Needs tag support or propagation across the path
Redirect / restricted	Endpoint sent to remediation or onboarding	Smooth user experience for non-compliant devices	Extra infrastructure; must not become a bypass

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Port-authentication rollout modes

Monitor mode

- Authentication runs but access is not restricted
- Reveals unknown devices and failed authentications
- Zero user impact — the right first phase

Low-impact mode

- Limited access before authentication (for example DHCP, DNS, boot services)
- Full access granted per authorisation result
- Balances security and compatibility for mixed estates

Closed mode

- No traffic until authentication succeeds
- Strongest posture
- Unforgiving: printers, phones and building systems must all be handled first

WATCH OUT · MAC-BASED BYPASS IDENTIFIES — IT DOES NOT AUTHENTICATE

Headless devices that cannot run 802.1X are often admitted by MAC address. MAC addresses are easy to spoof. Combine MAC-based access with profiling and give those devices a tightly restricted policy, so that a spoofed MAC gains almost nothing.

MACsec, IPsec and TLS

MACsec (IEEE 802.1AE)

- Encrypts Ethernet frames hop by hop at line rate in hardware
- Protects links: data-centre interconnect over fibre or wavelengths, campus uplinks
- Low latency, roughly 24–32 bytes of overhead per frame
- Traffic is clear inside each device; needs hardware support and, over a carrier, transparency for the key-agreement frames

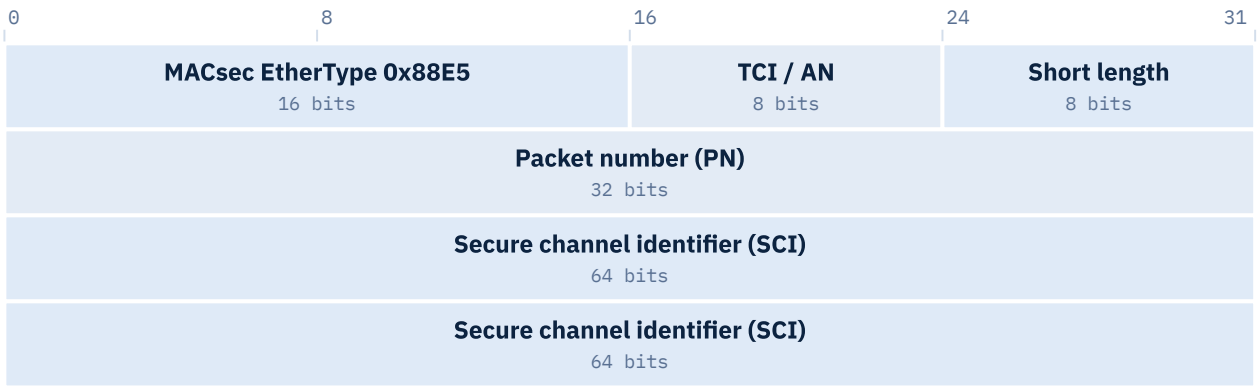
IPsec

- Encrypts IP packets between gateways across any routed network
- Protects WAN paths over the internet or untrusted carriers
- Overhead of roughly 50–80 bytes depending on mode, cipher, NAT traversal and tunnelling; MTU and MSS must be managed
- Throughput limited by gateway crypto capacity; adds a tunnel layer to operate

TLS / mutual TLS

- Encrypts application sessions end to end
- Protects data across every hop, including inside the data centre
- Closest fit to zero trust: service identity in certificates
- Reduces network visibility; inspection needs termination at a proxy or endpoint telemetry

The MACsec security tag (SecTAG) with SCI



The SecTAG follows the source MAC address and is 8 bytes, or 16 when the optional secure channel identifier is included; a 16-byte integrity check value follows the encrypted payload, giving the 24–32 bytes of overhead. Extended packet numbering (XPN) uses a 64-bit counter but still carries only the low 32 bits here.

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Encryption decision matrix

REQUIREMENT	PREFERRED CHOICE	WHY
Encrypt a 100G data-centre interconnect over a leased wavelength at line rate	MACsec	Hardware line-rate encryption with minimal latency on a point-to-point link
Encrypt branch traffic over the internet	IPsec (often within an SD-WAN overlay)	Works across any routed path and through NAT
Protect service-to-service calls regardless of network path	Mutual TLS	End-to-end with service identity; independent of network topology
Encrypt over a carrier Ethernet service	MACsec, if the service passes the required frames and tags	Line rate on a Layer 2 path; confirm transparency with the provider
Meet a policy of 'encrypt everything that leaves the building'	Layered: MACsec or IPsec on links, TLS for applications	No single layer covers every path and threat

WATCH OUT · PLAN FOR POST-QUANTUM CRYPTOGRAPHY NOW

Large quantum computers would break today's public-key algorithms (RSA and elliptic-curve key exchange and signatures), and data recorded now could be decrypted later — the 'harvest now, decrypt later' risk for long-lived secrets. NIST published its first post-quantum standards in August 2024 (FIPS 203 for key establishment, FIPS 204 and 205 for signatures) and has proposed retiring the vulnerable algorithms over the following decade; check the current timetable. Symmetric ciphers with 256-bit keys, as used for MACsec and IPsec bulk encryption, are not the weak point; key exchange and certificates are. Design for **crypto-agility**: inventory where keys are negotiated (IKE, TLS, MACsec key agreement, management access), prefer platforms that support hybrid post-quantum key exchange, and avoid hard-wired algorithms.

Encryption and path MTU

MEASURE	FORMULA	WHAT IT TELLS YOU
Effective payload MTU	`MTU_effective = path MTU − encapsulation overhead`	Every tunnel or tag reduces the space left for the original packet.
TCP MSS clamp (IPv4)	`MSS = MTU_effective − 40`	Clamp MSS on tunnel interfaces so TCP never sends segments that need fragmentation; IPv6 headers need 60 bytes instead of 40.
Worked example	`1500 − 73 (ESP tunnel with NAT traversal, illustrative) ≈ 1427; MSS ≈ 1387`	Many designs round down to an MTU of 1400 and an MSS of 1360 to leave headroom for additional encapsulation.

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MACsec on a data-centre interconnect

DCI-A · IOS-style CLI

```
key chain DCI-KC macsec
 key 01
  cryptographic-algorithm aes-256-cmac
  key-string <64-hex-character pre-shared CAK>
!
mka policy DCI-MKA
 macsec-cipher-suite gcm-aes-xpn-256
!
interface HundredGigE1/0/1
 description Wavelength to DC-B
 macsec network-link
 mka policy DCI-MKA
 mka pre-shared-key key-chain DCI-KC
```

Extended packet numbering (XPN) cipher suites avoid frequent rekeying on high-speed links. Key rotation and storage of the pre-shared connectivity association key must be part of the operational design.

SCENARIO · DECIDE · ENCRYPTING A DATA-CENTRE INTERCONNECT

Two data centres 40 km apart are joined by two leased 100G wavelengths carrying storage replication and application traffic. A new regulation requires encryption of all data leaving each site. Latency is critical for synchronous replication. The team proposes routing all inter-site traffic through the existing firewalls' IPsec engines, which are rated at 20 Gbps.

(A) Enable MACsec on the wavelength-facing interfaces of the interconnect switches or routers, with managed key rotation.

(B) Proceed with firewall IPsec and add more firewall pairs later if needed.

(C) Encrypt only the storage replication at the application layer and leave other traffic clear.

(D) Replace the wavelengths with an internet VPN service that is encrypted by default.

Circle your choice. Answer and feedback: module answer key.

Identity and encryption design checklist

☐ Every endpoint class (managed, unmanaged, headless, guest) has an access method and policy

☐ Port authentication rolls out monitor first, then low-impact, then closed where justified

☐ MAC-based access is paired with profiling and restricted policy

☐ Policy results use groups or tags where possible, not VLAN sprawl

☐ Encryption layer is chosen per path: link, network or application

☐ MTU and MSS are planned for every encapsulation

☐ Key and certificate lifecycle is automated and monitored

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REFLECT

Which endpoint class in your environment would break first if you turned on closed-mode authentication tomorrow? What would you need in place to make that safe?

Key terms

802.1X
An IEEE standard for port-based network access control using an authentication server.

Profiling
Identifying device type from observed attributes such as DHCP options, traffic and certificates.

MACsec
IEEE 802.1AE link-layer encryption that protects Ethernet frames hop by hop.

IPsec
A suite of protocols that authenticates and encrypts IP packets between peers or gateways.

MSS clamping
Rewriting the TCP maximum segment size so segments fit within a tunnel's reduced MTU.

MAC authentication bypass
Admitting a device based on its MAC address when it cannot perform 802.1X.

Change of authorisation
A message from the policy server that changes an active session's access without reconnecting.

MKA
MACsec Key Agreement: the protocol that establishes and rotates MACsec keys between peers.

Mutual TLS
TLS in which both client and server present certificates, giving both sides a verified identity.

Key takeaways

- Identity decides access at connection time; group tags make policy independent of IP addresses
- Roll out port authentication monitor first, then low-impact, then closed where justified
- MAC-based admission identifies but does not authenticate: pair it with profiling and tight policy
- Choose MACsec for links, IPsec for routed paths and mutual TLS for services — and manage MTU
- Build crypto-agility now: post-quantum risk sits in key exchange and certificates

KNOWLEDGE CHECK

11. A hospital wants to start 802.1X on 6,000 wired ports with many unknown medical and building devices. What is the safest first phase?

(A) Closed mode on all ports at once, with the help desk on standby
(B) MAC-based access for every device so that nothing ever breaks
(C) Dynamic VLAN assignment for every user and every device in a single change
(D) Monitor mode to discover endpoints and failures without blocking

12. Which encryption choice best protects service-to-service traffic in a way that is independent of the network path?

(A) MACsec on each data-centre link
(B) IPsec between the core routers
(C) Mutual TLS between the services
(D) A dedicated VLAN for service calls

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13. Why should MAC-based admission be combined with profiling and restricted policy?

(A) MAC-based admission cannot assign any authorisation result to the port

(B) MAC addresses are easily spoofed, so the policy must limit what a spoof gains

(C) Profiling is required by the 802.1X standard before any device may be admitted to a port

(D) Restricted policy improves the throughput available to headless devices

14. An IPsec overlay causes some large file transfers to stall while small requests work. What is the most likely design gap?

(A) MTU and TCP MSS were not adjusted for the encapsulation overhead

(B) The IPsec cipher suite is too weak for large file transfers to complete

(C) The tunnel lacks BFD, so failures are being detected far too slowly

(D) Large transfers need MACsec on the WAN instead of IPsec to complete

15. What is the main visibility trade-off of pervasive TLS?

(A) Every TLS session needs a dedicated hardware accelerator somewhere on the path

(B) TLS prevents flow records from being exported by network devices

(C) TLS forces all traffic to hairpin through the perimeter firewall

(D) Network sensors cannot inspect payloads without terminating sessions

16. A bank's inter-site traffic carries records that must stay confidential for 20 years. Which design step BEST addresses the post-quantum risk?

(A) Replace MACsec with IPsec, since IPsec uses newer symmetric ciphers

(B) Wait until large quantum computers exist before changing any design

(C) Shorten every key lifetime to one hour on all existing tunnels

(D) Adopt crypto-agile, hybrid post-quantum key exchange

☐ Lesson complete · Date _____ · Knowledge check score ____ / 6

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MODULE 6 · SECURE · LESSON 4 · 33 MIN

Compliance Zones & Security Visibility Design

Use segmentation to shrink regulatory scope, and design the network so security teams can see

- Design compliance zones that reduce the scope of payment-card, health and personal-data obligations
- Build a segmentation matrix with owners, enforcement points and evidence
- Design a layered visibility architecture for security operations

Regulation turns segmentation from good practice into a contractual obligation. Payment-card rules, health-data laws and privacy regulations all reward designs that **concentrate** sensitive data into small, well-controlled zones. Every system that can talk to a regulated zone tends to fall into scope too — so the network design directly changes audit cost, effort and risk.

> DEFINITION · SCOPE AND SCOPE REDUCTION
> Under the payment-card industry data security standard (PCI DSS, version 4.x — check the current revision), the **cardholder data environment** includes systems that store, process or transmit card data, plus systems connected to them or able to affect their security. Proper segmentation can take the rest of the network out of scope; the segmentation itself must be documented and validated by testing. Health and personal-data regimes differ in detail but reward the same pattern: know where the data is, isolate it and prove the isolation.

Compliance-driven zone design

DATA OR SYSTEM	TYPICAL OBLIGATIONS	NETWORK DESIGN RESPONSE
Payment-card data	Restricted access, strong segmentation, logging, regular testing	Small dedicated zone; tokenisation to shrink it; tested segmentation; tightly controlled admin path
Health records and clinical systems	Confidentiality, integrity, availability of care	Clinical zones separate from corporate and guest; medical devices isolated by type; availability weighed as heavily as confidentiality
Personal data under privacy law	Purpose limitation, data residency, breach notification	Region-aware placement of data; controlled cross-border paths; logging to support notification
Operational technology and safety systems	Safety and continuity of physical processes	Hierarchical zones and conduits; industrial DMZ; strictly brokered IT-to-OT flows
Government classified or sensitive data	Accredited boundaries, approved cryptography	Physically or cryptographically separated networks; cross-domain controls

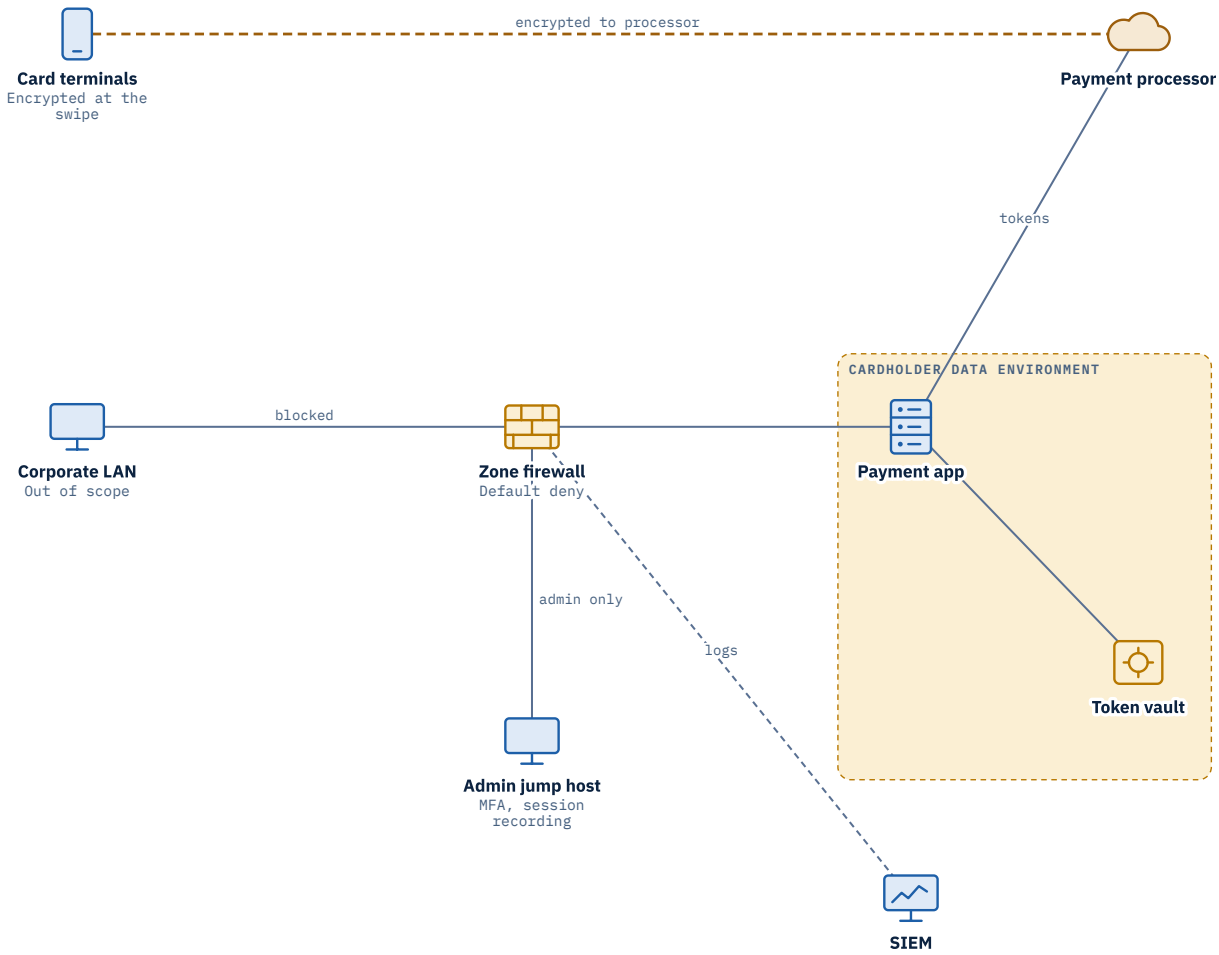
Scope-reduction method

1. **Find the data** Map where regulated data is stored, processed and transmitted, including logs, backups and support tools.
2. **Remove what you can** Tokenise, truncate or outsource processing so less of the estate ever touches the data.
3. **Isolate the rest** Place in-scope systems in a dedicated zone with default-deny policy and stateful enforcement.
4. **Control connected systems** Limit and document every system that can reach the zone — management, monitoring, backup, identity.

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5. **Prove it** Test that out-of-scope networks cannot reach the zone, and repeat the test after significant change.
6. **Keep it small** Treat any new flow into the zone as a design change with security review.

Shrinking card-data scope with encryption at the terminal and a small zone



Card data is encrypted at the terminal and travels to the processor without being readable on the store network. What remains in scope sits in a small default-deny zone, administered only through a recorded jump host and logged to the SIEM.

CASE STUDY · HOSPITAL SEGMENTATION PROJECT

A 900-bed teaching hospital runs one flat network for electronic health records, 4,000 medical devices (infusion pumps, imaging systems, patient monitors), building management, staff PCs, a research network and guest Wi-Fi. Many imaging systems run unsupported operating systems that the manufacturers will not let the hospital patch. Two hospitals in the region were hit by ransomware last year; one diverted emergency ambulances for three days. The board wants risk reduced within 12 months without any interruption to clinical care. The security team is five people.

1. Which macro zones would you define, and why?

2. How do you protect devices that cannot be patched or run agents?

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3. Where should enforcement sit, and in what order would you roll it out?

4. What visibility does the security team need, given its size?

Model answer: module answer key.

Compliance milestones that shape network design

MAR 2022
PCI DSS v4.0 published

MAR 2024
PCI DSS v3.2.1 retired

JUN 2024
PCI DSS v4.0.1 limited revision

OCT 2024
NIS2 transposition deadline
EU cyber-security duties for essential and important entities

JAN 2025
DORA applies
EU digital operational resilience for financial entities

MAR 2025
PCI DSS v4 future-dated requirements mandatory

Versions and dates move: confirm the current revision of each standard and the national transposition of each EU directive before quoting them in a design.

TEMPLATE Segmentation matrix

Record every permitted flow between zones or groups with its justification, owner, enforcement point and evidence, so policy can be reviewed and audited.

SOURCE ZONE / GROUP	DESTINATION ZONE / GROUP	ALLOWED SERVICES	ENFORCEMENT POINT	BUSINESS OWNER & JUSTIFICATION	LOGGING	REVIEW DATE
Imaging modalities	*Image archive*	*Image transfer and worklist ports only*	*Fabric edge group policy*	*Radiology — image storage*	*Flow + deny logs*	*Every 6 months*
Corporate users	*Card payment zone*	*None (default deny)*	*Data-centre zone firewall*	*Finance — scope reduction*	*Deny logs to security operations*	*Quarterly test*

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SOURCE ZONE / GROUP	DESTINATION ZONE / GROUP	ALLOWED SERVICES	ENFORCEMENT POINT	BUSINESS OWNER & JUSTIFICATION	LOGGING	REVIEW DATE

Security visibility architecture

Analytics and response	Security analytics and event management · Network detection and response · Automated response playbooks
Collection and brokering	Packet broker: aggregate, filter, deduplicate, load-balance to tools · Flow collectors · Log and telemetry pipeline
Sources	Optical taps on critical links · Port mirroring where taps are impractical · Flow export from switches, routers and cloud · Firewall, proxy, DNS and identity logs
Network	Zone boundaries and choke points · Data-centre fabric, campus, WAN, cloud

Taps, port mirroring and flow telemetry

Optical taps

- Passive copy of every packet, including errors
- No load on the switch; fails open on the production link
- Needs physical insertion and capacity planning for tools

Port mirroring

- Copy configured on the switch, no extra hardware
- Can drop copies under load; lower priority than forwarding
- Convenient for temporary or low-rate needs

Flow telemetry

- Records who talked to whom, when and how much
- Cheap to collect everywhere, including cloud flow logs
- No payload; sampling can hide short or small flows

INSIGHT · ENCRYPTION SHIFTS VISIBILITY, IT DOES NOT REMOVE THE NEED

Modern TLS with forward secrecy cannot be decrypted passively, even with the server's key. Visibility moves to points that terminate sessions (proxies, load balancers), to endpoints, to DNS and identity logs, and to analysis of encrypted-traffic metadata. Design those sources in deliberately rather than assuming a sensor in the middle will see everything.

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SCENARIO · DECIDE · THE SOC WANTS EVERYTHING

A new security operations lead asks for full packet capture on every link in the network, retained for 90 days. The network carries about 400 Gbps at peak across three data centres and 200 sites.

(A) Approve full capture everywhere; visibility is never wasted.

(B) Decline capture entirely and rely on firewall logs only.

(C) Mirror all traffic from every switch to a central analysis cluster over the production network.

(D) Design tiered visibility: flow telemetry and logs everywhere, packet capture at key choke points through a packet broker with short rolling retention, and on-demand capture elsewhere.

Circle your choice. Answer and feedback: module answer key.

Compliance and visibility checklist

☐ Regulated data locations are mapped, including logs, backups and support tools

☐ Scope has been reduced by tokenisation or outsourcing where possible

☐ Every flow into a regulated zone appears in the segmentation matrix with an owner

☐ Segmentation is tested and re-tested after significant change

☐ Flow telemetry covers every zone boundary, on premises and in cloud

☐ Packet capture is concentrated at choke points through a broker

☐ Visibility for encrypted traffic is designed at termination points and endpoints

REFLECT

Which system in your network is in regulatory scope only because it can reach a regulated zone? What would it take to remove that path?

Key terms

Cardholder data environment
The systems that store, process or transmit payment-card data, plus those connected to or able to affect them.

Tokenisation
Replacing sensitive data with a surrogate value that has no exploitable meaning outside a secure vault.

Packet broker
A device that aggregates, filters and distributes copies of traffic to security and monitoring tools.

Port mirroring
A switch feature that copies traffic from ports or VLANs to a monitoring port.

Scope reduction
Design measures that reduce the number of systems subject to a regulatory obligation.

Segmentation testing
Testing that proves out-of-scope networks cannot reach a protected zone.

Optical tap
A passive device that copies light from a fibre link for monitoring without affecting the link.

Network detection and response
Tools that analyse network traffic and metadata to detect and investigate threats.

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Forward secrecy
A property of key exchange that prevents past sessions from being decrypted even if long-term keys are later exposed.

Key takeaways

- Concentrate regulated data into small, well-controlled zones; everything that can reach a zone tends to join its scope
- Remove data first (tokenisation, encryption at the terminal), then isolate and test what remains
- Record every permitted flow with an owner, enforcement point and evidence
- Design visibility in tiers: flows and logs everywhere, packet capture at choke points
- Encryption moves visibility to termination points, endpoints, DNS and identity logs

KNOWLEDGE CHECK

17. A retailer wants to shrink the number of systems subject to payment-card audits. Which design step has the largest effect?

(A) Encrypt every link in the corporate network using MACsec on all of the uplinks
(B) Increase the logging retention on all servers so it covers at least one full audit cycle
(C) Tokenise card data and isolate the remaining in-scope systems in a tested zone
(D) Move every corporate system behind one larger and faster perimeter firewall

18. Why must segmentation used for scope reduction be tested?

(A) Auditors rely on proof that out-of-scope networks cannot reach the zone
(B) Testing is needed to calculate the bandwidth required between zones
(C) Untested segmentation automatically places the whole estate in scope for every audit
(D) Firewalls require periodic testing to keep their warranty valid

19. Which visibility source scales most cheaply across hundreds of sites and cloud networks?

(A) Optical taps on every access link
(B) Flow telemetry and cloud flow logs
(C) Full packet capture at each site
(D) Port mirroring from every switch

20. Modern TLS with forward secrecy is pervasive. Where should visibility be designed?

(A) At a passive sensor that holds a copy of each server's private key for decryption
(B) At the internet edge only, since internal traffic is not a source of threats
(C) Nowhere in particular, since encrypted traffic no longer needs to be monitored
(D) At session termination points, endpoints, DNS and identity logs, and metadata

21. In the hospital case, why were clinical device groups enforced after guest, research and building-management boundaries?

(A) Lower-risk boundaries build confidence and flow data before patient-critical systems
(B) Clinical devices do not need segmentation because they are already certified
(C) Guest and research networks carry more sensitive data than any of the clinical zones do
(D) Enforcement on clinical devices is not allowed by health-data regulation

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 6 · SECURE

Answer key

Knowledge checks

1. D Zero trust removes implicit trust from location, but enforcement points and telemetry in the network remain essential parts of the architecture.

2. A Macro zones contain movement between zones, but hosts inside the same zone can still reach each other unless micro-segmentation is added.

3. C Enforcing without flow discovery breaks legitimate traffic and undermines the programme. Observation and monitor mode make enforcement safe.

4. C The policy engine decides and the administrator instructs, but the enforcement point in the data path enables or terminates the connection.

5. A When the endpoint cannot be changed, enforcement must sit in the network around it.

6. B Stateful devices must see both directions. Symmetric routing solves the root cause; disabling inspection removes the control and the other options do not address asymmetry.

7. D Two independently managed layers mean one mistake does not expose internal zones, and brokering removes direct inbound paths.

8. B Distributed enforcement grows with the workloads and avoids hairpinning all east-west traffic through a central bottleneck.

9. C The problem is scope, not encryption. Application-level, brokered access limits each party to named resources.

10. A Edge filters remove obviously illegitimate packets cheaply; they complement, not replace, stateful inspection.

11. D Monitor mode reveals what is on the network and what would fail, with no clinical impact, before any enforcement is introduced.

12. C Mutual TLS travels with the session end to end and carries service identity; link and network encryption protect only the segments they cover.

13. B MAC-based access identifies but does not authenticate. Limiting access contains the damage from spoofing.

14. A Encapsulation reduces effective MTU. Without MSS clamping or working path-MTU discovery, large packets are dropped while small ones pass.

15. D Encrypted payloads are opaque to passive sensors. Visibility moves to proxies that terminate TLS, endpoint telemetry and metadata analysis.

16. D Recorded traffic could be decrypted later if the key exchange is broken, so long-lived data needs post-quantum protection of key establishment now. Symmetric bulk ciphers are not the weak point, and shorter key lifetimes do not change the vulnerable algorithm.

17. C Removing card data from most systems and isolating the rest takes the bulk of the estate out of scope. Link encryption and logging do not change what is in scope.

18. A Scope reduction depends on isolation actually working. Testing provides that evidence and catches drift after changes.

19. B Flow records are compact and widely supported, including in cloud. Packet-level sources are valuable but expensive to deploy everywhere.

20. D Forward secrecy prevents passive decryption with server keys. Visibility must come from points that see plaintext or from metadata and logs.

21. A Sequencing by impact protects care delivery while the team gains experience and refines the policy matrix.

Scenarios & case studies

> **Scenario · A flat manufacturing plant**
> **B** is best. Best. It delivers the largest risk reduction first, needs nothing on fragile controllers, and creates a foundation for finer segmentation later.

> **Scenario · Third-party access to plant systems**
> **C** is best. Best. Each vendor reaches only its own equipment, every session is attributed and recorded, and nothing on the plant network is exposed directly.

> **Scenario · Encrypting a data-centre interconnect**
> **A** is best. Best. Line-rate hardware encryption on point-to-point links meets the requirement with negligible latency and no new bottleneck.

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Case study · Hospital segmentation project

Macro zones: clinical systems (health records and clinical applications), medical devices (sub-grouped by type and manufacturer), imaging, building management and OT, corporate users, research (separate, with its own data-sharing controls), guest (internet only), and a management zone. **Unpatchable devices:** network-based containment — group them by type, allow only the flows each type needs (for example modality-to-archive and modality-to-worklist), block internet access, and put manufacturer remote support through a brokered jump service with session recording. **Enforcement:** inter-zone firewalls in the data centre for macro boundaries; group-based policy at the access and wireless edge for device groups. Roll out in **monitor mode** first, using several weeks of flow data to build the matrix; then enforce the lowest-risk boundaries (guest, research, building management) before clinical device groups, with clinical engineering signing off each step and an immediate rollback path. **Visibility:** with five people, favour high-signal sources — flow telemetry from all zone boundaries, firewall deny logs, DNS logs and passive device-profiling — feeding one analytics platform with tuned alerts, rather than full packet capture everywhere. Measure success as 'a compromised staff PC can reach no medical device and only the clinical services it needs'.

Scenario · The SOC wants everything

D is best. Best. It gives broad coverage cheaply, deep evidence where it matters most, and a mechanism to extend capture during investigations.

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07

MODULE 7 · CLOUD

Hybrid, Multi-Cloud & Automation-Driven Design

Cloud constructs, hybrid connectivity, addressing and DNS, automation, observability and cost

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Map cloud networking constructs to design decisions and choose transit topologies
- Select hybrid and multi-cloud connectivity patterns against availability, bandwidth and cost
- Build addressing and DNS designs that work across on-premises and several clouds
- Design networks for automation and observability from day 0
- Forecast and control the cost drivers of hybrid network designs

LESSONS

1. Cloud Constructs & Hybrid Connectivity · 33 min
2. Multi-Cloud, IP Addressing & Hybrid DNS · 33 min
3. Infrastructure as Code & Designing for Automation · 31 min
4. Observability & Cost-Aware Network Design · 31 min

MODULE 7 · CLOUD · LESSON 1 · 33 MIN

Cloud Constructs & Hybrid Connectivity

Translate familiar network design into cloud building blocks, then join them to the estate

- Map cloud networking constructs to their on-premises equivalents and design implications
- Choose between peering meshes, transit hubs and shared virtual networks
- Select hybrid connectivity patterns that meet availability, bandwidth and cost requirements

Public clouds expose networking as software constructs: virtual networks, route tables, gateways and security groups that appear in seconds through an API. The physics underneath has not changed, but the rules have. Some familiar behaviours disappear (there is usually no broadcast, no spanning tree, no multicast by default), some limits are hard quotas set by the provider, and every byte that crosses certain boundaries has a price.

Cloud constructs and what they mean for design

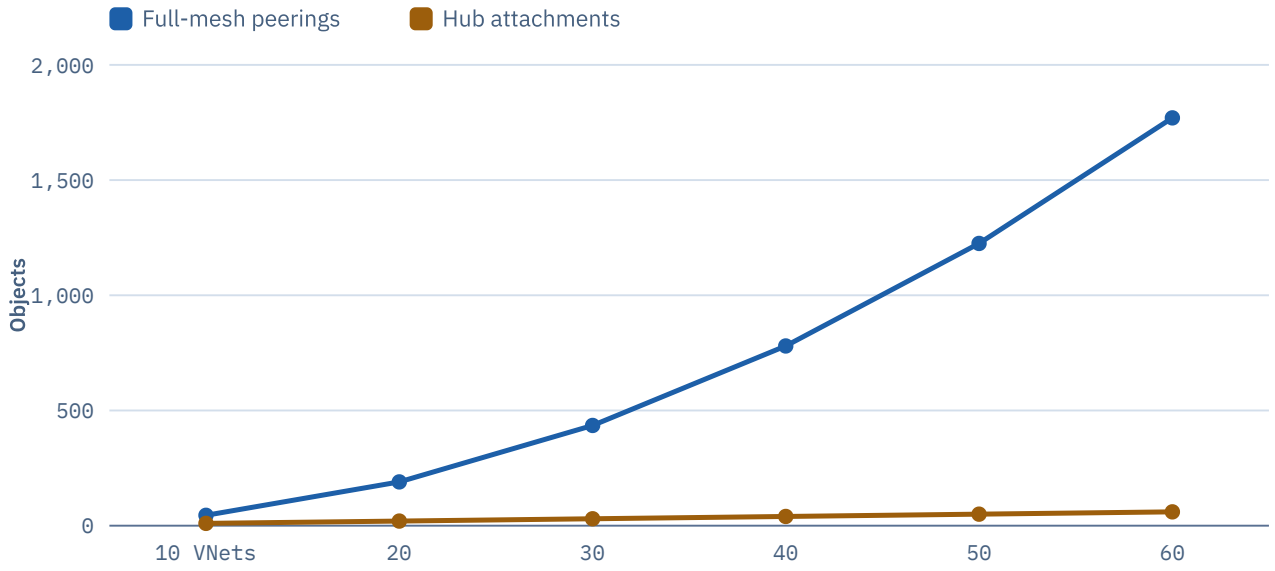
CONSTRUCT (GENERIC)	ON-PREMISES ANALOGUE	DESIGN NOTE
Virtual network	A routing domain or VRF with its own address space	Address range is hard to change later; plan size and non-overlap from day one
Subnet tied to an availability zone	A VLAN in one building or hall	Zone placement is a resilience decision — spread tiers across zones
Route table	Static routing per subnet	Controls whether traffic goes to a gateway, firewall appliance or transit hub
Security group / network ACL	Stateful host firewall / stateless access list	Primary micro-segmentation tool; manage as code to avoid sprawl
Peering	A direct routed link between two VRFs	Usually non-transitive: A–B and B–C does not give A–C
Transit hub (provider-managed router)	A core or WAN hub router	Gives transitive routing between many virtual networks and on-premises; often billed per attachment and per GB
Private endpoint for a provider service	A service published on an internal address	Keeps traffic to platform services off the internet; needs DNS design
Private dedicated connectivity	A leased line into the provider's edge	Predictable latency and bandwidth; lead time, port fees and colocation presence

Why meshes do not scale

MEASURE	FORMULA	WHAT IT TELLS YOU
Full-mesh peerings	`Peerings = n × (n − 1) ÷ 2`	Ten virtual networks need 45 peerings; fifty need 1,225 — each with routes and security rules to maintain.
Hub-and-spoke attachments	`Attachments = n`	Fifty virtual networks need fifty attachments to a transit hub, at the cost of a hub hop and per-GB processing charges.

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Objects to manage: full-mesh peering versus a transit hub



Computed from n(n – 1) ÷ 2 peerings against n hub attachments. Beyond a handful of virtual networks the mesh becomes unmanageable, which is why most estates choose a hub and keep direct peering for a few high-volume pairs.

Inter-network topology options

Full-mesh peering

- Lowest latency and no hub processing charges
- Non-transitive: every pair needs its own peering
- Becomes unmanageable beyond a handful of networks
- Fits small estates or high-volume pairs

Transit hub (hub-and-spoke)

- Transitive routing with central route tables
- Natural place for shared inspection and hybrid gateways
- Hub is a shared failure domain and a cost centre
- Fits most medium and large estates

Shared virtual network

- A central team owns one network shared into many accounts or projects
- Application teams consume subnets but do not manage routing
- Strong central control; blast radius of the shared network is large
- Fits organisations with a strong platform team

Private dedicated connectivity versus internet VPN

Private dedicated connectivity

- Predictable latency and throughput, from partner-provided sub-gigabit links up to 100 Gbps dedicated ports
- Lower per-GB transfer charges at many providers
- Weeks of lead time; needs presence at a colocation or exchange point
- Not encrypted by default — add MACsec or IPsec where required

Internet IPsec VPN

- Available in minutes, low fixed cost
- Throughput limited per tunnel; scale with multiple tunnels and multipath routing
- Latency and loss follow the internet path
- Encrypted by design; good as backup or for small sites

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INSIGHT · CLOUD-TO-CLOUD INTERCONNECTS ARE NOW A PRODUCT

Besides building your own routers in an exchange facility, several providers now sell managed private interconnects between their own clouds and those of other providers, and colocation operators offer virtual routers on their exchange fabrics. They remove hardware and lead time but add another provider's availability and pricing model to the design; compare them on the same diversity, capacity and cost criteria as any other path.

Hybrid connectivity patterns — decision matrix

PATTERN	AVAILABILITY	BANDWIDTH & LATENCY	COST	FITS
Single VPN	Low	Variable	Lowest	Proof of concept, non-critical workloads
Dual VPN from two on-premises sites	Medium	Variable	Low	Small production estates, backup path
One private connection with VPN backup	Medium to high, if the VPN can carry critical load	Predictable until failover	Medium	Moderate traffic where the VPN can absorb critical flows
Two private connections at one location	Medium — shared facility risk	Predictable	Medium to high	Bandwidth growth, not resilience
Private connections at two diverse locations	High	Predictable, survives a facility loss	High	Production estates with strict availability targets
Overlay extended into cloud (virtual appliances in the hub)	Depends on underlay	Uniform policy with branches	Appliance licences and compute	Large branch estates needing consistent segmentation

A typical hybrid path

01 On-premises data centre — Core routing and firewall

→

02 Colocation edge — Customer router at an exchange point

→

03 Private connection — Dedicated port into the provider edge

→

04 Provider gateway — BGP session terminates here

→

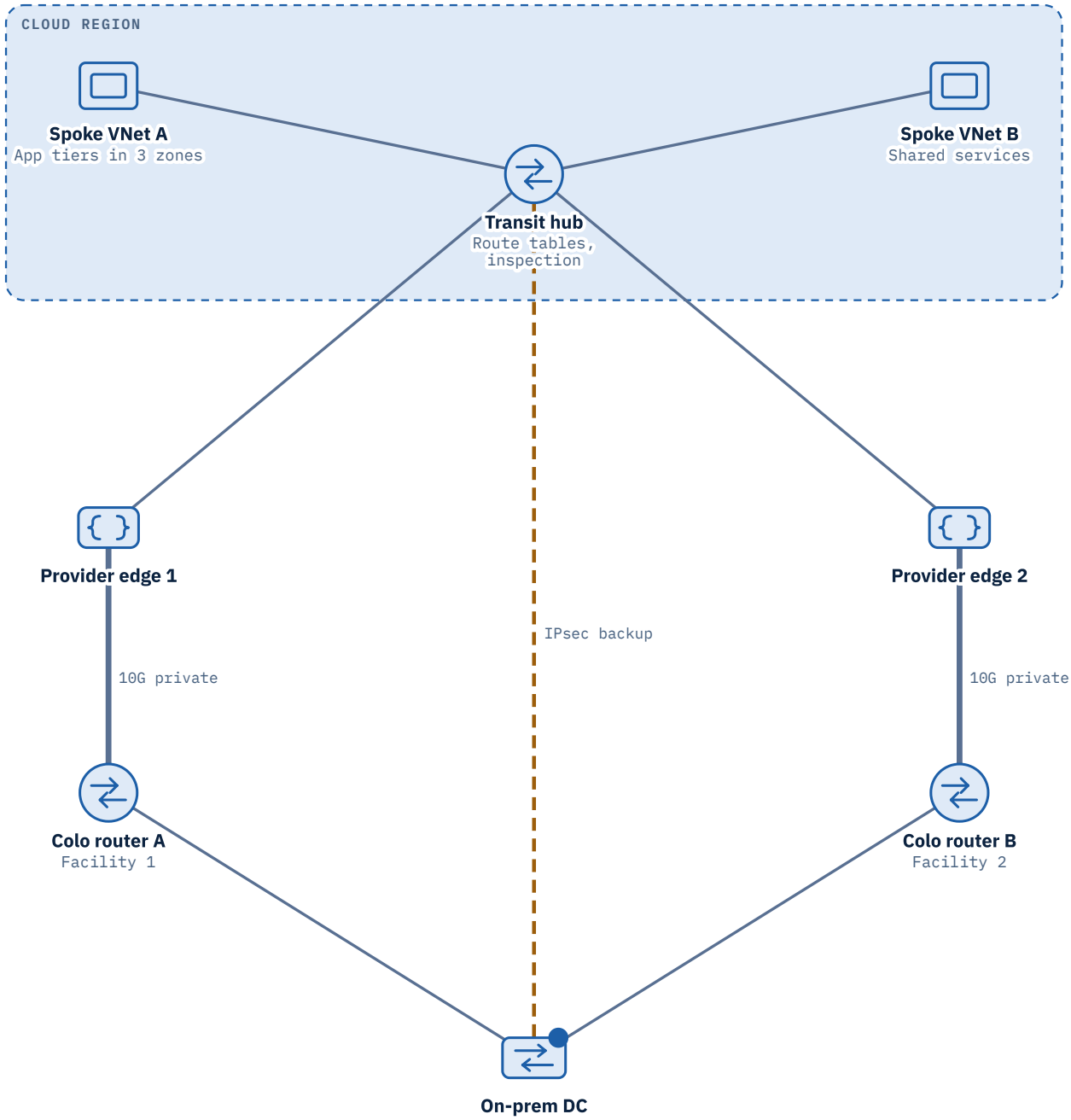
05 Transit hub — Routes to spoke networks, optional inspection

→

06 Spoke network — Workload subnets across zones

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Hybrid connectivity with diverse private connections and a VPN backup



Two private connections in different colocation facilities terminate on separate provider edges, so no single facility, device or port takes the path down. An IPsec tunnel from the data centre carries critical flows if both fail. The transit hub gives every spoke a route to on-premises.

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Preferring the private path with VPN as backup

COLO-R1 · IOS-style CLI

```
router bgp 64496
 neighbor 169.254.10.1 remote-as 64512
 neighbor 169.254.10.1 description Private connection
 neighbor 169.254.20.1 remote-as 64512
 neighbor 169.254.20.1 description IPsec backup tunnel
 address-family ipv4
  neighbor 169.254.10.1 activate
  neighbor 169.254.10.1 route-map PRIVATE-IN in
  neighbor 169.254.20.1 activate
  neighbor 169.254.20.1 route-map VPN-OUT out
!
route-map PRIVATE-IN permit 10
 set local-preference 200
route-map VPN-OUT permit 10
 set as-path prepend 64496 64496 64496
```

Local preference steers outbound traffic to the private path; prepending makes the VPN less attractive for return traffic. Check the provider's documented route-selection rules — some always prefer private connectivity over VPN for the same prefix regardless of path length, which may make prepending unnecessary.

SCENARIO · DECIDE · CONNECTING A RETAILER'S CLOUD PLATFORM

A retailer moves its order platform to a public cloud region. Steady traffic between stores, the warehouse system on premises and the cloud is about 3 Gbps, with peaks of 6 Gbps at seasonal sales. The availability target for the order path is 99.95%, and a single colocation facility hosts the retailer's current exchange presence. The finance team asks for the cheapest design that meets the target.

- (A) A pair of internet VPN tunnels from the head office.
- (B) Private connections at two diverse exchange locations, with a VPN as a tertiary path for critical flows only.
- (C) Two private connections in the existing colocation facility, on separate devices.
- (D) One private connection with an internet VPN backup sized for the full 6 Gbps.

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Cloud connectivity decision sheet**

Compare hybrid connectivity options against the workload's requirements so the choice and its trade-offs are explicit and reviewable.

WORKLOAD / TRAFFIC CLASS	BANDWIDTH (STEADY / PEAK)	LATENCY & AVAILABILITY TARGET	OPTION CHOSEN	DIVERSITY (LOCATIONS, CARRIERS)	ENCRYPTION	MONTHLY COST & RISK ACCEPTED
Order platform ↔ warehouse	*3 / 6 Gbps*	*< 15 ms, 99.95%*	*2 × 10G private at diverse locations + VPN tertiary*	*Two exchange points, two carriers*	*MACsec on private links*	*Port + carrier fees; VPN carries critical flows only*

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WORKLOAD / TRAFFIC CLASS	BANDWIDTH (STEADY / PEAK)	LATENCY & AVAILABILITY TARGET	OPTION CHOSEN	DIVERSITY (LOCATIONS, CARRIERS)	ENCRYPTION	MONTHLY COST & RISK ACCEPTED

Hybrid connectivity checklist

☐ Virtual network address ranges are allocated from the enterprise plan and never overlap

☐ Topology choice (mesh, hub, shared network) is justified by scale and ownership model

☐ Private connectivity is diverse by facility, not just by device

☐ Backup paths are sized for the traffic they must actually carry

☐ Encryption on private links is decided explicitly, not assumed

☐ Route preference between private and VPN paths matches the provider's selection rules

☐ Provider quotas (routes, attachments, tunnels) are checked against growth plans

REFLECT

Which of your cloud networks would be hardest to re-address if it turned out to overlap with a partner or an acquired company? What would you do differently today?

Key terms

Virtual network
A provider-hosted, isolated network with its own address space, subnets and route tables.

Transit hub
A provider-managed routing service that connects many virtual networks and hybrid links transitively.

Private endpoint
An interface with a private address that exposes a provider or partner service inside a virtual network.

Exchange point
A colocation facility where enterprises, carriers and cloud providers interconnect.

Availability zone
A physically separate location within a cloud region with independent power, cooling and networking.

Non-transitive peering
A peering model in which routes learned over one peering are not passed on to another.

Private dedicated connectivity
A dedicated physical or partner-provided connection between an enterprise network and a cloud provider's edge.

Security group
A stateful, cloud-native firewall applied to workloads or interfaces.

Key takeaways

- Map cloud constructs to familiar ones, but respect provider quotas, non-transitive peering and per-GB pricing
- Choose a transit hub for most estates; keep direct peering for a few high-volume pairs
- Diversity is by facility, not just device: two private connections in one facility share fate
- A backup path is only a backup for the traffic it can actually carry

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KNOWLEDGE CHECK

1. An organisation has 40 virtual networks that all need to reach each other and on-premises systems. Why is a transit hub usually preferred to full-mesh peering?

   (A) Peering cannot carry traffic between virtual networks in different availability zones

   (B) A transit hub removes all per-GB charges for traffic between the virtual networks

   (C) Full-mesh peering requires a separate internet VPN for every pair of the networks

   (D) A mesh needs 780 non-transitive peerings to maintain; a hub needs 40 attachments

2. Why does placing two private connections in the same colocation facility not deliver high availability on its own?

   (A) Providers do not allow two connections in a single facility to be active at the same time

   (B) The facility is a shared failure domain for power, fibre entry and operations

   (C) Private connections cannot be combined with BGP multipath load sharing

   (D) Two connections in one facility always share the same provider port

3. A design uses a private connection with an internet VPN as backup. What must be validated?

   (A) That the VPN uses the same autonomous system number as the provider's private gateway

   (B) That the private connection is encrypted with the same VPN cipher

   (C) That the VPN can actually carry the critical traffic during a failover

   (D) That the VPN terminates in the same availability zone as the workloads

4. Which statement about private dedicated connectivity is correct?

   (A) It always includes provider-managed encryption at no extra cost

   (B) It is not encrypted by default; add MACsec or IPsec if required

   (C) It carries traffic over the public internet with a reserved path

   (D) It removes the need for BGP between the enterprise and provider

5. What is the main trade-off of a centrally owned shared virtual network?

   (A) Application teams must manage all of their own routing and gateways independently

   (B) Every application must be deployed into the same availability zone

   (C) Shared networks cannot connect to on-premises sites through private links

   (D) Strong central control, but a large blast radius for mistakes in that network

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 7 · CLOUD · LESSON 2 · 33 MIN

Multi-Cloud, IP Addressing & Hybrid DNS

Choose multi-cloud for the right reasons, and get addresses and names right before they become migrations

- Evaluate when multi-cloud is justified and which interconnection pattern fits
- Build an IP addressing plan that scales across on-premises and several clouds
- Design hybrid DNS so private names and private endpoints resolve correctly everywhere
- Decide where IPv6-mostly or IPv6-only designs relieve IPv4 pressure

Multi-cloud is often presented as a resilience strategy. In practice it doubles the skills, tooling, security policy and connectivity an organisation must master. Expert designers ask what business driver justifies it — regulation, a capability only one provider offers, an acquisition, negotiating leverage — and then design the minimum interconnection that serves that driver. Addressing and DNS are where poor multi-cloud designs fail first.

Multi-cloud: reasons and costs

Valid drivers

- A regulatory or sovereignty requirement one provider cannot meet
- A unique service or capability on a second platform
- An acquisition that arrives with its own cloud estate
- A deliberate exit or concentration-risk strategy

Real costs

- Two sets of constructs, quotas, security models and skills
- Inter-cloud traffic usually pays egress on the sending side
- Consistent policy and observability across platforms is hard
- Failover between clouds is rarely tested and rarely works first time

INSIGHT · MULTI-REGION BEFORE MULTI-CLOUD

For availability alone, multiple regions or zones within one provider usually deliver more resilience per unit of effort than spreading one application across two providers. Use multi-cloud where the business driver is something other than availability, and be honest about the operational cost.

Inter-cloud connectivity patterns

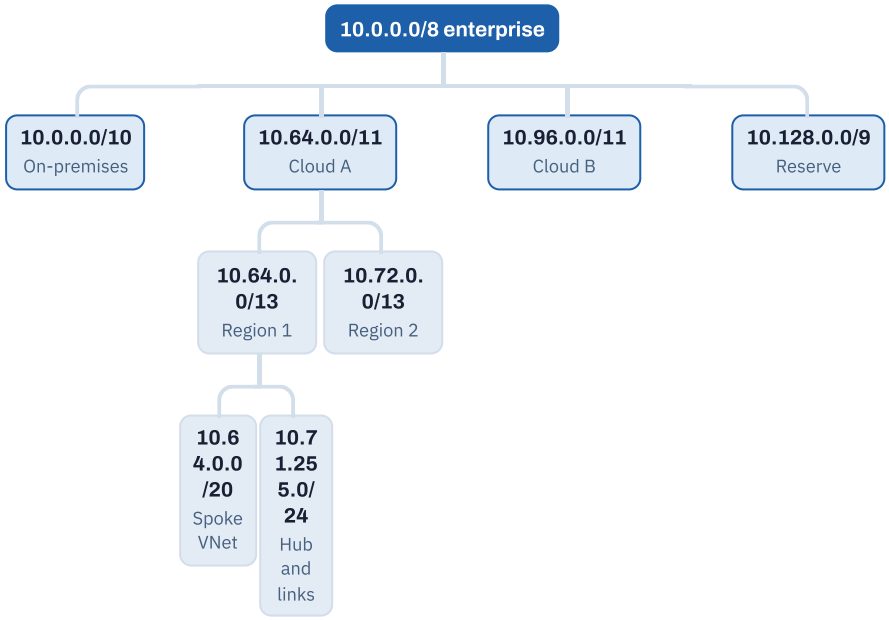
PATTERN	STRENGTHS	WEAKNESSES	FITS
Hairpin through on-premises data centre	Reuses existing private links and security stack	Latency and double egress; on-premises becomes a dependency	Low-volume inter-cloud traffic during transition
Interconnect at a cloud exchange point	Short, predictable paths; customer router or exchange fabric in the middle	Colocation presence and port fees	Steady, significant inter-cloud traffic
Direct IPsec VPN between clouds	Fast to build, encrypted	Per-tunnel throughput limits; internet variability	Small volumes and backup paths
Multi-cloud networking overlay platform	Uniform policy, routing and visibility across clouds	Another platform and licence; dependency on its controller	Large estates with several clouds and strong central teams

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IP addressing plan for hybrid and multi-cloud

1. **Reserve aggregates** Allocate one large, summarisable block per cloud and per region from the enterprise plan, leaving room for at least several years of growth.
2. **Never overlap** Keep cloud ranges unique across on-premises, other clouds and known partners; overlap forces NAT and breaks private connectivity.
3. **Size for zones** Divide each virtual network into per-zone subnets for each tier, so workloads can spread across zones without re-addressing.
4. **Account for platform consumption** Providers reserve a few addresses in every subnet, and container platforms can consume an address per pod — plan for both.
5. **Plan transit and link ranges** Set aside small ranges for hub attachments, tunnel interfaces and gateways.
6. **Plan IPv6** Use provider-assigned or enterprise IPv6 prefixes where supported to relieve IPv4 pressure, especially for large container estates.
7. **Record in IPAM** The address plan lives in an IPAM system that automation reads from; nobody allocates by hand.

A summarisable hybrid address plan



Each level summarises the one below: one /11 per cloud, one /13 per region, a /20 per virtual network and per-zone subnets inside it. Ranges are illustrative; the point is that nothing overlaps and every boundary aggregates.

Addressing arithmetic

MEASURE	FORMULA	WHAT IT TELLS YOU
Usable addresses in a cloud subnet	`Usable = 2^(32 − prefix length) − provider-reserved addresses`	Many providers reserve four or five addresses per subnet, so a /24 yields about 251–252 usable addresses, not 254.
Container address demand (one address per pod)	`Addresses needed ≈ nodes × maximum pods per node × growth factor`	200 nodes × 110 pods × 1.5 ≈ 33,000 addresses — more than a /17, which is why pod overlays, IPv6 or non-routed ranges behind NAT are common.

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WATCH OUT · OVERLAP IS A DEBT THAT COMPOUNDS

Every overlapping range eventually becomes NAT rules, special-case DNS and broken troubleshooting. When overlap is unavoidable — a merger, a partner with the same ranges — contain it at a defined boundary with explicit translation, and plan a re-addressing project rather than letting translation spread.

PRO TIP · IPV6-MOSTLY IS NOW A PRACTICAL DESIGN OPTION

IPv4 scarcity has a price: several large providers now charge for every public IPv4 address, and container estates exhaust private ranges quickly. An **IPv6-mostly** network gives IPv6-capable clients only IPv6 — DHCPv4 option 108 (RFC 8925) tells them they may skip IPv4 — while NAT64 and DNS64, plus 464XLAT on clients, keep IPv4-only destinations reachable; legacy devices still get dual stack. In cloud, IPv6-only subnets and dual-stack load balancers remove most private IPv4 demand. Plan IPv6 prefixes per region in the same hierarchy as IPv4, and test applications with hard-coded IPv4 literals before switching a segment.

Hybrid DNS building blocks

Private zones

- Name space visible only to associated virtual networks
- Holds internal names and private-endpoint records
- Association scope must be designed as carefully as routing

Resolver endpoints and forwarding

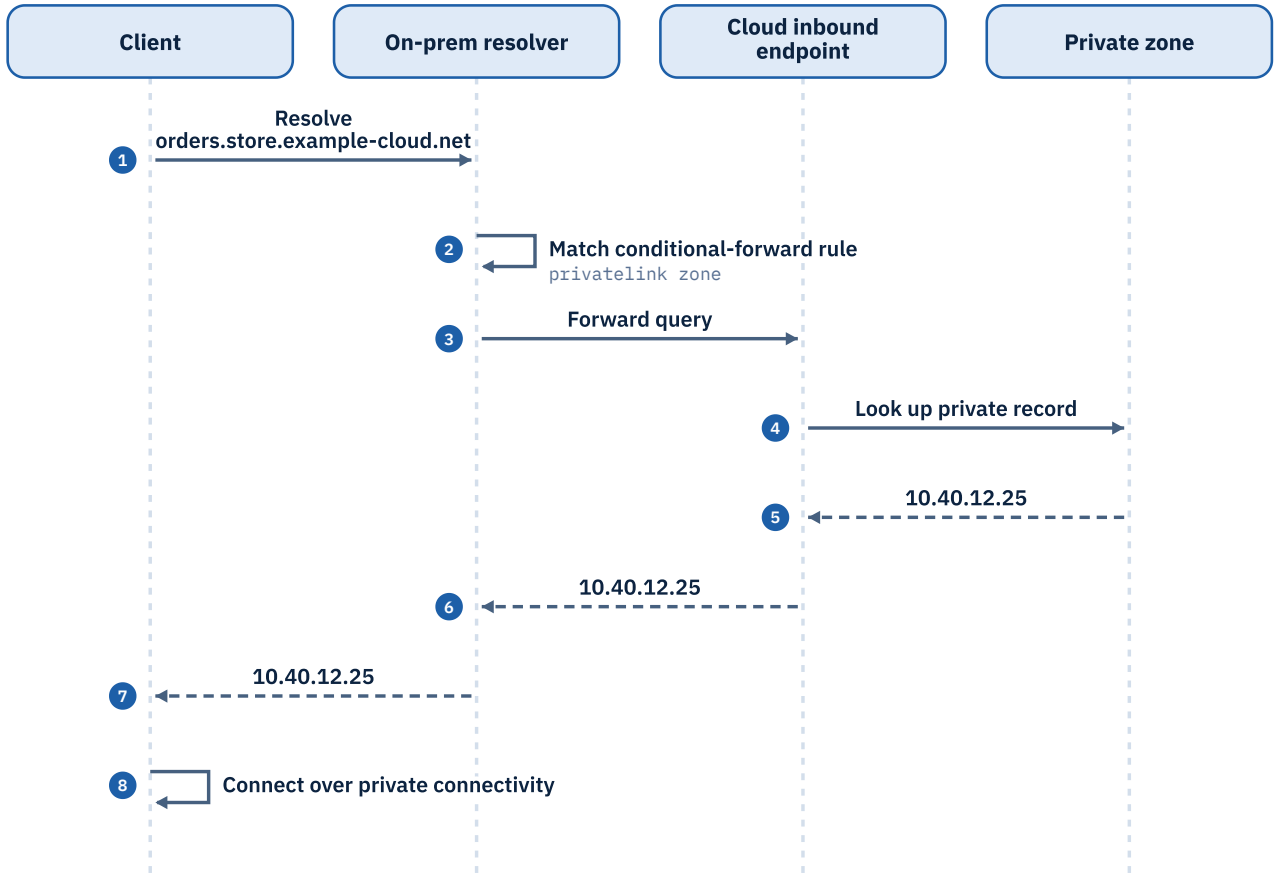
- Inbound endpoints let on-premises resolvers query cloud private zones
- Outbound rules forward cloud queries for on-premises domains to on-premises resolvers
- Conditional forwarding by domain, not 'forward everything'

Split-horizon

- The same name returns different answers inside and outside
- Lets internal clients reach private addresses for public service names
- Must be documented, or troubleshooting becomes guesswork

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On-premises client resolving a cloud private endpoint



Only the private-endpoint zone is forwarded, so every other name still resolves as before. The service name never changes; the resolver path decides whether the client gets the private address and stays off the internet.

Checking private-endpoint resolution from on premises

```
Linux host

# Before the conditional forwarder exists: public answer
$ dig +short orders.store.example-cloud.net
orders.privatelink.store.example-cloud.net.
198.51.100.44
# After forwarding the privatelink zone to the cloud inbound endpoint
$ dig +short orders.store.example-cloud.net
orders.privatelink.store.example-cloud.net.
10.40.12.25
```

The service name is the same in both cases; only the resolver path changes. Without the forwarder, on-premises clients either reach the public address or are refused if public access to the service has been disabled.

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SCENARIO · DECIDE · THE PRIVATE ENDPOINT THAT ONLY WORKS IN THE CLOUD

An application team moves a storage service behind a private endpoint and disables public access. Cloud workloads work fine. On-premises batch jobs now fail with 'access denied'. The on-premises resolvers forward all external names to the internet.

(A) Re-enable public access on the storage service for on-premises address ranges.

(B) Add hosts-file entries with the private address on each batch server.

(C) Add a conditional forwarder on the on-premises resolvers for the private-endpoint zone, pointing to the cloud inbound resolver endpoints.

(D) Move the batch jobs into the cloud so they use cloud resolvers.

Circle your choice. Answer and feedback: module answer key.

Multi-cloud, addressing and DNS checklist

☐ Each additional cloud has a written business driver beyond 'resilience'

☐ Inter-cloud traffic volumes and egress costs are estimated before the pattern is chosen

☐ Cloud address aggregates are reserved in IPAM and do not overlap anything known

☐ Container and platform address consumption is included in sizing

☐ Conditional forwarding exists in both directions between on-premises and cloud

☐ Private-endpoint zones resolve correctly from every location that needs them

☐ Split-horizon names are documented with both answers

REFLECT

Which of your cloud services would still resolve to a public address from on premises today? What would happen if public access to it were switched off tomorrow?

Key terms

Multi-cloud
Using two or more public cloud providers for production workloads.

IPAM
IP address management: the system of record for address allocations, ideally consumed by automation.

Conditional forwarding
Forwarding DNS queries for specific domains to designated resolvers.

Split-horizon DNS
Returning different answers for the same name depending on where the query comes from.

Concentration risk
The exposure created when a critical capability depends on a single provider.

Private zone
A DNS zone whose records are visible only to associated private networks.

Inbound resolver endpoint
An address inside a cloud network that accepts DNS queries from outside, typically from on-premises resolvers.

Egress charge
A fee charged by a cloud provider for data leaving its network or crossing certain boundaries.

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Key takeaways

- Multi-cloud needs a business driver beyond availability; multi-region in one provider is usually the stronger resilience answer
- Reserve summarisable, non-overlapping aggregates per cloud and region in IPAM before building
- Overlap becomes NAT, special-case DNS and broken troubleshooting — contain it and plan re-addressing
- Design hybrid DNS with conditional forwarding in both directions for private zones and endpoints
- IPv6-mostly and IPv6-only cloud subnets relieve IPv4 pressure; plan IPv6 in the same hierarchy

KNOWLEDGE CHECK

6. A company wants a second cloud provider purely to improve the availability of one application. What should the designer point out first?

   (A) Multiple regions in one provider usually give more resilience per unit of effort
   (B) Multi-cloud always halves egress charges by spreading traffic evenly across providers
   (C) A second provider removes the need for any on-premises disaster recovery site
   (D) Applications automatically fail over between providers if DNS is shared

7. A new virtual network was created with a range that overlaps an acquired company's data centre. What is the long-term cost?

   (A) No cost, because cloud virtual networks are isolated from all external routing
   (B) Higher egress charges, because overlapping ranges are billed at a premium rate
   (C) NAT, special-case DNS and harder troubleshooting at every point the ranges meet
   (D) Loss of availability-zone redundancy for every workload in the overlapping range

8. A cloud subnet is a /24. Roughly how many addresses can workloads use?

   (A) Exactly 256, as cloud subnets reserve no addresses
   (B) Exactly 254, as on a physical LAN
   (C) About 128, as half go to the gateway
   (D) About 251–252, after provider reservations

9. On-premises clients resolve a cloud private endpoint to its public address. What is the correct design fix?

   (A) Conditional forwarding of that zone to the cloud inbound resolver endpoint
   (B) Forwarding every on-premises DNS query to the public cloud resolvers
   (C) Creating hosts-file entries on every on-premises client that needs it
   (D) Assigning a public address to the private endpoint for use by on-premises clients

10. When does interconnecting clouds at an exchange point make most sense?

    (A) When inter-cloud traffic is occasional and small
    (B) When no colocation presence is available at all
    (C) When inter-cloud traffic is steady and significant
    (D) When the clouds share one virtual network range

11. A campus wants to stop handing IPv4 addresses to modern laptops and phones while legacy printers still need IPv4. Which design BEST fits?

    (A) Turn off IPv4 for every device on the campus in one change window
    (B) IPv6-mostly: option 108, NAT64/DNS64, legacy kept dual stack
    (C) Keep dual stack everywhere and enlarge every IPv4 subnet to a /16
    (D) Put all clients behind a second layer of IPv4 NAT inside the campus

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MODULE 7 · CLOUD · LESSON 3 · 31 MIN

Infrastructure as Code & Designing for Automation

Make automation a design requirement from day 0, not a project for year three

- Describe a layered automation architecture from intent to assurance
- Compare imperative scripting, declarative infrastructure as code and intent-based approaches
- Apply day-0 design principles that make a network automatable and safe to change

A network designed for manual operation can be automated later only with great effort: every site is slightly different, the real configuration lives on devices, and nobody can say what 'correct' looks like. A network designed for automation from day 0 has standard building blocks, a single source of truth and a pipeline that turns intent into configuration and verifies the result. The design work comes first; tooling follows.

Automation architecture (top first)

Intent and service catalogue	`Business requests: new site, new application segment, new cloud spoke` `Expressed as parameters, not device commands`
Source of truth	`IPAM and inventory` `Design data: site types, templates, policy matrix` `Version-controlled and reviewed`
Pipeline	`Version control and peer review` `Validation, simulation and plan/diff` `Approval gates and staged rollout`
Execution	`Infrastructure-as-code tools and device or cloud APIs` `Template rendering, idempotent apply`
Infrastructure	`Physical and virtual network devices` `Cloud networking constructs`
Assurance	`Telemetry and state validation` `Drift detection and reconciliation` `Feedback into intent and design`

Approaches to automation

Imperative scripts

- Step-by-step commands: do this, then this
- Quick to start, familiar to engineers
- Fragile: the outcome depends on the starting state; hard to make safe at scale

Declarative infrastructure as code

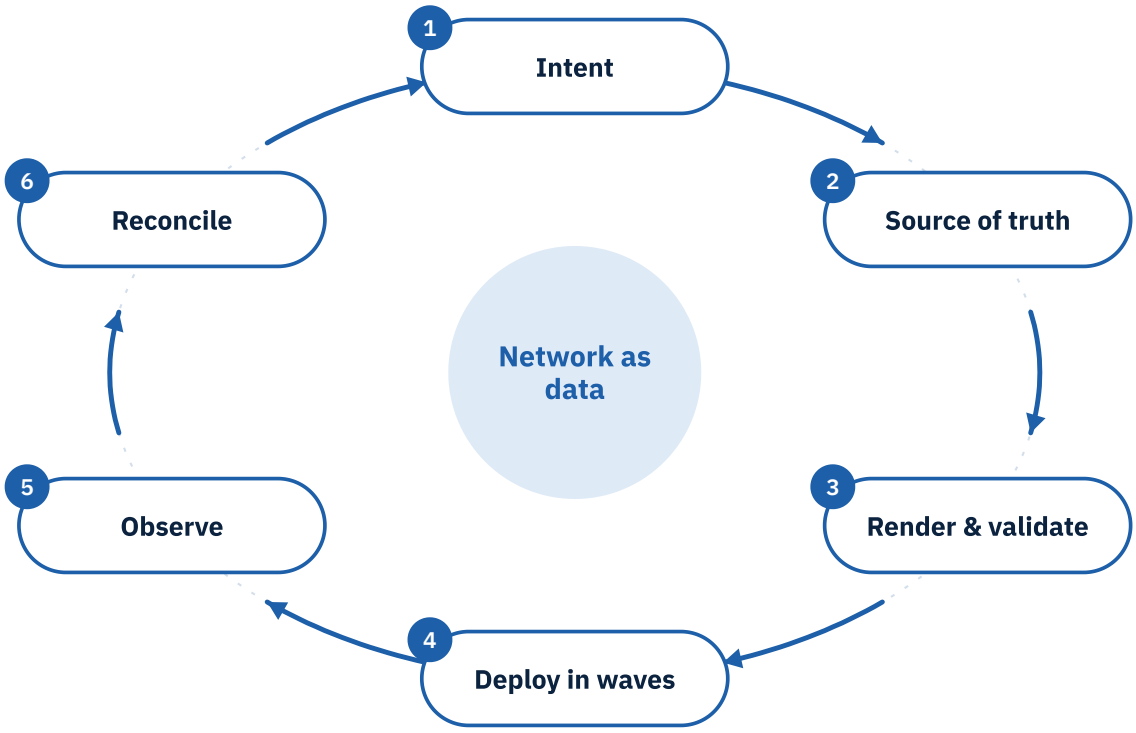
- Describe the desired end state; the tool computes the changes
- Plan/diff before apply; repeatable and reviewable
- Needs state management and discipline to avoid manual drift

Intent-based

- Describe business outcomes; the system translates, deploys and continuously verifies
- Closed loop: assurance checks the network still meets intent
- Requires mature data models and trust in the translation layer

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Closed-loop, intent-based network operations



1. **Intent** — Service request as parameters
2. **Source of truth** — IPAM, inventory, policy data
3. **Render & validate** — Templates, guardrails, twin
4. **Deploy in waves** — Canary, then domains
5. **Observe** — Streaming telemetry, SLOs
6. **Reconcile** — Detect drift, roll back or fix

Intent enters once, as data; the loop then keeps reality matching it. Drift or a failed objective sends the loop back to intent and design rather than to a manual fix on a device.

Day-0 design-for-automation principles

A1

Standard building blocks

A small number of site, pod and cloud-spoke types, each fully parameterised. Every exception is a cost.

A2

Data model first

Define the data that describes the network — sites, links, addresses, policies — before writing templates.

A3

Single source of truth

One authoritative store per data type; devices are rendered from it, never the reverse.

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A4

Idempotent changes

Applying the same intent twice produces the same result and no side effects.

A5

Programmable platforms

Select equipment and services with complete, stable APIs and structured telemetry as a design requirement.

A6

Guardrails as code

Policy checks (no overlapping ranges, no any-any rules, naming standards) run automatically in the pipeline.

A7

Drift is detected

Running state is compared with intended state continuously; differences are alerted or reconciled.

A8

Break-glass exists

A controlled, audited manual path exists for emergencies, and its use is reconciled afterwards.

A network change pipeline

01 **Commit**
Change to source-of-truth data or templates

→

02 **Review**
Peer review and automated linting

→

03 **Verify**
Model-based checks, simulation or a digital twin

→

04 **Plan**
Diff of exactly what will change on which device

→

05 **Deploy in waves**
Canary, then failure domains in order

→

06 **Assure**
Post-checks against intent; automatic rollback on failure

What to automate first — decision matrix

TASK	FREQUENCY	RISK IF WRONG	PRIORITY
Configuration backup and compliance audit	Daily	Low (read-only)	First — builds the inventory and baseline
New cloud spoke or branch from a standard type	Weekly	Medium	Early — high value, repeatable
Access-port and segmentation changes	Daily	Medium	Early, with guardrails
Software upgrades	Quarterly	High	After pre/post-checks and drain automation exist
Core routing policy changes	Rare	Very high	Last — automate verification before automating the change

WATCH OUT · AUTOMATION MULTIPLIES WHATEVER IT IS GIVEN

A pipeline that can update 2,000 devices in five minutes can also break 2,000 devices in five minutes. Tie automation to the change-wave and blast-radius design from the resilience module: canaries, failure-domain ordering, automatic halt on failed post-checks, and rate limits on how much can change at once.

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INSIGHT · CONSOLE CHANGES CREATE DRIFT

In cloud, a single change made by hand in a console silently diverges from the code. The next pipeline run either reverts it — surprising whoever made it — or fails. Design permissions so production network constructs can be changed only through the pipeline, with break-glass access audited and reconciled.

WATCH OUT · GENERATED CONFIGURATION IS STILL UNTRUSTED INPUT

AI tools that draft configuration, policy or automation code from a prompt can speed up routine work, but their output can be confidently wrong. Treat it exactly like a contribution from a new engineer: it enters through the source of truth and the pipeline, passes the same guardrails, model checks and peer review, and never goes straight to a device.

An automation roadmap for a small team

MONTHS 0–3
Read-only first
Backups, inventory, compliance reports

MONTHS 3–6
Standards and data
Site types, data model, source of truth

MONTHS 6–12
Pipeline deploys standard sites
Review, guardrails, staged rollout

YEAR 2
Closed-loop assurance
Drift detection, objective-driven checks

Illustrative phases. Each phase delivers value on its own and builds the data the next phase needs; skipping straight to deployment automation is how inconsistency gets automated.

SCENARIO · DECIDE · WHERE SHOULD THE MINISTRY'S AUTOMATION START?

A ministry's network team of eight engineers runs 300 district offices, two data centres and a growing cloud estate. Every district office is configured slightly differently. Leadership has approved budget for 'network automation' and a manager proposes buying an orchestration platform and pointing it at the existing devices.

(A) Define two or three standard office types and a data model, build a source of truth and read-only compliance checks, then automate deployment of the standard types.

(B) Buy the platform and import current device configurations as the starting templates.

(C) Write scripts to push urgent changes to all offices immediately, then think about structure later.

(D) Postpone automation until every office has been physically refreshed to new hardware.

Circle your choice. Answer and feedback: module answer key.

Automation-readiness checklist

☐ Site, pod and cloud-spoke types are standardised and parameterised

☐ A data model and source of truth exist for addresses, inventory and policy

☐ Platforms were selected with complete APIs and structured telemetry

☐ Changes flow through review, validation, plan and staged deployment

☐ Guardrails run automatically in the pipeline

☐ Drift detection compares running state with intended state

☐ Break-glass access is controlled, audited and reconciled

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REFLECT

If you had to rebuild one of your sites from data alone tomorrow, what information would be missing? That gap is where your source of truth should start.

Key terms

Infrastructure as code
Defining infrastructure in version-controlled files that tools apply to create or change resources.

Idempotent
Producing the same result no matter how many times an operation is applied.

Drift
A difference between a system's running configuration and its intended, codified state.

Digital twin
A model of the production network used to test changes before deployment.

Break-glass access
An emergency manual access path that bypasses normal automation under strict audit.

Declarative
Describing the desired end state and letting a tool determine the steps to reach it.

Source of truth
The authoritative data store from which network configuration is generated.

Intent-based networking
An approach in which business intent is translated into configuration and continuously verified.

Guardrail
An automated policy check that blocks changes violating agreed standards.

Key takeaways

- Design for automation from day 0: standard building blocks, a data model and one source of truth
- Declarative intent with plan/diff and guardrails makes change repeatable and reviewable
- Automation multiplies whatever it is given — tie it to change waves, canaries and automatic halts
- Restrict production change paths to the pipeline, with an audited break-glass route
- Start read-only, then standardise, then deploy, then close the loop with assurance

KNOWLEDGE CHECK

12. What distinguishes declarative infrastructure as code from imperative scripting?

(A) It runs commands in a strict order so the outcome depends on the starting state

(B) It describes the desired end state and computes the changes needed to reach it

(C) It removes the need for version control because the tool stores all history

(D) It can only manage cloud resources and never physical network devices

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13. A team wants to automate deployments across 300 inconsistent branch configurations. What should come first?

    (A) Standard site types and a data model in a source of truth
    (B) A script that pushes the same change to every branch at once
    (C) An import of all current configurations as the master templates
    (D) A hardware refresh so that every branch runs the same platform

14. Why should production cloud network constructs be changeable only through the pipeline?

    (A) Cloud consoles cannot modify route tables or security groups directly in production
    (B) Manual console changes cause drift that the next pipeline run reverts or fails on
    (C) Pipeline changes are billed at lower rates than changes made in the console
    (D) Providers require pipeline-only access for private connectivity to function

15. Which task is usually the safest first candidate for network automation?

    (A) Core routing policy changes across all of the regions
    (B) Fleet-wide software upgrades during business hours
    (C) Emergency changes made during a major live incident
    (D) Read-only configuration backup and compliance auditing

16. What does the assurance layer add in an intent-based design?

    (A) A faster way to render templates into device configuration files
    (B) A replacement for the peer review of changes before they are deployed
    (C) Continuous checks that the network still meets the declared intent
    (D) An approval step that removes the need for a source of truth

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MODULE 7 · CLOUD · LESSON 4 · 31 MIN

Observability & Cost-Aware Network Design

Design what you will measure and what you will pay before you build

- Design an observability architecture with service-level objectives for the network
- Estimate the main cost drivers of hybrid and cloud network designs
- Apply cost-aware patterns without weakening resilience or security

Two questions decide whether a cloud or hybrid network design survives its first year: **can the team see what is happening**, and **does the bill match the business case?** Both are design questions. Visibility that is not designed in is hard to retrofit, and network costs in cloud are dominated by traffic patterns that the architecture itself creates.

Observability stack

Objectives and alerting	`Service-level objectives per network service` `Alerts on objective burn, not on every metric`
Analysis	`Dashboards, correlation, anomaly detection` `Topology-aware root-cause views`
Collection	`Streaming telemetry and metrics` `Flow records and cloud flow logs` `Logs and events` `Synthetic tests and path tracing`
Sources	`Devices, appliances and cloud constructs` `Hosts, load balancers and applications`

Measurement methods

Polling (for example SNMP)

- Simple and universal
- Coarse intervals hide micro-bursts
- Load grows with scale

Streaming telemetry

- Devices push structured data at short intervals or on change
- Better resolution and scale
- Needs a collection pipeline and data model (for example gNMI with OpenConfig models)

Synthetic tests

- Active probes measure latency, loss and reachability along real paths
- Show user experience, including paths you do not own
- Measure only what you probe

Network signals worth objectives

SIGNAL	EXAMPLE OBJECTIVE	WHY IT MATTERS
Reachability	99.95% of synthetic probes between branches and the order platform succeed per month	The most direct measure of service availability
Latency	95th percentile branch-to-cloud latency under 40 ms	Drives application response time
Loss	Under 0.1% on the voice class	Real-time traffic degrades quickly with loss
Saturation	No redundant pair above its survivable ceiling for more than 15 minutes	Links failover risk to capacity

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SIGNAL	EXAMPLE OBJECTIVE	WHY IT MATTERS
Change health	Under 2% of automated changes rolled back	Measures the safety of the delivery process

Cost-aware design formulas

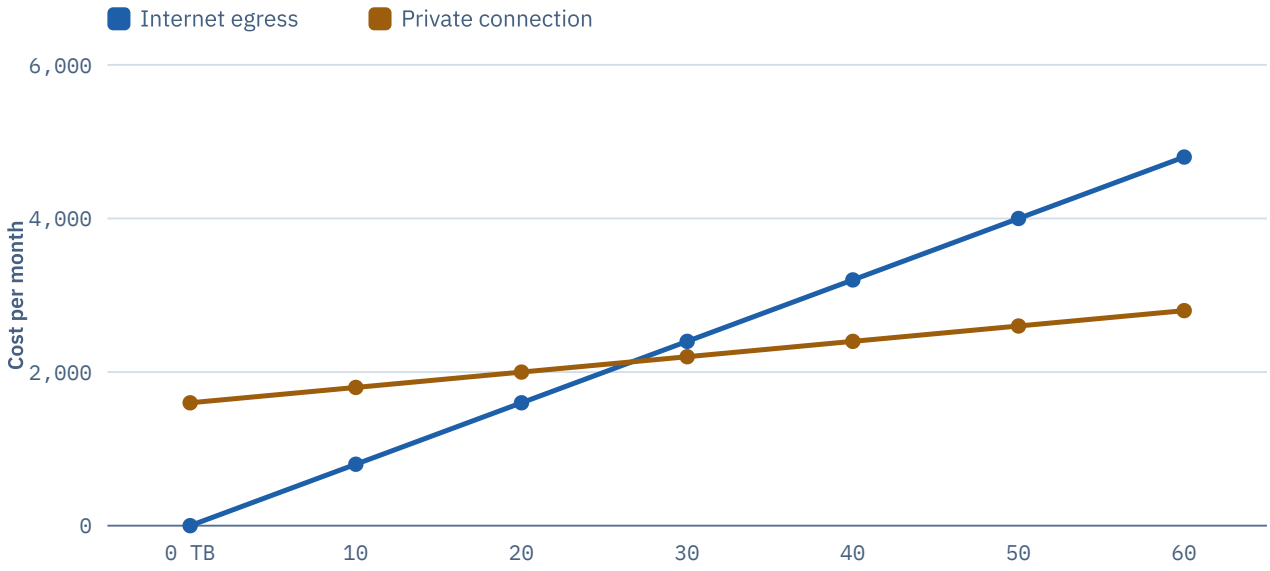
MEASURE	FORMULA	WHAT IT TELLS YOU
Monthly egress cost	`Cost = GB leaving the provider × egress rate per GB`	Internet egress and private-connectivity egress are often priced differently. Terms are changing: the EU Data Act, applicable since September 2025, requires providers to phase out switching charges, including egress when leaving, by January 2027, and some providers already waive egress for departing customers — record the pricing assumption and give it a review trigger.
Private connectivity break-even	`Break-even GB per month = fixed monthly port and circuit fees ÷ (internet rate − private rate)`	Above this volume, private connectivity is cheaper as well as more predictable.
Processing charges	`Cost = GB through a managed gateway × per-GB processing fee`	Managed NAT gateways, transit hubs and inspection services often charge per GB on top of hourly fees.

Worked example — is a private connection worth it? (illustrative prices)

1. **Inputs** 50 TB a month leaves the cloud to on premises. Illustrative internet egress is 0.08 per GB, private egress 0.02 per GB, and port plus circuit fees are 1,600 a month.
2. **Internet path cost** 50,000 GB × 0.08 = **4,000** a month.
3. **Private path cost** 50,000 GB × 0.02 + 1,600 = **2,600** a month — a saving of 1,400.
4. **Break-even** 1,600 ÷ (0.08 − 0.02) ≈ **26,700 GB**. Below about 27 TB a month, the VPN is cheaper on transfer cost alone; above it, the private connection wins — before counting its performance benefits.

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Monthly transfer cost: internet egress versus private connection



Computed from the illustrative prices in the worked example. The lines cross near 27 TB a month; below it the internet path is cheaper on transfer cost, above it the private connection wins.

Cost-aware patterns

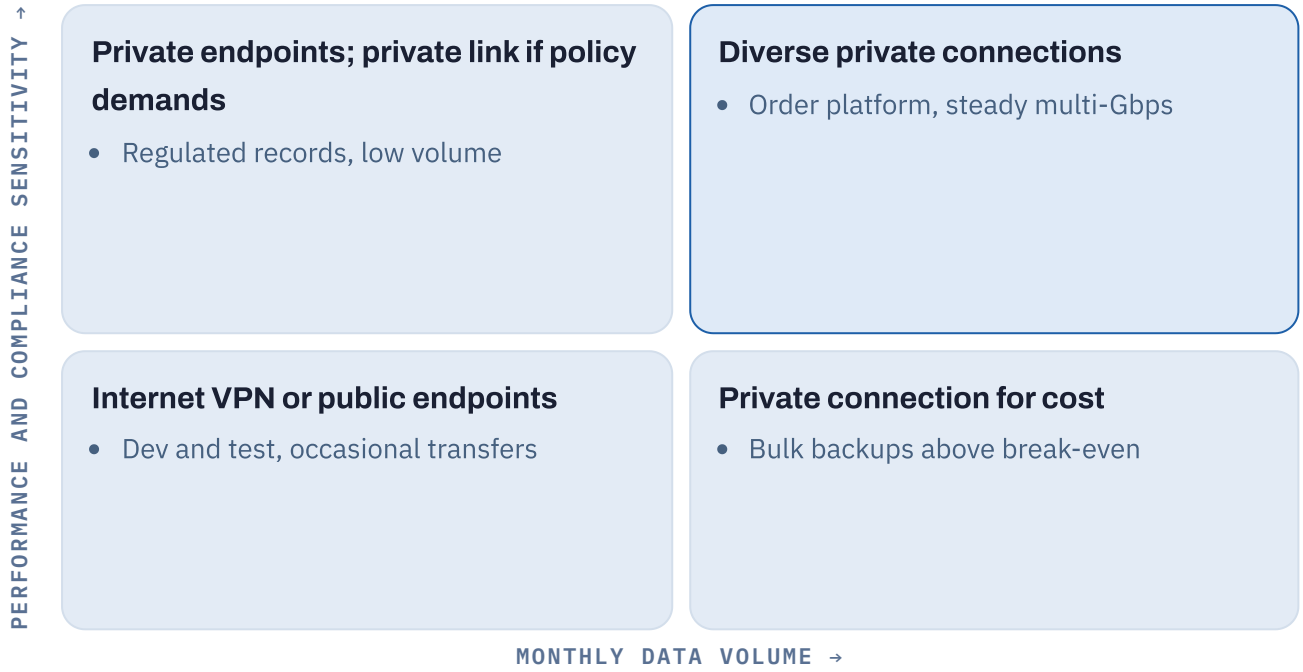
PATTERN	SAVES	WATCH FOR
Private endpoints or gateway endpoints for provider services	NAT gateway processing and internet egress	DNS design for private endpoints
Keep chatty tiers within a zone, replicate across zones deliberately	Cross-zone transfer charges	Do not trade away zone resilience for small savings
Inspect only what policy requires in the hub	Hub and appliance processing per GB	Document which flows skip inspection and why
Cache and compress at the edge, use content delivery for public content	Internet egress	Cache invalidation and data residency
Tag network resources and show cost per team	Unowned spend	Shared hubs need a fair allocation model
Right-size and power-manage capacity: idle ports, optics, appliances	Energy cost and carbon	Keep the headroom failover needs

WATCH OUT · THE CENTRALISED INSPECTION BILL

Sending every spoke-to-spoke and spoke-to-internet flow through a central inspection hub is attractive for security, but each traversal can pay transit attachment processing and appliance throughput charges. Decide flow by flow which inspection is required, and price the design before approving it.

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Choosing a cloud path by volume and sensitivity



Volume decides cost, sensitivity decides whether cost is allowed to decide. Re-check the quadrant when volumes cross the break-even point.

CASE STUDY · MINISTRY HYBRID-CLOUD MOVE

A government ministry with 120 district offices plans to move its citizen-services portal to a public cloud region inside the country, as required by data-residency law. The citizen records database must stay in the ministry's own data centre for the next three years. The portal makes many small queries to the records database for each citizen request. District offices reach the data centre over a national WAN. Peak portal traffic is about 1.5 Gbps; the availability target is 99.9%, with 99.95% required within two years. The finance ministry demands a monthly cost forecast and will scrutinise overruns. The network team is small and has no cloud experience.

1. How should the cloud region connect to the ministry data centre, and why?
2. What does the chatty portal-to-database pattern imply for the design?
3. How should addressing, DNS and automation be set up from day 0?
4. How will you forecast and control monthly network cost?

Model answer: module answer key.

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SCENARIO · DECIDE · THE SURPRISE NAT BILL

A team notices that its managed NAT gateway processing charges have tripled. Investigation shows nightly backups and container image pulls going from private subnets to the provider's own storage and registry services through the NAT gateway.

(A) Move the workloads to public subnets with public addresses so they bypass the NAT gateway.

(B) Schedule backups less often to reduce the volume.

(C) Replace the managed NAT gateway with a larger instance size.

(D) Add private or gateway endpoints for the storage and registry services so that traffic stays on the provider's private network.

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Network cost and observability forecast**

Forecast the monthly cost drivers of a hybrid design alongside the objectives that prove it works, so finance and operations review the same sheet.

COST OR SIGNAL	DRIVER / VOLUME	UNIT RATE OR OBJECTIVE	MONTHLY FORECAST	OWNER	ALERT THRESHOLD	NOTES
Private connectivity	*2 ports + circuits*	*Fixed fee*	*1,600*	*Network team*	*+10% vs forecast*	*Second location planned in year 2*
Portal-to-database latency	*Synthetic probe every 30 s*	*p95 < 10 ms*	*n/a*	*Platform team*	*Objective burn > 5% in 1 h*	*Triggers cache review if breached*

Observability and cost checklist

☐ Each network service has objectives for reachability, latency, loss and saturation

☐ Streaming telemetry, flow logs and synthetic tests cover hybrid and cloud paths

☐ Alerts fire on objective burn rather than on every metric

☐ Egress, processing and port costs are modelled before the design is approved

☐ Provider services are reached through private or gateway endpoints

☐ Inspection is applied per flow according to policy, not by default to everything

☐ Resources are tagged and cost is reported per owner

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REFLECT

Which flow in your current or planned cloud design is the most expensive per gigabyte, and is that cost buying you something the business actually requires?

Key terms

Observability
The ability to understand a system's internal state from the telemetry it produces.

Streaming telemetry
Push-based, structured export of device data at high frequency or on change.

Flow log
A record of network flows produced by a cloud platform or device, without payload.

Gateway endpoint
A route-table-based path to a provider service that keeps traffic off the internet and NAT.

Service-level objective
A measurable target for a service, such as latency or availability, over a defined period.

Synthetic test
An active probe that measures reachability and performance along a path.

Processing charge
A per-GB fee for traffic passing through a managed cloud network service.

Showback
Reporting cost to the teams that cause it, without necessarily charging them.

Key takeaways

- Design observability in: objectives for reachability, latency, loss, saturation and change health
- Streaming telemetry and synthetic tests see what polling and device counters miss
- Cloud network cost is created by traffic patterns — model egress, processing and port fees before approving
- Use private endpoints, zone-aware placement and selective inspection without trading away resilience
- Record pricing assumptions and review them; egress terms and regulation are changing

KNOWLEDGE CHECK

17. Monthly fixed fees for a private connection are 1,600 and the egress price difference versus internet is 0.06 per GB. What is the approximate break-even volume?

(A) About 27 TB per month
(B) About 2.7 TB per month
(C) About 96 TB per month
(D) About 270 TB per month

18. NAT gateway charges have tripled because workloads reach the provider's storage service through it. What is the best fix?

(A) Give the workloads public addresses and move them into public subnets
(B) Move to a larger NAT gateway with more throughput
(C) Reduce backup frequency so less traffic is sent
(D) Use private or gateway endpoints for the provider service

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19. Why are synthetic tests valuable in hybrid designs?

(A) They replace flow logs as the main source of security evidence in cloud estates

(B) They remove the need for any telemetry from network devices

(C) They measure real path performance, including segments you do not own

(D) They reduce egress costs by caching data near the users

20. An alerting design pages engineers on every interface over 70% utilisation. What is the better approach?

(A) Lower the alert threshold to 50% so that problems are caught even earlier

(B) Alert on objective burn and on pairs above their survivable ceiling

(C) Remove utilisation monitoring because cloud capacity is unlimited

(D) Page only when an interface has been completely down for an hour

21. A chatty portal in the cloud makes many small queries to an on-premises database. What is the main design risk?

(A) Bandwidth exhaustion on the private connection to the provider

(B) Round-trip latency multiplied across many queries per request

(C) IP address exhaustion in the portal's virtual network subnets

(D) Excessive DNS queries overwhelming the cloud resolver service

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MODULE 7 · CLOUD

Answer key

Knowledge checks

1. D 40 × 39 ÷ 2 = 780 peerings, each needing routes and rules. A hub provides transitive routing with far fewer objects, at the cost of a hub hop and processing charges.

2. B Device redundancy protects against device faults, but a facility event removes both. Diversity of location is needed for high availability.

3. C A backup path that cannot carry the required load is not a backup for that load. Size it for the critical flows it must protect.

4. B Private connectivity provides isolation and predictability, not confidentiality by itself. Encryption must be added where policy requires it.

5. D Central ownership simplifies governance but concentrates risk: a fault in the shared network affects every consumer.

6. A Multi-cloud doubles operational complexity and inter-cloud failover is hard. Multi-region within one provider is often the stronger availability answer.

7. C Overlap forces translation wherever the networks connect, and the operational debt grows with every new service.

8. D Providers commonly reserve four or five addresses per subnet for gateway, DNS and internal use.

9. A Conditional forwarding sends only the relevant zone to the cloud resolvers, which return the private address. The other options are unscalable or reopen public exposure.

10. C Exchange-point interconnection has fixed costs that pay off with volume; small or occasional traffic suits VPN or hairpinning.

11. B IPv6-mostly lets capable clients run IPv6-only while translation keeps IPv4 destinations reachable and legacy devices keep dual stack. A big-bang IPv4 shutdown breaks legacy devices; larger subnets and extra NAT preserve the scarcity problem.

12. B Declarative tools reconcile current state with desired state and show a plan before applying. Imperative scripts execute steps whose outcome depends on where you start.

13. A Automation needs a clean model of what correct looks like. Without standards, tools simply reproduce inconsistency at speed.

14. B Drift breaks the link between code and reality. Restricting change paths keeps the source of truth authoritative.

15. D Read-only tasks build inventory and baselines with no risk of outage, and they reveal how far the estate is from standard.

16. C Assurance closes the loop by checking running state and behaviour against intent, detecting drift and failures after deployment.

17. A 1,600 ÷ 0.06 ≈ 26,700 GB, or about 27 TB a month. Above that volume, private connectivity is cheaper on transfer cost.

18. D Endpoints keep provider-service traffic off the NAT path and private. The other options weaken security, do not change per-GB charges, or trade away recovery objectives.

19. C Active probes show reachability and latency end to end across provider and carrier networks that you cannot instrument directly.

20. B Objective-based alerting focuses attention on user impact and real risk instead of noisy thresholds.

21. B Each query pays the hybrid round trip; dozens per request quickly add up. Caching or an API layer near the portal reduces the number of crossings.

Scenarios & case studies

> **Scenario · Connecting a retailer's cloud platform**
> **B** is best. Best. Diverse facilities remove the shared-location risk, private capacity carries the peaks predictably, and the VPN covers a double failure for the most critical traffic at low cost.

> **Scenario · The private endpoint that only works in the cloud**
> **C** is best. Best. On-premises clients receive the private address and use private connectivity, and every future private endpoint in that zone works the same way.

> **Scenario · Where should the ministry's automation start?**
> **A** is best. Best. It creates the standard and the data that any tool needs, delivers early value through visibility, and makes later automation safe.

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Case study · Ministry hybrid-cloud move

Connectivity: start with private connectivity from the ministry data centre to the in-country region, plus an IPsec VPN as backup sized for critical portal traffic, which fits 99.9%. Plan a second private connection at a diverse location to reach 99.95% within two years. **Chatty pattern:** many small queries per request make latency, not bandwidth, the main risk — measure the round trip, and if it is too high, add a read cache or API layer in the cloud near the portal rather than routing every query across the hybrid link; keep a synthetic test on that path with a latency objective. **Addressing and DNS:** allocate a dedicated, non-overlapping aggregate for the cloud region from the ministry's IPAM; design conditional forwarding in both directions so the portal resolves the database by its internal name and on-premises staff resolve cloud private endpoints correctly. **Automation:** build the cloud network as code from day 0 with a standard spoke type and pipeline guardrails; restrict console changes; use the small team's time on standards rather than hand-built resources. **Cost:** model egress (mostly responses to citizens over the internet and database replies, which are on-premises egress), private-connectivity port fees, transit and NAT processing; use private endpoints for provider services; tag all resources; publish a monthly forecast with alerts when actual spend exceeds forecast by an agreed margin.

Scenario · The surprise NAT bill

D is best. Best. Traffic to provider services no longer traverses the NAT gateway, reducing cost and keeping workloads private; update DNS so names resolve to the endpoints.

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08

MODULE 8 · SCENARIO

Design Scenarios, Migration & Capstone

Staged scenarios, constraint reading, migration and rollback, executive communication and the capstone

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Read staged design scenarios for hard constraints and adapt decisions as information changes
- Choose migration strategies and design rollback, go/no-go and point-of-no-return controls
- Communicate designs to executives in terms of decisions, cost and residual risk
- Apply all eight sheets of the ZCDE Design Blueprint to a multi-stage scenario
- Prepare for scenario-based assessment and plan the ZCDE capstone

LESSONS

1. How Expert Design Scenarios Unfold · 29 min
2. Migration Strategies, Risk & Rollback Design · 33 min
3. Communicating Designs to Executives · 27 min
4. Mock Scenario: Kestrel Freight · 43 min
5. Exam Strategy & Capstone Introduction · 29 min

MODULE 8 · SCENARIO · LESSON 1 · 29 MIN

How Expert Design Scenarios Unfold

Requirements arrive in pieces and change underfoot — learn to read, decide and adapt

- Describe how real design engagements and scenario assessments reveal information in stages
- Read a brief for hard constraints, soft preferences and hidden signals
- Absorb new information without discarding sound earlier decisions

No design engagement hands you a complete, stable specification. The first meeting reveals the obvious goals; the second reveals the budget; the third reveals that the regulator, the union or the acquired company has its own ideas. Expert designers are distinguished less by what they know than by how they behave as information arrives: they extract constraints quickly, commit to decisions with stated assumptions, and adjust only what the new information actually touches.

How a design scenario unfolds

01 Brief — Background, business goals, current estate

→

02 Extract — Requirements, constraints, assumptions, risks

→

03 Decide — Initial design choices with rationale

→

04 New information — A constraint changes or a stakeholder speaks up

→

05 Re-evaluate — Revisit only the decisions the change touches

→

06 Plan the move — Migration, rollback, sequencing

→

07 Communicate — Explain decisions and trade-offs to each audience

Reading a brief for what matters

R1 Business driver
Why the project exists — revenue, regulation, risk, cost. Every design choice must serve it.

R2 Hard constraints
Budget ceilings, legal and regulatory rules, fixed dates, existing contracts, skills that cannot be hired in time.

R3 Soft preferences
Familiar technologies, preferred suppliers, a manager's favourite architecture. Respect them, but they yield to hard constraints.

R4 Brownfield reality
What already exists and cannot be replaced at once: addressing, protocols, contracts, staff habits.

R5 Signal words
Words like must, only, currently, planned, next year and without interruption often carry the key constraint.

R6 Stakeholder tension
Conflicting goals between security and operations, finance and availability, headquarters and regions.

Signal words and what they usually mean

PHRASE IN THE BRIEF	LIKELY MEANING	DESIGN IMPLICATION
'within the existing budget'	No new capital	Favour reuse, phased investment, operating-cost options

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PHRASE IN THE BRIEF	LIKELY MEANING	DESIGN IMPLICATION
'the team is familiar with…'	Operational skills constraint	Prefer designs the team can run; budget training if not
'the regulator requires…'	Hard, non-negotiable constraint	Design to the rule first; optimise everything else around it
'must not be interrupted'	Zero-downtime migration	Parallel run or phased cut-overs with tested rollback
'we plan to acquire…' or 'expand into…'	Future scale and integration	Address space, routing policy and segmentation that absorb growth
'currently uses…'	Brownfield dependency	Coexistence and migration path are part of the design

Hard versus soft constraints

Hard constraints

- Cannot be traded away without the sponsor's explicit decision
- Examples: law, regulation, contract terms, budget ceiling, safety
- A design that violates one is wrong, however elegant

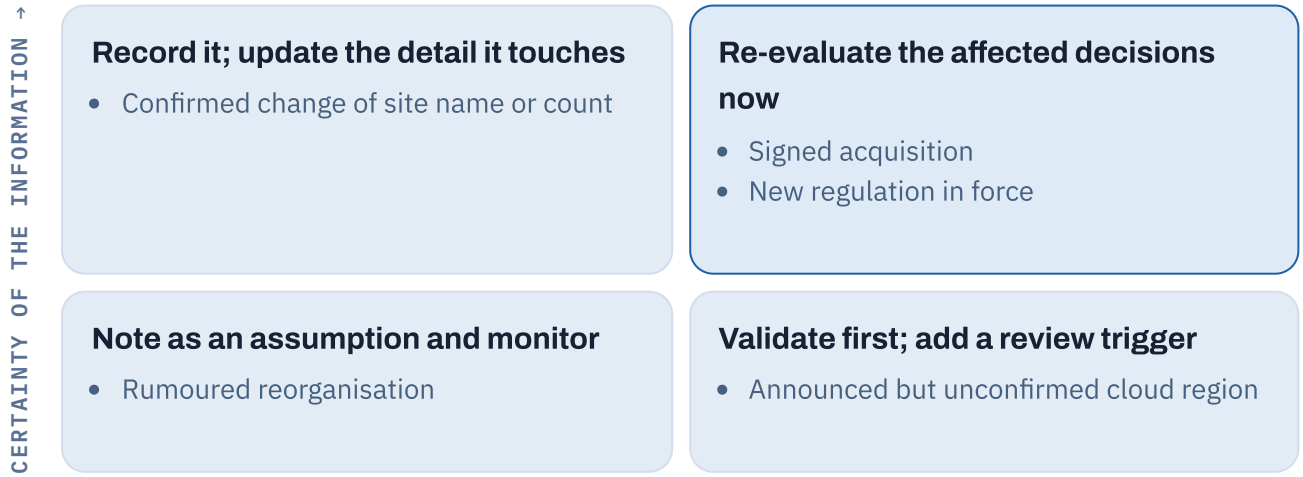
Soft constraints and preferences

- Can be traded when they conflict with something more important
- Examples: preferred supplier, familiar protocol, aesthetic consistency
- Trade them openly and record why

Absorbing new information

1. **Read the delta** Identify exactly what is new or changed — do not reread the whole brief through a new lens.
2. **Classify it** Is it a new requirement, a changed constraint, a clarification or a risk that has materialised?
3. **Trace the impact** Which earlier decisions depend on the changed item? Use the decision log to find them.
4. **Keep what still holds** Decisions untouched by the change stay. Redesigning everything signals panic, not rigour.
5. **Update and record** Change the affected decisions, note the reason and the new assumptions.
6. **Communicate** Tell stakeholders what changed, what it costs and what risk is now accepted.

Triage for new information



Place each new fact by how certain it is and how many earlier decisions it touches. Only the top-right quadrant justifies reopening decisions immediately; uncertain news becomes an assumption with a validation action or a review trigger.

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WATCH OUT · TWO CLASSIC FAILURE MODES

Anchoring: defending the first design after a constraint has changed, because changing feels like admitting error. **Thrashing**: redesigning everything every time something changes, so no decision ever stabilises. The expert path lies between: decisions are firm but traceable, and each one names the assumptions that would reopen it.

SCENARIO · DECIDE · THE BUDGET CUT

Earlier in an engagement you recommended private connections at two diverse exchange locations to reach a 99.95% target for a cloud-hosted service. The sponsor now reports a 40% cut in the network budget for this year. The availability target is unchanged for next year, but this year's target is relaxed to 99.9%.

(A) Keep the original design and ask the sponsor to find the money elsewhere.

(B) Phase the connectivity: one private connection plus a VPN backup sized for critical traffic this year, with the second diverse connection planned for next year and the interim risk recorded.

(C) Restart the design from scratch with a cheaper technology family.

(D) Drop private connectivity and run everything over internet VPN permanently.

Circle your choice. Answer and feedback: module answer key.

PRO TIP · KEEP A DECISION LOG

For each decision record: the decision, the options considered, the reason, the assumptions it depends on and the requirement it serves. When new information arrives, search the log for affected assumptions. This turns a confusing change into a short, defensible list of updates.

Scenario-reading checklist

☐ The business driver is written in one sentence

☐ Every hard constraint is listed with its source

☐ Soft preferences are recorded separately from hard constraints

☐ Brownfield dependencies and coexistence needs are identified

☐ Stakeholder tensions are named, with who decides

☐ Each design decision lists the assumptions that would reopen it

☐ New information is classified before any decision is changed

REFLECT

Recall a project where requirements changed midway. Did the team anchor, thrash or adapt cleanly? What would a decision log have changed?

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Key terms

Hard constraint
A limit that cannot be traded away without an explicit decision by the sponsor, such as law or a budget ceiling.

Soft constraint
A preference that can be traded when it conflicts with more important requirements.

Brownfield
An environment with existing systems that a new design must coexist with or migrate from.

Decision log
A record of design decisions with options considered, rationale and dependent assumptions.

Anchoring
Holding on to an initial design or estimate despite new information that should change it.

Thrashing
Repeatedly redesigning in response to each change so that no decision stabilises.

Assumption
A condition treated as true for design purposes, which must be validated or monitored.

Signal word
A word or phrase in a brief that indicates a constraint or priority, such as must or without interruption.

Key takeaways

- Information arrives in stages; commit to decisions with stated assumptions
- Separate hard constraints from soft preferences and brownfield realities
- When something changes, classify it, trace its impact through the decision log and change only what it touches
- Avoid both anchoring and thrashing

KNOWLEDGE CHECK

1. A brief says the regulator requires transaction data to remain in-country, and the IT director prefers a particular cloud provider that has no in-country region. How should the designer treat these?

   (A) The preference wins because the IT director is also the project's sponsor
   (B) Both are soft constraints to be balanced against the overall cost
   (C) Both are hard constraints, so the project cannot go ahead at all
   (D) The regulatory rule is a hard constraint; the preference yields to it

2. Midway through a design, the sponsor announces an acquisition of a company with overlapping address space. What should the designer do first?

   (A) Restart the whole design so that the acquisition is properly reflected throughout
   (B) Ignore it until the acquisition completes and the networks are joined
   (C) Classify the change and trace which earlier decisions it actually affects
   (D) Add NAT everywhere so that every overlap is handled automatically

3. What does 'must not be interrupted' in a brief most often imply?

   (A) Parallel running or phased cut-overs with tested rollback
   (B) A requirement for 99.999% availability on every network component
   (C) A ban on any configuration changes until the project has finished
   (D) A need to use the newest hardware to avoid any risk of failure

4. Which behaviour best describes anchoring in a design engagement?

   (A) Recording assumptions alongside each design decision
   (B) Redesigning everything whenever a stakeholder speaks
   (C) Defending the original design after a constraint has changed
   (D) Phasing investments to match a reduced annual budget

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5. What is the main value of listing the assumptions behind each design decision?

(A) When information changes, you can quickly find which decisions to revisit

(B) Assumptions transfer the design risk from the designer to the sponsor

(C) Listing them removes the need to validate them with the stakeholders later on

(D) It allows the design to be approved without any review by operations

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 8 · SCENARIO · LESSON 2 · 33 MIN

Migration Strategies, Risk & Rollback Design

Getting from here to there is part of the design — and often the riskiest part

- Compare phased, parallel-run and big-bang migration strategies
- Design rollback points, go/no-go decisions and a point of no return
- Plan coexistence for merged or brownfield networks

A target design that cannot be reached safely is not a design; it is a picture. Migration is where outages, overruns and lost confidence happen, and where an expert designer earns trust. The question is not only 'what is the end state?' but 'what are the intermediate states, how long do we live in each, and how do we get back if one fails?'

Migration strategies

Phased

- Move in waves — by site, service or failure domain
- Each wave is small; lessons feed the next
- Long coexistence period between old and new
- Default choice for large estates

Parallel run

- Old and new run side by side; traffic or users move gradually
- Easy rollback: move traffic back
- Double cost and effort while both run
- Fits critical services with zero-interruption requirements

Big bang

- Everything moves in one cut-over
- Shortest coexistence; simplest end state
- Highest risk; rollback is all or nothing
- Fits small estates, or cases where coexistence is impossible

Choosing a strategy — decision matrix

FACTOR	FAVOURS PHASED	FAVOURS PARALLEL RUN	FAVOURS BIG BANG
Estate size	Large	Any, for critical services	Small
Interruption tolerance	Short per-wave windows acceptable	Near zero	One longer window acceptable
Coexistence feasibility	Old and new can interoperate	Both can serve the same users	Old and new cannot interoperate
Budget for double running	Moderate	High	Low
Team experience with the new design	Building	Building	High, with rehearsal

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A phased WAN migration programme

MONTH 1–2
Prepare and interconnect
Hubs join old and new WANs; coexistence tested

MONTH 3
Pilot
Two representative, low-impact sites

MONTHS 4–8
Waves by tier
Each wave: pre-check, cut-over, validate, hypercare

MONTH 9
Go/no-go for decommission
Formal decision by the sponsor

MONTH 10
Point of no return
Old circuits ceased; fix forward only

MONTH 11
Decommission and close
Records and source of truth updated

Illustrative durations for a 400-site estate. The point of no return — ceasing the old circuits — comes only after every wave has been through hypercare and the business has signed off.

Migration risk controls

MR1
Pilot
Migrate a representative but low-impact target first and measure everything.

MR2
Coexistence design
Design the interop boundary — routing, addressing, policy — as carefully as the end state.

MR3
Rollback points
Define, per step, how to go back, how long it takes and what triggers it.

MR4
Point of no return
Identify the step after which rollback is no longer possible, and gate it with a formal decision.

MR5
Change freeze
Stop unrelated changes during migration windows so failures are attributable.

MR6
Hypercare
Staff a heightened support period after each wave with clear escalation.

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One migration wave

01 Prepare
Pre-stage configuration, confirm prerequisites

→

02 Pre-check
Capture baseline routing, reachability, application health

→

03 Cut-over
Execute scripted steps with timestamps

→

04 Validate
Compare against baseline; run application tests

→

05 Go / no-go
Commit or roll back by the decision time

→

06 Hypercare
Monitor closely; resolve issues fast

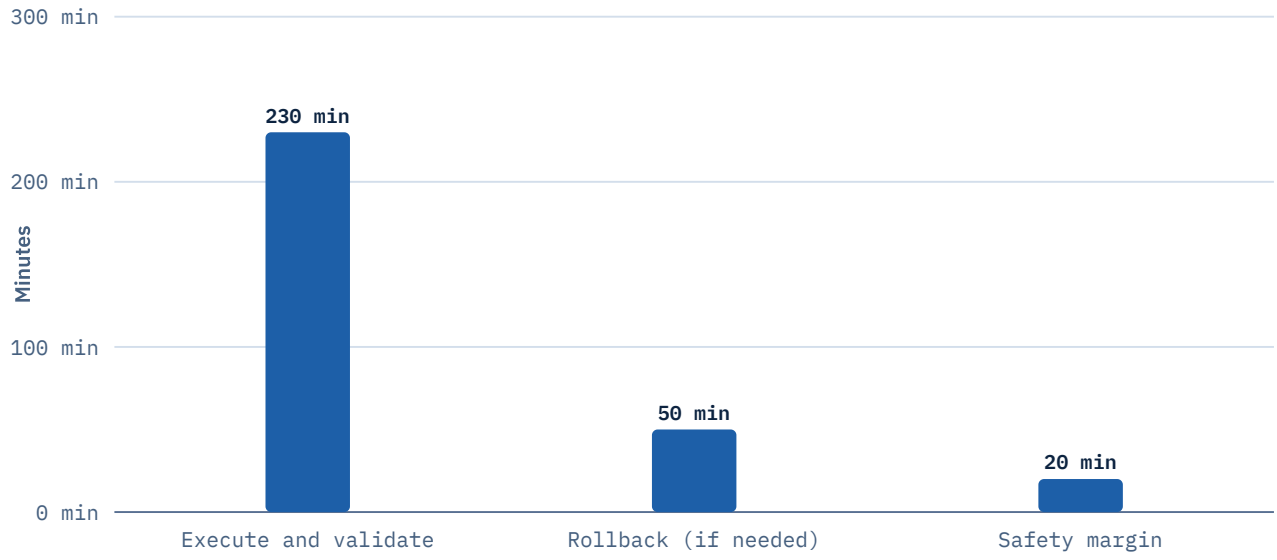
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07 Decommission
Remove the old path once stable

Rollback timing

MEASURE	FORMULA	WHAT IT TELLS YOU
Latest go/no-go time	`T_decision = window end − rollback duration − safety margin`	If validation is not complete by this time, roll back. Do not negotiate with the clock.
Worked example	`05:00 − 50 min − 20 min = 03:50`	With a window ending at 05:00, a rehearsed 50-minute rollback and a 20-minute margin, the decision must be made by 03:50.
Risk exposure	`Exposure = probability × impact`	Compare per wave to choose sequencing: low-exposure waves first build evidence for high-exposure ones.

The cut-over night as a time budget



Computed from the worked example: a window from 00:00 to 05:00, a 50-minute rollback and a 20-minute margin leave a go/no-go decision at 03:50. Everything to the left of that point is time to execute and validate.

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SCENARIO · DECIDE · 03:40 AND STILL FAILING

A data-centre core migration starts at 00:00 with a window that ends at 05:00. The rehearsed rollback takes 50 minutes, and the team agreed a 20-minute margin. At 03:40, two of twelve application validation tests are still failing; the engineers believe they can fix them in 'about half an hour'.

- (A) Start the rollback now, as agreed, and analyse the failures before rescheduling.
- (B) Continue troubleshooting until 04:30, then decide.
- (C) Commit the migration and fix the two applications during business hours.
- (D) Extend the maintenance window by phoning the duty manager.

Circle your choice. Answer and feedback: module answer key.

Keeping an old AS number visible during a merger

PE1 · IOS-style CLI

```
! PE1 now runs the merged AS 64500; customer still peers with legacy AS 64510
router bgp 64500
 neighbor 198.51.100.9 remote-as 64499
 neighbor 198.51.100.9 description Customer C - unchanged config
 neighbor 198.51.100.9 local-as 64510 no-prepend replace-as
```

The customer keeps its configuration pointing at AS 64510. no-prepend stops the local AS being added to routes received from the customer, and replace-as hides the real AS 64500 in routes sent to it, so AS-path lengths stay as they were. The feature buys time; plan when each customer will move to the new AS.

CASE STUDY · TELECOM MERGER MIGRATION

Two regional operators merge. Operator A (AS 64500) runs IS-IS and an MPLS core across 60 points of presence; Operator B (AS 64510) runs OSPF and an MPLS core across 35. Both use 10.0.0.0/8 for infrastructure and management, with many overlaps. Together they serve 900 business customers with BGP sessions, mobile backhaul, and an emergency-services network contract that allows no outage longer than one second on the emergency path. The board wants a single network and NOC within 18 months. Operator B's staff know only OSPF.

1. Which migration strategy fits, and why?
2. How would you consolidate the autonomous systems without coordinating 900 customer changes at once?
3. How would you unify the interior routing protocols?
4. How do you handle the overlapping infrastructure addressing?
5. How will you protect the emergency-services path during migration?

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Model answer: module answer key.

TEMPLATE

Migration plan sheet

Plan each wave with prerequisites, steps, validation, rollback and the point of no return, so that every cut-over is rehearsable and every decision is timed.

WAVE / SCOPE	PREREQUISITES	CUT-OVER STEPS & OWNER	VALIDATION TESTS	ROLLBACK TRIGGER & DURATION	GO/NO-GO TIME	POINT OF NO RETURN
Wave 3 — PoPs North-2 and North-3 to IS-IS	*IS-IS adjacencies up and verified for 7 days; staff trained*	*Raise OSPF distance router by router (network team)*	*FIB comparison, LSP tests, customer VPN probes*	*Any VPN probe fails > 2 min; 15 min*	*02:45*	*OSPF process removed (next wave)*

Migration design checklist

☐ Strategy (phased, parallel, big bang) is justified against the decision matrix
☐ The coexistence state is designed and documented like an end state
☐ Every step has a rollback method, duration and trigger
☐ Go/no-go times are calculated and agreed before the window
☐ The point of no return is identified and gated by a formal decision
☐ A pilot wave precedes high-impact waves
☐ Unrelated changes are frozen during migration windows
☐ Hypercare staffing and escalation are arranged for each wave

REFLECT

Think of a migration you have seen fail or overrun. Which risk control — pilot, coexistence design, rollback point, go/no-go timing or hypercare — was missing?

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Key terms

Phased migration
Moving to a new design in a sequence of smaller waves.

Big bang migration
Moving everything to the new design in a single cut-over.

Rollback
Returning to the previous known-good state after a failed change.

Go/no-go decision
A timed decision to commit or roll back a change based on agreed criteria.

Ships in the night
Running two routing protocols in parallel, with one preferred, to migrate between them.

Parallel run
Operating old and new systems side by side while traffic or users move between them.

Coexistence
The intermediate state in which old and new designs interoperate during migration.

Point of no return
The step after which rollback is no longer possible or practical.

Hypercare
A period of heightened support immediately after a migration wave.

Key takeaways

- The intermediate states are part of the design: plan coexistence as carefully as the end state
- Phased waves suit large estates, parallel run suits zero-interruption services, big bang suits small or non-interoperable cases
- Every step needs a rollback method, duration and trigger; compute the go/no-go time and keep to it
- Gate the point of no return with a formal decision

KNOWLEDGE CHECK

6. A window ends at 06:00, the rehearsed rollback takes 40 minutes and the agreed margin is 20 minutes. When is the latest go/no-go decision?

   (A) 05:20
   (B) 05:40
   (C) 06:00
   (D) 05:00

7. Which migration strategy best suits a critical service that cannot tolerate interruption and whose old and new platforms can both serve users?

   (A) Big bang over a long holiday weekend
   (B) Parallel run with gradual traffic movement
   (C) Phased by alphabetical site name order
   (D) Freeze changes until a new budget year

8. During an AS consolidation, what does local-as with no-prepend and replace-as achieve?

   (A) Customers are moved to the new AS automatically without being told
   (B) Customers keep peering with the legacy AS while path lengths are unchanged
   (C) The legacy AS is advertised to the internet in place of the new AS
   (D) Routes from customers are filtered so that the address overlap is hidden from peers

9. Why is a 'ships in the night' approach used for interior routing migrations?

   (A) Both protocols run together so forwarding can be verified before switching preference
   (B) It lets the two protocols share a single link-state database across the merged core
   (C) It removes the need to train the operations staff on the new protocol before the cut-over
   (D) It converts OSPF configuration into IS-IS configuration automatically

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10. What is the purpose of identifying a point of no return in a migration plan?

(A) To show that rollback planning is no longer needed anywhere in the plan

(B) To mark the moment hypercare support can be ended

(C) To start decommissioning the old platform at once

(D) To gate the irreversible step with a formal decision

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 8 · SCENARIO · LESSON 3 · 27 MIN

Communicating Designs to Executives

Turn architecture into decisions, money and risk that leaders can act on

- Structure a one-page design brief for an executive audience
- Translate technical properties into business consequences
- Quantify the value of resilience and security investments

Executives do not approve designs; they approve **decisions** — spend this, accept that risk, change this process. A brilliant design that cannot be explained in their terms will lose to a weaker one that can. Communication is therefore a design skill, and the expert designer prepares for the executive conversation as carefully as for the technical review.

Two audiences, two conversations

Engineering review

- How it works, protocols, failure behaviour
- Detailed diagrams and configuration standards
- Evidence: tests, calculations, references
- Goal: technical correctness

Executive decision

- What it costs, what it protects, what it risks
- One page, one picture, two or three options
- Evidence: business impact, comparisons, outcomes
- Goal: a clear decision and an owner

The one-page design brief

B1
Context
Two sentences: what is happening in the business and why a decision is needed now.

B2
Decision required
The specific question the executive must answer, and by when.

B3
Options
Two or three real options, including 'do nothing', with cost, benefit and risk.

B4
Recommendation
Your recommended option and the single most important reason.

B5
Risk and cost
What remains at risk, what it costs over its lifetime, what it saves.

B6
Ask
Exactly what you need: approval, budget, a policy decision, a named owner.

Translating technical statements

TECHNICAL STATEMENT	BUSINESS TRANSLATION
99.99% availability target	About 53 minutes of outage a year, compared with nearly 9 hours today
Single carrier for all branches	One road-works incident can close every branch at once
Sub-second convergence with BFD and fast reroute	Calls and card payments continue when a link fails
Flat network, no segmentation	One infected laptop can reach every payment and patient system
RPO zero, RTO 15 minutes	No lost transactions and service back within 15 minutes if a site is destroyed

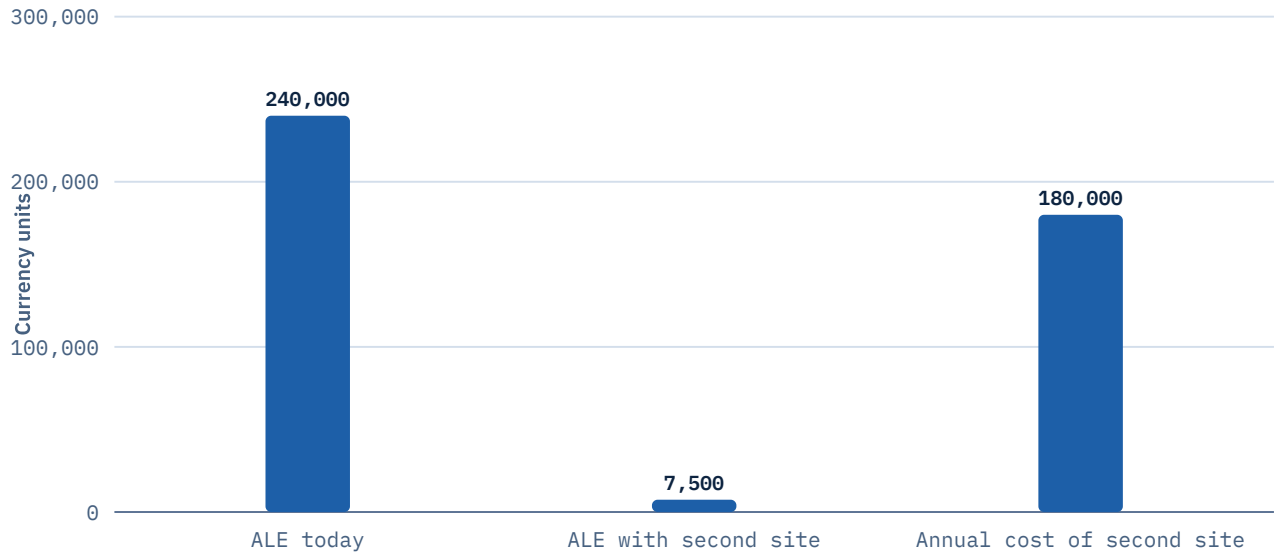
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TECHNICAL STATEMENT	BUSINESS TRANSLATION
Private connectivity break-even at 27 TB a month	Above our current volume, the dedicated link is cheaper as well as faster

Quantifying risk reduction

MEASURE	FORMULA	WHAT IT TELLS YOU
Single loss expectancy	`SLE = cost per hour of outage × expected outage duration`	The cost of one occurrence of the event.
Annualised loss expectancy	`ALE = SLE × ARO`	ARO is the expected number of occurrences per year.
Return on a resilience investment	`ROI = (ALE before − ALE after − annual cost) ÷ annual cost`	Positive means the investment pays for itself in avoided losses.
Worked example	`Before: 120,000 × 8 h × 0.25 = 240,000. After: 120,000 × 0.25 h × 0.25 = 7,500. Annual cost 180,000 → ROI ≈ 29%`	A second site reduces a site-loss outage from 8 hours to 15 minutes and pays for itself, before counting reputation and regulatory exposure.

Expected annual loss with and without a second site



Computed from the worked example. The second site costs 180,000 a year but removes about 232,500 of expected annual loss, which is the comparison a finance leader wants to see.

WATCH OUT · COMMON WAYS TO LOSE THE ROOM

Opening with topology diagrams; using acronyms without translation; presenting one option as if there were no alternative; hiding the cost in an appendix; or claiming 'zero risk'. Executives trust designers who show trade-offs and name residual risk.

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Presenting a trade-off

1. **State the decision** 'We need to decide whether to fund a second data centre this year.'
2. **Show the options** Do nothing, warm standby, or active/active — each with cost, outage exposure and time to deliver.
3. **Recommend** 'We recommend warm standby: it removes 97% of expected outage cost for 60% of the active/active price.'
4. **Name residual risk** 'A regional disaster affecting both sites remains; we cover it with off-site backups and a 24-hour recovery target.'
5. **Say what would change your mind** 'If transaction volume doubles, active/active becomes the better option — we will review in 12 months.'
6. **Ask** 'We need approval for 180,000 a year and a business owner for DR testing.'

Options on cost and risk reduction



An executive view of the options in the trade-off example. The recommendation sits where most of the expected loss is removed for a moderate cost; active/active becomes the better buy only if volumes or penalties grow.

SCENARIO · DECIDE · THE CFO'S QUESTION

You are presenting a proposal for a second data centre for a payment service. The CFO interrupts: 'We have had one serious outage in five years. Why should we spend 180,000 a year on this?'

(A) Explain the failure-domain analysis and show the redundancy diagram in detail.

(B) Point out that every other payment provider has two data centres.

(C) Quantify the expected annual loss from a site outage with and without the second site, compare it with the cost, and mention the regulatory exposure.

(D) Reassure the CFO that the current site is very reliable and the investment is just a precaution.

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Executive decision brief**

Summarise a design decision on one page for an executive audience, with options, recommendation, cost and residual risk.

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DECISION REQUIRED	OPTIONS (INCL. DO NOTHING)	RECOMMENDATION & MAIN REASON	BUSINESS IMPACT	LIFETIME COST	RESIDUAL RISK	OWNER & DATE
Fund a second site for the payment service?	*Do nothing / warm standby / active-active*	*Warm standby — removes most outage cost at 60% of active-active price*	*Site-loss outage cut from ~8 h to 15 min*	*180,000 a year*	*Regional disaster; covered by off-site backup with 24 h RTO*	*COO, decision by 30 June*

Executive communication checklist

☐ The brief fits on one page and states the decision required

☐ At least two real options are shown, including doing nothing

☐ Technical properties are translated into business consequences

☐ Costs are shown over the asset's lifetime, not just the first year

☐ Residual risk is named, with how it is managed

☐ The ask is explicit: money, policy, owner or date

☐ Supporting technical detail is available but kept out of the main page

REFLECT

Take a design decision you are working on. Write the 'decision required' sentence and the single most important reason for your recommendation in under 40 words.

Key terms

Design brief
A short document that frames a design decision for decision-makers.

Annualised rate of occurrence
The expected number of times a risk event occurs per year.

Residual risk
The risk that remains after a design or control is in place.

Single loss expectancy
The expected cost of one occurrence of a risk event.

Annualised loss expectancy
The expected yearly cost of a risk: single loss expectancy multiplied by annual rate of occurrence.

Total cost of ownership
The full cost of a solution over its life, including operation, support and refresh.

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Do-nothing option
The baseline option of making no change, against which other options are compared.

Decision owner
The person accountable for making and standing behind a decision.

Key takeaways

- Executives approve decisions — cost, risk accepted, owner — not designs
- One page: context, decision required, options including do nothing, recommendation, residual risk, the ask
- Translate technical properties into business consequences
- Quantify with expected annual loss against annual cost, and name what would change your mind

KNOWLEDGE CHECK

11. Outage cost is 50,000 per hour, a site-loss event causes a 10-hour outage, and such events are expected once every five years. What is the ALE?
    - (A) 50,000
    - (B) 250,000
    - (C) 100,000
    - (D) 500,000

12. What should come first on a one-page executive design brief?
    - (A) A detailed topology diagram of the proposed design
    - (B) The list of protocols and technologies to be used
    - (C) A history of all past incidents in the network
    - (D) The context and the specific decision required

13. Why should a 'do nothing' option be included?
    - (A) It gives a baseline to compare the cost and risk of every other option
    - (B) Executives always choose it, so it must be visible on the page
    - (C) It is required for any design proposal to pass a formal technical review board
    - (D) It shows that the designer has no preference among the options

14. Which translation of 'single carrier for all branches' works best for executives?
    - (A) The design lacks provider diversity at both Layer 1 and Layer 2
    - (B) Both branch circuits share an SRLG in the access network
    - (C) One incident at the carrier can close every branch at once
    - (D) The WAN availability is calculated as a serial dependency

15. What is the best way to respond when an executive asks for a guarantee of zero risk?
    - (A) Agree, since a strong design eliminates all meaningful risk anyway
    - (B) Explain the residual risk and how it is managed and monitored
    - (C) Refuse to proceed until the executive accepts more budget
    - (D) Redirect the conversation to the technical design details

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MODULE 8 · SCENARIO · LESSON 4 · 43 MIN

Mock Scenario: Kestrel Freight

A full multi-stage design across all eight blueprint sheets

- Apply every sheet of the ZCDE Design Blueprint to one evolving scenario
- Make and revise decisions as requirements arrive in stages
- Produce a coherent, defensible design and migration path

This lesson walks through a complete mock scenario in stages, as an expert design engagement or a scenario-based assessment would. Read each stage, decide before revealing the feedback, and keep your own decision log. Kestrel Freight is fictional; the constraints are typical of real logistics, retail and public-sector projects.

Stage 1 — The brief

1. **Business** Kestrel Freight runs 140 depots, two automated sorting hubs, a 1,500-person head-office campus and one owned data centre. Business customers want real-time parcel tracking through an API within 12 months.
2. **Targets** The tracking API must reach 99.95% availability. Depot operations must continue for at least four hours if the WAN fails. The network team has 12 engineers.
3. **Current estate** Depots use a single broadband line with a 4G backup. The data centre has one internet edge. The campus and hubs share one flat network with the sorting-hub control systems.
4. **Constraints** Capital budget is fixed for this year. A logistics regulator requires shipment records to stay in-country. The board has asked for a plan in six weeks.

Stage 1 — extracting what matters

ITEM	TYPE	DESIGN CONSEQUENCE
Tracking API at 99.95% in 12 months	Business driver, measurable target	At least two independent failure domains for the API path
Depots survive four hours without WAN	Hard operational requirement	Local survivability: local breakout, cached services, local scanning
Shipment records in-country	Hard regulatory constraint	Cloud region and DR site selection limited to in-country options
Flat network with sorting-hub controls	Brownfield risk	Segmentation of OT is a priority, not an option
Fixed capital budget, 12 engineers	Hard financial and skills constraints	Phased design; prefer operating-cost services and automation

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How the Kestrel scenario unfolds

WEEK 0
Brief
API at 99.95%, depot survivability, in-country data

WEEK 2
API placement
In-country cloud region across zones

WEEK 4
Depots and WAN
Local buffering, dual-transport overlay

MONTH 6
Acquisition of Swiftline
Overlapping addresses, four-month target

MONTH 7
Insurer conditions
OT segmentation, MFA access, monitoring

MONTH 9
Budget cut of 25%
Re-sequence; protect launch and insurer items

Information arrives in stages, as it does in real engagements. Note which earlier decisions each stage leaves untouched — that is the discipline the scenario tests.

SCENARIO · DECIDE · STAGE 2 — WHERE SHOULD THE TRACKING API LIVE?

The single owned data centre has no second site. The in-country cloud region offers three availability zones. The tracking API is new, stateless at the front end, and reads from the shipment database, which stays on premises for now. The budget is fixed this year.

(A) Host the API in the owned data centre and build a second data centre next year.

(B) Host the API in a cloud region abroad with better pricing and replicate data back home nightly.

(C) Host the API front end in the in-country cloud region across two or more zones, replicate a read-only copy of tracking data to the cloud, and connect privately to the data centre.

(D) Host the API across two public cloud providers for maximum resilience.

Circle your choice. Answer and feedback: module answer key.

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SCENARIO · DECIDE · STAGE 3 — DEPOTS AND THE WAN

Depots must keep scanning parcels for four hours without the WAN. Scanners post events to the tracking platform. Today the depot broadband and 4G backup both terminate on one small router, and depot staff reach all applications through the data centre.

(A) Add a local event buffer at each depot so scanners keep working offline, use an overlay across broadband and 4G with application-aware path selection, and break out to the cloud API directly.

(B) Replace each depot's broadband with a private MPLS circuit.

(C) Keep the design and ask depots to record parcels on paper during outages.

(D) Add a second broadband line from the same provider at each depot.

Circle your choice. Answer and feedback: module answer key.

SCENARIO · DECIDE · STAGE 4 — NEW INFORMATION: AN ACQUISITION

Six months in, Kestrel announces the purchase of Swiftline, a courier with 60 depots. Swiftline uses the same 10.0.0.0/8 ranges, its own BGP AS and an OSPF network. The board wants Swiftline depots visible in tracking within four months of closing.

(A) Re-address all Swiftline depots before connecting anything.

(B) Merge Swiftline's OSPF into Kestrel's routing domain immediately to get full reachability.

(C) Restart the whole Kestrel design to include Swiftline from scratch.

(D) Connect Swiftline at a defined boundary with BGP and controlled translation for overlapping ranges, integrate tracking through the API rather than the network, then re-address Swiftline in phases.

Circle your choice. Answer and feedback: module answer key.

SCENARIO · DECIDE · STAGE 5 — SECURITY PRESSURE

A competitor suffers a ransomware attack that stops its sorting hubs for a week. Kestrel's insurer demands, as a condition of renewal in five months, segmentation of operational technology, multi-factor remote access and evidence of monitoring. Depots also accept card payments for cash-on-delivery parcels.

(A) Buy a next-generation firewall for every depot and enable all features.

(B) Separate hub OT from IT with an industrial DMZ, put card terminals in a small isolated payment zone with tokenisation, replace remote VPN access with brokered application access, and add flow telemetry at every zone boundary.

(C) Encrypt all traffic between depots and the data centre.

(D) Wait for the renewal and negotiate a higher premium instead.

Circle your choice. Answer and feedback: module answer key.

The design across the blueprint

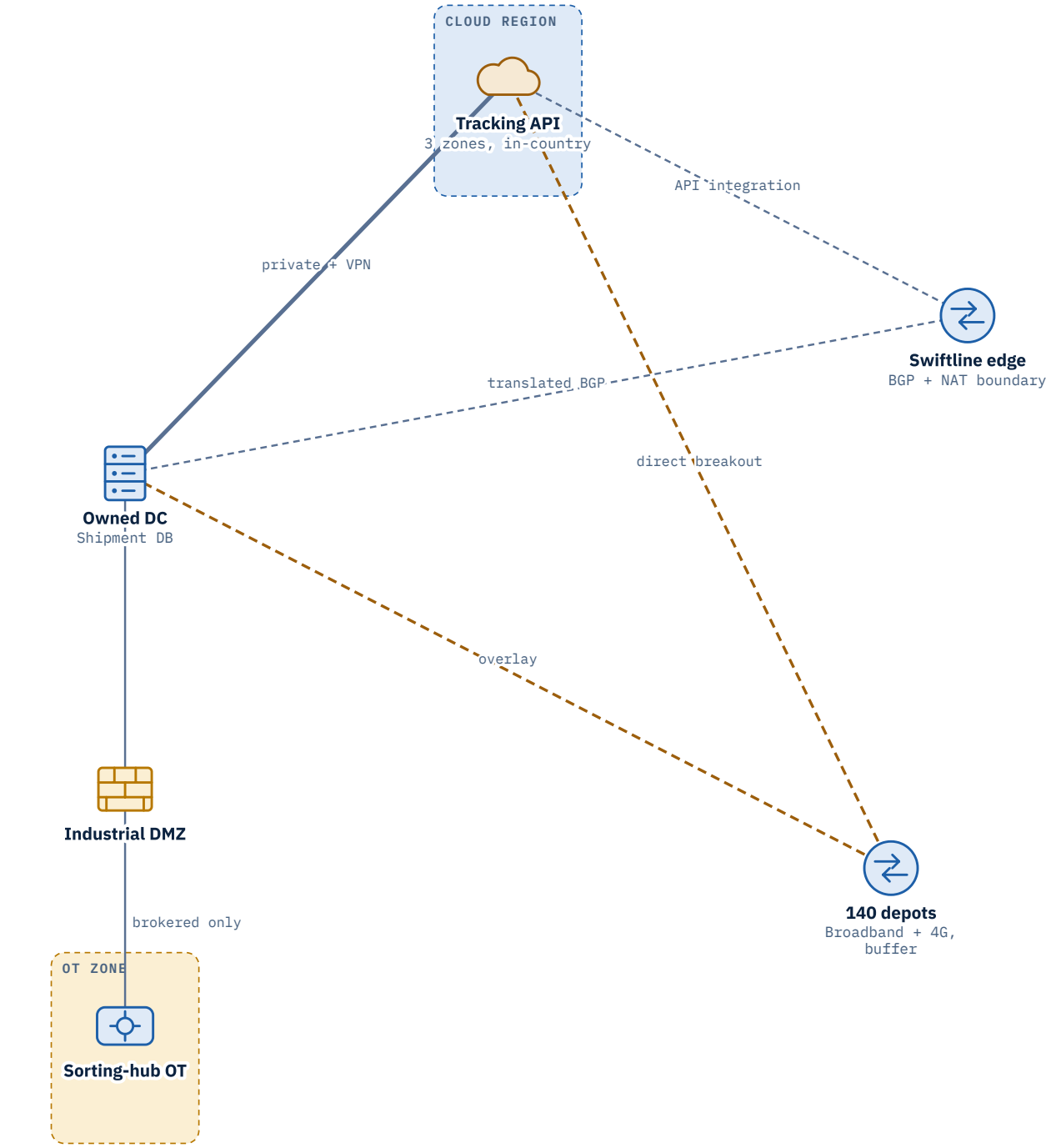
BLUEPRINT SHEET	KEY QUESTION	KESTREL DECISION
Requirements	What must be true, and what cannot change?	99.95% API, four-hour depot survivability, in-country data, fixed capital, small team
Campus	How do campus and hubs contain faults and threats?	Routed access, separate hub OT zones, industrial DMZ

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BLUEPRINT SHEET	KEY QUESTION	KESTREL DECISION
WAN	How do depots stay up and reach the cloud?	Dual-transport overlay, local buffering, direct cloud breakout
Routing	How are domains joined and policies enforced?	BGP boundary to Swiftline, controlled translation, phased re-addressing
Resilience	Where are the failure domains and recovery targets?	API across cloud zones, read replica, DR plan for the data centre
Secure	What can a compromised host reach?	Zoned OT, isolated payment zone, brokered access, boundary telemetry
Cloud	How are cloud resources connected, named and automated?	Private connectivity, in-country region, IaC from day 0, hybrid DNS
Scenario	How do we get there safely and explain it?	Phased waves, rollback per wave, one-page brief to the board

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Kestrel Freight target design



The API runs in an in-country cloud region across zones with a read copy of tracking data; depots buffer locally and break out directly to the API; hub OT sits behind an industrial DMZ; Swiftline joins at one translated BGP boundary.

CASE STUDY · STAGE 6 — THE FINAL BRIEF

Nine months in, the board cuts the remaining network budget by 25% because of a fuel-price shock, and asks whether the tracking API can still launch on time. The API front end is built in the cloud across zones; private connectivity to the data centre is one connection with a VPN backup; depot overlays are deployed at 90 of 140 Kestrel depots; Swiftline is connected at a translated boundary; hub OT segmentation is designed but not yet enforced.

1. Which remaining work items protect the launch, and which can be deferred?

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2. What residual risks would you present to the board?

3. How would you structure the one-page brief?

Model answer: module answer key.

TEMPLATE **Blueprint decision log**

Track each design decision across the blueprint with its rationale and the assumptions that would reopen it, so staged changes can be absorbed cleanly.

SHEET	DECISION	OPTIONS CONSIDERED	RATIONALE	ASSUMPTIONS THAT WOULD REOPEN IT	STAGE DECIDED / REVISED
Cloud	*API front end in in-country region across zones*	*Own DC; second DC; foreign region; two providers*	*Meets 99.95% at operating cost, in-country*	*Region loses in-country status; budget for second DC appears*	*Stage 2*

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Design review checklist

☐ Every requirement and constraint traces to at least one design decision
☐ Every design decision traces back to a requirement or constraint
☐ Failure domains and blast radius meet stated targets
☐ Availability has been calculated, including common-cause effects
☐ A compromised host's reach has been assessed for each zone
☐ Cloud connectivity, addressing and DNS have been designed together
☐ The migration path has rollback points and timed go/no-go decisions
☐ Residual risks are written down with owners
☐ The executive brief fits on one page

REFLECT

At which stage of the Kestrel scenario would you most likely have made a different decision? Write down the assumption behind your alternative and what evidence would settle the difference.

Key terms

Local survivability
The ability of a site to keep operating its critical functions when its WAN connection fails.

Translated boundary
A defined interconnection point where overlapping addresses are translated, keeping overlap contained.

Re-sequencing
Changing the order of work items to protect priorities when constraints change.

Industrial DMZ
A buffer zone between IT and OT networks through which all cross-boundary traffic is brokered.

Read replica
A copy of data kept close to consumers to serve reads without crossing slower or less reliable paths.

Direct cloud breakout
Sending traffic from a site straight to a cloud service rather than through a central data centre.

Traceability
The ability to link each requirement to design decisions and each decision back to requirements.

Decision log
A running record of design decisions, options, rationale and assumptions.

Key takeaways

- Extract drivers, hard constraints and brownfield risks before choosing anything
- Use operating-cost services and local survivability when capital and staff are tight
- Absorb new stages by tracing impact: most earlier decisions should survive
- Under budget pressure, protect hard constraints and the largest risks, and make deferred risk explicit

KNOWLEDGE CHECK

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16. In the Kestrel scenario, why was the tracking data replicated read-only into the cloud?

(A) To move the master shipment database to the cloud immediately

(B) To keep the hybrid link out of the request path while data stays in-country

(C) To allow the owned data centre to be fully shut down before the API launch date

(D) To meet the four-hour depot survivability requirement directly

17. Why was local event buffering the key element of the depot design?

(A) It reduces the bandwidth needed on the depot's broadband line

(B) It allows depots to remove their 4G backup circuits entirely

(C) It satisfies the four-hour offline requirement regardless of the WAN

(D) It is required by the logistics regulator for all shipment records kept at depots

18. When the acquisition was announced, which earlier decisions needed to change?

(A) Mainly addressing and interconnection at the new boundary

(B) Every decision, because the estate size had changed

(C) The cloud region choice for the tracking API front end

(D) The depot local-buffering and overlay design choices

19. Facing a 25% budget cut, why did the recommended plan keep OT segmentation and defer depot overlays?

(A) Depot overlays had already been shown to add no real value to the overall design

(B) OT segmentation was the cheapest item left in the remaining plan

(C) Depot overlays could not be deployed until after the API launch

(D) OT segmentation was an insurer condition and the largest continuity risk

20. Which design-review test best confirms that a design is not gold-plated?

(A) Every component in the design is duplicated at least once

(B) Every site uses the newest available platform from one single supplier

(C) Every link is sized for double its forecast peak traffic

(D) Every design decision traces back to a requirement or constraint

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MODULE 8 · SCENARIO · LESSON 5 · 29 MIN

Exam Strategy & Capstone Introduction

Think like an assessor, answer like an architect, and prepare your capstone design

- Apply a repeatable method to scenario-based design questions
- Recognise common distractor patterns in design options
- Plan the ZCDE capstone deliverables and how they will be reviewed

Scenario-based design questions do not test recall; they test judgement under constraints. Most options are technically valid in some network. Only one is the best answer for **this** organisation, at **this** stage, with **these** constraints. The method below works for exam items, for the capstone and for real design reviews.

Answering a design question

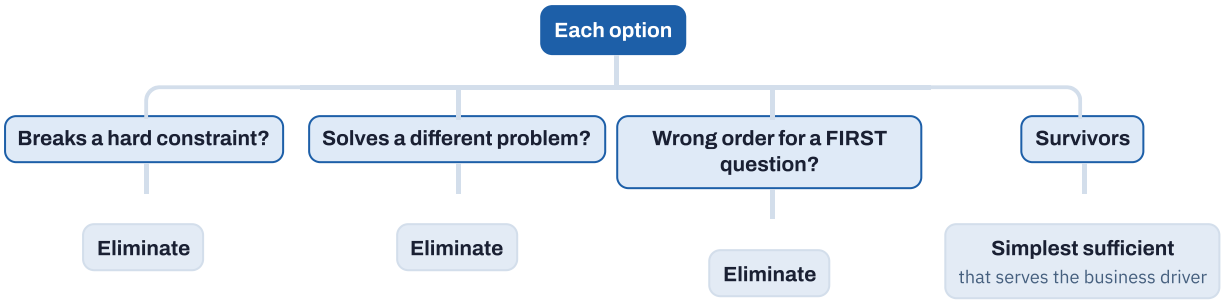
1. **Identify the role and stage** Who is asking, and where are we — requirements, design, migration or operation?
2. **Find the binding constraint** Underline the hard constraint or the new information in the stem. It usually decides the answer.
3. **Eliminate violators** Strike any option that breaks a hard constraint, however attractive it is technically.
4. **Check what is asked** 'Best', 'first', 'most cost-effective' and 'lowest risk' are different questions.
5. **Prefer the simplest sufficient option** Among options that meet requirements, choose the one with less complexity, cost and operational risk.
6. **Confirm against the business driver** Read your choice back against why the project exists.

Common distractor patterns

PATTERN	WHAT IT LOOKS LIKE	WHY IT IS WRONG
Technically superior, constraint-breaking	The most advanced design, but over budget or against regulation	Hard constraints come first
Solves a different problem	A good fix for latency when the issue is availability	Does not address what was asked
Right idea, wrong order	Enforce segmentation before discovering flows	Sequence matters in design and migration
Over-engineered	Five nines for a service that needs three	Adds cost and complexity without business value
Product-led	'Buy platform X' without design work	Tools do not replace standards, data and decisions
Operationally unrealistic	A design the team cannot run	Skills and operations are constraints too

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Eliminating distractors



Apply the tests in order; most items leave one option standing after the first three. If two survive, the business driver breaks the tie.

Exam-day habits

E1
Read the last sentence first
Know what is being asked before reading the scenario detail.

E2
Keep a mini log
In staged scenarios, jot down constraints and decisions as they appear.

E3
Do not re-fight earlier stages
Accept the scenario's earlier decisions unless the new information changes them.

E4
Watch the qualifiers
First, best, least, most and without interruption change the answer.

E5
Budget your time
Do not let one hard item consume time needed for several easier ones.

E6
Trust business logic
When unsure between two technical answers, pick the one that better serves the stated business driver.

SCENARIO · DECIDE · PRACTICE ITEM

A hospital's design brief states that clinical systems must never be interrupted, the network team is four people, and the budget allows one of four projects this year. Ransomware has hit two nearby hospitals. Which project should the designer recommend FIRST?

(A) Segment medical devices and clinical systems from corporate and guest networks, starting in monitor mode.
(B) Replace the campus core with a new high-capacity fabric.
(C) Move all clinical systems to a public cloud region.
(D) Deploy full packet capture on every link for forensic readiness.

Circle your choice. Answer and feedback: module answer key.

INSIGHT · THE CAPSTONE IS A DESIGN DEFENCE

The ZCDE capstone asks you to take one realistic scenario from brief to executive decision and defend your choices. Reviewers look for traceability, quantified trade-offs, honest residual risk and a safe migration path — not for the most elaborate topology.

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Capstone deliverables

Design pack

- Requirements register: business, technical, constraints, assumptions
- Weighted decision matrices for three major choices
- Campus and data-centre design with scaling limits
- WAN/SD-WAN and internet-edge design
- Routing design (IGP and BGP) with convergence goals

Delivery and defence

- Resilience design with availability calculations and RPO/RTO
- Security and segmentation design
- Hybrid/multi-cloud connectivity and automation-driven operations
- Phased migration plan with rollback and risk register
- Ten-slide executive design presentation, then defend your three hardest trade-offs

Capstone process

1. **Choose a scenario** Use a provided ZCDE scenario or an anonymised real one with enough complexity to touch all eight sheets.
2. **Frame requirements** Build the requirements, constraints and assumptions register; agree it before designing.
3. **Design sheet by sheet** Work through Campus, WAN, Routing, Resilience, Secure and Cloud, logging each decision.
4. **Stress-test** Inject at least one change — budget, acquisition, regulation — and show how the design adapts.
5. **Plan the migration** Sequence waves with rollback and timing.
6. **Brief and defend** Present the ten-slide executive presentation and answer reviewers' challenges.

A suggested capstone schedule



WEEKS 1–2
Scenario and requirements register
Agree constraints and assumptions

WEEKS 3–5
Design sheet by sheet
Decision log and matrices

WEEK 6
Stress-test
Inject a change; trace its impact

WEEK 7
Migration plan
Waves, rollback, go/no-go timing

WEEK 8
Brief and defend
Ten slides and three hardest trade-offs

Illustrative eight-week plan for a part-time candidate. Agreeing the requirements register before designing, and leaving time to stress-test, matter more than the exact weeks.

TEMPLATE Capstone review scorecard

Self-assess and peer-review a capstone design against the criteria reviewers use, with evidence for each score.

CRITERION	WEIGHT	EVIDENCE IN THE DESIGN PACK	SCORE (1–5)	REVIEWER COMMENT
Traceability of requirements to decisions	*20%*	*Requirements register cross-referenced in every decision table*	*4*	*Two cloud decisions lack a requirement reference*
Quantified resilience	*15%*	*Availability sheet with common-cause factors; DR targets*	*3*	*β values are assumed — state the evidence*

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CRITERION	WEIGHT	EVIDENCE IN THE DESIGN PACK	SCORE (1–5)	REVIEWER COMMENT

Capstone readiness checklist

- ☐ The scenario touches all eight blueprint sheets
- ☐ Requirements, constraints and assumptions are agreed before design starts
- ☐ Every decision table lists options considered and rationale
- ☐ Availability and cost figures are calculated, not asserted
- ☐ At least one injected change is absorbed with a traced impact
- ☐ The migration plan has timed go/no-go points and rollback
- ☐ The executive brief fits on one page and names residual risk
- ☐ You can defend your three hardest trade-offs in two minutes each

REFLECT

Which of the six distractor patterns are you personally most likely to fall for? Write one question you will ask yourself on exam day to catch it.

Key terms

Binding constraint
The constraint in a scenario that most directly determines the correct design choice.

Gold-plating
Adding capability beyond what requirements justify.

Capstone
A culminating design project that applies the whole programme to one scenario.

Distractor
An answer option that is plausible but not the best answer for the scenario.

Qualifier
A word such as first, best or least that changes what a question is asking.

Design defence
Explaining and justifying design decisions under questioning by reviewers.

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Requirements register
A structured list of requirements, constraints and assumptions with sources and owners.

Scorecard
A weighted set of criteria used to assess a design consistently.

Key takeaways

- Scenario questions test judgement under constraints, not recall
- Find the binding constraint, eliminate violators, check the qualifier, then choose the simplest sufficient option
- Distractors are often technically superior, out of order or operationally unrealistic
- The capstone is a design defence: traceability, quantified trade-offs, honest residual risk and a safe migration

KNOWLEDGE CHECK

21. A question asks for the FIRST thing a designer should do after a new regulation is announced mid-project. Three options are good long-term designs. What does the qualifier suggest?

(A) Choose the step that establishes impact before committing to changes
(B) Choose the most complete long-term design among all of the options given
(C) Choose the option with the most security features listed in it
(D) Choose the option that the operations team is most familiar with

22. An option proposes the most advanced architecture in the question but exceeds the stated budget. How should it be treated?

(A) Selected, because it gives the strongest technical result
(B) Eliminated, because it violates a hard constraint
(C) Kept as a tie-breaker if two other options seem equal
(D) Selected only if the scenario mentions future growth

23. Two options both meet every requirement. One uses three technologies the team does not know; the other uses one they run today. Which is usually best?

(A) The option with three new technologies
(B) Whichever option is newer in the market
(C) The simpler option the team can operate
(D) Neither, since both need further study

24. What do capstone reviewers primarily look for?

(A) The largest number of redundant devices and links shown in the final topology
(B) Use of the newest technologies available from leading suppliers in 2026
(C) Traceability, quantified trade-offs, honest residual risk, safe migration
(D) A design that requires no changes to be made to the existing network

25. In a staged scenario, new information arrives that affects only addressing. What is the best exam behaviour?

(A) Revise only the affected decisions and keep the rest
(B) Re-answer every earlier stage in light of the change
(C) Ignore the new information to keep answers consistent
(D) Choose the option that changes the most components

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MODULE 8 · SCENARIO

Answer key

Knowledge checks

1. D Regulation is non-negotiable without a formal decision; supplier preference is a soft constraint that gives way when they conflict.

2. C Tracing impact keeps unaffected decisions stable and focuses effort on addressing, routing and segmentation decisions that change.

3. A The phrase usually signals a zero-downtime migration constraint, which shapes the migration strategy more than component choice.

4. C Anchoring is clinging to the first answer. Redesigning everything is thrashing, while recording assumptions and phasing are healthy practices.

5. A Assumptions are the hooks that connect new information to affected decisions, enabling targeted change rather than wholesale redesign.

6. D 06:00 − 40 minutes − 20 minutes = 05:00. After that, a rollback could not complete inside the window with margin.

7. B Parallel running lets traffic move gradually and back again, which suits zero-interruption requirements when both platforms can serve users.

8. B The feature presents the old AS to the customer and hides the extra AS from paths, so customer configuration can move later.

9. A Running both with one preferred allows validation of the new protocol's results and a controlled, reversible switch.

10. D Once rollback is impossible, failure must be fixed forward. A formal gate ensures the organisation consciously accepts that risk.

11. C SLE = 50,000 × 10 = 500,000. ARO = 0.2. ALE = 500,000 × 0.2 = 100,000.

12. D Executives need to know quickly why they are reading and what they must decide; detail supports the decision, it does not lead.

13. A Doing nothing has its own cost and risk. Showing it makes the value of change explicit.

14. C The executive translation states the business consequence plainly. The others are accurate but technical.

15. B No design is risk-free. Naming residual risk and its management builds credibility and supports an informed decision.

16. B A read copy near the API removes per-request dependence on the hybrid connection, raising availability, while the region keeps data in-country.

17. C Only local operation meets a requirement to keep working without any WAN. More transport paths reduce the chance of outage but cannot guarantee it.

18. A Tracing the change shows that cloud and depot decisions still hold; only addressing, routing interconnection and integration were affected.

19. D Re-sequencing protects hard constraints and the largest risks first; the deferred depots still have a working, if less resilient, design.

20. D A decision with no requirement behind it is cost without purpose. Traceability in both directions catches gaps and gold-plating.

21. A 'First' asks about sequence. Understanding the regulation's impact on existing decisions comes before redesign.

22. B A design that breaks a hard constraint is wrong for that scenario, regardless of technical merit.

23. C Among sufficient options, lower complexity and operational risk usually win; skills are a real constraint.

24. C The capstone is a design defence. Elaborate topologies without traceability or risk honesty score poorly.

25. A Targeted revision mirrors expert practice: absorb the change where it applies and keep sound decisions stable.

Scenarios & case studies

> **Scenario · The budget cut**
> **B** is best. Best. It meets this year's target within budget, keeps the path to next year's target, and makes the accepted risk explicit.

> **Scenario · 03:40 and still failing**
> **A** is best. Best. The decision time is 03:50; at 03:40 the agreed criteria are not met. Rolling back preserves the service and the team's credibility.

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Case study · Telecom merger migration
Strategy: phased by point of presence and service, after an initial interconnection phase — big bang is impossible at this scale and the one-second constraint rules out uncontrolled cut-overs. **Interconnect first:** join the cores at two or more locations using inter-AS techniques so services can span both networks during coexistence. **AS consolidation:** choose AS 64500 as the target; migrate Operator B's PE routers to it while using **local-as with no-prepend and replace-as** so customers keep peering with 64510 unchanged; then move customers to the new AS gradually in agreed windows. **Interior routing:** run IS-IS alongside OSPF on B's routers ('ships in the night') with IS-IS less preferred, verify that both produce the same forwarding, then change preference router by router and remove OSPF; train B's staff before the change, not after. **Addressing:** allocate a new, unique infrastructure range for the merged network; re-address loopbacks and management area by area, using temporary translation only at the management boundary, never in the forwarding core. **Emergency path:** migrate it last, with fast-reroute protection verified in the new core, BFD on all affected links, a rehearsed sub-second failover test, and a cut-over that moves traffic by make-before-break tunnels so the old path remains until the new one is proven. **Governance:** a single change calendar and freeze across both NOCs from the first day of interconnection.

Scenario · The CFO's question
C is best. Best. It answers in the CFO's terms — expected loss versus cost — and adds the non-financial factor that affects the decision.

Scenario · Stage 2 — Where should the tracking API live?
C is best. Best. Multiple zones give independent failure domains for the API at operating cost, the read copy removes the hybrid link from the request path, and data stays in-country.

Scenario · Stage 3 — Depots and the WAN
A is best. Best. Local buffering meets the survivability requirement by design, two transport paths improve availability, and direct cloud breakout avoids hairpinning through the data centre.

Scenario · Stage 4 — New information: an acquisition
D is best. Best. It meets the four-month goal through application integration, contains overlap at one boundary, and keeps a planned path to a clean addressing model. Most earlier decisions stand unchanged.

Scenario · Stage 5 — Security pressure
B is best. Best. It addresses each insurer demand, reduces payment-card scope, and fits the team through a small number of well-placed controls.

Case study · Stage 6 — The final brief
Protect the launch: the API path (cloud zones, read replica, private connection with VPN backup) is already in place, so the launch date holds. Complete hub OT segmentation enforcement because it is an insurer condition and the largest business-continuity risk. **Defer:** the remaining 50 depot overlays move to next year's budget — those depots keep broadband with 4G backup, and local event buffering is prioritised for them because it is cheap and protects tracking accuracy; Swiftline re-addressing stays behind its translated boundary for now. **Residual risks:** the 50 depots without the overlay have lower WAN resilience; the single private connection keeps the API's dependency on the data centre path for writes; translation at the Swiftline boundary adds operational complexity. **One-page brief:** context (budget cut, launch at risk?); decision required (approve re-sequenced plan); options (keep scope and slip launch; re-sequence and launch on time; cut security work); recommendation (re-sequence: launch on time, keep insurer conditions, defer depot overlays); cost and residual risk; ask (approval and a named business owner for depot operations risk).

Scenario · Practice item
A is best. Best. It addresses the dominant risk, fits a small team when phased, and can be introduced without interrupting clinical systems.

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REFERENCE

Glossary

2

2N A redundancy model with two complete, independent systems, each able to carry the full load. (Resilience)

8

802.1X An IEEE standard for port-based network access control using an authentication server. (Secure)

9

95th percentile The utilisation value exceeded only 5% of the time in a sample; a common basis for sizing that ignores brief spikes but captures sustained load. (Requirements)

A

Access layer The campus layer that connects endpoints and wireless access points and applies edge policy such as authentication and QoS marking. (Campus)

Active-active A design in which multiple sites serve production traffic simultaneously. (Campus)

Active-standby A design in which one site serves production while another is held ready to take over after failover. (Campus)

Active/active A redundancy model in which all members carry traffic in normal operation. (Resilience)

Active/standby A redundancy model in which one member carries traffic and another waits to take over. (Resilience)

ADD-PATH A BGP capability allowing more than one path for the same prefix to be advertised to a peer. (Routing)

Address plan A hierarchical allocation of address blocks to regions, sites and functions that makes summarisation possible. (Routing)

Administrative preference A router-local value (often called administrative distance) that ranks routes to the same prefix from different sources. (Routing)

Anchoring Holding on to an initial design or estimate despite new information that should change it. (Scenario)

Annualised loss expectancy The expected yearly cost of a risk: single loss expectancy multiplied by annual rate of occurrence. (Scenario)

Annualised rate of occurrence The expected number of times a risk event occurs per year. (Scenario)

Anti-spoofing filter A filter that drops packets whose source addresses could not legitimately arrive on that interface. (WAN)

Anycast gateway The same gateway IP and MAC address configured on every edge switch so endpoints find their gateway locally wherever they connect. (Campus)

Application-aware routing Selecting a path per application based on measured loss, latency and jitter against policy thresholds. (WAN)

Area border router (ABR) An OSPF router connecting the backbone to one or more other areas, where inter-area summarisation is configured. (Routing)

AS-path prepending Adding extra copies of your AS number to an advertisement to make a path look longer and less attractive. (WAN)

Assumption A statement treated as true for planning but not yet verified; each one is a potential risk until validated. (Requirements)

Asymmetric routing A condition in which the two directions of a flow take different paths. (Secure)

Asynchronous replication Replication in which writes are acknowledged locally and copied to the remote site afterwards. (Resilience)

Attached bit An IS-IS flag set by L1/L2 routers that tells L1 routers to use them as the default exit from the area. (Routing)

Availability zone A physically separate location within a cloud region with independent power, cooling and networking. (Cloud)

B

Back-to-back firewalls A DMZ placed between an outer and an inner firewall tier for defence in depth. (WAN)

Backbone area OSPF area 0, to which all other areas must connect and through which inter-area traffic passes. (Routing)

Bandwidth-delay product The amount of data in flight required to keep a path full: bandwidth multiplied by round-trip time. (WAN)

Baseline A measured snapshot of the current network — inventory, traffic, health and operations — used as the reference for design and comparison. (Requirements)

Beta factor (β) The fraction of a component's failures that are common-cause and affect redundant members simultaneously. (Resilience)

BFD Bidirectional Forwarding Detection: a lightweight protocol that detects forwarding failures between neighbours in milliseconds. (Routing)

BGP community A tag attached to routes that providers and peers use to apply agreed policy, such as changing preference or blackholing. (WAN)

BGP-only fabric A data-centre design using eBGP on every fabric link instead of an IGP. (Routing)

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Big bang migration Moving everything to the new design in a single cut-over. (Scenario)

Binding constraint The constraint in a scenario that most directly determines the correct design choice. (Scenario)

Black hole A condition in which traffic is attracted to a router that cannot deliver it, usually because a summary hides a failed component. (Routing)

Blast radius The business-visible scope of a failure: users, sites, transactions or functions affected and for how long. (Resilience)

Border leaf A leaf that connects the fabric to external networks and services rather than to servers. (Campus)

Break-glass access An emergency manual access path that bypasses normal automation under strict audit. (Cloud)

Brownfield An environment with existing systems that a new design must coexist with or migrate from. (Scenario)

BUM traffic Broadcast, unknown-unicast and multicast traffic that must be flooded in a layer 2 segment. (Campus)

Business requirement An outcome or capability the organisation needs, expressed in business terms and owned by a business stakeholder. (Requirements)

C

Campus block A repeatable unit of access and distribution switches serving a building or zone. (Campus)

Canary The first small target of a change, monitored closely before the change spreads. (Resilience)

Capstone A culminating design project that applies the whole programme to one scenario. (Scenario)

Cardholder data environment The systems that store, process or transmit payment-card data, plus those connected to or able to affect them. (Secure)

Cell architecture A design that partitions a service into independent, self-contained units so a fault affects only one unit. (Resilience)

Change of authorisation A message from the policy server that changes an active session's access without reconnecting. (Secure)

Change wave An ordered group of devices or domains that receives a change together. (Resilience)

Clos network A multistage switching topology that builds large, non-blocking capacity from many smaller switches. (Campus)

Cluster ID An identifier used by route reflectors to detect and discard reflected routes that loop back to their cluster. (Routing)

Coexistence The intermediate state in which old and new designs interoperate during migration. (Scenario)

Collapsed core A two-tier design where core and distribution functions run on the same pair of switches. (Campus)

Common-mode failure A single cause, such as a software defect or a bad change, that defeats all redundant members at once. (Resilience)

Concentration risk The exposure created when a critical capability depends on a single provider. (Cloud)

Conditional forwarding Forwarding DNS queries for specific domains to designated resolvers. (Cloud)

Constraint A fixed limit on the design — budget, policy, regulation, existing assets, skills or time — that removes options rather than weighting them. (Requirements)

Containment Design choices that stop a fault from propagating beyond its origin, such as routed boundaries or summarisation. (Resilience)

Control-plane policing Rate-limiting traffic destined to a device's control plane so floods cannot exhaust its CPU. (Resilience)

Convergence The process by which all routers reach a consistent, correct forwarding state after a change. (Routing)

Core layer The high-speed transport layer that interconnects distribution blocks with minimal policy and state. (Campus)

Coverage factor The proportion of failures that are detected and recovered automatically by the redundancy mechanism. (Resilience)

D

Data centre interconnect (DCI) The network connecting two or more data centres for replication, workload mobility and resilience. (Campus)

Decision log A record of design decisions with options considered, rationale and dependent assumptions. (Scenario)

Decision owner The person accountable for making and standing behind a decision. (Scenario)

Declarative Describing the desired end state and letting a tool determine the steps to reach it. (Cloud)

Dedicated internet access (DIA) An internet circuit with guaranteed bandwidth and an SLA up to the provider edge. (WAN)

Default deny A policy stance in which anything not explicitly permitted is blocked. (Secure)

Design authority The person or board that reviews and approves design decisions and their exceptions. (Requirements)

Design brief A short document that frames a design decision for decision-makers. (Scenario)

Design decision record A short, structured document capturing one design decision: context, options, choice, rationale, consequences and review trigger. (Requirements)

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Design defence Explaining and justifying design decisions under questioning by reviewers. (Scenario)

Design exception A documented, approved deviation from a standard design, recorded with its owner and justification. (Requirements)

Digital twin A model of the production network used to test changes before deployment. (Cloud)

Direct cloud breakout Sending traffic from a site straight to a cloud service rather than through a central data centre. (Scenario)

Discard route A route to a null interface installed for a summary, so packets for unused parts of it are dropped rather than looping. (Routing)

Distractor An answer option that is plausible but not the best answer for the scenario. (Scenario)

Distribution layer The layer that aggregates access switches into a block, often hosts the L2/L3 boundary and summarises routing information. (Campus)

DMZ A network zone for services reachable from untrusted networks, separated by filtering from both the internet and internal networks. (WAN)

Do-nothing option The baseline option of making no change, against which other options are compared. (Scenario)

Drain Gracefully moving traffic away from a device or link before work so that removing it causes no loss. (Resilience)

Drift A difference between a system's running configuration and its intended, codified state. (Cloud)

Dynamic tunnel An overlay tunnel created on demand between two sites when traffic needs a direct path, and removed after it is idle. (WAN)

E

East-west traffic Traffic between servers, services or storage inside the data centre. (Campus)

ECMP Equal-cost multipath: forwarding flows across several equal-cost routes at once, typically by hashing packet headers. (Campus)

Egress charge A fee charged by a cloud provider for data leaving its network or crossing certain boundaries. (Cloud)

Elicitation The structured activity of drawing out requirements from stakeholders, documents and data. (Requirements)

Exchange point A colocation facility where enterprises, carriers and cloud providers interconnect. (Cloud)

F

Failure domain The part of the network affected when a single component fails or is misconfigured. (Requirements)

Fast reroute Pre-computed backup paths installed in advance so a router can switch locally as soon as a failure is detected. (Routing)

First-hop redundancy protocol A protocol, such as VRRP, that lets two or more routers share a virtual gateway address for endpoints. (Campus)

Flooding domain The set of routers that receive each other's link-state advertisements and share one topology database. (Routing)

Flow log A record of network flows produced by a cloud platform or device, without payload. (Cloud)

Forward secrecy A property of key exchange that prevents past sessions from being decrypted even if long-term keys are later exposed. (Secure)

G

Gateway endpoint A route-table-based path to a provider service that keeps traffic off the internet and NAT. (Cloud)

GEO satellite Geostationary satellite service with wide coverage but long round-trip delay because of orbital distance. (WAN)

Global server load balancing Health-aware DNS that directs clients to the best available site. (Resilience)

Go/no-go decision A timed decision to commit or roll back a change based on agreed criteria. (Scenario)

Gold-plating Adding capability beyond what requirements justify. (Scenario)

Graceful restart A mechanism in which neighbours keep forwarding through a restarting router while its control plane recovers. (Resilience)

Graceful shutdown community A well-known BGP community (65535:0) that asks receivers to lower the preference of routes before a session closes. (Resilience)

Group tag A label carried with traffic or bound to an endpoint that lets policy be written by group rather than by IP address. (Secure)

Guardrail An automated policy check that blocks changes violating agreed standards. (Cloud)

H

Hairpinning Sending traffic to a central device and back, often across the same links, to apply a control. (Secure)

Hard constraint A limit that cannot be traded away without an explicit decision by the sponsor, such as law or a budget ceiling. (Scenario)

Hidden priority The requirement in a scenario that outranks the others — often regulatory, deadline-driven or skill-limited — and decides between otherwise valid options. (Requirements)

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Hierarchy Assigning clear roles to layers and aggregating at their boundaries to control state, traffic paths and failure domains. (Requirements)

High-level design (HLD) A document describing the architecture, principles, major decisions and their rationale for sponsors, architects and operations leads. (Requirements)

Hub site A data centre or regional site where many spoke tunnels terminate and shared services or inspection are provided. (WAN)

Hypercare A period of heightened support immediately after a migration wave. (Scenario)

I

iBGP full mesh The requirement that every iBGP speaker peer with every other, because routes learned via iBGP are not re-advertised to iBGP peers by default. (Routing)

Idempotent Producing the same result no matter how many times an operation is applied. (Cloud)

IGP Interior gateway protocol: a routing protocol used within one administrative domain, such as OSPF or IS-IS. (Routing)

Inbound resolver endpoint An address inside a cloud network that accepts DNS queries from outside, typically from on-premises resolvers. (Cloud)

Industrial DMZ A buffer zone between IT and OT networks through which all cross-boundary traffic is brokered. (Scenario)

Infrastructure as code Defining infrastructure in version-controlled files that tools apply to create or change resources. (Cloud)

Intent-based networking An approach in which business intent is translated into configuration and continuously verified. (Cloud)

IPAM IP address management: the system of record for address allocations, ideally consumed by automation. (Cloud)

IPsec A suite of protocols that authenticates and encrypts IP packets between peers or gateways. (Secure)

IS-IS Intermediate System to Intermediate System: a link-state IGP that runs directly over layer 2 and uses two levels of hierarchy. (Routing)

J

Jitter Variation in packet delay; real-time applications absorb it with a jitter buffer that adds delay. (WAN)

L

L2/L3 boundary The point where switched (layer 2) forwarding ends and routing begins, usually where the endpoint's default gateway lives. (Campus)

Lateral movement An attacker's progression from an initially compromised host to other systems inside the network. (Secure)

LEO satellite Low-Earth-orbit satellite service with much lower latency than geostationary systems, typically tens of milliseconds. (WAN)

LFA Loop-free alternate: a precomputed backup next hop that will not loop traffic back through the failed element. (Resilience)

Link-state protocol A protocol in which routers flood descriptions of their links so every router builds the same topology map and computes shortest paths. (Routing)

Local internet breakout Sending internet-bound traffic directly from a branch rather than backhauling it to a central site. (WAN)

Local preference A BGP attribute, local to an AS, that sets the preferred exit; higher is preferred. (WAN)

Local survivability The ability of a site to keep operating its critical functions when its WAN connection fails. (Scenario)

Loop-free alternate (LFA) A neighbour that can be used as a backup next hop because its shortest path to the destination does not loop back through the failing router. (Routing)

Low-level design (LLD) A detailed document specifying configuration standards, values and procedures for those who build and operate the network. (Requirements)

M

MAC authentication bypass Admitting a device based on its MAC address when it cannot perform 802.1X. (Secure)

Macro-segmentation Separation of traffic into isolated routing domains, such as VRFs or virtual networks. (Campus)

MACsec IEEE 802.1AE link-layer encryption that protects Ethernet frames hop by hop. (Secure)

Max-metric An OSPF technique that advertises the highest metric on transit links so other routers avoid the sender. (Resilience)

MED Multi-exit discriminator: a BGP attribute suggesting the preferred entry among several links to the same neighbouring AS. (WAN)

Micro-loop A transient forwarding loop caused by routers updating their FIBs at slightly different times. (Resilience)

Micro-segmentation Policy between groups of endpoints within the same routing domain, often based on group tags or identity. (Campus)

MKA MACsec Key Agreement: the protocol that establishes and rotates MACsec keys between peers. (Secure)

Modularity Building the network from repeatable, self-contained blocks so change and failure are contained and growth is predictable. (Requirements)

MoSCoW A prioritisation scheme dividing requirements into Must, Should, Could and Won't-have-this-time. (Requirements)

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MPLS L3VPN A provider service that carries customer routes in a private routed network, keeping each customer's traffic separate and supporting QoS end to end. (WAN)

MSS clamping Rewriting the TCP maximum segment size so segments fit within a tunnel's reduced MTU. (Secure)

MTBF Mean time between failures: the average operating time between failures of a repairable component. (Resilience)

MTTR Mean time to restore: the average time from failure to restored service, including detection, travel, repair and verification. (Resilience)

Multi-cloud Using two or more public cloud providers for production workloads. (Cloud)

Multihoming Connecting a network to more than one provider or provider location for resilience or traffic control. (WAN)

Mutual redistribution Redistributing in both directions between two routing domains. (Routing)

Mutual TLS TLS in which both client and server present certificates, giving both sides a verified identity. (Secure)

N

N+1 A redundancy model with one spare unit beyond what peak load requires. (Resilience)

Network detection and response Tools that analyse network traffic and metadata to detect and investigate threats. (Secure)

Non-blocking A design whose uplink capacity equals or exceeds downlink capacity (1:1), so any traffic pattern can be carried at full rate in theory. (Campus)

Non-functional requirement A requirement about how well the network operates — availability, manageability, security, scalability — rather than what it connects. (Requirements)

Non-stop routing A mechanism in which a standby route processor keeps routing sessions alive so neighbours never see a restart. (Resilience)

Non-transitive peering A peering model in which routes learned over one peering are not passed on to another. (Cloud)

North-south traffic Traffic entering or leaving the data centre, to users, the WAN or the internet. (Campus)

NSSA Not-so-stubby area: a stub-like OSPF area that may inject its own external routes, which the ABR translates for the rest of the domain. (Routing)

O

Observability The ability to understand a system's internal state from the telemetry it produces. (Cloud)

Operability How easily and safely a network can be monitored, changed and repaired by the team that runs it. (Requirements)

Optical tap A passive device that copies light from a fibre link for monitoring without affecting the link. (Secure)

OSPF Open Shortest Path First: an IP-based link-state IGP using areas around a backbone area 0. (Routing)

Overlay A logical network built on top of the underlay using encapsulation, such as VXLAN. (Campus)

Overload bit An IS-IS flag that tells other routers not to use the sender for transit traffic. (Resilience)

Oversubscription ratio The ratio of total downstream port capacity to total upstream capacity at an aggregation point. (Campus)

P

Packet broker A device that aggregates, filters and distributes copies of traffic to security and monitoring tools. (Secure)

Parallel run Operating old and new systems side by side while traffic or users move between them. (Scenario)

PE-CE routing The routing protocol or static routing between a customer edge router and the provider edge in a provider VPN. (WAN)

Phased migration Moving to a new design in a sequence of smaller waves. (Scenario)

Physical diversity Separation of circuits along their entire physical route — ducts, building entries and exchanges — not just different providers. (WAN)

Playback Presenting the analysed requirements back to stakeholders to confirm understanding before design begins. (Requirements)

Point of no return The step after which rollback is no longer possible or practical. (Scenario)

Policy decision point The logical component that evaluates policy and decides whether access is granted, denied or revoked. (Secure)

Policy enforcement point The component in the traffic path that enables, monitors and terminates connections as instructed by the decision point. (Secure)

Port mirroring A switch feature that copies traffic from ports or VLANs to a monitoring port. (Secure)

Prefix-independent convergence A FIB structure in which many prefixes share a next-hop object, so one update reroutes them all. (Resilience)

Private dedicated connectivity A dedicated physical or partner-provided connection between an enterprise network and a cloud provider's edge. (Cloud)

Private endpoint An interface with a private address that exposes a provider or partner service inside a virtual network. (Cloud)

Private zone A DNS zone whose records are visible only to associated private networks. (Cloud)

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Processing charge A per-GB fee for traffic passing through a managed cloud network service. (Cloud)

Profiling Identifying device type from observed attributes such as DHCP options, traffic and certificates. (Secure)

Provider-independent addressing Address space assigned directly to an organisation so it can be announced through any provider. (WAN)

Q

Qualifier A word such as first, best or least that changes what a question is asking. (Scenario)

R

Re-sequencing Changing the order of work items to protect priorities when constraints change. (Scenario)

Read replica A copy of data kept close to consumers to serve reads without crossing slower or less reliable paths. (Scenario)

Recovery point objective (RPO) The maximum acceptable amount of data loss, measured in time, after a failure. (Campus)

Redistribution Injecting routes learned from one routing protocol or source into another. (Routing)

Representative period A measurement window long enough to include the business cycles, such as month-end or seasonal peaks, that drive demand. (Requirements)

Requirement owner The stakeholder accountable for a requirement, who can confirm, change or trade it off. (Requirements)

Requirements register A structured list of requirements, constraints and assumptions with sources and owners. (Scenario)

Requirements traceability The ability to link each design decision back to the requirement or constraint that justified it, and forward to how it is verified. (Requirements)

Residual risk The risk that remains after a design or control is in place. (Scenario)

Reverse proxy A server that accepts client connections on behalf of back-end applications and relays requests to them. (Secure)

Review trigger A condition recorded with a decision that, if met, requires the decision to be revisited. (Requirements)

Rollback Returning to the previous known-good state after a failed change. (Scenario)

Rolling upgrade Upgrading members of a redundant system one at a time while the others carry traffic. (Resilience)

Route leaking Advertising selected prefixes from the IS-IS level 2 backbone into a level 1 area so L1 routers choose better exits. (Routing)

Route origin authorisation (ROA) A signed record stating which AS may originate a prefix, used by others to validate route origins. (WAN)

Route reflector An iBGP speaker allowed to re-advertise iBGP routes to its clients, removing the need for a full mesh. (Routing)

Route summarisation Advertising one aggregate prefix in place of many more specific prefixes it contains. (Routing)

Route tag A value attached to a route, often at redistribution, used by policy to recognise its origin. (Routing)

Routed access A campus design in which access switches route, using layer 3 point-to-point uplinks instead of trunked VLANs. (Campus)

Routing domain A set of routers running one routing protocol instance under a single administrative policy. (Routing)

RPO Recovery point objective: the maximum tolerable data loss, expressed as a period of time. (Resilience)

RTBH Remotely triggered blackholing: signalling a provider, typically with a BGP community, to discard all traffic to a specific prefix. (WAN)

RTO Recovery time objective: the maximum tolerable time to restore service after a disaster. (Resilience)

S

Scope creep Uncontrolled growth of requirements after the baseline is agreed, usually because exclusions were never written down. (Requirements)

Scope reduction Design measures that reduce the number of systems subject to a regulatory obligation. (Secure)

Scorecard A weighted set of criteria used to assess a design consistently. (Scenario)

Scrubbing Filtering attack traffic out of a traffic stream in a provider or cloud network and forwarding the clean traffic. (WAN)

SD-WAN A software-defined overlay that builds encrypted tunnels across multiple transports and steers applications by policy and measured path quality. (WAN)

Security group A stateful, cloud-native firewall applied to workloads or interfaces. (Cloud)

Segmentation testing Testing that proves out-of-scope networks cannot reach a protected zone. (Secure)

Sensitivity analysis Varying weights or scores to see whether the preferred option changes; a fragile result needs explicit discussion. (Requirements)

Serialisation delay The time needed to place a frame's bits onto a link; it depends on frame size and link rate. (WAN)

Service-level objective A measurable target for a service, such as latency or availability, over a defined period. (Cloud)

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Shared fate A condition in which supposedly independent components fail together because of a common dependency. (Resilience)

Shared risk link group (SRLG) A set of links that share a physical resource, such as a duct or fibre, and therefore fail together. (Resilience)

Ships in the night Running two routing protocols in parallel, with one preferred, to migrate between them. (Scenario)

Showback Reporting cost to the teams that cause it, without necessarily charging them. (Cloud)

Signal word A word or phrase in a brief that indicates a constraint or priority, such as must or without interruption. (Scenario)

Single loss expectancy The expected cost of one occurrence of a risk event. (Scenario)

Site tier A standard site design defining edge count, transports, resilience target and policy for a class of locations. (WAN)

Soak time A deliberate waiting period after a change, under production load, before proceeding. (Resilience)

Soft constraint A preference that can be traded when it conflicts with more important requirements. (Scenario)

Source of truth The authoritative data store from which network configuration is generated. (Cloud)

SPF throttling Timers that delay and back off shortest-path computations to balance fast reaction against stability. (Resilience)

Spine-leaf A two-tier folded Clos topology in which every leaf connects to every spine, giving equal-cost paths between any two leaves. (Campus)

Split brain A failure in which two sites or cluster members each believe they own the same resource after losing contact with each other. (Campus)

Split-brain A condition in which isolated sites both act as primary, producing conflicting state. (Resilience)

Split-horizon DNS Returning different answers for the same name depending on where the query comes from. (Cloud)

Stakeholder Any person or group affected by, or able to influence, the design and its outcome. (Requirements)

Streaming telemetry Push-based, structured export of device data at high frequency or on change. (Cloud)

Stub area An OSPF area that receives no external routes, using a default route to reach them. (Routing)

Stub site A site with a single path or exit to the rest of the network, where a default or static route is sufficient. (Routing)

Success measure The metric a stakeholder uses to judge the project, such as cost per site, mean time to repair or audit findings. (Requirements)

Super-spine An additional tier that interconnects multiple spine-leaf pods to scale beyond one spine's port count. (Campus)

Superseded decision A decision record replaced by a newer one; kept for history and linked to the decision that replaced it. (Requirements)

Survivable ceiling The maximum utilisation per member that still lets the survivors carry the load after the tolerated failures. (Resilience)

Synchronous replication Replication in which a write completes only after the remote copy acknowledges it, so distance adds directly to write latency. (Campus)

Synthetic test An active probe that measures reachability and performance along a path. (Cloud)

T

Technical requirement A measurable property of the network derived from a business requirement, such as an availability target or a delay budget. (Requirements)

Thrashing Repeatedly redesigning in response to each change so that no decision stabilises. (Scenario)

Three-legged firewall A single firewall (or pair) with separate interfaces for outside, DMZ and inside zones. (WAN)

TI-LFA Topology-independent loop-free alternate: a segment-routing technique that provides precomputed backup paths along the post-convergence path. (Resilience)

TLV Type-length-value encoding that lets a protocol carry new kinds of information without changing its structure. (Routing)

Tokenisation Replacing sensitive data with a surrogate value that has no exploitable meaning outside a secure vault. (Secure)

Total cost of ownership All costs over the life of a solution — capital, operating, support, staff and exit — not just the purchase price. (Requirements)

Totally stubby area An OSPF area that receives only intra-area routes and a default route. (Routing)

Traceability The ability to link each requirement to design decisions and each decision back to requirements. (Scenario)

Trade-off analysis Comparing options by what each gains and gives up across cost, complexity and risk, rather than by features alone. (Requirements)

Traffic tromboning Inefficient traffic that crosses the DCI and back because a workload and its gateway, service or storage are at different sites. (Campus)

Transit Carrying traffic between two other networks; an enterprise should not provide it between its providers. (WAN)

Transit hub A provider-managed routing service that connects many virtual networks and hybrid links transitively. (Cloud)

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Translated boundary A defined interconnection point where overlapping addresses are translated, keeping overlap contained. (Scenario)

Trust boundary The point at which the network starts trusting QoS markings, usually the access port or the endpoint. (Campus)

U

Underlay The physical routed network that provides reachability between fabric nodes. (Campus)

Uplink A link from a lower layer to a higher one, such as access to distribution, whose capacity sets the oversubscription ratio. (Campus)

V

Virtual network A provider-hosted, isolated network with its own address space, subnets and route tables. (Cloud)

Volumetric DDoS An attack that aims to exhaust link or network capacity with sheer traffic volume. (WAN)

VXLAN A MAC-in-UDP encapsulation that carries layer 2 frames across a routed network, identifying segments with a 24-bit network identifier. (Campus)

W

Warm standby A DR pattern with a fully built but scaled-down secondary environment ready to take load. (Resilience)

Web application firewall A filter that inspects traffic to published web applications for attacks such as injection and automated abuse. (WAN)

Weighted decision matrix A table that scores options against criteria multiplied by agreed weights to support a transparent choice. (Requirements)

Witness / quorum A third-party vote that decides which site remains active when the sites lose contact. (Resilience)

Z

Zero trust A security model that grants access per session based on identity, device state and context, with no implicit trust from network location. (Secure)

Zero-touch provisioning Automatic onboarding in which a new edge device obtains its configuration from the orchestrator when first connected. (WAN)

Zero-trust network access Brokered access that connects verified users and devices to specific applications rather than to a network. (Secure)

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CAPSTONE

Capstone: Full design package for a national-scale organisation

Produce a complete, defended network design for a fictional organisation (for example a national university system or a mobile financial service), covering every blueprint sheet, with a phased migration plan and an executive presentation.

Deliverables

☐ Requirements register: business, technical, constraints, assumptions

☐ Weighted decision matrices for three major choices

☐ Campus and data-centre design with scaling limits

☐ WAN/SD-WAN and internet-edge design

☐ Routing design (IGP and BGP) with convergence goals

☐ Resilience design with availability calculations and RPO/RTO

☐ Security and segmentation design

☐ Hybrid/multi-cloud connectivity and automation-driven operations

☐ Phased migration plan with rollback and risk register

☐ Ten-slide executive design presentation

Self-assessment rubric

CRITERION	DEVELOPING	PROFICIENT	EXCELLENT
Requirements	○ Generic	○ Captured	○ Prioritised, traceable to decisions
Decisions	○ Asserted	○ Justified	○ Weighed against alternatives and constraints
Campus/DC	○ Template copied	○ Fits needs	○ Scales with known limits
WAN & routing	○ Single option	○ Sound design	○ Optimised for cost, risk and convergence
Resilience	○ Assumed	○ Redundant	○ Quantified availability and DR
Security	○ Afterthought	○ Segmented	○ Zero-trust aligned and compliant
Cloud & ops	○ Ignored	○ Connected	○ Automation-ready, observable, cost-aware
Migration & communication	○ Big bang	○ Phased plan	○ Low-risk plan, clear executive story

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Planning pages

Sketch your design, timeline or architecture here.

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EXAM PREPARATION

Practice exam · 60 questions

The official online exam draws 45 questions at random from a separate, secret bank that tests the same skills, with a 90-minute limit and a 85% pass mark. Practise with these questions first: time yourself, mark your answers, then check the key and read every explanation, including for the questions you got right.

Requirements — Design Methodology

1. A logistics firm wants a new WAN. The sponsor states the budget is fixed, the operations lead says the team cannot support new protocols this year, and the CIO "would prefer" to keep the current router vendor. How should the architect treat these three inputs?

   (A) Budget and team capability as constraints; vendor as a weighted preference

   (B) All three as hard constraints that remove options from further consideration

   (C) All three as weighted criteria in a single decision matrix

   (D) Budget as a constraint; team capability and vendor as assumptions

2. During discovery for a hospital network refresh, interface data shows 22% annual traffic growth, but the imaging department plans to move all scans to a cloud archive next year. What is the BEST basis for capacity planning?

   (A) The measured 22% trend alone, because historic data is objective evidence

   (B) Measured trend plus a separately modelled, validated imaging uplift

   (C) A blanket doubling of capacity to absorb any unknown future change

   (D) Waiting until the imaging move is live, then measuring the new demand

3. A weighted decision matrix ranks option A first by 0.05 points. A sensitivity check shows that moving 10 points of weight from cost to operability makes option B the winner. What should the architect do?

   (A) Recommend option A because it scored highest under the agreed weights

   (B) Recommend option B because operability is usually underweighted

   (C) Discard the matrix and select the option the team is most familiar with

   (D) Present both, showing the choice hinges on cost versus operability

4. A stakeholder asks for "a stretched VLAN between both data centres" during requirements gathering. What should be recorded in the requirements register?

   (A) A technical requirement for a stretched VLAN, owned by that stakeholder

   (B) A constraint that layer 2 must be available at both data centres

   (C) The confirmed underlying need, such as keeping addresses during migration

   (D) Nothing yet, because proposed solutions are discussed only after the design work starts

5. A scenario describes a retailer with a nine-person network team, a mandated go-live in four months, and a security requirement for segmentation of payment systems. Which option should generally be eliminated FIRST when comparing designs?

   (A) One that meets the segmentation requirement using familiar technology at moderate cost

   (B) One that uses firewall policy between VRFs instead of an overlay fabric

   (C) One needing skills and tooling the team cannot obtain within four months

   (D) One that delivers payment segmentation first and other changes later

6. Which content BEST belongs in the high-level design rather than the low-level design?

   (A) The VLAN-to-subnet table and default gateway addresses for every access switch in the campus

   (B) OSPF hello and dead timer values for each link type in the campus

   (C) The step-by-step cut-over procedure for migrating each floor

   (D) The reasons routed access was chosen over layer 2 access and its consequences

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7. A university requires that research transfers avoid the central stateful firewall, while the security office requires all external traffic to be inspected. What is the architect's BEST action?

   (A) Route research traffic through the firewall because security outranks research

   (B) Document the conflict and options with risks; let the owners decide

   (C) Bypass the firewall for research because the throughput need is quantified

   (D) Buy a larger firewall so both requirements appear to be met without discussion

8. A design decision record states: "Chose a collapsed core for the headquarters; revisit if a second building is approved." What does the second clause provide?

   (A) A review trigger that tells future designers when to reopen the decision

   (B) A constraint that prevents the company from approving or building a second building

   (C) An assumption that a second building will definitely never be approved

   (D) A consequence describing the cost of the collapsed-core decision

Campus — Campus & Data Centre Design

9. A single-building office has 9 access stacks, a three-person team and no growth plans. A proposal adds a dedicated core pair above two distribution switches. Which design BEST fits?

   (A) The proposed three-tier design so the campus is ready for any future growth

   (B) A collapsed core, with the access stacks uplinked to one switch pair

   (C) A full mesh of links between all access stacks with no aggregation layer

   (D) An overlay campus fabric with distributed anycast gateways on every stack

10. A data hall can supply 600 kW to IT equipment. The proposed GPU training cluster needs about 900 kW at full build, and the sponsor wants the full cluster this year. What should the architect do FIRST?

    (A) Raise oversubscription in the GPU fabric so that fewer switches draw power

    (B) Swap every 400G optic for a 100G optic to cut the power the fabric draws

    (C) Record power as a hard constraint and agree a phased or smaller build

    (D) Proceed with the full build and assume facilities will add supply later

11. A leaf has 48 × 25 Gbps server ports. The requirement is no worse than 3:1 oversubscription even after one spine fails. Which uplink design is the MINIMUM that meets it?

    (A) 4 × 100 Gbps uplinks to 4 spines

    (B) 3 × 100 Gbps uplinks to 3 spines

    (C) 8 × 50 Gbps uplinks to 8 spines

    (D) 5 × 100 Gbps uplinks to 5 spines

12. A two-tier spine-leaf fabric uses 64-port spines with one link to each leaf. The business needs 90 leaves within two years. What is the BEST design response?

    (A) Plan a super-spine tier or higher-radix spines before leaf count reaches 64

    (B) Connect the extra leaves to existing leaves using spare server ports

    (C) Add more spines to the fabric, since the number of spines limits how many leaves can connect

    (D) Use two links from each leaf to each spine to increase capacity

13. Two data centres 40 km apart host 300 VLANs. Only 10 VLANs carry clustered applications that require the same subnet at both sites. Which DCI design BEST fits?

    (A) Stretch all 300 VLANs so any workload can move between the sites

    (B) Pure routed DCI, requiring all ten clustered applications to be redesigned before migration

    (C) Native layer 2 trunks for the 10 VLANs with a single shared gateway

    (D) Routed DCI plus an overlay extending only the 10 segments, with local gateways

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14. An application owner states that synchronous writes can tolerate at most 1 ms of added round-trip latency. The fibre route between candidate sites is 140 km. What is the BEST conclusion?

(A) The sites are too far apart for synchronous replication within the budget

(B) The sites are suitable, because 140 km adds only about 0.7 ms round trip

(C) The sites are suitable if the DCI uses dark fibre instead of a carrier

(D) The budget is irrelevant because synchronous replication is asynchronous at the network layer

15. A stock exchange must deliver market data with no gaps if any single network device fails. Which design BEST meets this requirement?

(A) One network with fast-reroute protection pre-computed on every multicast link and router

(B) Two independent distribution networks carrying duplicate A and B feeds

(C) One spine-leaf fabric with eight spines and BFD on all links

(D) One network with redundant supervisors in every switch chassis

16. A large campus needs consistent group-based policy for 30,000 users and devices that roam across 40 buildings. The team has strong automation skills. Which design direction BEST fits?

(A) A collapsed core in each building with per-building VLANs and ACLs

(B) Layer 2 access with VLANs spanning every building on the campus

(C) An overlay campus fabric with anycast gateways and group-based policy

(D) Routed access with per-subnet ACLs maintained manually on each switch

WAN — WAN, Internet Edge & Provider Design

17. A bank's 150 branches must keep ATM service through any single failure, and the regulator requires encryption over any network the bank does not control. Budget favours internet transports. Which branch design BEST fits standard branches?

(A) Two broadband circuits from different providers with no encryption

(B) An encrypted overlay over broadband plus cellular, with physical path diversity

(C) A single MPLS circuit with a cellular backup used only for ATMs

(D) Two broadband circuits from the same provider, both carried under an encrypted overlay

18. A company with 250 sites in a hub-and-spoke overlay is adding branch-to-branch video for about 20 concurrent sessions. Which change BEST meets the need?

(A) Enable dynamic, on-demand tunnels between branches for real-time traffic

(B) Build a permanent full mesh of tunnels across all transports between all sites

(C) Upgrade both hubs to carry the additional video traffic through them

(D) Send branch-to-branch video over cellular so it avoids the hub entirely

19. An e-commerce company's 10 Gbps internet links were saturated by a 60 Gbps attack. It has two ISPs and a public web storefront. Which mitigation design BEST fits?

(A) A larger on-premises DDoS appliance placed behind the edge routers

(B) Upgrading both internet links to 40 Gbps to absorb future attacks

(C) Upstream cloud scrubbing or an always-on proxy in front of the storefront

(D) Remotely triggered blackholing of the storefront's address with both ISPs during attacks

20. A multihomed enterprise wants most inbound traffic to arrive via ISP-A, using ISP-B as backup. ISP-A and ISP-B both prefer customer routes with local preference. Which approach is MOST likely to achieve this?

(A) Prepend the enterprise AS several times on every advertisement sent to the backup ISP-B

(B) Send a higher MED to ISP-B than to ISP-A on all advertisements

(C) Set a higher local preference on routes received from ISP-A

(D) Use ISP-B's customer communities to lower its local preference for the routes

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21. An enterprise edge router peers with two ISPs and redistributes all BGP-learned routes into BGP advertisements towards both. Which risk is MOST serious?

(A) Inbound traffic will always prefer the ISP with the shorter AS path

(B) The router will learn a default route from each ISP at once

(C) The ISPs will reject the enterprise's own prefixes as too specific

(D) The enterprise can become transit between the two ISPs

22. A mining site has no fixed-line service. Requirements: good voice quality and no single failure isolating the site. Cellular coverage is reliable. Which design BEST fits?

(A) GEO satellite primary with a second GEO terminal from the same provider

(B) Two LEO terminals from the same provider for full redundancy

(C) LEO satellite primary with 4G/5G as an independent secondary path

(D) 4G/5G primary with no secondary because coverage is reliable

23. Branch users need low-latency SaaS access, and regulation requires all internet access to be logged centrally. The security team is small. Which design BEST fits?

(A) Local breakout through a cloud security service that logs centrally

(B) Backhaul all internet traffic to the data centre over the private WAN

(C) Local breakout with a separately managed firewall at every branch

(D) Local breakout without inspection, relying on the SaaS provider's controls

Routing — Routing Protocol Design

24. A national retailer runs one OSPF area covering its core and 600 flapping broadband branches, each dual-homed to one of four regional hubs. Core CPU spikes during SPF runs. Which design BEST fixes the root cause?

(A) Longer SPF throttle timers on all core routers to reduce recalculations

(B) Regional totally stubby areas with hub ABRs that summarise to the backbone

(C) A separate OSPF area for each of the 600 branches attached to area 0

(D) Static default routes on all branches, removing OSPF from the WAN edge

25. An OSPF network of 300 routers meets all current requirements. The team is expert in OSPF; a vendor recommends IS-IS. There are no IPv6 or segment-routing plans. What should the design record state?

(A) Migrate to IS-IS because it is preferred by large service providers

(B) Run IS-IS in parallel with OSPF to prepare for an eventual migration

(C) Keep OSPF, with a review trigger tied to new IPv6 or traffic-engineering needs

(D) Migrate to BGP everywhere because it is the most scalable protocol

26. Two routing domains are joined at two routers with mutual redistribution for resilience. After failures, some routes loop between the domains. Which design change BEST prevents this while keeping both links?

(A) Tag routes at redistribution and deny the other domain's tag on the way back

(B) Remove redistribution from one router and accept the resulting single point of failure

(C) Set the maximum metric on every redistributed route in both directions

(D) Redistribute only connected routes between the two domains at both points

27. Core routers advertise a summary for a region. When one regional subnet's only router fails, traffic for that subnet is dropped at the core. What explains this behaviour?

(A) The core withdraws the summary as soon as any component route fails

(B) The summary still attracts the traffic, and the discard route drops it

(C) The failed subnet is still being advertised as a specific route

(D) OSPF discards summaries whenever a single component is unreachable

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28. An enterprise runs iBGP on 80 routers in a full mesh to carry partial internet and VPN routes. Adding routers is slow and error-prone. Which design BEST fits?

(A) A single route reflector, which halves the number of client sessions compared with using two
(B) Redistribution of all BGP routes into the IGP to remove iBGP altogether
(C) A confederation with one sub-AS per router to keep the full-mesh behaviour
(D) Two redundant route reflectors in separate failure domains, all routers as clients

29. Clients of a route reflector have two internet exits, but after an exit fails they lose traffic for several seconds while the reflector sends a new path. What BEST addresses this?

(A) Move the reflector into the forwarding path of the core routers
(B) Advertise additional paths so clients already know a backup
(C) Increase the BGP keepalive and hold timers on all client sessions
(D) Replace the two internet exits with a single higher-capacity exit

30. An application requires recovery from any single link failure within 100 ms. Which combination BEST meets this convergence budget?

(A) Default protocol hello timers with an aggressive SPF throttle setting
(B) BFD for detection with a full SPF and FIB rebuild on every failure
(C) BFD for fast detection plus pre-computed loop-free backup paths
(D) Longer SPF throttle timers so that routers batch topology changes

Resilience — High Availability & Resilience Design

31. A stock exchange's market-data service must reach 99.99%. The current path is a single firewall cluster (99.95%) in series with redundant routers and diverse carriers. The team proposes a third carrier. What should the designer recommend instead?

(A) Accept the third carrier, as adding more parallel paths always raises end-to-end availability
(B) Upgrade the routers to higher-MTBF models to lift the overall chain
(C) Lower the target to 99.95% so that the current design is compliant
(D) Address the firewall cluster, the serial element capping the chain near 99.95%

32. Two leaf switches in a data-centre pod each carry 55% of their capacity in an active/active pair. The application owner asks whether the pod can survive a leaf failure without degradation. What is the best answer?

(A) Yes — active/active pairs always fail over without degradation
(B) Yes — the second leaf will throttle background traffic at once
(C) No — at 55% each, the surviving leaf would be overloaded
(D) No — active/active pairs cannot fail over without a manual step

33. A real-time control application tolerates 200 ms of loss. Core links are point-to-point dark fibre with no intermediate devices. Convergence today is 1.5 s, dominated by SPF and FIB programming. What should be prioritised?

(A) Precomputed loop-free alternates, so traffic switches before SPF and FIB finish
(B) BFD on every core link to speed up detection of failures on the dark fibre
(C) Graceful restart on all core routers so that forwarding continues on the old state
(D) Longer SPF throttle timers so the control plane remains stable during failure

34. A service provider edge router is the only attachment point for 300 single-homed business customers. It has dual route processors. Which control-plane resilience feature fits this role best?

(A) Aggressive BFD timers so customers detect the failure within 150 ms
(B) Non-stop routing, so a processor switchover is invisible to customers
(C) Removing the standby route processor to simplify the software design
(D) Fast reroute on the customer links, using backup paths via other PEs

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35. A payment processor's two data centres are 400 km apart. The business asks for an RPO of zero and an RTO of 30 minutes for a site loss. Write latency budgets cannot absorb more than 2 ms. What design best meets the requirements?

(A) Run synchronous replication between the two existing sites 400 km apart

(B) Use asynchronous replication to the far site and document an RPO of zero

(C) A nearby metro site for synchronous replication, with the far site asynchronous

(D) Stretch a VLAN between the two existing sites so the database can fail over instantly

36. A university core pair will be upgraded. Each router runs at 35% utilisation. Policy requires that no planned work removes all redundancy for the core. Which plan best meets this?

(A) Upgrade both routers in one window with the supplier's engineer on the bridge

(B) Use in-service upgrade on both routers at once, since forwarding continues throughout

(C) Defer the upgrade indefinitely to avoid any reduction in redundancy at all

(D) Drain, upgrade and soak one router, then repeat for the other in a later window

37. A retailer's 1,200 stores depend on one central policy controller cluster for their WAN overlay. A bad policy push last month disconnected every store for an hour. What design change most reduces the blast radius of a repeat?

(A) Add a third node to the existing controller cluster to improve its redundancy

(B) Regional cells with separate controller instances and staged change waves

(C) Increase the controller's hardware capacity so pushes complete much faster

(D) Require two engineers to approve every policy change before it is pushed

38. An MTBF of 100,000 hours and an MTTR of 48 hours apply to the only access switch at 400 small branches. Budget allows either a second switch per branch or an on-site spare and remote-hands contract cutting MTTR to 4 hours. Branches tolerate a few hours' outage a year. What is the better choice?

(A) The spares contract, which cuts expected downtime about twelvefold at lower cost

(B) The second switch, because any redundancy always beats a faster repair time

(C) Neither, since a 48-hour MTTR already meets a requirement of a few hours

(D) The second switch, because the spares contract does not change availability

Secure — Security & Segmentation Design

39. A bank wants to stop ransomware moving from user laptops to core banking servers. Users and servers are in separate VRFs with a firewall between them, but servers in the same application tier can talk freely. What should the next design step target?

(A) A second perimeter firewall at the internet edge from a different supplier

(B) Moving all of the users into a single larger VRF with stricter rules

(C) Encrypting all traffic between the servers with IPsec in tunnel mode

(D) Micro-segmentation inside the server zones, built from observed flows

40. A university must give 40,000 students and staff access to internal applications from unmanaged personal devices. Today a remote-access VPN places them on the campus network. What design best reduces exposure?

(A) A larger VPN concentrator cluster with more licences and stronger ciphers

(B) A separate VPN pool per faculty, each with its own campus network address range

(C) Brokered, application-level access connecting verified users to named apps

(D) Blocking all remote access and requiring users to come to campus instead

41. Two data centres 20 km apart are linked by dark fibre carrying 400 Gbps. A new policy requires encryption of all inter-site traffic with minimal latency. What is the most suitable design?

(A) IPsec tunnels terminated on the existing pair of data-centre firewalls

(B) MACsec on the inter-site links of the interconnect switches or routers

(C) TLS added to every application, leaving the links themselves unchanged

(D) A carrier internet VPN service replacing the dark fibre entirely

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42. A merchant processes card payments at 300 stores. Auditors currently treat the whole corporate network as in scope. What design change most reduces audit scope?

(A) Tokenise card data at the terminal and isolate what remains in a tested zone

(B) Encrypt all store WAN links with IPsec so card data is protected in transit

(C) Install host-based antivirus on every corporate system to meet the standard's controls

(D) Add a second firewall at the data-centre perimeter facing the internet

43. A hospital is introducing 802.1X across 8,000 ports. Clinical engineering warns that some infusion pumps cannot authenticate and any interruption is unsafe. What rollout design is most appropriate?

(A) Monitor mode, then low-impact mode with profiled MAC access and tight policy for pumps

(B) Closed mode on every port at once, with a help-desk team and clinical engineers ready to respond

(C) Exempting all clinical floors from network access control permanently

(D) Dynamic VLANs for all devices, enforced on the first day of the rollout

44. An active/active pair of data centres each has its own stateful firewall cluster. Users report intermittent session resets. Captures show requests entering through Site A and replies leaving through Site B. What is the best design correction?

(A) Disable stateful inspection for the affected applications at both sites

(B) Increase bandwidth between the two sites so replies arrive much sooner

(C) Merge both firewall clusters into one stretched cluster across sites

(D) Make traffic symmetric per site, using per-site egress and source NAT

45. A security operations team of six must monitor 250 sites, three data centres and two cloud estates. What visibility design gives the best coverage for the team's capacity?

(A) Full packet capture on every link with 90 days of retention for every site

(B) Flows and key logs everywhere, plus brokered packet capture at choke points

(C) Firewall logs from the data-centre perimeter only, as the internal network is trusted

(D) Port mirroring from every access switch into one central analysis cluster

Cloud — Hybrid, Multi-Cloud & Automation-Driven Design

46. An enterprise has 60 virtual networks in one cloud region, plus two on-premises data centres. Most traffic flows between a few shared-services networks and the rest. What topology best balances manageability and cost?

(A) Full-mesh peering between all 60 virtual networks plus both data centres

(B) A separate VPN from each virtual network to each on-premises data centre

(C) A transit hub, with direct peering only for the few highest-volume pairs

(D) A single large virtual network holding every application team's workloads

47. A manufacturer needs 4 Gbps steady between its plant and a cloud region, a 99.95% target and predictable latency. Its only colocation presence is one facility. What is the best connectivity design?

(A) Two private connections in the existing single facility, terminating on separate routers

(B) Four internet VPN tunnels from the plant with multipath load balancing

(C) One private connection in the existing facility with no backup path

(D) Private connections at two diverse facilities, plus a VPN for critical flows

48. After moving a database service behind a private endpoint, cloud applications work but on-premises reporting tools connect to the public address and are refused. What is the correct design fix?

(A) Conditional forwarding of the endpoint zone to the cloud inbound resolver

(B) Allow the on-premises public egress addresses on the database firewall

(C) Publish the private endpoint address in the organisation's public DNS zone

(D) Forward all on-premises DNS queries to the cloud provider's resolvers

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49. A container platform will run 300 nodes with up to 110 pods each, and every pod needs an address. The enterprise has little routable IPv4 space left. What addressing approach is most appropriate?

(A) Allocate a routable /15 to the pods so every one is reachable directly

(B) Pod overlay or IPv6, keeping routable IPv4 for nodes and services that need it

(C) Reuse a range already used on premises, since cloud is separate anyway

(D) Limit the platform to ten pods per node to fit the remaining space

50. A team runs a centralised inspection hub that every spoke-to-spoke and internet flow traverses. Monthly hub processing charges now exceed the cost of the workloads in some spokes. What should the designer do?

(A) Remove the inspection hub entirely and rely only on security groups

(B) Move every workload into a single spoke to avoid the hub altogether

(C) Apply inspection per flow as policy requires; route other flows directly

(D) Double the size of the inspection appliances to reduce the processing charges

51. A ministry with a small network team wants to start automating its 250 offices, which are all configured slightly differently. What should be the first deliverable?

(A) An orchestration platform that imports today's configurations as templates

(B) Scripts that push urgent changes to all 250 offices at the same time

(C) A full hardware refresh so that every office uses the same platform

(D) Standard office types, a data model and read-only compliance checks

52. A company is considering a second public cloud provider. The stated reason is 'resilience' for an application that already runs in one region. What is the best initial recommendation?

(A) Adopt the second provider at once, since two providers cannot fail together

(B) Run the application in both providers with a shared stretched virtual network

(C) Evaluate multi-region in the current provider and test cross-region failover

(D) Keep one region but buy the provider's highest support tier for the application

53. A cloud portal makes 40 sequential queries per user request to an on-premises database across a hybrid link with a 12 ms round trip. Users complain of slow pages. What design change helps most?

(A) Upgrade the private connection from 1 Gbps to 10 Gbps of dedicated capacity

(B) A cache or API layer near the portal so far fewer queries cross the link

(C) Add a second private connection to share the query load evenly

(D) Enable compression on all traffic between the portal and database

Scenario — Design Scenarios, Migration & Capstone

54. Halfway through a design, the sponsor reports that a regulator now requires all customer data to stay in-country. Earlier decisions chose a cloud region abroad for analytics only. What should the designer do FIRST?

(A) Trace which decisions depend on data location and assess the impact on each

(B) Restart the whole design, as the regulatory change invalidates the approach

(C) Continue as planned, since analytics is a separate workload from the core

(D) Move every workload, including non-customer systems, back on premises

55. A core migration window runs from 01:00 to 05:00. The rehearsed rollback takes 45 minutes and the agreed safety margin is 15 minutes. At 03:55 validation is incomplete but the engineers are confident. What should the change lead do?

(A) Continue until 04:45, since the engineers are confident about finding the fix

(B) Roll back now, as the 04:00 go/no-go time is reached without success

(C) Commit the change and fix any remaining issues during business hours

(D) Ask the service owner to extend the window until about 07:00 instead

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56. Two operators merge. 700 enterprise customers have BGP sessions configured to the acquired operator's AS. The target is a single AS within a year. What approach avoids coordinating every customer change at once?

(A) Move routers to the new AS while presenting the legacy AS via local-AS features

(B) Ask all 700 customers to change their BGP configuration on one agreed night together

(C) Keep both AS numbers permanently and run two separate core networks

(D) Renumber the customers' own AS numbers to match the new operator's AS

57. A CFO asks why a second data centre costing 200,000 a year is needed. An outage costs 150,000 per hour; a site loss would take 12 hours to recover today and 30 minutes with the second site; such an event is expected once every four years. What is the most persuasive answer?

(A) Every major competitor already operates two data centres for this service

(B) The current data centre is old and will need replacing within ten years

(C) Expected annual loss falls from 450,000 to about 19,000, exceeding the 200,000 cost

(D) Industry guidance states that critical services should have two sites

58. A national retailer must migrate 900 stores to a new WAN design. The old and new designs can interoperate through the data-centre hubs. Stores tolerate short overnight outages. Which strategy fits best?

(A) Big bang across all 900 stores during a single weekend window

(B) Parallel run with two complete networks at every store indefinitely

(C) Wait until every store is refurbished and migrate during the refits

(D) Phased waves of stores after a pilot, using the hubs as the coexistence point

59. In a staged scenario, stage 3 adds that the operations team will shrink from ten to six engineers next year. The design so far includes three different overlay technologies. What is the best response?

(A) Reduce technology diversity to suit the smaller team; keep other decisions

(B) Keep the design unchanged, as staffing levels are an operations concern only

(C) Add a new monitoring platform so that six engineers can watch all three overlays

(D) Outsource the entire network so team size no longer affects the design

60. A design review finds that the proposed network includes 99.999% availability for an internal reporting service whose requirement is 99.9%. What is the most appropriate review finding?

(A) Commendable: higher availability is always better value for the business overall

(B) Gold-plating: the extra cost and complexity serve no stated requirement

(C) Neutral: availability has no effect on cost or complexity once designed

(D) Critical: 99.999% is not technically achievable for a reporting service

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EXAM PREPARATION

Practice exam answers & explanations

1. A A fixed budget and a team that cannot support new protocols are hard limits. "Would prefer" signals a preference that should be weighted, not used to exclude options. Treating everything as constraints over-restricts; treating everything as criteria lets a non-viable option win; skills are a known fact, not an assumption.

2. B Data anchors the baseline, while a known future driver must be modelled explicitly and confirmed with its owner. Ignoring it, guessing a blanket figure, or waiting all leave the design without a defensible sizing basis.

3. D A fragile result means the decision hinges on a business judgement. The architect should make the dependency visible to the owners. Picking either option silently, or abandoning the analysis, hides the trade-off.

4. C A stretched VLAN is a proposed solution. The register should capture the confirmed need it serves, so the design can consider alternatives. Recording the solution as a requirement or constraint locks in risk; ignoring the request loses information.

5. C Options that break a constraint — here time and skills — are eliminated before scoring. The other options are viable approaches that would then be compared on their merits.

6. D The HLD records architecture and rationale for sponsors, architects and operations leads. Exact values and procedures are implementation detail for the LLD.

7. B Conflicting requirements belong to their owners. The architect informs the decision with options such as a research path protected by filtering and monitoring. Arbitrary precedence or silently spending budget removes the business from a business decision.

8. A The clause names the condition under which the decision should be revisited. It does not restrict the business, assert an assumption, or describe a cost.

9. B With one building, few blocks and no growth, a collapsed core meets the needs at the lowest cost and complexity. A separate core, mesh or fabric adds devices or complexity without a requirement.

10. C Power and cooling cap what the hall can host, so the gap is a constraint the owners must resolve — by phasing, resizing or funding new supply. Oversubscribing or slowing the fabric damages the workload while saving little, and assuming new supply hides a delivery risk.

11. D Downstream is 1,200 Gbps. With 5 × 100 Gbps, one spine failure leaves 400 Gbps, giving exactly 3:1. With 4 uplinks a failure leaves 300 Gbps (4:1); with 3, 200 Gbps (6:1); with 8 × 50 Gbps, 400 Gbps normally (3:1) but 350 Gbps after a failure (about 3.4:1).

12. A Spine port count caps leaf count in a two-tier fabric. More spines add bandwidth, not leaf capacity; daisy-chaining leaves breaks the Clos design; double links halve the possible leaf count.

13. D This meets the validated need while keeping most of the estate in separate failure domains. Stretching everything merges the sites; routed-only ignores a real requirement; plain trunks with one gateway keep storm, split-brain and tromboning risks.

14. A At about 5 µs per km one way, 140 km is 0.7 ms one way and 1.4 ms round trip from propagation alone, exceeding the 1 ms budget. Dark fibre removes some equipment delay but not propagation.

15. B Duplicating the feed over independent networks lets receivers arbitrate without any loss. Fast reroute and redundancy within one network shorten outages but cannot guarantee zero loss during failover.

16. C Policy scale and mobility across many buildings, plus the skills to operate it, justify a fabric. Spanning VLANs create huge failure domains, and subnet-based ACLs are hard to keep consistent at this scale.

17. B Two transports with independent failure causes meet the single-failure requirement, and the overlay provides encryption. Unencrypted public transport breaks the regulation; a single primary or same-provider pair leaves shared failure points.

18. A Dynamic tunnels give direct low-latency paths only where needed. A full mesh creates tens of thousands of idle tunnels; hub upgrades do not remove hairpin delay; cellular-by-default adds variability and cost.

19. C Volumetric attacks must be absorbed upstream, before they reach the links. On-site appliances cannot help once links are full, bigger links are still smaller than the attack, and blackholing takes the storefront offline.

20. D Provider communities change preference inside ISP-B, where prepending alone is overridden by its customer-route preference. MED is not compared across different ASes, and local preference on received routes affects only outbound traffic.

21. D Re-advertising one provider's routes to the other invites transit traffic that can saturate the enterprise's links. An outbound filter permitting only the enterprise's own prefixes prevents it.

22. C LEO latency suits voice and cellular is a physically different path. GEO delay hurts voice, same-provider terminals share failure domains, and a single path breaks the no-isolation requirement.

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23. A A cloud security service gives short SaaS paths, central logging and a single policy point suitable for a small team. Backhaul adds latency; per-branch firewalls multiply management; no inspection fails the logging requirement.

24. B Regional areas contain branch churn and give the core stable summaries, while hubs provide the default exit. Longer timers slow all convergence; 600 areas create sprawl; static defaults lose dual-hub failover.

25. C No unmet requirement justifies the risk of migration. A review trigger ensures the decision is revisited if requirements change. Parallel IGPs and BGP-everywhere add complexity without benefit.

26. A Tagging stops routes re-entering their source domain, the cause of the loop, while preserving both points. Removing one point loses resilience; metrics do not prevent feedback; connected-only breaks reachability.

27. B The summary remains advertised while any component exists, so traffic for the failed subnet still arrives and the null route drops it — the intended trade-off that prevents loops.

28. D Redundant reflectors remove the full-mesh burden without creating a single point of failure. A single reflector is fragile; redistributing large route sets into the IGP harms it; one sub-AS per router is unmanageable.

29. B The reflector hides the second path, so clients wait for a new one. Advertising multiple paths lets them switch immediately. Moving the reflector does not change what it advertises, longer timers slow detection, and one exit removes resilience.

30. C Sub-second detection with locally pre-installed backups avoids waiting for propagation and computation. Default hellos detect in seconds; full recomputation may exceed the budget; longer throttles slow convergence.

31. D In a serial chain the weakest element dominates. With carriers and routers already redundant, a third carrier adds almost nothing while the single firewall cluster caps the result. Lowering the target avoids the design problem rather than solving it.

32. C With two members and one tolerated failure, each must stay under (2 − 1) ÷ 2 = 50%. At 55% each, the survivor would need 110% capacity. Throttling is not a design guarantee, and active/active failover is normally automatic.

33. A Detection on dark fibre is already fast through loss of signal, so BFD adds little. The bottleneck is computation and FIB update, which precomputed backups bypass. Graceful restart does not help link failures, and longer timers make things slower.

34. B Single-homed customers have no alternative path, so preserving sessions and forwarding through a processor switchover is the goal. Fast detection only makes customers notice faster, and there are no alternative PEs for single-homed links.

35. C 400 km of fibre adds at least about 4 ms of round trip, breaking the 2 ms budget. A metro site gives zero data loss for site events, while the far site protects against regional disasters with an honest non-zero RPO. A stretched VLAN does not change physics and joins the failure domains.

36. D A rolling, drained upgrade leaves one known-good router carrying 70% of a single router's capacity, within limits. Touching both at once creates a shared operational failure domain, and indefinite deferral leaves known vulnerabilities in place.

37. B The failure was common-mode: one change reached everything. Cells with staged rollout cap any single push to a fraction of stores. More nodes or capacity do not stop a bad policy replicating, and dual approval helps but does not contain impact.

38. A Availability rises from 100,000 ÷ 100,048 (about 4.2 hours of downtime a year) to 100,000 ÷ 100,004 (about 21 minutes). That meets the tolerance at lower cost than duplicating 400 switches. MTTR is a design lever, not just an operational detail.

39. D Macro separation already exists between users and servers; the remaining lateral path is inside server zones. Micro-segmentation built from observed flows closes it. Perimeter changes and encryption do not stop east-west movement.

40. C Network-level VPN access grants broad reach from unmanaged devices. Application-level brokering limits each user to specific applications and hides the rest. More capacity or faculty pools do not change the exposure model.

41. B MACsec provides line-rate, low-latency hardware encryption on point-to-point links. Firewall IPsec would bottleneck 400 Gbps, application TLS cannot be guaranteed for all traffic, and replacing fibre with an internet service reduces performance.

42. A Scope follows the data and anything that can reach it. Tokenisation removes card data from most systems, and a tested isolated zone keeps the rest out of scope. Encryption, antivirus and perimeter firewalls do not change which systems are connected to card data.

43. A Monitor mode reveals what would fail without impact. Low-impact mode with profiled MAC-based access and tight policy brings pumps under control safely. Closed mode risks patient safety, and permanent exemptions leave the riskiest zone uncontrolled.

44. D Stateful devices must see both directions. Symmetric routing fixes the root cause. Disabling inspection removes the control, bandwidth is irrelevant, and a stretched cluster enlarges the failure domain to both sites.

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45. B Flows and logs scale cheaply across sites and cloud; packet capture concentrated at choke points provides depth where it matters. Capturing everything is unaffordable and unusable for six people, and perimeter-only logging ignores lateral movement.

46. C A hub provides transitive routing with few attachments; selective peering for heavy pairs avoids hub processing charges where volume is high. A 60-network mesh means 1,770 peerings, per-network VPNs multiply tunnels, and one giant network enlarges the blast radius.

47. D A single facility is a shared failure domain. Diverse facilities provide the availability, private capacity provides predictable latency for 4 Gbps, and a VPN covers a double failure for critical traffic. Internet VPNs cannot guarantee predictability at this rate.

48. A The resolver path is wrong, not the routing. Conditional forwarding sends only the relevant zone to the cloud resolvers, which return the private address. Reopening public access defeats the purpose, publishing private addresses leaks internal data, and forwarding everything creates a broad dependency.

49. B About 33,000 or more addresses would exhaust scarce routable space. Overlay or IPv6 pod addressing conserves routable IPv4. Reusing an on-premises range creates overlap, and limiting pods cripples the platform.

50. C Inspection should be applied where policy requires it, not by default to everything. Removing the hub drops necessary controls, a single spoke loses isolation, and bigger appliances do not reduce per-GB processing fees.

51. D Automation needs a defined 'correct' state and data to render from. Read-only compliance checks deliver early value with no risk. Importing inconsistency or scripting against it amplifies problems, and refresh is not a prerequisite.

52. C Multi-region within one provider usually delivers more resilience for less complexity. Multi-cloud doubles skills and tooling and is rarely tested well. Stretched networks across providers are impractical, and support tiers do not change architecture.

53. B Forty sequential queries at 12 ms each add nearly half a second of waiting. The problem is round trips, not bandwidth. Reducing crossings fixes it; more capacity, a second link or compression do not shorten the round trip.

54. A Tracing impact focuses change where it is needed — here, analytics placement and any flows carrying customer data. Restarting is thrashing, ignoring it violates a hard constraint, and moving everything overreacts.

55. B 05:00 − 45 − 15 = 04:00. With criteria unmet at the decision time, the agreed plan is to roll back. Continuing, committing with failures or extending under pressure all trade a controlled change for an uncontrolled outage risk.

56. A Local-AS techniques let the provider move first while customers keep their configuration, then migrate customers gradually. A single night for 700 changes is unmanageable, permanent dual ASes defeat the merger goal, and customers' own AS numbers are not the provider's to change.

57. C ALE before = 150,000 × 12 × 0.25 = 450,000; after = 150,000 × 0.5 × 0.25 = 18,750. The avoided loss of about 431,000 exceeds the 200,000 cost. This answers the CFO in financial terms; the other answers are weaker arguments.

58. D A large estate, feasible coexistence and tolerance for short windows point to phased waves after a pilot. Big bang concentrates risk, indefinite parallel running is expensive, and tying migration to refits delays benefits unpredictably.

59. A Operational capacity is a design constraint. Reducing technology diversity directly lowers operational load while other decisions stand. Ignoring it, adding tools or wholesale outsourcing do not address the core mismatch.

60. B Decisions must trace to requirements. Designing two nines beyond the need adds cost and complexity without business value. Higher availability is not free, and five nines is achievable but not warranted here.

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NOTES

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